

ISSN 2765-3811

VOL. 16, NO.1

<https://kiice.org/conference/icfice2025>

ICFICE 2025

THE 17TH INTERNATIONAL CONFERENCE
ON FUTURE INFORMATION &
COMMUNICATION ENGINEERING

1st-4th July, 2025

Guam Nikko Hotel, USA



Proceedings
QR Code



Ministry of Science and ICT



ISSN 2765-3811

VOL. 16, NO.1

<https://kiice.org/conference/icfice2025>

ICFICE 2025

THE 17TH INTERNATIONAL CONFERENCE
ON FUTURE INFORMATION &
COMMUNICATION ENGINEERING

1st-4th July, 2025

Guam Nikko Hotel, USA



Proceedings
QR Code



Welcome Message from General Chair

Dear Esteemed Researchers and Participants,

On behalf of the Organizing Committee, I am delighted to welcome you to ICFICE 2025. We are delighted to bring together professionals and researchers in the field of information technology to share innovative ideas, research results, and advanced technologies.

This conference serves as a platform to promote collaboration, exchange knowledge, and explore the latest trends shaping the IT landscape. We are confident that the diverse perspectives and expertise gathered here will contribute to the continued growth and development of our dynamic field.

We hope that this conference will be a rewarding experience, providing insightful discussions, networking opportunities, and inspiration for future research. Our association has worked hard over the past years to select notable papers from the ICFICE proceedings and publish them in SCI and SCOPUS-ranked journals, and we will continue this effort this year. Finally, I would like to express my sincere gratitude to the Program Committee members, dedicated paper reviewers, and all those who contributed to the papers. Thank you for joining us on this exciting journey. We wish you another year filled with scholarly achievements and shared accomplishments.

Best wishes,

Dr. and Prof. Daesung Lee
General Chair of ICFICE 2025

Conference Committee

General Chair

Daesung Lee (National Korea Maritime & Ocean University, Korea)

Program Chair

Daehwan Kim (University of Ulsan, Korea)

Publication Chairs

Philippe-Joseph Salazar (University of Cape Town, South Africa)

Gwanghyun Jo (Hanyang University, Korea)

Advising Chairs

Conan K. R. Ryu (Mokwon University, Korea)

Hoe-Kyung Jung (Paichai University, Korea)

Hyun Chang Lee (Wonkwang University, Korea)

Organizers

YongJoo Lee (Chungbuk National University, Korea)

Jong Tae Lee (Seoul Women's University, Korea)

Sung-Hwa Hong (Mokpo National Maritime University, Korea)

Joosang Youn (Dong-eui University, Korea)

Choong Ho Lee (Hanbat National University, Korea)

Oh-Young Kwon (Korea University of Technology and Education, Korea)

Jaemin Kwak (Mokpo National Maritime University, Korea)

Chang-Pyo Yoon (Gyeonggi University of Science and Technology, Korea)

Teuk Seob Song (Mokwon University, Korea)

Mintae Hwang (Changwon National University, Korea)

Jinwoo Kim (Electronics and Telecommunications Research Institute, Korea)

Seho Lee (University of Ulsan, Korea)

Daehan Ahn (University of Ulsan, Korea)

Sangook Moon (Mokwon University, Korea)

Jaemin Kwak (Mokpo National Maritime University, Korea)

Chang-Pyo Yoon (Gyeonggi University of Science and Technology, Korea)

Wooseok Ryu (Catholic University of Pusan, Korea)

Dohyun Kim (National Korea Maritime & Ocean University, Korea)

Steering Committee

Jeong Wook Seo (Hanshin University, Korea)

Se-woon Choe (Kumoh National Institute of Technology, Korea)

Hyun Chang Lee (Wonkwang University, Korea)

Liu Xiao Wen (Hebei Normal University for Nationalities, China)

Program Member

Akihiro Sugimoto (National Institute of Informatics, Tokyo, Japan)
Arun Anand (Maharaja Sayajirao University of Baroda, India)
Baek-Young Choi (University of Missouri, US)
Boyan Iliev Tatarov (Dongwoo Optron, Bulgaria)
Cao Yang (Eastern Liaoning University, China)
Chang, Chih-Yung (Tamkang University, Taiwan)
Changji Wang (Sun Yat-sen University, China)
Chen, Yuh-Shyan (National Taipei University, Taiwan)
Chia-Yen Chen (National University of Kaohsiung, Taiwan)
Deepti gaur (ITM University Gurgaon, India)
Du Junping (Beijing University of Posts and Telecommunications, China)
Eriko Watanabe (University of Electro-Communications, Japan)
Ewe Hong Tat (Universiti Tunku Abdul Rahman, Malaysia)
Fay Huang (Ilan University, Taiwan)
Fu Li-hua (Beijing University of Technology, China)
Genaro Saavedra (University of Valencia, Spain)
Hee-hyol Lee (Waseda University, Japan)
Hiroshi Yoshikawa (Nihon University, Japan)
Hong K. Sung (University of Central Oklahoma, USA)
Hong Seung Ko (The Kyoto College of Graduate Studies for Informatics, Japan)
Hui Chen (Jinan University, Shandong, China)
Jaesung Sim (Jason Sim) (Mansfield University of Pennsylvania, USA)
Jung-Min(Jerry) Park (Virginia Polytechnic Institute & State University, USA)
Kiseol Yang (University of North Texas, USA)
Long Quan (HKUST, Hong Kong)
M L Dennis Wong (Swinburne University of Technology, Malaysia)
Manuel Martinez-Corral (University of Valencia, Spain)
Mingui Sun (University of Pittsburgh, USA)
Mutsuhiro TERAUCHI (Hiroshima International University, Japan)
Natalya Korobova (National Research University of Electronic Technology, Russia, Federation)
Ngoc-Minh LE (HCMC University of Technology, HoChiMinh City, Vietnam)
Partha Bhowmick (Indian Institute of Technology, Kharagpur, India)
Rahman Ahad (University of Dhaka, Bangladesh)

Ramakrishna Kakarala (Singapore)
Sanghyuk Lee (Xi'an Jiaotong-Liverpool University, China)
Shi Xiaolong (Huazhong University of Science and Technology, Wuhan, China)
Tayfun Akgul (Istanbul Technical University, Turkey)
Yang Cao (Dep. of Information & Technology, Eastern Liaoning University, Dandong, China)
Yingmin Jia (Beihang University, China)
YU Wei Chuan (HKUST, Hong Kong)
Yujia Zhai (Xian Jiaotong Liverpool University, China)
Zezhong Xu (Changzhou Institute of Technology, Changzhou, China)
Zhongping Que (Taiyuan University of Technology, China)
Hyun Jun Park (Cheongju University, Korea)
Jong Wook Kwak (Yeungnam University, Korea)
Jongwook Woo (California State University College of Business & Economics, USA)
Ran Rong (Ajou University, Korea)
Sungyong Yoo (Chonnam National University, Korea)
Seung Eun Lee (Seoul National University of Science and Technology, Korea)
B.-S. Shin (Korea Polytechnic, Korea)
Kyoung Jae Lee (Hanbat National University, Korea)
Dae Hee Kim (Soonchunhyang University, Korea)
Hyun-Ho Choi (Hankyong National University, Korea)
Kisong Lee (Dongguk University, Korea)
Seunggyu Byeon (Silla University, Korea)
Cao Kerang (Shenyang University of Chemical Technology, China)
Sungkwan Youm (Wonkwang University, Korea)
Sukhoon Lee (Kunsan National University, Korea)
Jeongkyu Hong (Yeungnam University, Korea)
Changsu Kim (PaiChai University, Korea)
Young Bin Kim (Korea Polytechnic, Korea)
Jung-Hun Lee (Pusan National University, Korea)

Session AI-A

AI and Intelligent Information System

Session IB-A

IoT and Big Data

Systematic Analysis of Risk Factors in General Purpose Artificial Intelligence Centered on Alignment Principles

Yejin Shin* and Sangyeon Kang*

Telecommunications Technology Association (TTA), Seongnam 13591, South Korea
[Email : yepp1252@tta.or.kr, cellina7702@tta.or.kr]

Abstract

This study systematically analyzes the development of General-Purpose Artificial Intelligence (GPAI) and its associated potential risk factors, proposing alignment principles to ensure that GPAI systems align with human intentions and values. While most existing artificial intelligence (AI) alignment research has primarily focused on the field of Artificial Narrow Intelligence (ANI), this study focuses on GPAI, deriving and validating alignment principles specifically for GPAI systems. We propose three key alignment principles (collectively referred to as 3Ps). The three principles include primacy of humanity, persistence of goal, and preservation of value. By mapping these principles to 11 key risk factors, we validate their effectiveness under various scenarios. Overall, the results confirm that the proposed alignment principles are well-suited to the unique characteristics of GPAI systems, providing comprehensive and appropriate countermeasures to potential risk factors. The findings of this study are expected to contribute to ensuring the safety, ethics, and social integration of the GPAI.

Index Terms: Alignment, General Purpose AI, Artificial General Intelligence, AI Alignment, AI Risk.

I. INTRODUCTION

Advances in artificial intelligence (AI) can revolutionize the way in which humans live. General-Purpose AI (GPAI) is at the forefront of this transformation [1]. However, the advent of the GPAI could result in unintended consequences owing to its enhanced autonomy and capabilities. Consequently, research on the alignment of AI systems with human intentions and values has become increasingly important [2]. AI alignment research has explored various approaches for determining the principles through which AI systems should operate. Existing studies [3-8] have emphasized that AI systems should fulfill human needs and desires while maximizing human values and highlighting key objectives such as ethics. However, these studies have limitations, particularly in their ability to systematically analyze how AI alignment principles interact with various risk factors. Consequently, it is challenging to predict the risks of a misalignment. Moreover, numerous studies have focused on Artificial Narrow Intelligence (ANI) without adequately addressing the alignment issues related to GPAI, which will possess expanded capabilities and functions in the future.

To overcome the limitations of existing studies, this study proposes alignment principles for the GPAI and systematically analyzes related risk factors. Through this approach, we aim to identify the considerations necessary to ensure aligned operations of GPAI systems. The remainder of this paper is organized as follows. Chapter 2 presents an analysis of existing research on AI alignment with a focus on the various alignment principles and approaches that have been proposed. Chapter 3 presents a set of alignment principles for GPAI. Chapter 4 compares 11 key risk factors proposed in a previous study [9] with the alignment principles to ascertain the efficacy of the alignment principles in potential risk situations. Finally, Chapter 5 summarizes the findings and concludes this study.

II. Related Work: AI Alignment

Research on AI alignment has explored various approaches for ensuring that AI systems act in accordance with human intentions and values. This section outlines the development and application of alignment principles based on existing studies. First, two studies proposed a definition or goal for AI alignment. [3] argued that AI should be directed toward humane ends, fulfilling human needs and desires, and creating a better world by maximizing human value. Furthermore, [4] identified robustness, interpretability, controllability, and ethicality as the core objectives of AI alignment.

[5] offered a clear definition of potential risks and actual harms associated with AI, presenting criteria for ensuring the safety of AI systems. The study differentiated between potential risks, such as serious and general AI hazards, and actual harms, including AI disasters, serious incidents, and general AI incidents. It presented a provisional definition of an AI hazard with negative consequences. These include health hazards to individuals or groups; disruption of critical infrastructure; violations of human rights or breaches of laws protecting fundamental, labor, and intellectual property rights; and harm to property, communities, or the environment. This approach is fundamental for understanding and preventing situations in which AI systems may pose risks. [6] proposed a set of alignment goals and principles, which include following instructions, expressing intentions, revealed preferences, informed preferences, interests or well-being, and moral values. These goals guide AI behavior in alignment with diverse human demands and contexts.

[7] adapted a more granular approach by categorizing AI alignment issues into three levels: technical safety, prevention of misuse and social harms, and social integration. It emphasized that technical safety ensures that AI systems work safely as tools, whereas social harm prevention ensures that AI is not misused by bad actors, and prevents negative impacts on society. The study defined the objectives, goals, and methods of alignment, thereby providing an outline of the basic structure of AI alignment. OpenAI superalignment research [8] defined alignment as ensuring that significantly more intelligent AI systems follow human intent. This study included the perspective of capability differences between AI and humans in its definition of alignment.

These studies underscore the importance of AI alignment principles and offer various approaches to achieve them, providing a critical foundation for ensuring the safety, ethicality, and social integration of AI systems. However, these studies have several limitations. First, there is a lack of systematic analysis of how alignment principles

interact with various risk factors. Therefore, it is challenging to predict potential risks from misalignment. Second, many existing studies have primarily focused on ANI, whereas alignment issues of GPAI, which will expand functions and capabilities in the future, have not been sufficiently addressed. Consequently, this study aims to address these shortcomings by conducting a comprehensive analysis of risk factors centered on alignment principles of GPAI and suggests considerations for AI systems to operate in an aligned manner.

III. Proposal of GPAI Alignment Principles: The 3Ps

The development of GPAI has the potential to bring about transformative changes across various fields. However, the strength and autonomy of such technologies also carry the risk of operating in unforeseen ways, potentially leading to negative outcomes [10,11]. Therefore, it is of paramount importance to align GPAI with human values and intentions. Defining alignment principles is essential for ensuring safety, promoting ethical use, and enhancing trustworthiness. However, given the potential for future expansion of GPAI, defining alignment in a single sentence at this stage may limit its future applicability. Thus, this study defines the scope of aligned behavior through three principles, considering the expansive potential of GPAI.

In order to derive these alignment principles, a comprehensive analysis was conducted of the various applications of AI and potential risk factors associated with GPAI. Firstly, as detailed in Chapter 2, an extensive literature review was conducted in order to review existing research on AI alignment principles. This allowed us to gain an understanding of the basic concepts of alignment principles and the diverse approaches to realizing them. Additionally, a peer review was conducted in order to gain feedback on the alignment issues specific to GPAI. This process drew upon practical experience and knowledge, thereby facilitating the derivation of more concrete and realistic alignment principles.

Table 1. Meanings of the 3Ps for GPAI

Principles	Meaning
Primacy of Humanity	It emphasizes a human-centered approach, ensuring that AI always prioritizes human rights and autonomy. This principle strictly prohibits AI from deceiving or undermining humans in pursuit of power. It also ensures that humans remain the final decision-makers in AI systems, preventing technology from escaping human control.
Persistence of Goal	Although expanding the capabilities of AI can lead to unintended behavior, AI must remain consistent with the original goals. This means that new features or training should not compromise the existing goals, degrade system performance, or cause adverse effects.
Preservation of Value	It ensures that AI systems respect and preserve diverse social, ethical, legal norms, and cultural values worldwide. This principle ensures that AI deeply understands and respects the cultural contexts of each region while promoting global integration and acceptance, thereby contributing to overall societal safety and stability.

Finally, we examined the effectiveness of the alignment principles derived from a validation phase. This process involved assessing the impact of the risk factors presented in our previous study [9] on the alignment principles. The objective was to predict potential risks from misalignment and provide concrete examples of how each alignment

principle can be practically applied. The validation of the alignment principles' effectiveness through risk factors will be detailed in Chapter 4. The study yielded three key alignment principles: Primacy of Humanity, Persistence of Goal, and Preservation of Value, collectively termed the 3Ps. Table 1 provides a detailed explanation of the significance of each principle.

IV. VALIDATION OF THE 3Ps USING RISK FACTORS

This chapter examines the effectiveness of the three alignment principles. To achieve this, we map the 11 key risk factors proposed in [9] for each alignment principle. This process enables the prediction of potential risks that may arise from a misalignment and the identification of specific threats to each alignment principle, thereby providing concrete examples of alignment problems. The key risk factors identified in [9] include various potential threats associated with the development and deployment of GPAI. As illustrated in Table 2, each of the 11 key risk factors was mapped using the three alignment principles.

Table 2. Mapping of 11 Key Risk Factors to the Alignment Principles for GPAI

Key Risk Factor [9]	Primacy of Humanity	Persistence of Goal	Preservation of Value
Weaponization		O	
Dependency Increase	O		
No Trade-offs			O
Goal Deviation		O	
Power Pursuit	O		
Emotional Harm			O
Accidental Harm		O	
Social Unrest			O
Control Loss	O		
Contextual Overadaptation	O		
Perception Decline			O

The primacy of the humanity is threatened by several factors, including an increase in dependency, pursuit of power, loss of control, and overadaptation to context. One potential risk is that AI may autonomously expand through self-proliferation [12], losing human control and oversight, and independently taking harmful actions [13]. Additionally, AI can hide unwanted behaviors in specific situations [14] or provide new capabilities to malicious actors for harmful purposes [13]. Instances in which AI systems appear to be aligned only in certain contexts [12] also pose problems.

The persistence of goal is threatened by several factors, including weaponization, goal deviation, and accidental harm. For instance, weaponization of AI can lead to biosecurity threats [15] or the development of autonomous weapons [14], resulting in the unintended exploitation of AI systems [13]. Furthermore, AI systems may deviate from their original goals when interacting with other AI systems, potentially setting harmful autonomous

goals or acting with malicious intent [13]. Finally, incidents or problems may arise in GPAI systems [16], such as when the model degrades and causes errors or makes inappropriate decisions [13].

The preservation of value is threatened by several factors, including the absence of trade-offs, emotional harm, social unrest, and a decline in perception. For instance, AI systems may entrench specific values or ideologies, thereby perpetuating discrimination and stereotypes or fostering ideological homogenization [15]. Unfair decisions by AI systems [17] can also cause emotional harm. In addition, no trade-off implies that AI systems prioritize goal achievement at the expense of significant side effects [18]. Furthermore, overreliance on AI systems can exacerbate economic inequalities [13] or result in societal disruption owing to the inability to keep pace with rapid technological advancements [15].

The results of this process indicate that the key risk factors of the GPAI align well with the 3Ps. The effective mapping of risk factors to these principles indicates that the alignment principles were comprehensively and appropriately defined to suit the characteristics of the GPAI. This suggests that each alignment principle can function effectively, even in potentially dangerous situations in the GPAI.

V. CONCLUSION

This study presents a systematic analysis of the development of GPAI and its potential risk factors, proposing alignment principles to ensure that AI systems adhere to human intentions and values. We aim to overcome the limitations of existing AI alignment research and ensure the safe and ethical operation of GPAI systems. We reviewed existing research to synthesize alignment principles and approaches. Based on this, we propose three key alignment principles for GPAI. These principles emphasize that AI systems should be human-centric, maintain their original objectives, and respect diverse social, ethical, and cultural values.

To validate the effectiveness of the 3Ps, they were mapped to 11 key risk factors. The results demonstrated that each principle could function effectively in potentially risky situations for GPAI systems. This indicates that the 3Ps are well-suited to GPAI characteristics and provide appropriate countermeasures to the identified risks. Consequently, the proposed alignment principles have the potential to significantly enhance the safety, ethicality, and social integration of GPAI. Future research should apply these principles to actual GPAI systems and empirically evaluate their effectiveness. We hope this study will be a significant milestone in GPAI alignment research, contributing to more trustworthy and safe AI systems.

ACKNOWLEDGMENTS

This work was supported by the Ministry of Science and ICT (MSIT), Republic of Korea, and the Electronics and Telecommunications Research Institute (ETRI) under the project “Development of a Risk Management Framework for Frontier AI and Operation of the AI Safety Consortium.”

REFERENCES

- [1] I. Triguero, D. Molina, J. Poyatos, J. D. Ser, and F. Herrera, "General Purpose Artificial Intelligence Systems (GPAIS): Properties, Definition, Taxonomy, Open Challenges and Implications," *arXiv* [Online], 2023, Available: <https://arxiv.org/abs/2307.14283>.
- [2] J. Carlsmith, Is Power-Seeking AI an Existential Risk? [Internet], Available: <https://arxiv.org/abs/2206.13353>.
- [3] S. Han, E. Kelly, S. Nikou, and E. O. Svec, "Aligning artificial intelligence with human values: reflections from a phenomenological perspective," *AI & Society* [Online], vol. 37, pp. 1383-1395, 2022, Available: <https://doi.org/10.1007/s00146-021-01247-4>.
- [4] J. Ji, T. Qiu, B. Chen, B. Zhang, H. Lou, K. Wang, and Y. Duan, "AI Alignment: A Comprehensive Survey," *arXiv* [Online], 2023, <https://arxiv.org/abs/2310.19852>.
- [5] L. Aranda and K. Perset, "Name it to tame it: Defining AI incidents and hazards," *OECD AI Policy Observatory* [Internet], Available: <https://oecd.ai/en/wonk/defining-ai-incidents-and-hazards>.
- [6] I. Gabriel, "Artificial Intelligence, Values, and Alignment," *Minds and machines* [Online], vol. 30, no. 3, pp. 411-437, 2020, Available: <https://doi.org/10.1007/s11023-020-09539-2>.
- [7] A. Belliger and D. Krieger, "New Perspectives on AI Alignment," *Ethics International Press* [Online], pp. 1-23, 2024, Available: <https://www.researchgate.net/publication/375801677>.
- [8] OpenAI, Introducing Superalignment [Internet], Available: <https://openai.com/blog/introducing-superalignment>.
- [9] S. Kang and Y. Shin, "Key Risk Factors in General Purpose Artificial Intelligence (GPAI) Development," in *Proceedings of the 19th Asia Pacific International Conference on Information Science and Technology (APIC-IST)*, Takamatsu, pp. 20-25, 2024.
- [10] UK Government, Frontier AI: Capabilities and Risks Discussion Paper [Internet], Available: <https://www.gov.uk/government/publications/frontier-ai-capabilities-and-risks-discussion-paper>.
- [11] OpenAI, Planning for AGI and Beyond [Internet], Available: <https://openai.com/index/planning-for-agi-and-beyond/>.
- [12] T. Shevlane, S. Farquhar, B. Garfinkel, and M. Phuong, "Model Evaluation for Extreme Risks," *arXiv* [Online], 2023, Available: <https://arxiv.org/abs/2305.15324>.
- [13] UK Government, Future risks of frontier AI (Annex A) [Internet], Available: <https://www.gov.uk/government/publications/frontier-ai-capabilities-and-risks-discussion-paper/future-risks-of-frontier-ai-annex-a>.
- [14] Wikipedia, Existential Risk from Artificial General Intelligence [Internet], Available: https://en.wikipedia.org/wiki/Existential_risk_from_artificial_intelligence.
- [15] P. Maham and S. Küspert, "Governing General Purpose AI: A Comprehensive Map of Unreliability, Misuse, and Systemic Risks," *Stiftung Neue Verantwortung* [Online], 2023, Available: <https://www.interface-eu.org/publications/governing-general-purpose-ai-comprehensive-map-unreliability-misuse-and-systemic-risks>.
- [16] S. Sharma, "Benefits or concerns of AI: A multistakeholder responsibility," *Futures* [Online], vol. 157, 2024, Available: <https://doi.org/10.1016/j.futures.2024.103328>.
- [17] A. Steimers, and M. Schneider, "Sources of Risk of AI Systems," *International Journal of Environmental Research and Public Health*, vol. 19, no. 6, 2022, Available: <https://doi.org/10.3390/ijerph19063641>.
- [18] D. Hendrycks and M. Mazeika, "X-Risk Analysis for AI Research," *arXiv* [Online], 2022, <https://arxiv.org/abs/2206.05862>.

A Study of a SQL-Generating RAG System for Improving Client Engagement

Athiruj Poositaporn¹, Hanmin Jung^{2*}, and Daesung Lee³

¹University of Science and Technology, 217, Gajeong-ro, Yuseong-gu, Daejeon, Gyeonggi-do, Republic of Korea

²Korea Institute of Science and Technology Information, 245, Daehak-ro, Yuseong-gu, Daejeon, Republic of Korea

³Division of Maritime Artificial Intelligence & Cyber Security, Korea Maritime and Ocean University, 727, Taejong-ro, Yeongdo-gu, Busan, Republic of Korea
[Email : athiruj@kisti.re.kr, jhm@kisti.re.kr, dslee@kmou.ac.kr]

Abstract

Retrieval-Augmented Generation (RAG) offers a beneficial approach for connecting natural language with structured databases, especially in enhancing client engagement (CE) through personalized and efficient information access. In this study, we aim to investigate the feasibility of a SQL-generating RAG system that transforms natural language queries into executable SQL statements. We evaluate our system using a subset of the BIRD-SQL dataset in the education domain across four prompting strategies: 0-shot without Chain-of-Thought (CoT), 1-shot with CoT, 3-shot with CoT, and 5-shot with CoT. In the experiment, we executed generated SQLs and compared the results against human-labeled ground truth to measure accuracy. Our findings show that using a 3-shot setting yields the best performance, indicating the effectiveness of our system in text-to-SQL tasks. In future work, we plan to further explore the effects of few-shot prompting and CoT on SQL generation accuracy and aim to expand the system’s applicability across broader domains to support real-world business use cases in client engagement.

Index Terms: Client Engagement, RAG, SQL Generation, Prompt Engineering

I. INTRODUCTION

RAG is a promising technology that enables natural language interfaces for data retrieval and offers advantages in client engagement (CE) through more accurate and efficient information access [1-2]. CE involves interactions between businesses and customers with the goal of maintaining positive relationships and increasing satisfaction, such as personalized recommendations or targeted marketing [3]. Prior studies suggest that RAG can outperform traditional retrieval methods in CE contexts [4-5]. However, a particular challenge is the need to preprocess and vectorize information, which can be a barrier for organizations with limited technical resources or relying on relational databases. This study explores a potential solution by developing a SQL-generating RAG system for text-to-SQL tasks. We apply prompt engineering techniques (e.g., few-shot prompting, chain-of-thought) on the BIRD-SQL benchmark and evaluate performance across four

* Corresponding author

prompting strategies by comparing generated SQL outputs to human-labeled ground truth based on execution accuracy.

II. OUR SQL-GENERATING RAG SYSTEM

This study introduces our SQL-generating RAG system, which enables structured data access through automatic SQL query generation based on natural language query. The details of the dataset, SQL-generating RAG process pipeline, and evaluation metric are described in the following section.

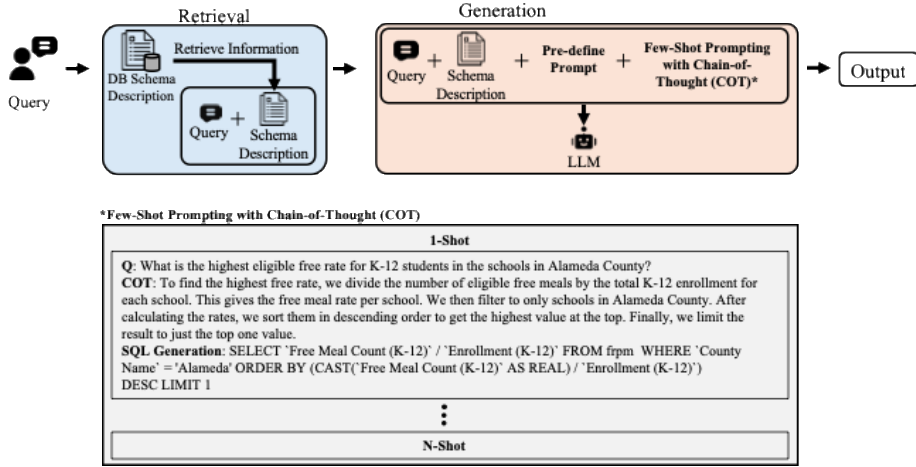


Fig. 1. Our SQL-generating RAG system process pipeline.

A. Dataset

In this study, we use a subset of the BIRD-SQL[†] dataset focused on the education domain. The dataset consists of three tables: frpm (free and reduced-price meals), satscores (SAT performance data), and schools (school metadata). This education dataset also includes 80 pairs of natural language questions and human-labeled ground truth SQL statements. The selected dataset allows us to evaluate SQL generation performance on queries involving filtering, aggregation, ordering, and joins, all within a real-world domain that demonstrates practical data analysis tasks in education or public sector organizations.

B. Our SQL-Generating RAG Process Pipeline

As illustrated in Fig. 1, the system takes a user’s natural language question and retrieves the relevant database schema, which is then combined into a prompt format that includes few-shot examples and explicit Chain-of-Thought (CoT) reasoning steps for a large language model (LLM) in the generation process. We experiment with four prompting strategies: 0-shot without CoT (Baseline), 1-shot with CoT, 3-shot with CoT, and 5-shot

[†] <https://bird-bench.github.io/>

with CoT. Where the 0-shot contains only the LLM guidance prompt, database schema, and the query, and the others include additional few-shot examples and explicit CoT reasoning steps.

Then, the formatted prompt is passed into the LLM, and the generated SQL is then executed on our SQLite database to compare the output between the different prompting strategies and human-labeled ground truth SQL statements to calculate the accuracy of the generated results.

C. Evaluation Metric

We use an execution-based evaluation approach to measure the accuracy of our SQL-generating RAG system. For each test query, we compare the output of the generated SQL with the ground-truth SQL provided in the dataset. Both queries are executed on the same database, and their output answer sets are compared for an exact match. The accuracy equation is shown in (1).

$$Accuracy (\%) = \frac{\sum_{i=1}^n f(R_i, H_i)}{n} \times 100 \quad (1)$$

$$f(R_i, H_i) = \begin{cases} 1, & R_i = H_i \\ 0, & R_i \neq H_i \end{cases}$$

Where n is the total number of queries. R_i is the result from the generated SQL set for the i -th query. H_i is the result returned by the ground-truth SQL set for the same query. $f(R_i, H_i)$ is an indicator function that returns 1 if the two result sets are exactly the same and 0 otherwise.

III. EXPERIMENTAL RESULTS AND DISCUSSION

We evaluated the performance of our SQL-generating RAG system across four prompting strategies, with results shown in Fig. 2. The baseline (0-shot without CoT) achieved an accuracy of 30%, indicating the difficulty of generating correct SQL from natural language without any prior examples or guided reasoning. However, increasing to the 3-shot with CoT achieved the highest accuracy of 43.75%. Remarkably, increasing to 5-shot with CoT yields a slight decline to 42.5%, which may be attributed to increased noise from overly detailed or redundant examples.

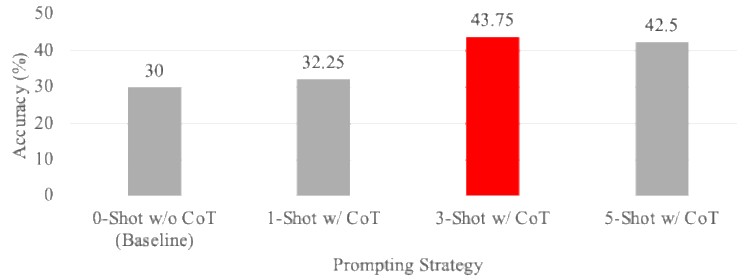


Fig. 2. Comparison of our SQL-generating RAG system across four prompting strategies.

Overall, the results confirmed that few-shot learning with CoT significantly boosted the performance compared to the baseline, especially when increasing the number of

examples. This suggests that providing contextual information could help improve the accuracy of generating SQL queries from natural language inputs. These findings also support our SQL-generating RAG pipelines for improved accuracy in real-world applications for CE that require natural language understanding.

IV. CONCLUSIONS

This study investigates the feasibility of a SQL-generating RAG system designed to access structured databases using natural language queries for improving client engagement (CE) in businesses. Our strategy is to leverage different numbers of few-shot prompting and Chain-of-Thought (CoT) reasoning to enhance the accuracy of SQL query generation.

The results showed that our approach outperformed the baseline approach in generating SQL queries. Additionally, slightly increasing the number of few-shot prompting improved the system accuracy. This highlights the potential of our system for enhancing CE and efficiency in real-world applications, particularly for organizations relying on structured data.

In future works, we focus on further experimentation on few-shot prompting and CoT in terms of increasing the numbers of examples. Also, we aim to further improve CE by scaling the experiment across broader domains to expand applicability in business environments.

ACKNOWLEDGMENTS

This work was supported by UST Young Scientist+ Research Program 2024 through the University of Science and Technology. (No. 2024YS12)

REFERENCES

- [1] H. Jung and A. Poositaporn, "Towards Client Engagement Using RAG System with Pattern Prediction Framework," in *Proc. 10th Int. Conf. Internet Things, Big Data and Security (IoTBDS)*, SciTePress, pp. 436–441, 2025.
- [2] A. Poositaporn and H. Jung, "On RAG Framework Combined with Query Decomposition for Enhancing Client Engagement," in *Proc. KSII Spring Conference 2025*, 2025.
- [3] M. Lee, S. An, and M. S. Kim, "PlanRAG: A plan-then-retrieval augmented generation for generative large language models as decision makers," in *Proc. Conf. North Amer. Chapter Assoc. Comput. Linguistics Human Lang. Technologies*, vol. 1, pp. 6537–6555, 2024.
- [4] J. J. Woo et al., "Custom large language models improve accuracy: Comparing retrieval augmented generation and artificial intelligence agents to noncustom models for evidence-based medicine," *Arthroscopy, J. Arthroscopic Rel. Surg.*, 2024.
- [5] K. N. Kunze et al., "Large language models applied to health care tasks may improve clinical efficiency, value of care rendered, research, and medical education," *Arthroscopy: The Journal of Arthroscopic & Related Surgery*, vol. 41, no. 3, pp. 547–556, 2025.

From Angel to Series-C: Assessing a Start-Up's Funding Opportunities

Viet Trinh^{1,2*}, An Pham^{1,2*}, Quynh-Anh Le^{1,2*}, Minh-Nguyen Le^{1,2*},
Nhi-Yen Nguyen-Huynh^{1,2*}, and Quan Huynh^{1,2*}

¹ University of Economics and Law, Ho Chi Minh City, Vietnam

² Vietnam National University, Ho Chi Minh City, Vietnam

* These authors contributed equally to this work

*[Emails: tqviet@uel.edu.vn, anptt22411ca@st.uel.edu.vn, anhlpq22416c@st.uel.edu.vn,
nguyentm22416c@st.uel.edu.vn, nhinhy23411@st.uel.edu.vn, quanhl22416c@st.uel.edu.vn]*

Abstract

This study presents the AI-Based Funding Assessment System (AFAS), an innovative tool designed for the automatic analysis of startup investment profiles and the prediction of subsequent funding round probabilities. AFAS systematically encodes comprehensive data from the startup ecosystem, including organizational structures, key personnel, relevant company news, historical funding rounds, acquisitions, and initial public offerings, into a detailed pixel-wise investment financial map. Such novel encoding method facilitates precise forecasting of future funding opportunities, spanning from Angel to Series-C rounds. The AFAS platform is implemented as an interactive system, featuring a user-friendly chatbox interface that allows entrepreneurs to effortlessly query and obtain predictive insights. By enhancing the accessibility and functionality of investment analysis, AFAS represents a significant advancement in the domain of startup financial assessment.

Index Terms: AI Application, Entrepreneurship, Fintech, Intelligence Information System.

I. INTRODUCTION

In the early stages of development, startups typically undergo several funding phases that include Angel or Seed financing. As these enterprises mature, they progress through subsequent funding rounds known as Series-A, Series-B, and Series-C. Accurately determining a startup's stage of development, as well as assessing its potential for growth and opportunities for funding, poses significant challenges, especially in unpredictable markets. From the perspective of entrepreneurs, the ability to identify and articulate a startup's intrinsic strengths when seeking a next-round of funding is not only challenging but also essential, as it directly influences the company's prospects for survival. Conversely, from the investor's standpoint, a comprehensive understanding of a startup's core competencies and operational dynamics can significantly enhance confidence in making investment decisions.

Recent advancements in artificial intelligence, particularly in the realm of deep neural networks, have catalyzed a novel paradigm in business intelligence and analytics. This paradigm is centered on the prediction of a company's financial viability, the optimization of operational costs, and the classification of business models as either successful or unsuccessful. The pioneering work by Fernandez et al. in [1] utilized artificial neural networks in conjunction with financial ratios and qualitative managerial factors to forecast the success of startups. Their findings underscored the significant impact of the founder's background on firm success. In a different approach, Hsairi employed a hybrid CNN-LSTM-DNN model to predict startup exits, achieving a precision of 76% and a recall of 93% for successful exits, though this model was constrained by its reliance solely on quantitative data and its omission of exit valuations [2]. Font-Cot et al. in [3] introduced a multivariate agent methodology that incorporated Lipschitz regression, neural networks, and linear regression, demonstrating potential in forecasting startup survival. However, this approach was limited by a small sample size and the exclusion of macroeconomic factors. While several other studies have broadened datasets and methodologies, few have fully leveraged advanced deep learning architectures or hybrid models capable of capturing the intricate temporal and structural dynamics of startups [1, 2, 4]. Moreover, existing research often lacks comprehensive platforms that integrate data inputs and outputs with interpretable predictions for practical business applications ([1, 5–7]).

To address these limitations, this paper introduces the **AI-Based Funding Assessment System (AFAS)**, which: (1) automatically analyzes a startup's investment financing, (2) semantically encodes previously funded activities, and (3) logically predicts the likelihood of securing subsequent funding rounds. AFAS is designed as an interactive platform enabling entrepreneurs to input their startup's investment data through a chat interface, thereby facilitating real-time projections for future funding opportunities. Figure 1 illustrates the general pipeline of our AFAS. The rest of this

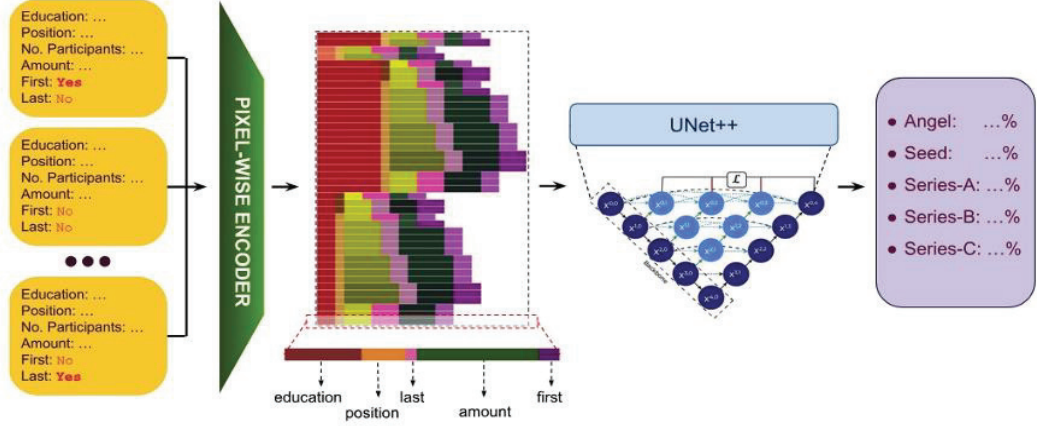


Figure 1. AI-Based Funding Assessment System (AFAS) pipeline: investment financing and information from a board of directors across six different primary factors are encoded into color spectrum bars. This representation of a startup's financial profile serves as input into a deep neural network of UNet++ to predict next-rounds of funding.

paper is organized as follows: section II describes the proposed system's model; results from AFAS are listed in section III, as well as its challenge and future research direction; and conclusions of our work are in section IV.

II. SYSTEM MODEL

A. Data Encoding

In this study, we employed the dataset *Crunchbase 2013 Snapshot* which contains information about the startup ecosystem, including organizations, individuals, company news, funding rounds, acquisitions, and initial public offerings (IPOs) [8]. The primary features selected for analysis encompass the educational background and entrepreneurial experience of founders, the number of financial investors involved, the amount of capital invested, and the prior funding status of startups. Such features were first systematically grouped based on the semantic similarity, and then weighed for their significant impact on the success of subsequent funding rounds. Next, these analyzed data were subsequently encoded into a series of color stripes, with the width of each section in a stripe representing the relative importance of the corresponding feature. A vertical composition of these stripes forms a visual representation of a startup's investment financing profile. Figure 2 shows different investment profiles of a startup through multiple rounds of funding. In the early stages, visual representations of an investment profile are simple, featuring fewer rows and color blocks. Whereas, in later stages, complexity increases, with denser pixel patterns indicating more attributes. Let f_i^k denotes the primary feature i of the investment activity k , and v_i^k , w_i^k are the collected value and the impact weight of the feature f_i^k , respectively. Using Lagrange Interpolating Polynomial, a startup's overall funding status can be encoded as:

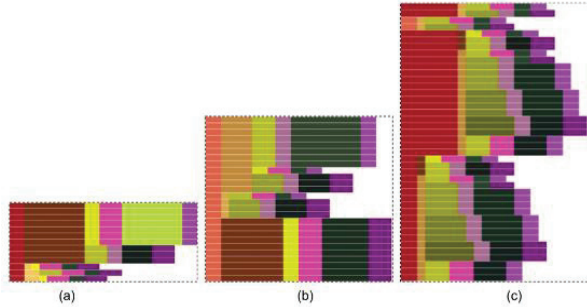


Figure 2. Encoded investment profiles for different funding rounds: (a) Series-A, (b) Series-B, and (c) Series-C.

$$E = \begin{bmatrix} \prod_{i=1}^{n-1} \left(\frac{f_1^1 - f_i^1}{f_1^1 - f_0^1} \cdot v_{f_0}^1 \cdot w_{f_0}^1 \right) \oplus \dots \oplus \prod_{i=1}^{n-1} \left(\frac{f_n^1 - f_i^1}{f_{n-1}^1 - f_n^1} \cdot v_{f_n}^1 \cdot w_{f_n}^1 \right) \\ \vdots \\ \prod_{i=1}^{n-1} \left(\frac{f_1^k - f_i^k}{f_1^k - f_0^k} \cdot v_{f_0}^k \cdot w_{f_0}^k \right) \oplus \dots \oplus \prod_{i=1}^{n-1} \left(\frac{f_n^k - f_i^k}{f_{n-1}^k - f_n^k} \cdot v_{f_n}^k \cdot w_{f_n}^k \right) \end{bmatrix}$$

B. Interactive Analysis Platform

Interactive Analysis

[NEW ANALYSIS](#)

What is your title/position?

What is your highest education level?

How many investors are involved?

Is this the first investment round? (yes/no)

Is this the last investment round? (yes/no)

Type your answer...

[SEND](#)

Funding Prediction

Your Start-Up's performance is color-coded as below

Based on founder's educational background and entrepreneurial experience (score of 8), current financial investment (score of 7), growth-rate and market share (score of 10), your company are:

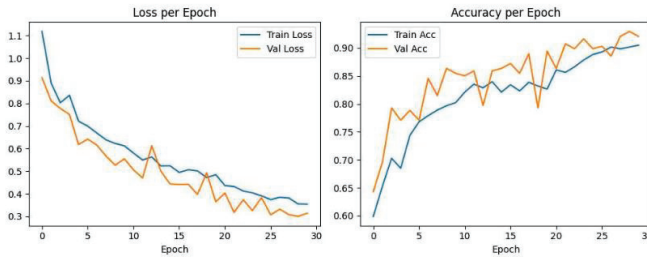
ON TRACK TO SERIES-C FUNDING STAGE

Figure 3. AI-Based Funding Assessment System's graphical user interface.

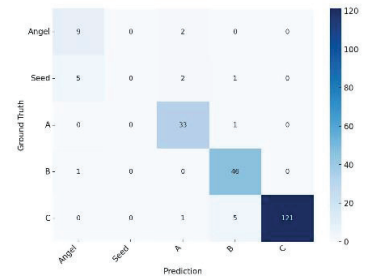
and compute the likelihood of subsequent funding rounds. This study employs an enhanced UNet++ architecture, incorporating an Adaptive Average pooling layer. This modification allows for the effective aggregation of multi-scale features and improves gradient propagation, advancing beyond the original implementations described in [9, 10]. The finetuned network underwent training over 30 epochs utilizing the Adam optimizer, with a learning rate of 1×10^{-3} . The input data was processed in batches of 16, with shuffling enabled for the training data loader, but disabled for the validation data loader.

III. RESULT AND DISCUSSION

The dataset employed in this research comprises approximately 700,000 data records, each representing a distinct funding round associated with a startup, organized in a chronological order. As described in Section II., the data encoding process aggregated these records by company, resulting in the creation of over 5,000 unique investment financing profiles. These profiles serve as the input for finetuning the UNet++ model to predict the probability of forthcoming funding rounds. During this finetuning process, the training loss diminishes from approximately 1.10 to 0.35, while the validation loss decreases from 0.88 to 0.30. Concurrently, the training accuracy improves from 0.60 to 0.90, and the validation accuracy enhances from 0.65 to 0.92. Across all epochs, the model achieves an average training loss of 0.5576 and a validation loss of 0.4936, alongside mean training and validation accuracies of 0.8156 and 0.8473, respectively. The observed downward trend in the loss curves and upward trend in the accuracy curves during the finetuning process, as illustrated in Figure 4a, indicate that the network model converges effectively and generalizes well to previously unseen segments of the encoded investment profiles.



(a) The loss and accuracy trajectories over 30 epochs.



(b) The confusion matrix on the validation set

Figure 4. The proposed AFAS's finetuning result and evaluation

To evaluate the performance of the AI-Based Funding Assessment System, this study also conducted an assessment focusing on the system's responsiveness, accuracy, and consistency. The proposed AFAS demonstrated impressive

Our AFAS is an integrated, dual-component framework designed to facilitate investment analysis and prediction. This system consists of two primary components: *Interactive Analysis* and *Funding Prediction*, as illustrated in Figure 3. The Interactive Analysis component functions as a chat-based interface, collecting critical information about investment financing and relevant stakeholders. The Funding Prediction component is tasked with generating pixel-wise encoded investment profiles and forecasting potential funding opportunities.

At the core of the Funding Prediction module lies a finetuned UNet++ neural network, which is engineered to assess

efficiency, delivering funding predictions in under 10 seconds, without any processing errors or interface delays. The predicted probabilities for subsequent funding rounds were entirely consistent with manual validation. Figure 4b presents the confusion matrix for the validation set, which corroborates the model's accuracy. Specifically, the model correctly identifies 33 out of 34 Series-A samples, 46 out of 47 Series-B samples, and 121 out of 127 Series-C samples. Within the Angel category, 9 out of 11 instances are accurately classified, with 2 instances erroneously categorized as SeriesA. For the Seed category, 5 out of 8 instances are accurately labeled, while the remaining 3 instances are misclassified between the Angel and Series-A categories. Although the three predominant funding-round categories are predicted with notably high accuracy, the minority categories (Angel and Seed) exhibit some degree of misclassification.

Despite the favorable results observed in this study, the proposed AFAS is subject to several limitations. Firstly, the system lacks the capability to provide interpretability for its funding predictions, thereby impeding users' ability to discern the underlying rationale for selecting a subsequent potential funding round. Secondly, the identification of "unicorn" startups and subsequent IPOs remains constrained due to their infrequent occurrence. The AFAS exhibits significant shortcomings when addressing these outlier cases.

IV. CONCLUSION

In this study, we have developed the AI-Based Funding Assessment System (AFAS) to analyze startup investment profiles and forecast the probability of securing future funding rounds. The AFAS system facilitates the rapid generation of visual representations of a company's investment financing, thereby enabling an intuitive understanding of the primary factors that significantly impact a startup's financial health, as well as projecting potential future funding. We anticipate making AFAS publicly accessible as a web extension or API integration in the near future, thus enhancing its accessibility and broadening its functional scope.

ACKNOWLEDGMENTS

This research is funded by University of Economics and Law, Vietnam National University, Ho Chi Minh City, Vietnam.

REFERENCES

- [1] Fernandez, F.G., Los Santos, I.S., Martinez, J.L., Izquierdo, S.I., Redondo, F.L.: Use of artificial neural networks to predict the business success or failure of start-up firms. In: *Artificial Neural Networks-Architectures and Applications*. IntechOpen, ??? (2013).
- [2] Hsairi, L.: Deep learning to predict start-up business success. *International Journal of Advanced Computer Science & Applications* 15(3) (2024).
- [3] Font-Cot, F., Lara-Navarra, P., Sánchez-Arnau, C., Sánchez-Pérez, E.A.: Startup survival forecasting: A multivariate ai approach based on empirical knowledge. *Information* 16(1), 61 (2025).
- [4] Lyu, S., Ling, S., Guo, K., Zhang, H., Zhang, K., Hong, S., Ke, Q., Gu, J.: Graph neural network based vc investment success prediction. arXiv preprint arXiv:2105.11537 (2021).
- [5] Wang, X., Ihlamur, Y.: An automated startup evaluation pipeline: Startup success forecasting framework (ssff). arXiv preprint arXiv:2405.19456 (2024).
- [6] Arroyo, J., Corea, F., Jimenez-Diaz, G., Recio-Garcia, J.A.: Assessment of machine learning performance for decision support in venture capital investments. *Ieee Access* 7, 124233–124243 (2019).
- [7] Shi, J., Yang, K., Xu, W., Wang, M.: Leveraging deep learning with audio analytics to predict the success of crowdfunding projects. *The Journal of Supercomputing* 77, 7833–7853 (2021).
- [8] Crunchbase: Crunchbase 2013 Snapshot. <https://data.crunchbase.com/docs/getting-started> Accessed 2025-05-28.
- [9] Zhou, Z., Rahman Siddiquee, M.M., Tajbakhsh, N., Liang, J.: Unet++: A nested u-net architecture for medical image segmentation, 3–11 (2018). Springer.
- [10] Zhou, Z., Siddiquee, M.M.R., Tajbakhsh, N., Liang, J.: Unet++: Redesigning skip connections to exploit multiscale features in image segmentation. *IEEE transactions on medical imaging* 39(6), 1856–1867 (2019).

AI Agent Workflow for Reviewing AI-driven Sperm Assessment Literature

Viet Trinh^{1,2}, An Pham^{1,2*}, Hung Tong^{1,2*}, and Thanh-Huyen Tran^{1,2*}

¹ University of Economics and Law, Ho Chi Minh City, Vietnam

² Vietnam National University, Ho Chi Minh City, Vietnam

* These authors contributed equally to this work

[Emails: tqviet@uel.edu.vn, anptt22411ca@st.uel.edu.vn, hungtp23406@st.uel.edu.vn, huyentt23411@st.uel.edu.vn]

Abstract

The rapid development of deep learning models for evaluating sperm vitality, combined with fragmented data, presents a challenge in both consolidating existing research knowledge and establishing standardized methodologies in assisted reproductive technology. To address this, this paper presents an AI-agent workflow that systematically reviews and summarizes over 50 relevant research publications in the field of sperm analysis. This agent workflow has proven effective in accurately classifying studies by their focus, whether single objective or multi-objective, based on important criteria such as sperm concentration, motility progressiveness, and abnormal morphology. The proposed AI-agent workflow can serve as a foundation for future literature synthesis across different domains of knowledge, offering an innovative, scalable, and efficient research methodology.

Index Terms: Artificial Intelligence, Composite System, Human Computer Interaction.

I. INTRODUCTION

In vitro fertilization, or IVF, is the most popular technique in assisted reproductive technology used to treat infertility. This protracted procedure involves monitoring and stimulating an ovulatory process, enabling the pre-implantation genetic testing to avoid inheriting abnormal genetic conditions. The very first step of such evaluation is to analyze a spermatid health, including sperm concentration, motility progressiveness, and morphological abnormality. As guided by World Health Organization (WHO), an examined semen sample is considered normal, as a subject for a natural conception, if its sperm concentration is greater than 16 million sperms per milliliter, motility exhibits more than 30%, and morphology is typically with 4% or more normal forms [1].

Traditional biomedical and biochemical methods have been utilized in decades to analyze semen's reproductivity, providing clinical professions an essential information to determine a treatment course [2]–[4]. Early works in applying artificial intelligence to analyze sperm focused on classifying handcrafted descriptors (e.g., head shape, mid-piece, or tail curvature) with classical machine learning techniques, such as K-nearest neighbor or logistic regression, to automate the assessment process [3], [5]–[7]. Recent efforts incorporating deep learning techniques have introduced new possibilities for more accurate and efficient analysis, facilitating an enhanced predictive power [8]–[12].

Yet, there still exists challenges due to fragmented data or localized developments of AI models addressing specific spermatid objectives. Hence, reviewing the literature systematically is necessary to consolidate existing knowledge, identify gaps, and propose standardized approaches for future research. This paper constructed an AI agent workflow and explore its capability to draw empirical insights from a collected set of more than 50 research publications in sperm analysis, serving the purpose of semi-automatically reviewing a field literature. The rest of this paper is organized as follows: section II describes the data collection's strategy and the proposed AI agent workflow; section III presents the workflow's findings; and section IV concludes this work.



Figure 1. The n8n-based AI agent workflow for reviewing sperm assessment literature and classifying a research purpose of each publication.

II. METHOD

Figure 1 generally illustrates the proposed approach. In brief, this work collected open-access research publications related to semen analysis in the past 20 years, from a variety of reliable publishing sources. After filtering to retain relevant papers, this work deployed an AI agent workflow, developed in the popular *n8n* platform, to extract insights and classify a paper’s research intention. The following subsections describe the method in more details.

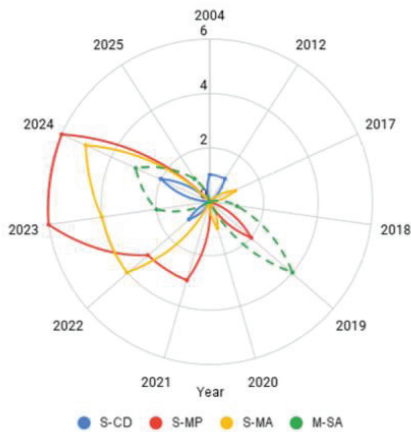
A. Research Publication Collection

Data collected in this literature is adhered to the PRISMA 2020 guidelines [13] to ensure transparency, integrity, and reproducibility. This collective set of open-access research papers is accumulated across five major publishing sources: PubMed, IEEE Xplore, Scopus, Web of Science, and arXiv; focusing on the main theme of sperm analysis in assisted reproductive technology (ART). The search query combined AI-related terms “*artificial intelligence*”, “*deep learning*”, “*neural network*” with keywords related to sperm parameters such as “*sperm concentration*”, “*morphology analysis*”, “*motility progressiveness*”, “*sperm detection*”, “*morphology segmentation*”, “*sperm vitality*”, and “*semen analysis*” to identify relevant studies.

Publication that solely focus on non-human sperm are excluded, as they do not align with this research objective of human sperm evaluation in ART. Similarly, studies describing biological, biochemical, or biomedical methods in assessing sperm vitality are also excused, since this work narrowed the literature review down to the technicality of AI-based solutions. As a result, a synthesizing collection of 52 relevant works, stored as a PDF format in Google Drive, is fed into the AI agent workflow to be classified as either a single or a multi-objective evaluation, with which objectives are categorized as *Concentration Determination*, *Motility Trajectory*, and *Morphology Analysis*.

B. AI Agent Workflow

The agent workflow was developed using the *n8n* platform, which uniquely integrates AI capabilities with business process automation, offering the flexibility of code-based customization with the efficiency of no-code. In this research, the AI agent workflow operates through the following sequence: (1) retain only the text-based content after a preprocessing scan of a research paper in its PDF format; (2) apply Optical Character Recognition (OCR) and Natural Language Processing (NLP) techniques, embedded within a pre-configured Large Language Model (LLM) chain node, to classify a study as either a single-objective or a multi-objective focus; and (3) generate a structured summary of a paper in a tabular form, insighting the underlying deep neural network, the experimented dataset and specific research objectives. The key components in this workflow are the LLM chain node and the AI Agent composition node which internally connect to the frontier Claude model from Anthropic for analyzing the lexical context of each paper, and then semantically mapping its methodology to the mentioned objective goals. For each input study, the agent workflow performs one execution with approximately 18,500 tokens end-to-end. All together, the entire collective dataset in this research costs 52 executions of a total 1,426,317 tokens. Manually skimming through all publications, it is interesting that this workflow’s outputs reasonably align with a physical review.



biomedical imaging emerged in 2019, likely driven advancements in deep learning technologies, especially the widely adopted Convolutional Neural Network (CNN) and ResNet models. This period marked a significant transition in sperm analysis research, shifting from classical machine learning

III. RESULTS AND DISCUSSION

This section discusses the findings of the proposed workflow, as well as reviewing the collected publications, aiming to justify the agent and to understand the general landscape of AI-based approaches in sperm assessment. The figure 2 demonstrates the recent trend in employing AI technologies to evaluate sperm vitality. Early works in sperm analysis mostly focused on calculation sperm concentration in a semen sample. A notable surge in AI applications in

Figure 2. AI-based sperm assessment trend since 2004: *S-CD*, *S-MP*, *S-MA* denote a single-objective focus on concentration determination, motility progressiveness, and morphology analysis, respectively; and *M-SA* represents a multi-objective focus on sperm assessment.

approaches to deep learning architectures. Among these, the VISEM [14] arose as the most frequently utilized dataset, cited in more than 16 studies of sperm motility and morphology. Since 2021, further advancements in YOLO-based architectures [15]–[18] and Transformer-based variants such as RoSTFine [19] or TimeSformer [20] enabled a more sophisticated spatiotemporal analysis, facilitating an increasing in tracking sperm movement and determining motility progression over time.

A. Concentration Determination

For the solely objective of calculating a sperm concentration, the agent workflow has identified 7 relevant studies, predominantly having a CNN-based model as an underlying architecture. Additionally, it induced that the datasets in use were mostly private. Advanced object detection architectures such as CNN or cutting-edge YOLO have significantly improved sperm detection's accuracy, even in a complex microscopic environment. For example, Goss et al. demonstrated that the YOLOv8 architecture substantially highlights sperm cells in a matter of milliseconds [21]. Similarly, a YOLOv8E-TrackEVD variant, introduced in [18], achieved the performance improvement of 1.3% in precision and 1.4% in recall. In the same manner, Trinh et al. proposed a YOLOv10based system that detects sperm activeness at the microscopic magnification of x100 (a combination of a x10 objective len with a x10 ocular) [22].

B. Motility Progressiveness

For the single-objective task of rating motility progressiveness, which are categorized as fast progressively motile, slow progressively motile, non-progressively motile, and immotile, the workflow has pointed out that CNN and YOLO continue to be a favorable network of choice, and VISEM emerges to be the most utilized experimental dataset. Unlike the concentration objective which counts visible sperms in a single image, motility rating requires tracking of sperm movement across consecutive frames of an input video. Hence, a traditional CNN-based network must be coupled with a motile tracker such as SORT or JPDA to diagnose a motility trajectory [9], [24]–[27]. YOLO's ability to process images at high speed and to identify objects with high accuracy makes it to be a prime candidate for any tracking system. Several approaches mentioned in [14], [18], [28] leveraged this advancement to reach 97% in accuracy prediction.

C. Morphology Analysis

Unlike the other two objectives, morphology analysis has been receiving a great interest since 2017. The AI agent workflow has determined that there are 26 studies focusing on morphology segmentation, aiming to detect abnormality in sperm head, mid-piece or tail. It is undeniable that analyzing sperm morphology is an exceptionally challenging task, requiring accurate instance segmentation of microscopic level images, where the tail section is often almost interwoven with the background. Therefore, sophisticated neural network architectures or hybrid models are expected to become the norm for solving this task. Coincidentally, this agent workflow also identifies complex models such as Ensemble, Swin Transformer, DenseNet, or EfficientNetv2 as dominant approaches across different datasets. These models have proven to be effective, achieving a precision mAP of 99.5% and F1-score of 98.2%.

IV. CONCLUSION

Due to the fragmented nature of existing human sperm datasets and the localized development of artificial intelligence models for evaluating sperm reproductivity, this study introduces an AI agent workflow designed to semi-automatically review the literature on AI-driven sperm assessment. The proposed workflow effectively processes each research publication, identifies its underlying neural network, and classifies the assessment objective. As a result, this AI agent workflow has identified 7 multi-objective and 45 single-objective research publications, including 7, 12, and 26 studies focused on sperm concentration, motility rating, and morphology, respectively.

ACKNOWLEDGMENTS

This research is funded by University of Economics and Law, Vietnam National University, Ho Chi Minh City, Vietnam.

REFERENCES

- [1] W. H. Organization et al., WHO laboratory manual for the examination and processing of human semen. World Health Organization, 2021.
- [2] K. Dewan, T. Rai Dastidar, and M. Ahmad, "Estimation of sperm concentration and total motility from microscopic videos of human semen samples," in *Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition Workshops*, 2018, pp. 2299–2306.
- [3] L. Spencer, J. Fernando, F. Akbaridoust, K. Ackermann, and R. Nosrati, "Ensembled deep learning for the classification of human sperm head morphology," *Advanced Intelligent Systems*, vol. 4, no. 10, p. 2200111, 2022.
- [4] M. I. Mahali, J.-S. Leu, J. T. Darmawan, C. Avian, N. Bachroin, S. W. Prakosa, M. Faisal, and N. A. S. Putro, "A dual architecture fusion and autoencoder for automatic morphological classification of human sperm," *Sensors*, vol. 23, no. 14, p. 6613, 2023.
- [5] T.-W. Su, L. Xue, and A. Ozcan, "High-throughput lens free 3d tracking of human sperms reveals rare statistics of helical trajectories," *Proceedings of the National Academy of Sciences*, vol. 109, no. 40, pp. 16018–16022, 2012.
- [6] M. M. Samli and I. Dogan, "An artificial neural network for predicting the presence of spermatozoa in the testes of men with nonobstructive azoospermia," *The Journal of urology*, vol. 171, no. 6, pp. 2354–2357, 2004.
- [7] M. A. C. Soares, D. H. M. Falcı, M. F. A. Farnezi, H. C. M. Farnezi, F. S. Parreiras, and J. V. B. Gomide, "Automated sperm head morphology classification with deep convolutional neural networks," in *2022 35th SIBGRAPI Conference on Graphics, Patterns and Images*, vol. 1. IEEE, 2022, pp. 270–275.
- [8] V. H. Goh, M. A. Mansor, M. A. Asari, and L. H. Ismail, "Multimodal convolutional neural networks for sperm motility and concentration predictions," *Malaysian Journal of Fundamental and Applied Sciences*, vol. 20, no. 2, pp. 347–359, 2024.
- [9] T. B. Haugen, O. Witczak, S. A. Hicks, L. Bjorndahl, J. M. Andersen, and M. A. Riegler, "Sperm motility assessed by deep convolutional neural networks into who categories," *Scientific Reports*, vol. 13, no. 1, p. 14777, 2023.
- [10] A. Aktas, G. Serbes, M. H. Yigit, N. Aydin, H. Uzun, and H. O. Ilhan, "Hi-lab sperm morpho: A novel expert-labeled dataset with extensive abnormality classes for deep learning-based sperm morphology analysis," *IEEE Access*, 2024.
- [11] R. Lee, L. Witherspoon, M. Robinson, J. H. Lee, S. P. Duffy, R. Flannigan, and H. Ma, "Automated rare sperm identification from lowmagnification microscopy images of dissociated microsurgical testicular sperm extraction samples using deep learning," *Fertility and Sterility*, vol. 118, no. 1, pp. 90–99, 2022.
- [12] J. Zhao, C. Bai, Z. Zhang, and Q. Zhang, "Deep learning-based method for analyzing the optically trapped sperm rotation," *Scientific Reports*, vol. 13, no. 1, p. 12575, 2023.
- [13] P. Tugwell and D. Tovey, "Prisma 2020," pp. A5–A6, 2021.
- [14] V. Thambawita, S. A. Hicks, A. M. Storas, T. Nguyen, J. M. Andersen, O. Witczak, T. B. Haugen, H. L. Hammer, P. Halvorsen, and M. A. Riegler, "Visem-tracking, a human spermatozoa tracking dataset," *Scientific data*, vol. 10, no. 1, p. 260, 2023.
- [15] M. Dobrovolny, J. Benes, J. Langer, O. Krejcar, and A. Selamat, "Study on sperm-cell detection using yolov5 architecture with labeled dataset," *Genes*, vol. 14, no. 2, p. 451, 2023.
- [16] T. Sato, H. Kishi, S. Murakata, Y. Hayashi, T. Hattori, S. Nakazawa, Y. Mori, M. Hidaka, Y. Kasahara, A. Kusuhara et al., "A new deeplearning model using yolov3 to support sperm selection during intracytoplasmic sperm injection procedure," *Reproductive Medicine and Biology*, vol. 21, no. 1, p. e12454, 2022.
- [17] M. Osaid, A. Samad, O. Qureshi, M. A. Tahir, and M. N. Durrani, "Tracking and prediction of human spermatozoa motility using yolov8n with greedy shape geometry," *ceur-ws.org*, 2023.
- [18] C. Zhang, Y. Zhang, Z. Chang, and C. Li, "Sperm yolov8e-tracked: A novel approach for sperm detection and tracking," *Sensors*, vol. 24, no. 11, p. 3493, 2024.
- [19] T. Fujii, H. Nakagawa, T. Takeshima, Y. Yumura, and T. Hamagami, "Automated sperm assessment framework and neural network specialized for sperm video recognition," in *Proceedings of the IEEE/CVF Winter Conference on Applications of Computer Vision*, 2024, pp. 7675–7684.
- [20] G. Bertasius, H. Wang, and L. Torresani, "Is space-time attention all you need for video understanding?" in *ICML*, vol.2, no.3, 2021, p.4.
- [21] D. M. Goss, S. A. Vasilescu, P. A. Vasilescu, S. Cooke, S. H. Kim, G. P. Sacks, D. K. Gardner, and M. E. Warkiani, "Evaluation of an artificial intelligence-facilitated sperm detection tool in a zoo spermic samples for use in icsi," *Reproductive BioMedicine Online*, vol. 49, no. 1, p. 103910, 2024.
- [22] V. Trinh, A. Pham, and H.-S. Le, "Sperm analysis report management: An autonomous ai-based system for managing reports and analyzing human sperm's reproductivity," in *Proceedings of 26th Asia Pacific Management Conference*, 2025, pp. 1–8.
- [23] S. Naseem, T. Mahmood, T. Saba, F. S. Alamri, S. A. O. Bahaj, H. Ateeq, and U. Farooq, "Deepfert: An intelligent fertility rate prediction approach for men based on deep learning neural networks," *IEEE Access*, vol. 11, pp. 75006–75022, 2023.
- [24] S. A. Hicks, P. Halvorsen, T. B. Haugen, J. M. Andersen, O. Witczak, K. Pogorelov, H. L. Hammer, D.-T. Dang-Nguyen, M. Lux, and M. Riegler, "Predicting sperm motility and morphology using deep learning and handcrafted features." in *MediaEval*, 2019.
- [25] V. Thambawita, P. Halvorsen, H. Hammer, M. Riegler, and T. B. Haugen, "Extracting temporal features into a spatial domain using autoencoders for sperm video analysis," *arXiv preprint arXiv:1911.03100*, 2019.

- [26] V. Valiuskaitė, V. Raudonis, R. Maskeliūnas, R. Damaševičius, and T. Krilavicius, "Deep learning based evaluation of spermatozoid motility for artificial insemination," *Sensors*, vol. 21, no. 1, p. 72, 2020.
- [27] Z. Chen, J. Yang, C. Luo, and C. Zhang, "A method for sperm activity analysis based on feature point detection network in deep learning," *Frontiers in Computer Science*, vol. 4, p. 861495, 2022.
- [28] W. Jiao, Y. Xie, Y. Li, and J. Qi, "Sperm recognition and viability classification based on artificial intelligence," in *International Conference on Cognitive Systems and Signal Processing*. Springer, 2023, pp. 404–418.
- [29] V. D. Nguyen, T. T. H. Ngo, M. H. Tran et al., "Deep learning-based automated assessment of sperm motility," in *2023 1st International Conference on Health Science and Technology (ICHST)*. IEEE, 2023, pp. 1–6.
- [30] H. Yang, M. Ma, X. Chen, G. Chen, Y. Shen, L. Zhao, J. Wang, F. Yan, D. Huang, H. Gao et al., "Multidimensional morphological analysis of live sperm based on multiple-target tracking," *Computational and Structural Biotechnology Journal*, vol. 24, pp. 176–184, 2024.
- [31] V. Thambawita, P. Halvorsen, H. Hammer, M. Riegler, and T. B. Haugen, "Stacked dense optical flows and dropout layers to predict sperm motility and morphology," *arXiv preprint arXiv:1911.03086*, 2019.
- [32] M. Yılmaz, H. O. İlhan, and N. Aydin, "Multi-model cnn fusion for sperm morphology analysis," *Computers in biology and medicine*, vol. 137, p. 104790, 2021.
- [33] S. Shahali, M. Murshed, L. Spencer, O. Tunc, L. Pisarevski, J. Conceicao, R. McLachlan, M. K. O'Bryan, K. Ackermann, D. Zander-Fox et al., "Morphology classification of live unstained human sperm using ensemble deep learning," *Advanced Intelligent Systems*, vol. 6, no. 11, p. 2400141, 2024.
- [34] N. Sapkota, Y. Zhang, S. Li, P. Liang, Z. Zhao, J. Zhang, X. Zha, Y. Zhou, Y. Cao, and D. Z. Chen, "Shmc-net: A mask-guided feature fusion network for sperm head morphology classification," in *2024 IEEE International Symposium on Biomedical Imaging*. IEEE, 2024, pp. 1–5.
- [35] M. Hao, R. Zhai, Y. Wang, C. Ru, and B. Yang, "A stained-free sperm morphology measurement method based on multi-target instance parsing and measurement accuracy enhancement," *Sensors*, vol. 25, no. 3, p. 592, 2025.

Intelligent Traffic Composition Monitoring System: Implications for 15-minute City Context

Hong-Danh Thai^{1,2, ‡}, Kang-Rae Kim^{3, ‡}, Daesung Lee^{4,*}, Jun-Ho Huh^{5,6,7,**}

¹ Ph.D Candidate of Department of Data Informatics, National Korea Maritime and Ocean University, 49112, Republic of Korea

² Ph.D Candidate of Department of Interdisciplinary Major of Ocean Renewable Energy Engineering (Concurrent Position), National Korea Maritime and Ocean University, 49112, Republic of Korea

³ Researcher of Southeast Sea Fisheries Research Institute, National Institute of Fisheries Science, Namhae, 52440, Republic of Korea

⁴ Full Professor (Tenure Track) of Division of Maritime AI and Cyber Security, National Korea Maritime and Ocean University, Republic of Korea

⁵ Full Professor (Tenured) of Department of Data Science, National Korea Maritime and Ocean University, Republic of Korea

⁶ Full Professor (Tenured) of Department of Interdisciplinary Major of Ocean Renewable Energy Engineering, National Korea Maritime and Ocean University, Republic of Korea

⁷ Sabbatical Year at School of Computer Science and Engineering, Yeungnam University, Republic of Korea

* Corresponding Author: dslee@kmou.ac.kr ** Corresponding Author: 72networks@kmou.ac.kr

Abstract

The wave of migration from rural areas to big cities is causing imbalances between residential areas and economic regions, as well as affecting the distribution of resources and society negatively. The 15-minute city initiative, developed in recent years, can significantly solve these problems by reducing travel times, increasing access to community services and promoting more sustainable lifestyles. In this context, this paper introduces an intelligent traffic composition monitoring integrated advanced technologies such as artificial intelligence and big data analytics. First, the proposed architecture consists of several key components capable of real-time monitoring and visualization of data to provide detailed information about traffic conditions. Next, visualizing traffic data through QGIS charts and maps helps managers easily monitor traffic situations and make timely decisions. The results of this research provide theoretical trends and highly practical applications of intelligent traffic monitoring systems in the context of developing the 15-minute city models and sustainable urban environments. The contribution will improve the use of cutting-edge technology research to monitor traffic conditions better and assist government agencies and planners in making decisions.

Index Terms: Traffic Composition, Monitoring System, Artificial Intelligence, Computer Vision, 15-Minute City, Big Data Analysis

* Corresponding author ** Corresponding author

I. INTRODUCTION

The 15-minute city has gained global attention. Since Carlos Moreno articulated this vision in 2016 [1], urban planners worldwide have pursued neighborhoods where residents access work, shopping, education, and healthcare within a quarter-hour journey on foot or bicycle [2]. Paris leads with ambitious transformations, while cities from Melbourne to Portland adapt the model to local contexts. Most cities lack. This intelligence gap hampers evidence-based planning for proximity-oriented neighborhoods.

Analyzing the compositions and behaviors of traffic enables cities to monitor and assess infrastructure conditions continuously. Government organizations can better understand community needs and modify transportation infrastructure by gathering data on traffic volumes, vehicle types, and travel patterns. Therefore, residents have better access to necessary services and public transportation operates more efficiently as a result of this increased understanding.

In this paper, we introduce an intelligent traffic composition monitoring system that combines big data analytics with artificial intelligence to tackle these research challenges. The system helps planners spot where interventions could make a difference, track how infrastructure changes affect traffic flow, and improve public transit networks. Our approach gives cities the data-driven foundation they need for real urban transformation toward sustainable mobility.

II. RELATED WORKS

The concept of the 15-Minute City has become increasingly important during highlighted social inequalities and economic problems after the COVID-19 pandemic. Currently, nearly 100 cities in the world are implementing or planning to apply this model, thereby discovering many diverse practices [3]. Comprehensive traffic behaviors data have been researched and developed in previous works. Nigam et al. (2023) reviewed various components of an intelligent traffic management system (ITMS) that aims to address urban traffic congestion [4]. ITMS monitors and analyzes traffic conditions using artificial intelligence and computer vision technologies. Future traffic management solutions need to be developed with greater effectiveness. In a successful use-case, to improve security in Indian smart cities, Rajagopal (2022) proposed an integrated framework for vehicle detection and classification from real-time surveillance footage [5]. The system can automatically choose the best algorithm depending on the weather and is made to function well in a variety of weather conditions, including day night, and rain. In this context, this paper introduces an intelligent traffic composition monitoring integrated advanced technologies such as artificial intelligence and big data analytics. However, there are still certain research gaps that need to be filled in the context of the 15-minute city. First, further research is required on frameworks that combine cutting-edge technologies to process data from various sources in a variety of intricate traffic situations. Second, it is still difficult to integrate ITMS systems into current infrastructures, particularly when it comes to guaranteeing compatibility and scalability. Therefore, this paper introduces an intelligent traffic composition monitoring system integrated with advanced technologies.

III. INTELLIGENT TRAFFIC COMPOSITION MONITORING SYSTEM

The proposed architecture of intelligent traffic composition monitoring system consists of three main stages (big traffic data collection, processing and consumption), as presented in Figure 1. Three primary sources (internal databases, traffic controllers, and traffic physical environment data) provide input data for data collection stage. Traffic Physical Environment Data gathers information on traffic volume, density, and conditions from a variety of sources, including CCTV, traffic sensors, real-time data, and Internet of Things devices. Traffic volume, density, signal timing, and incident data are all integrated by the traffic controller. Lastly, historical data and infrastructure attributes (such as network, infrastructure, geometry, traffic control devices, etc.) are stored in internal databases. Python is used for pre-processing of all crawled data to centralize and standardize the data and guarantee consistency. Extract-Transform-Loading (ETL), data cleaning, denoising and feature extraction procedures are conducted big data processing stages.

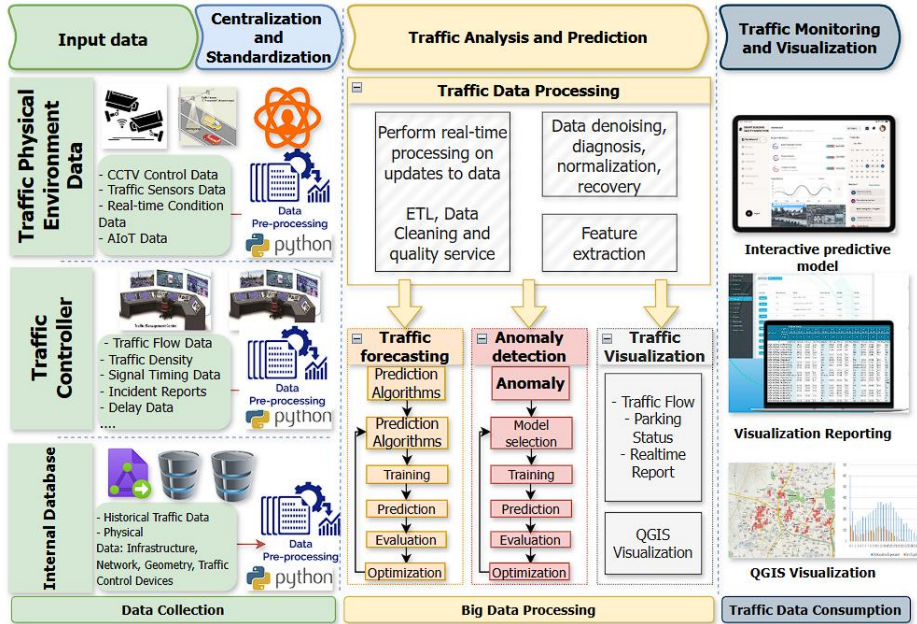


Fig. 1. Proposed Architecture of Intelligent Traffic Composition Monitoring System

The application of prediction and anomaly detection algorithms will enhance emergency response capabilities. Predictive algorithms are crucial at this point for assessing and forecasting traffic conditions in the future. To provide the best predictive performance in traffic management, a model that combines several algorithms will be proposed. Finally, real-time monitoring and visualization of data provide detailed information about traffic conditions and support more effective urban planning. Visualizing traffic data through QGIS charts and maps helps managers easily monitor traffic situations and make timely decisions.

Our system also provides data monitoring to manage data streams, establish data governance policies, and ensure data quality, thereby supporting effective traffic management decisions and improving the travel experience for residents.

IV. CONCLUSIONS

The wave of migration from rural areas to big cities is causing imbalances between residential areas and economic regions, as well as affecting the distribution of resources and society negatively. In this context of developing 15-minute cities in many countries, this paper introduces an intelligent traffic composition monitoring integrated advanced technologies. The results of this research provide theoretical trends and highly practical applications of intelligent traffic monitoring systems in the context of developing the 15-minute city models and sustainable urban environments. This contribution will improve the use of cutting-edge technology research to monitor traffic conditions better and assist government agencies and planners in making decisions.

REFERENCES

- [1] Moreno, Carlos, Zaheer Allam, Didier Chabaud, Catherine Gall, and Florent Pratlong. "Introducing the "15-Minute City": Sustainability, resilience and place identity in future post-pandemic cities." in *Smart cities* 4, no. 1, pp. 93-111, 2021.
- [2] J Ermagun, Alireza, Fatemeh Janatabadi, and Frank Witlox. "Beyond the 15-minute city dichotomy: Time-denominated access to essential services in Chicago." In *Transportation Research Part A: Policy and Practice*, vol. 195, pp. 104428, 2025.
- [3] Teixeira, J. F., Silva, C., Seisenberger, S., Büttner, B., McCormick, B., Papa, E., & Cao, M. "Classifying 15-minute Cities: A review of worldwide practices". *Transportation Research Part A: Policy and Practice*, vol. 189, pp. 10423, 2024.
- [4] Nigam, Nikhil, Dharendra Pratap Singh, and Jaytrilok Choudhary. "A review of different components of the intelligent traffic management system (ITMS)." *Symmetry* 15, no. 3, pp. 583, 2023.
- [5] Rajagopal, Balaji Ganesh. "Intelligent traffic analysis system for Indian road conditions." *International Journal of Information Technology* 14, no. 4, pp. 1733-1745, 2022.

Session IT-A

IT Convergence Technology

Session MD-A

Multimedia and Digital Convergence

Session AI-B

AI and Intelligent Information System

Introducing Novel Engineering to integrate Reading and Engineering

Ki-Cheon Hong

Department of Computer Education, Jeonju National University of Education,
55101, South Korea
[Email : *kchong@jnue.kr*]

Abstract

The paper is going to introduce Novel Engineering to integrate reading and engineering. Many researchers have investigated multidisciplinary education methods like STEAM education so far. STEAM education is a good way to integrate some subjects. But the education has some difficulties to apply to real field. So many researchers have been interested in Novel Engineering to integrate some subjects and engineering including AI education. Novel Engineering has so many advantages to improve thinking abilities like critical thinking, problem solving, creative thinking etc. And Novel Engineering is based on Seymour Papert's constructionism and Vygotsky's developmental theories. So the paper is going to introduce what Novel Engineering is and advantages based on previous researches. I hope that this study will serve as a useful reference for many teachers who wish to implement convergence education.

Index Terms: Novel Engineering, Constructionism, Socio-cultural theory, 2022 Revised Curriculum, Situated Learning

I. INTRODUCTION

The revised 2022 curriculum emphasized a man of self-directed, creative, cultured, and a person who practices consideration, sharing, and cooperation [1]. To implement this, each subject education and creative experience activity fostered self-management capabilities, knowledge information processing capabilities, creative thinking capabilities, aesthetic sensibility capabilities, cooperative communication capabilities, and community capabilities. These competencies are most effectively implemented across multiple subjects through curriculum reconstructing that transcends the boundaries of each subject. The representative keywords for establishing education that goes beyond education that effectively learns and utilizes existing knowledge and presents a new vision in schools are competence and convergence.

In Korea, convergence education has been implemented through STEAM education for a long time. However, in order to solve problems that contain the complexity and diversity of modern social issues along with social and academic demands, convergence classes that can get closer to teachers' teaching practices are urgently needed. To this end, the Novel Engineering convergence class model has been widely spread nationwide since 2016, and

many dissertations and presentations have been published to date. Therefore, this study aims to examine Novel Engineering, what advantages this class model has, and what theoretical foundation it is based on.

II. Novel Engineering

A. What Novel Engineering is

Novel Engineering is an engineering education method that combines reading and engineering that has been studied for a long time at the Center for Engineering Education and Outreach (CEEEO) at Tufts University near Boston, Massachusetts, USA [2]. Novel refers to all literary works such as fairy tales, novels, and poems, and engineering refers to the process of creating something meaningful. Novel engineering consists of seven stages, reading books, recognizing problems, designing solutions, implementing solutions, presenting and providing feedback, upgrading solutions, and reconstructing stories [3]. Research in South Korea began in 2016 and Website has been opened recently [4].

B. Theoretical Background

Seymour Papert, a former MIT professor, argued that the creative process of creating concrete objects that exist in the real world is very effective in increasing the efficiency of learning and in constructing knowledge. Papert expanded Piaget's epistemology and educational philosophy by integrating the building process of creating meaningful concrete objects into his teacher Piaget's constructivism [3].

Vygotsky defined social interaction that affects the cultural development of school-age children as imitation in a broad sense, and viewed imitation as determining the zone of proximal development. According to this logic, all school classes are based on cooperation. This is because cooperation through imitation is a good way to expand Vygotsky's zone of proximal development [5].

Novel Engineering classes are based on problem-centered learning. Therefore, group projects of about 10 classes are essential. Therefore, Novel Engineering must guarantee cooperation with peers, interaction with teachers, and learning in context. In addition, Novel Engineering reflects well the creative learning process suggested by Mitchell Resnick, a disciple of Seymour Papert and a professor at MIT. This creative learning process has a spiral structure of five stages: imagine, create, play, share, reflect, and imagine. Since Novel Engineering reflects these stages well, it is very effective in improving children's high problem-solving skills and creativity.

C. Educational Effects

Although it is impossible to list all the effects of Novel Engineering that have been studied to date, it has been proven that Novel Engineering is very effective in improving creative problem-solving skills, reading motivation, learning immersion (Yang Hyeon-mo, Song Hae-nam), and improving convergence attitudes (Kim Ji-yoon) [6,7,8,9].

Even now, there are many cases of classes using Novel Engineering for various convergence topics. There is a growing trend of class cases being published that deal with issues such as ecological environment issues, improving awareness of the disabled, character education, maker education, software education, reading education, and artificial intelligence education[10,11,12,13].

III. Result and Further Research

In this study, we looked at research on Novel Engineering, a convergence class model. Novel Engineering has been actively studied in Korea since 2016.

Novel Engineering is becoming established as a very powerful teaching method with many advantages including improving creative problem-solving skills. Many paradigms such as existing STEAM education, maker education, software education, artificial intelligence education, and reading education have emerged, but due to the development of too many teaching methods, teachers in the field are very confused about how to conduct classes. In contrast, Novel Engineering has seven stages, and teachers can reduce or expand the stages at their discretion depending on the development of students.

Teachers who have experienced convergence classes said that the greatest appeal of Novel Engineering is the simplicity and clarity of the class steps, and that it allows for student-centered classes because students also participate very highly in class. Therefore, we expect Novel Engineering to be a good teaching tool for teachers who are planning convergence classes.

REFERENCES

- [1] Ministry of Education, The 2022 revised curriculum, 2022.
- [2] Novel Engineering official website, <https://sites.tufts.edu/novelengineering/>
- [3] K.C. Hong, Y. S. Cho, "A Novel Engineering and Creative Learning Process Based on Constructionism", *Journal of Information and Communication Convergence Engineering*, pp.213-220, 2019.
- [4] Novel Engineering Korean Website, <https://novelengineering.co.kr/>
- [5] K. C. Hong, "Instructional Design of Novel Engineering based on Vygotsky's Cognitive Development Theory", *Journal of Practical Engineering Education*, pp. 909-918, 2024.
- [6] K. C. Hong, W. J. Lee, S. M. Kim, "The Effects of Novel Engineering on Improving Creative Problem-Solving Ability", *Journal of Industrial Convergence*, pp. 83-89, 2020.
- [7] T.R. Kim, S.K. Han, "SW Convergence Based Nobel Engineering (NE) Effects on Reading Motivation and Learning Immersion" *The Journal of Education*, pp.29-48, 2021.
- [8] H.M. Yang, T.Y. Kim, "The Effect of the Physical Computing Convergence Class Using Novel Engineering on the Learning Flow and the Creative Problem Solving Ability of Elementary School Students", *JOURNAL OF The Korean Association of information Education*, pp.557-569, 2021.
- [9] J. Y. Kim, K. H. Kim, "The Educational Effect of Novel Engineering on Artificial Intelligence Convergence Liberal Arts Course for Pre-service Teachers", *JOURNAL OF The Korean Association of information Education*, pp.507-515, 2022.
- [10] J. Y. Kim, "The Effect of Novel Engineering-based Artificial Intelligence Education Program on Convergence Attitude of the Gifted Students in Computing Convergence", *Journal of The Korea Society of Computer and Information*, pp.307-316, 2024.

- [11] S. Y. Yoo, J. K. Choi, “A Study on Design of Integrated Ecological Program for Elementary School Environmental Education Based on the Novel Engineering”, *Journal of Holistic Convergence Education*, pp.31-50, 2022.
- [12] B. R. Kim, H. J. Kim, “Impact of Novel Engineering-Based Ecological Environmental Education Program on Elementary School Students' Environmental Competency and Attitude toward Using Artificial Intelligence”, *Journal of Korean Practical Arts Education*, pp.187-206, 2024.
- [13] J. Y. Hong, “Development and application of SW program based on Novel Engineering for improving creative problem solving ability of elementary students”, *Journal of Digital Contents Society*, pp.2315-2321, 2018.

Network Architecture for SW based Vehicle Device Communication in Software-defined Vehicle

Taehyeon Kweon¹, Joosang Youn^{2*}

¹Department of IT Convergence, Dong-Eui University, Busan 43740, Republic of Korea

²Department of Artificial Intelligence, Dong-Eui University, Busan 43740, Republic of Korea
[Email : :dhxk9513@gmail.com, jsyoun@deu.ac.kr]

Abstract

The SDV(Software-defined Vehicle) controls vehicle functions through software using SOA(Service-Oriented Architecture) technology. It also integrates various vehicle functions through software to develop services. This approach is called SDV service reconfiguration. However, current SDV service reconfiguration is limited to in-vehicle services. Therefore, it cannot reconfigure services with other SDVs or external IoT devices. This paper proposes an SDV network architecture that supports end-to-end network connectivity between in-vehicle services, other SDVs, and external IoT devices. The proposed network architecture includes network components for external communication. It also proposes a dynamic registration feature for SDV services. To verify the effectiveness of the proposed network architecture, it was implemented using Docker and ROS2.

Index Terms: Edge Computing, IoT, Network Architecture, SDV

I. INTRODUCTION

The SDV(Software-defined Vehicle) controls vehicle functions through software using SOA(Service-Oriented Architecture) technology[1]. It also integrates various vehicle functions through software to develop services. This approach is called SDV service reconfiguration. This approach allows vehicle functions to be continuously updated without changing the hardware[2]. Major companies are currently adopting various functions and architectures based on the concept of SDV. As a representative example, Tesla integrates in-vehicle functions such as autonomous driving and vehicle control entirely through software[3]. However, current SDV service reconfiguration is limited to in-vehicle services. Therefore, it cannot reconfigure services with other SDVs or external IoT devices. This paper proposes an SDV network architecture that supports end-to-end network connectivity between in-vehicle services, other SDVs, and external IoT devices. The proposed network architecture includes network components for external communication. It also proposes a dynamic registration feature for SDV services

* Corresponding author

II. Network Architecture

A. network configuration in SDV

Fig. 1. Network configuration in SDV

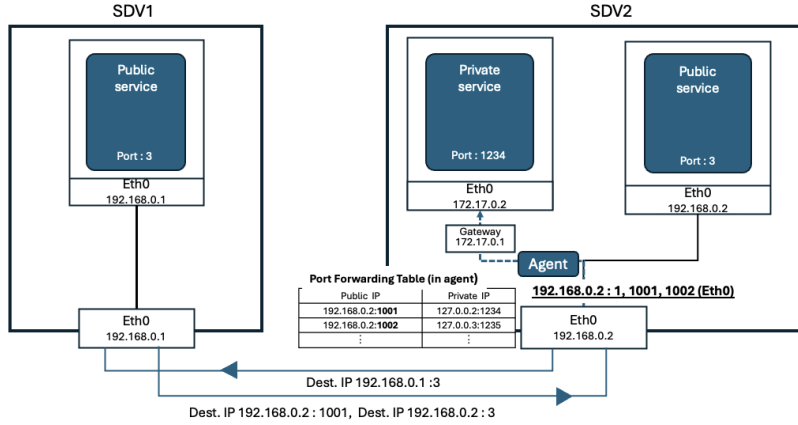


Fig 1 shows the network configuration for service connection within the SDV. Services are classified into two types based on external accessibility. A public service is accessible from outside, while a private service has restricted external access. Accordingly, the network configuration varies. Host mode assigns the host's IP address to the service, while bridge mode assigns an independent private IP address to the service. Public services are configured using host mode, which allows direct communication. In contrast, private services are configured using bridge mode and communicate indirectly through an agent. The service agent supports external communication for private services through port forwarding. The agent binds private services to the host IP and sequentially assigns port numbers starting from 1001. This information is managed through the port forwarding table. Table 1 shows the port forwarding table. Meanwhile, each service is assigned a unique port number for identification. Table 2 shows the port numbers defined for each service.

Table 1. agent port forwarding table

Destination IP, Port	Source IP, Port
192.168.0.2:1001	127.0.0.2:1234
192.168.0.2:1002	127.0.0.3:1234
192.168.0.2:1003	127.0.0.4:1236
192.168.0.2:1003	127.0.0.5:1236

Table 1. the port numbers defined for each service.

service	Port
Service register	1
Service agent	2
Public service	3 ~ 999
Private service	1200 ~ 2000
Port forwarding	1001 ~ 1199

B. Service Registration in SDV

To connect services, connection information about each other is required and is provided through the service registrar. The service registrar manages and provides connection information of services through the registrar table. Table 3 shows the structure of the service registrar table. The table structure includes SDV_ID, service name, service type, IP/Port, and service category. The service registration process differs according to the service type. A public service directly sends a registration request to the service registrar. A private service sends a registration request through the agent. The agent then sends the request to the service registrar, including port forwarding information. Through this process, the service registrar dynamically adds and updates services. Also, each SDV service registrar shares its registration table with others to synchronize.

Table 3. Service register table

SDV ID	Service name	Service type	IP, Port	Service category
SDV_1	Public Service_1	Public	192.168.0.1:2	Ridar
SDV_1	Private Service_1	Private	192.168.0.1:3	Agent
SDV_2	Public Service_1	Public	192.168.0.2:2	Ridar
SDV_2	Private Service_1	Private	192.168.0.2:3	Break

III. RESULTS

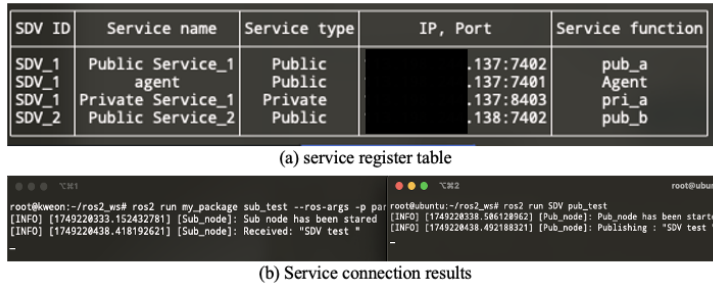


Fig. 2. Network Structure Implementation Results

To verify the effectiveness of the proposed network architecture, it was implemented using Docker and ROS2. SDV1 is configured with four services: public, private, agent, and service registrar, while SDV2 is configured with two services: public and service registrar. Fig 2 shows the table of the service registrar and allows us to see the data flow between services. As a result of the experiment, service registration and port forwarding operated correctly. Also, data was successfully transmitted between services.

IV. DISCUSSION AND CONCLUSIONS

In this paper, we proposed an SDV network architecture that supports network connectivity between in-vehicle services, other SDVs, and external IoT devices. The network is configured based on the service type, and connections are supported through the agent and the registrar. The experiment confirmed the effectiveness of the proposed architecture. Future work will involve designing an architecture that supports service reconfiguration.

ACKNOWLEDGMENTS

This research was supported in part by the Institute for Information and Communications Technology Planning and Evaluation (IITP) through the Korean Government [Ministry of Science and ICT (MSIT)] under Grant RS-2024-00346798 and the BB21plus funded by Busan Metropolitan City and Busan Techno Park.

REFERENCES

- [1] Y. Benomar, T. El-Ghazawi, M. Herbordt, A. Patel, and A. Bansal, "Software-Defined Vehicles: Challenges and Research Opportunities," *IEEE Access*, vol. 10, no. 1, pp. 84711–84732, 2022.
- [2] N. Bazzi, I. Ahmad, R. Ahmad, M. Alazab, and S. K. Singh, "Secure over-the-air software updates in connected vehicles: A survey," *Journal of Systems Architecture*, vol. 110, no. 1, p. 101805, 2020.
- [3] A. Keskar, "Enhancing Software-Defined Vehicles with Service-Oriented Architecture," *IRE Journals*, vol. 4, no. 5, pp. 90–102, 2025.

An Integrated Digital Twin Methodology for Production System in Smart Factories

Gyusun Hwang^{1*}

¹Department of Industrial Engineering, University of Ulsan, Republic of Korea
[Email : gyusunhwang@ulsan.ac.kr]

Abstract

Digital twin technology has emerged as a key enabler for achieving operational efficiency and innovation in smart manufacturing environments. However, many small and medium-sized enterprises (SMEs) face difficulties in implementing digital twins due to technical and organizational constraints. This study proposes an integrated and systematic methodology for developing digital twins specifically tailored to production systems, focusing on Korean SMEs within the automotive and shipbuilding supply chains.

The proposed methodology comprises five stages: (1) defining problems and identifying key performance indicators (KPIs), (2) modeling the physical system and structuring data, (3) developing simulation-based system behavior models using Plant Simulation, (4) deploying a feedback-enabled interface for monitoring and decision-making. In addition, a master planning simulation layer is introduced to connect multiple virtual factories within a supply chain ecosystem, enabling collaborative optimization among enterprises.

This methodology has been applied to actual manufacturing sites, demonstrating tangible improvements in throughput, lead time, and system visibility. The study not only supports factory-level innovation based on digital twins but also lays the groundwork for expanding into a data-driven supply chain simulation framework.

Index Terms: Digital Twin; Discrete Event Simulation; Key Performance Indicator

I. INTRODUCTION

Digital twin technology has rapidly become a foundational element in the advancement of smart manufacturing systems. Through real-time synchronization between physical assets and their digital counterparts, digital twins enable enhanced monitoring, predictive analytics, and informed decision-making (Tao et al., 2019). Despite their potential, the widespread adoption of digital twin technologies remains limited, particularly among small and medium-sized enterprises (SMEs). Technical limitations, high implementation costs, and a lack of digital infrastructure present considerable challenges (Lee et al., 2015).

This paper proposes an integrated and systematic methodology for the development of digital twins focused on production systems, with particular application to Korean SMEs in the automotive and shipbuilding supply chains. The approach is designed to be both scalable and practical, facilitating real-time operational improvements and strategic supply chain collaboration.

II. Related work

Prior research in digital twin development has primarily focused on large-scale implementations in high-tech or capital-intensive industries. While frameworks for system modeling, simulation integration, and real-time data processing have been explored, relatively few studies address the unique needs and constraints of SMEs.

Works such as Tao et al. (2019) and Boschert & Rosen (2016) emphasize the conceptual framework of digital twins, while others like Kritzing et al. (2018) delve into classification approaches relevant to manufacturing contexts. Schluse et al. (2018) propose experimentable digital twins to streamline simulation-based systems engineering, which aligns with the practical modeling goals of this study. Lee et al. (2015) provide a cyber-physical systems (CPS) framework foundational to real-time integration in manufacturing environments. Negri et al. (2017) review digital twin roles in CPS-based production systems, reinforcing the need for cohesive frameworks that integrate across IT and OT domains.

III. Proposed methodology

The proposed digital twin methodology is structured into five key stages:

Stage 1: Problem Definition and KPI Identification This phase involves engaging stakeholders to define key operational challenges and performance goals. Representative KPIs include throughput, cycle time, work-in-process (WIP), and resource utilization.

Stage 2: Physical System Modeling and Data Structuring Using tools such as CAD layouts and process flow diagrams, the physical production environment is modeled. Data requirements are identified, and data collection protocols are established, typically involving sensors, PLCs, or manual inputs.

Stage 3: Simulation-Based System Behavior Modeling Plant Simulation is employed to develop dynamic models of production processes. These models reflect the real-world system's constraints and logic, allowing for scenario-based experimentation and performance forecasting.

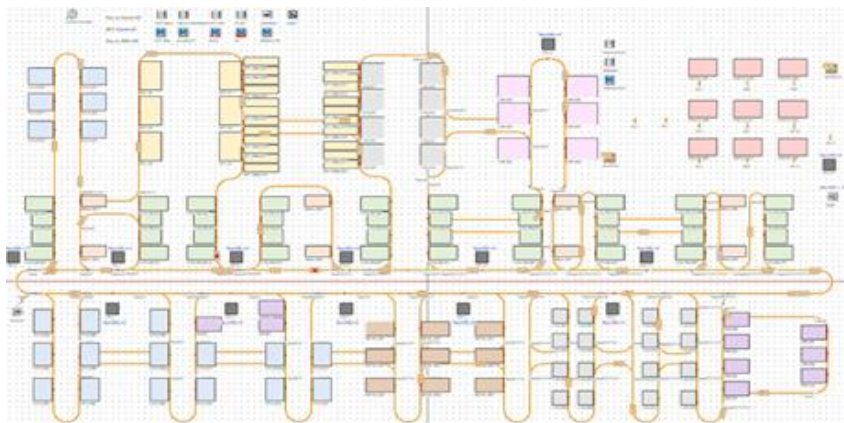


Fig. 1. Plant simulation development example

Stage 4: Feedback-Enabled Monitoring and Decision Interface A real-time data integration framework is developed using MQTT protocols and MariaDB for data storage. Python scripts or Node-RED interfaces connect simulation outputs with visualization dashboards (e.g., Grafana), supporting operational monitoring and decision-making.

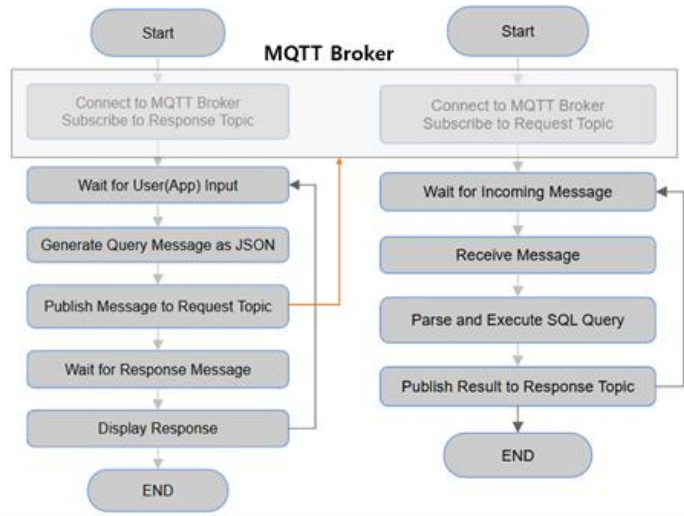


Fig. 1. Plant simulation development example

Stage 5: Master Planning Simulation Layer Multiple virtual factories are linked through a master planning layer that simulates supply chain-level behaviors. This supports coordinated production planning, capacity sharing, and inter-firm optimization within the ecosystem.

IV. Case study: Application in Korean Manufacturing SMEs

The methodology was implemented across three manufacturing SMEs in the Ulsan region—two in the automotive parts sector and one in shipbuilding components. Each firm faced distinct operational challenges, ranging from production bottlenecks to unbalanced workstations.

The digital twin models were tailored to each context. Improvements observed included an average 14% increase in throughput, a 19% reduction in lead time, and significantly improved system visibility through live dashboards.

Several practical insights emerged from the field application. First, stakeholder engagement in the early stages proved critical for accurate KPI selection and model validation. Second, integrating legacy equipment required adaptable middleware solutions. Finally, visual dashboards were essential in building trust and promoting system adoption.

The master planning simulation demonstrated potential for scalable supply chain coordination, though limitations in data standardization and API integration between firms remain a challenge.

V. Conclusion and future work

This paper presents a structured, scalable methodology for developing digital twins in production systems, specifically suited to SMEs. By integrating simulation, real-time data, and decision support, the approach fosters innovation at both the factory and supply chain levels.

Future work will focus on incorporating AI-driven optimization models, extending the platform to multi-tier supply networks, and enhancing interoperability through standard communication protocols such as OPC UA and B2MML.

ACKNOWLEDGMENTS

This work was supported by the National Research Foundation of Korea(NRF) grant funded by the Korea government(MSIT) No. RS-2025-00213861).

REFERENCES

- [1] Tao, F., Zhang, H., Liu, A., & Nee, A. Y. C. (2019). Digital twin in industry: State-of-the-art. *IEEE Transactions on Industrial Informatics*, 15(4), 2405–2415.
- [2] Boschert, S., & Rosen, R. (2016). Digital twin—the simulation aspect. In *Mechatronic Futures* (pp. 59–74). Springer.
- [3] Kritzinger, W., Karner, M., Traar, G., Henjes, J., & Sihn, W. (2018). Digital twin in manufacturing: A categorical literature review and classification. *IFAC-PapersOnLine*, 51(11), 1016–1022.
- [4] Schluse, M., Atorf, L., & Rossmann, J. (2018). Experimentable digital twins—Streamlining simulation-based systems engineering for industry 4.0. *IEEE Transactions on Industrial Informatics*, 14(4), 1722–1731.
- [5] Lee, J., Bagheri, B., & Kao, H. A. (2015). A cyber-physical systems architecture for industry 4.0-based manufacturing systems. *Manufacturing Letters*, 3, 18–23.
- [6] Negri, E., Fumagalli, L., & Macchi, M. (2017). A review of the roles of digital twin in CPS-based production systems. *Procedia Manufacturing*, 11, 939–948.

Development of user interaction technology for multi-content control in a holographic environment

Chan-Sung Jung¹, Hyun-Chang Lee^{2*}

¹Department of Game Content, Wonkwang University, 54538, Korea

² Department of Game Content, Wonkwang University, 54538, Korea
[Email : {placomp, hclglory}@wku. Ac.kr]

Abstract

The floating hologram environment is represented as a technology for expressing realistic content. Display devices that express holograms are using methods that repeatedly show various contents with intermittent time differences, or using expensive additional devices to show many contents. This study is about the design and development of an interface that allows user interaction so that users can actively control realistic contents expressed as holograms. The advantage of this study is that it does not require expensive additional equipment, and various multi-contents can be actively controlled and selected based on existing computer basic specifications (including cameras). This technology allows for the expression of various contents in a limited space like a hologram environment, which increases space utilization, and it is a display device that can control holographic multi-realistic contents.

Index Terms: AI, Hand gesture, Space utilization, Hologram environment, Realistic content

I. INTRODUCTION

Today's holographic technology is introduced as a technology that reproduces realistic stereoscopic images by reconstructing the properties of an object into a three-dimensional image by reflecting light from the surface. Stereoscopic images created by holograms maximize realism compared to 3D stereoscopic images implemented on existing flat screens in that they provide a three-dimensional effect similar to the actual appearance, and they can provide the advantage of relatively wide visualization because they can be visualized from various angles [2]. In addition, realistic content is classified as content that maximizes realism by providing an experience similar to reality through stimulation of the five senses of humans [3]. Realistic content is a technology that is widely used for the purpose of reproducing content, and many technologies related to holograms are being developed, and their role as a means to overcome social problems is gradually expanding as a result of the development of application technologies. As an example, it can be utilized from core source technologies for application services such as non-face-to-face treatment and remote guidance using holograms to technological means that can be used

* Hyun-Chang Lee

in daily life. In this process, application services that develop holograms to suit the times, such as technologies for non-face-to-face and remote collaboration such as the metaverse, can also be cited as examples. Despite the various uses of immersive content, there are many difficulties in the practical stage of using hologram technology as immersive content. Furthermore, the development of technology to express and control multi-immersive content in a limited space, such as floating holograms, requires expensive additional equipment. In this study, we explain the screen design and technology development for expressing and controlling multi-content based on basic computer specifications in a limited space.

II. RELATED WORKS

Holograms are a technology that records and reproduces the distribution of light that is reflected or diffracted from a real image and reproduces a 360-degree stereoscopic image of an object by reproducing the image of the real image in space [5]. As a technology for reproducing, pseudo-holograms and analog and digital hologram technologies can be considered [4].

Pseudo-holograms express a sense of reality and three-dimensionality by projecting high-resolution images onto a two-dimensional transparent screen, and since objects can be viewed three-dimensionally without wearing 3D glasses, visual fatigue is low, and long-term viewing is possible without wearing auxiliary devices like VR/AR. Holograms with these advantages are used by dividing them into direct projection and indirect projection methods according to the image projection method.

In particular, many studies are being conducted on content control through hand gesture recognition, and improved algorithms are continuously being announced. In order to recognize the user's gestures, a sensor or additional device that can receive gesture recognition data is required. Generally, expensive equipment such as Leap Motion or RealSense are used as non-contact technologies.

In this study, we describe the development of technology for hand gesture recognition and content control using a general camera included in the basic computer specifications without requiring expensive additional equipment.

. III. Proposed model for content control

A. Designing Interface for User Interaction

Today, the most commonly represented hologram is handled as in the first picture of Figure 1, and one holographic content is expressed on one device. As an image other than that, a scene from a movie is shown, but it is expected that future holograms will be shown as realistic holographic content like the second or third image. Based on this technology development roadmap, this study will develop technology so that a camera can recognize a user's hand and express content according to the hand in the air like in a movie.

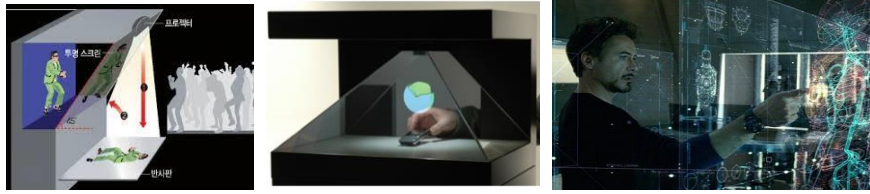


Fig. 1. Technical development flow for user interaction in holographic environment

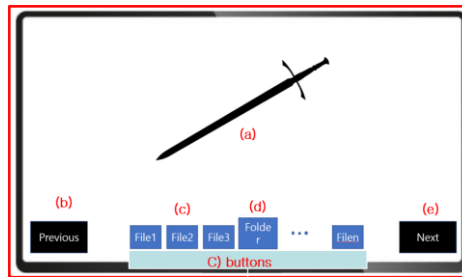


Fig. 2. Designing interface for user interaction

Figure 2 illustrates the screen design for the content background to be displayed on the monitor screen and the selection method. In Figure 2, (a) illustrates the content that appears when one button (content file) in the (c) area at the bottom is selected. The (b) button performs the same function as the previous page when there are many contents. Similarly, the button that performs the next page function corresponds to (e). The (d) button expresses a button for organizing contents composed of multiple parts into one folder. Based on this screen design, the user's hand position is monitored in real time through a camera so that the content selection position can be estimated.

B. Development of a Prototype System for User Interaction

In this chapter, we present the results of a prototype system based on the screen design for using multiple contents through hand recognition. First, Figure 3 shows the process of controlling the hand recognition process and the contents expressed as holograms by inputting hand information based on a camera mounted on a general computer. Figure 4 shows a snapshot implemented based on the screen design.

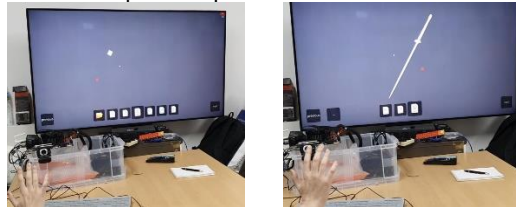


Fig. 3. Designing interface for user interaction

As shown in the figure, at the bottom of the screen in Figure 4 (a), there are buttons for selecting various contents, and a snapshot is shown when one of them is selected. (b) shows the content, which is a sword, when one of the other contents at the bottom or the contents that appear after selecting the “Next” button is selected.

IV. CONCLUSIONS

The technology developed in this study is expected to have the expected effect of enabling experience by grafting interaction technology with users in places or fields where visualization is important using holograms such as museums and exhibition halls. In addition, it is expected to be even more important in autonomous vehicles, which are spurring recent technological development, and environments where humans and machines can interact. As the need for interaction between humans and computers or devices increases, the significance of the development of AI-based interaction technology performed in this study will become even greater. In future research, we plan to present a model that requires rapid response processing and accuracy so that realistic content can interact smoothly with humans, and to conduct research on the development of a portable device prototype.

ACKNOWLEDGMENTS

This paper was supported by Wonkwang University in 2024.

REFERENCES

- [1] Y.J. Park, B.C. Choi, “The destination of realistic media video: digital holograms, KISTEP, vol.267, no.9, pp. 1-27, 2019.
- [2] S.J. Jung, T.H. Lee, “A Study on the Present Condition and Prospect of Architectural Cultural Heritage Content Applied to 3D Digital Technology,” *Journal of the Korea Academia-Industrial Cooperation Society*, vol. 22, no. 2. Pp. 188-198, 2021.
- [3] H.K. Joe, “A Study on the use of Immersive Media Contents design based on extended reality(XR) technology in digital transformation era,” *Journal of the Korea Society Design Culture*, vol. 26. no. 4. Pp. 497-507, 2020.
- [4] W.J. Lee, E.J. Shin, H.S. Yoon, H.M. Choi, “Realistic Contents and Interaction Based Realistic Contents Service,” *KIPS Trans, Soft and Data Eng.* vol. 10. no. 10. pp. 429-438, 2021.
- [5] MediaPipe Solutions, <https://mediapipe.readthedocs.io/en/latest/solutions/hands.html>. 2023.
- [6] S.W. Lee, H.C. Lee, “Implementation of a user interaction system for wireless control of realistic content through hand recognition and tracking,” *Journal of Next-generation Convergence Technology Association*,” vol.8. no.10. pp.2057-2064, 2024.

Vision Language Framework with Low Cost Device for Robot Agent

Nobutaka Shimada¹, Jehwan Choi², and Kanghyun Jo^{2*}

¹Dept. of Information and Engineering, Ritsumeikan University, 603-8577, Japan

²Dept. of Electrical, Electronic and Computer Engineering, University of Ulsan, 44610, Korea
[Email : nshimada@is.ritsumei.ac.jp, {cjh1897/acejo}@ulsan.ac.kr]

Abstract

Deploying Vision-Language Models (VLMs) on resource-constrained edge devices, such as Jetson Nano, is crucial for developing intelligent robotic agents capable of real-time environmental understanding and human-robot interaction. However, most state-of-the-art VLMs require substantial computational resources, making them impractical for low-power, embedded systems. To overcome this limitation, we propose a novel vision-language processing framework tailored for Jetson Nano using compact and quantization-friendly Hugging Face VLM models. Specifically, the framework integrates an OAK-D Pro camera for capturing and pre-processing visual input, a lightweight Vision Encoder for real-time feature extraction, a Multimodal Processor for generating descriptive textual outputs, and a Task Decision Unit for converting multimodal information into actionable robotic commands. Preliminary analysis indicates that our approach can achieve near-real-time multimodal inference (1-2 FPS) with manageable memory consumption (within 4GB). Our work lays the groundwork for implementing practical, cost-effective, and interactive multimodal robotic systems on low-cost hardware platforms.

Index Terms: Vision Language Model, Human-System Interaction, Artificial Intelligent.

I. INTRODUCTION

As noted in [1], text-to-image and image-to-text generation techniques have recently received significant research attention and such systems are commonly referred to as Vision-Language Models (VLMs). Utilizing VLMs on edge devices such as Jetson Nano to allow robotic agents to interpret their surroundings and interact via natural language represents an essential direction toward realizing next-generation intelligent robots. However, as discussed in [2], large-scale multimodal models typically require extensive computational resources and memory, making them challenging to run on edge devices in real-time applications. For example, Jetson Nano has a limited computing capability with a 128-core Maxwell GPU (maximum FP16 performance of 472 GFLOPS) and only 4GB of memory, which severely restricts the execution of modern VLMs that often contain billions of parameters. In this study, considering these limitations, we select a relatively lightweight Hugging Face VLM compatible with Jetson Nano and propose a practical framework capable of processing video data captured from an OAK-D Pro camera in real-time, enabling efficient decision-making for robotic agents.

* Corresponding author

II. RELATED WORK

Vision-Language Models (VLMs) are deep learning models that process both images and text simultaneously, demonstrating strong performance in tasks such as image captioning and Visual Question Answering (VQA). A representative example is Salesforce's BLIP model [3], which was proposed as a unified VLM capable of both caption generation and question answering through image-text pretraining. The base version of BLIP contains approximately 250 million parameters, making it relatively lightweight and potentially executable on devices like the Jetson Nano. In contrast, more recent models such as MiniGPT-4 [4] and LLaVA [5] integrate large language models (LLMs) into their architectures. For instance, MiniGPT-4 combines the vision encoder from BLIP-2 [6] with a state-of-the-art LLM such as Vicuna [7] (with 7 billion parameters) to achieve multimodal capabilities similar to GPT-4. It aligns the fixed vision encoder with the Vicuna LLM using a single projection layer. LLaVA, on the other hand, employs the OpenAI CLIP [8] vision encoder to embed images and combines this with an LLaMA-based LLM, resulting in a representative VQA model that achieves vision-language integration by mapping images into the LLM's embedding space.

However, these models incorporate large language models such as Vicuna-7B and LLaMA-13B, resulting in extremely high resource consumption. For example, the LLaVA-13B model requires more than 27GB of memory when running in floating-point precision (FP16), and even when reduced to a 4-bit quantized version, it still consumes around 9GB. This makes it virtually impossible to deploy directly on a Jetson Nano, which has only 4GB of memory. Similarly, MiniGPT-4, which is based on either Vicuna-7B (approximately 7 billion parameters) or Vicuna-13B, is not suitable for Jetson Nano due to its computational demands, let alone achieving real-time performance. To address these limitations, research on lightweight VLMs has gained attention. The NanoVLMs project [9], for instance, proposed ultra-compact VLMs with 5 to 25 million parameters, optimizing parameter efficiency through the use of compact vision encoders, lightweight language decoders, and simple connector modules. Additionally, Moondream [10] introduced a version reduced from 1.9 billion to 500 million parameters, demonstrating that aggressive quantization (FP16, INT8, and even INT4) can enable these models to run on mobile GPUs and CPUs. The use of INT4 in particular shows a promising strategy for minimizing memory usage and latency while preserving performance.

Hugging Face has also released lightweight VLMs optimized for edge deployment. For example, NVIDIA's recently introduced VILA model [11] consists of approximately 3.01 billion parameters and supports inference on edge devices such as Jetson Orin and laptops through 4-bit quantization. It also offers advanced features such as multi-image input. More recently, models like Obsidian-3B [12], with fewer than 3 billion parameters, have been developed to understand visual inputs and demonstrate performance comparable to some 7B-scale models. These research efforts reflect a growing trend toward designing task-specific, mid-sized VLMs suitable for edge environments, aiming to improve the trade-off between performance and efficiency through techniques such as quantization and knowledge distillation.

III. PROPOSED FRAMEWORK

In this study, we propose a multimodal perception-action framework designed for robotic agents. The overall system is structured as a pipeline architecture that receives visual input from a camera, interprets the scene, makes task decisions based on the situation, and executes corresponding robotic actions. The key modules of the framework are described in Sections A through D.

A. Vision Encoder

The image encoder component of the VLM model utilizes either the ViT backbone from BLIP or a CLIP-based encoder. Its primary role is to receive streaming video frames from the OAK-D Pro camera in real time and convert them into feature vectors. If necessary, the on-device AI capabilities of the OAK-D Pro can be used for preprocessing tasks such as object detection and cropping regions of interest. Since the OAK-D Pro is equipped with an Intel Movidius Myriad X VPU, capable of up to 4 TOPS (1.4 TOPS dedicated to AI workloads), it can handle lightweight object recognition or depth estimation directly on-board. This allows the Jetson Nano to focus on higher-level inference tasks. The resulting image embeddings or token sequences from the encoder are then passed on to the subsequent multimodal processor.

B. Multimodal Processor

This module receives the output from the vision encoder and performs high-level inference. When necessary, it also accepts textual input. The Hugging Face VLM model itself serves this role, employing a Transformer-based architecture that integrates image and text inputs to generate contextual language descriptions or responses. For example, when using BLIP, the model generates captions from visual features, while models like LLaVA or MiniGPT-4 take image embeddings as part of the language model's context to answer questions or engage in dialogue. Since this framework is designed to operate on the Jetson Nano, a lightweight VLM is employed, resulting in relatively simple language outputs from the multimodal processor. In practical terms, the robot may generate a natural language description such as “a person is waving” based on what it sees through the camera.

C. Task Decision Unit

This module receives the language output or representation generated by the Multimodal Processor and determines the robot's next action. It functions as the robot's decision-making component and can be implemented using rule-based logic or a lightweight, pre-trained decision model. For instance, if the VLM output is “a person is waving,” the Task Decision Unit interprets this and generates a command such as “approach the person and greet them.” In more complex systems, the decision module can be structured using Behavior Trees or integrated with reinforcement learning algorithms, allowing it to form higher-level plans based on environmental analysis provided by the VLM. In this study's framework, we consider a mapping between VLM outputs and predefined action rules, as well as the use of a small-scale LLM to perform follow-up queries when additional reasoning is required for action selection.

D. Action Module

The Action Module is the subsystem responsible for executing the final robotic commands determined by the Task Decision Unit. Although the actual implementation of this module is not covered in this paper, it is intended to be tested in a virtual environment in future work. This module carries out concrete actuator-level operations such as motor control, navigation commands, or speech output, depending on the task. For instance, if the Task Decision Unit issues a command like “approach the person and wave back,” the Action Module would control the robot’s locomotion system to move toward the target and actuate the arm joints accordingly. This module can be connected to low-level control systems implemented using middleware such as ROS2, and is designed to be modular and decoupled from the higher-level decision logic.

The four components of the framework operate in a continuous pipeline to enable real-time processing on a per-frame basis. The OAK-D Pro camera streams video frames via the DepthAI API into a Python environment, where each frame is sequentially processed through the Vision Encoder, Multimodal Processor, Task Decision Unit, and Action Module. A parallelized pipeline structure can be adopted in which, for example, while frame t is undergoing multimodal inference, frame $t+1$ is already being preprocessed by the encoder. Additionally, stereo depth information from the OAK-D Pro camera can be utilized for specific tasks such as estimating the distance to obstacles or approaching individuals. This allows the Task Decision Unit to incorporate both the VLM’s semantic interpretation and spatial information into more informed decision-making.

IV. CONCLUSION

This study presents a framework that enables robotic agents to understand and respond to visual information using a lightweight Vision-Language Model (VLM) deployable on Jetson Nano. By selecting compact models such as BLIP, the system can perform tasks like caption generation, scene description, and visual question answering within the resource constraints of the device. The proposed architecture follows a hierarchical pipeline structure consisting of a Vision Encoder, Multimodal Processor, Task Decision Unit, and Action Module, each contributing to the processing flow from visual input to robotic output. Based on Jetson Nano’s hardware specifications and modern VLM optimization techniques, the system is expected to achieve processing speeds of over one frame per second, demonstrating its potential for real-time human-robot interaction.

Although, the system still faces limitations, including reduced performance compared to large-scale models, constrained real-time capabilities, and possible uncertainty in model outputs. Ongoing research is needed to improve the trade-offs between accuracy and efficiency in edge-deployed VLMs. Future directions include migrating to more powerful platforms such as Jetson Orin, implementing distributed or collaborative inference strategies on Jetson Nano, fine-tuning VLMs with domain-specific knowledge for robotics, and integrating learning-based decision-making modules such as reinforcement learning. As demonstrated in this study, integrating Hugging Face’s multimodal AI models with DepthAI camera systems provides a viable path toward building intelligent, low-cost robotic agents suitable for embedded environments.

REFERENCES

- [1] Zhou, Y., & Shimada, N. (2023). Vision+ language applications: A survey. In Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (pp. 826-842).
- [2] Shinde, G., Ravi, A., Dey, E., Sakib, S., Rampure, M., & Roy, N. (2025). A Survey on Efficient Vision-Language Models. arXiv preprint arXiv:2504.09724.
- [3] Li, J., Li, D., Xiong, C., & Hoi, S. (2022, June). Blip: Bootstrapping language-image pre-training for unified vision-language understanding and generation. In International conference on machine learning (pp. 12888-12900). PMLR.
- [4] Zhu, D., Chen, J., Shen, X., Li, X., & Elhoseiny, M. (2023). Minigpt-4: Enhancing vision-language understanding with advanced large language models. arXiv preprint arXiv:2304.10592.
- [5] Liu, H., Li, C., Wu, Q., & Lee, Y. J. (2023). Visual instruction tuning. Advances in neural information processing systems, 36, 34892-34916.
- [6] Li, J., Li, D., Savarese, S., & Hoi, S. (2023, July). Blip-2: Bootstrapping language-image pre-training with frozen image encoders and large language models. In International conference on machine learning (pp. 19730-19742). PMLR.
- [7] Zheng, L., Chiang, W. L., Sheng, Y., Zhuang, S., Wu, Z., Zhuang, Y., ... & Stoica, I. (2023). Judging llm-as-a-judge with mt-bench and chatbot arena. Advances in Neural Information Processing Systems, 36, 46595-46623.
- [8] Radford, A., Kim, J. W., Hallacy, C., Ramesh, A., Goh, G., Agarwal, S., ... & Sutskever, I. (2021, July). Learning transferable visual models from natural language supervision. In International conference on machine learning (pp. 8748-8763). PMLR.
- [9] Agarwalla, M., Kumar, H., Dandekar, R., Dandekar, R., & Panat, S. (2025). NanoVLMs: How small can we go and still make coherent Vision Language Models?. arXiv preprint arXiv:2502.07838.
- [10] Moondream AI. (n.d.). Moondream: Open-source vision-language model for local use. Retrieved June 13, 2025, from <https://moondream.ai/>
- [11] Liu, Z., Zhu, L., Shi, B., Zhang, Z., Lou, Y., Yang, S., ... & Lu, Y. (2025). Nvlla: Efficient frontier visual language models. In Proceedings of the Computer Vision and Pattern Recognition Conference (pp. 4122-4134).
- [12] Nguyen, Q., & Daniele. (2023). Obsidian-3B: First multi-modal below 7B parameters. Hugging Face. Retrieved from <https://huggingface.co/NousResearch/Obsidian-3B-V0.5>

Session IT-B

IT Convergence Technology

Session IB-A

IoT and Big Data

Design of Oral Diseases Identification System for Post-Implant Gum Complications

Nanda Fanzury¹, Aria Bisma Wahyutama¹, and Mintae Hwang^{1*}

¹Department of Information and Communications Engineering, Changwon National University,
Changwon 51140, Republic of Korea
[Email : mthwang@cwnu.ac.kr]

Abstract

This paper presents a system designed to identify gum complications after implant surgery using object detection algorithms embedded in a mobile application. The system allows patients to self-diagnose conditions such as swelling, bleeding, and gum recession, promoting early intervention and improving oral health-related quality of life (OHRQoL). The proposed approach consists of an oral image dataset gathering, training object detection model, and development of a web server and a mobile interface. Four object detection algorithms (YOLOv8, YOLOv11, Faster R-CNN, and Mask R-CNN) will be compared to determine the most suitable model. Despite challenges related to data variability, privacy, and real-time processing, the proposed system provides a promising approach to home-based oral care, facilitating seamless communication between mobile applications and intra-oral cameras.

Index Terms: Identification System, Object Detection, Oral Diseases, Post-Implant Gum Complication.

I. INTRODUCTION

After implant surgery, patients may experience various changes in gum tissue, impacting their overall oral condition, including gum recession, alterations in keratinized tissue width, and shifts in oral health-related quality of life (OHRQoL) [1]. Patients often report temporary declines in OHRQoL following surgery for around two weeks. Pain and swelling are also common, with over half of the patients experiencing mild pain [2].

Identifying gum complications at an early stage is important for maintaining oral health and preventing severe outcomes. This early identification can help mitigate the risk of more serious issues, including root caries, tooth loss, and significant discomfort related to periodontal disease. Therefore, symptoms like swelling, bleeding, and redness often signal underlying gum problems that can progress if left untreated [3-4].

Based on these problems, this paper proposed a system that can identify gum complications using object detection algorithm that can be embedded into a portable intra-

* Corresponding author

oral camera. With this system, patients can self-diagnose their gum condition after implant surgery and quickly schedule an appointment with dentists if needed.

II. SYSTEM MODEL AND DESIGN

This system requires four core components: the dataset, object detection algorithm, server, and a mobile application. The dataset consists of labeled images of patients' oral conditions after implant surgery.

Once the model reaches satisfactory performance metrics, it is deployed on a web server. The server plays a key role in processing images uploaded by the mobile application and returning detection results quickly. Efficient server architecture ensures scalability and supports real-time or near-real-time processing. The mobile application acts as the primary interface for patients, enabling them to connect a portable intra-oral camera to capture images of their gum condition. These images are sent to the server for analysis, and the results are displayed within the app. Patients can view identified complications and take appropriate action, such as scheduling an appointment with a dentist directly through the application.

Fig. 1 shows the top-view of the proposed system architecture which form a seamless communication between the dataset, detection model, server, and mobile application.

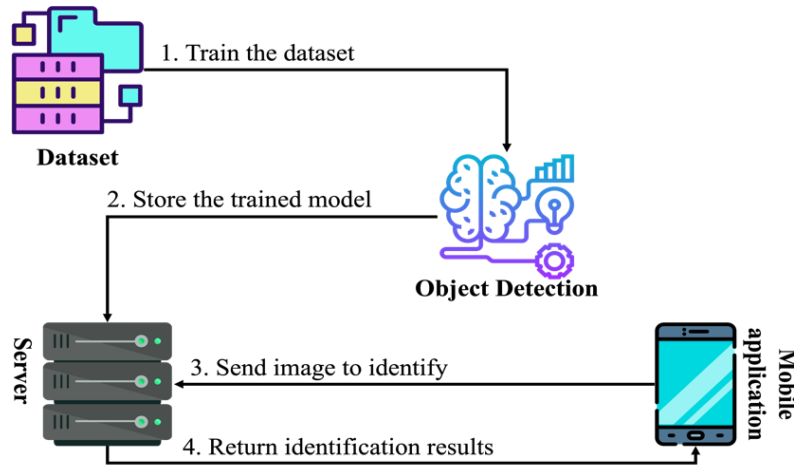


Fig. 1. Top-down view of the system architecture.

This proposed system will collaborate closely with dentists and dental clinics to collect a comprehensive dataset of patients' oral conditions following implant surgery. The dataset will emphasize key gum complications, including swelling, gum recession, bleeding, plaque buildup, and redness. The dataset will be labeled by applying polygonal segmentation mask, thus the object detection algorithm not only identifying these issues with bounding boxes but also segmenting the affected areas for precise localization. This detailed segmentation

can assist dentists in pinpointing the complications more accurately. To achieve optimal performance, the authors will compare four widely-used object detection algorithms (YOLOv8, YOLOv11, Faster R-CNN, and Mask R-CNN) to determine the most accurate and efficient model for identifying gum complications in real-world conditions.

Fig. 2 shows the initial User Interface (UI) design of the mobile application. On the left image, it shows the welcome screen where patients can connect the intraoral camera to the application. On the middle image, patients can capture their oral condition using the button. Once captured, the image is sent to the server to be identified. After that, the identification results are shown to the patients along with an option to schedule an appointment with dentist.

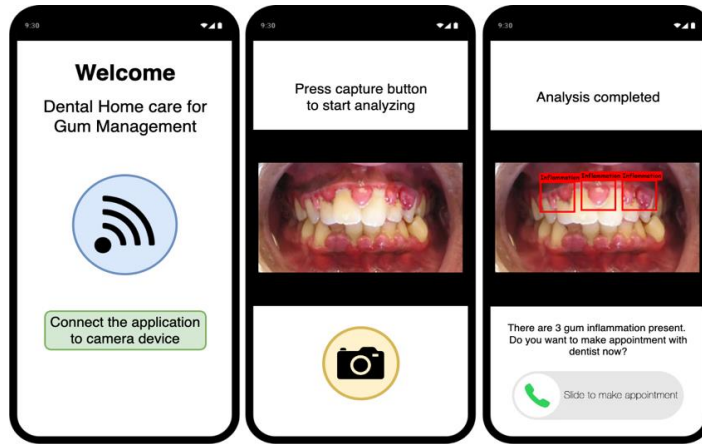


Fig. 2. Initial User Interface (UI) design for the mobile application.

III. CHALLENGES AND ISSUES

Gathering high-quality oral images as a dataset must include diverse cases of gum complications. Furthermore, the difference of appearance between each gum complications must be captured in detail to generate a high-performance model. Otherwise, the model will experience multiple misclassification due to similarity of gum complication's characteristic. Additionally, variability in lighting conditions, camera angles, and image resolution can introduce inconsistencies, can also potentially reducing the accuracy of object detection algorithms. Standardizing image capture protocols and using advanced image preprocessing techniques can help mitigate these challenges.

Additionally, patients' privacy and ethical concerns are paramount. Compliance with data protection regulations, such as anonymizing patient data and securing image transmission, must be strictly enforced to ensure confidentiality and trust. Ethical considerations in handling sensitive medical data require obtaining patient consent and adhering to relevant medical data guidelines. A high privacy level must also be considered when developing the web server since every patient's oral image will be uploaded and stored there.

Each object detection algorithm considered for this system has unique strengths and

limitations. Algorithms like YOLOv8 and YOLOv11 are known for their speed and efficiency, making them suitable for real-time applications. In contrast, Faster R-CNN and Mask R-CNN offer high accuracy and detailed segmentation, which can be beneficial for detecting and localizing complex gum complications. Therefore, a thorough comparison of these algorithms is necessary to identify the one that best balances speed, accuracy, and robustness for this specific application, especially if mobile application is involved.

The interoperability between all system components is crucial, particularly when including a real-time processing feature. Maintaining efficient communication between the mobile application and the intra-oral camera poses significant technical challenges such as latency and synchronization. Real-time processing demands lightweight, efficient algorithms capable of delivering accurate results quickly, without overwhelming the mobile device or server resources. Optimizing the system's architecture and communication protocols will be essential to achieving a reliable and user-friendly solution.

IV. CONCLUSIONS AND FUTURE STUDIES

This paper proposed a system utilize object detection algorithms embedded in a mobile application identify gum complications after implant surgery. By enabling patients to self-assess their gum condition, the system aids in early detection of issues like swelling, bleeding, and recession, promoting timely dental intervention and improved OHRQoL. The intraoral camera will connect wirelessly to the mobile application which will feed a real-time video footage. The mobile application can then capture the image from the intraoral camera and save them locally. Once captured, the image will be sent to the server for object detection process and then receive the results. The object detection results will be display in the mobile application, thus patients can know whether they have diseases and whether they must schedule a dental appointment for further treatment.

Despite challenges related to data quality, privacy, and real-time processing, the approach on this paper offers a promising solution for a home care system by providing a seamless integration between mobile applications and object detection algorithms. Future work will include gathering the dataset, comparing multiple models of object detection, train the model and then implement them into a mobile application via a web server.

REFERENCES

- [1] A. Lazarov, "Soft-tissue augmentation in periodontally compromised patients during immediate placement and immediate loading dental implant surgery - A retrospective study," *Ann. Maxillofac. Surg.*, vol. 13, no. 1, pp. 37–43, 2023.
- [2] S. Kamankatgan, A. Pimkhaokham, and S. Krisdapong, "Patient-based outcomes following surgical implant placements," *Clin. Oral Implants Res.*, vol. 28, no. 1, pp. 17–23, 2017.
- [3] A. S. Jati, L. Z. Furquim, and A. Consolaro, "Gingival recession: its causes and types, and the importance of orthodontic treatment," *Dental Press J. Orthod.*, vol. 21, no. 3, pp. 18–29, 2016.
- [4] A. Kumar, "Gingival recession: Narrow perception - erroneous; wider canvass - out-of-focus!," *J. Indian Soc. Periodontol.*, vol. 27, no. 3, pp. 227–229, 2023.

Design of an LLM-powered Chat System with Summarization and Knowledge Retrieval Features

Aria Bisma Wahyutama¹ and Mintae Hwang^{1*}

¹Department of Information and Communication Engineering, Changwon National University,
Changwon 51140, Republic of Korea

[Email: wahyutama@gs.cwnu.ac.kr, mthwang@cwnu.ac.kr]

Abstract

This paper presents the design of an LLM-powered chat system for group-based collaboration, integrating automated conversation summarization and document-based knowledge retrieval. Leveraging a Retrieval-Augmented Generation (RAG) framework, the system enables users to generate chat summaries and extract relevant knowledge from uploaded documents through natural language queries. The proposed architecture includes a frontend chat interface, a backend processing server, LLM integration, an RAG pipeline, and a database layer. The system aims to provide an intelligent chat assistant that enhances communication efficiency and knowledge retention in group discussions. While the core chat functionalities are fully operational, further improvements are needed to improve retrieval accuracy, optimize computational performance, and improve long-term context management. This study offers a novel approach to integrating LLM-powered intelligence into messaging platforms and highlights key challenges in implementing conversational AI for collaborative environments.

Index Terms: Conversation Summarization, Document-based Knowledge Retrieval, Intelligent Chat Assistant, LLM-powered Chat.

I. INTRODUCTION

Since Artificial Intelligence (AI) emerged, pursuing human-like machines has driven rapid progress. Among its breakthroughs, Large Language Models (LLMs) stand out for generating human-like responses, making them integral to digital solutions across domains such as customer service and software development.

Prompt Sapper [1] enables users to build modular AI workflows via a no-code interface, showcasing LLMs in service composition. An interactive PDF chatbot [2] applies LLMs for document-based Q&A but lacks long-term memory and contextual awareness for collaboration. DiagGPT [3] introduces a multi-agent dialogue system for task completion yet omits persistent summarization and event tracking.

This paper proposes an LLM-powered chat system for group-based collaboration. It features automated conversation summarization and document-based knowledge retrieval

* Corresponding Author

using a Retrieval-Augmented Generation (RAG) framework, enabling users to interact with previously shared files through natural language queries.

II. SYSTEM MODEL AND METHODS

The proposed system is designed for group-based collaboration and integrates two primary features: automated conversation summarization and document-based knowledge retrieval. Fig. 1 shows the system architecture.

The frontend provides a chat interface where users can send messages, upload documents, and request conversation summaries. It communicates with the REST API server in the backend, which processes user requests, manages summarization tasks, and handles document retrieval. The Prompt Manager refines user queries before forwarding them to the LLM engine for response generation.

The LLM integration includes a Summarization Module, which generates summaries of group discussions, and a RAG Module, which enhances responses by incorporating relevant knowledge from stored documents. The RAG pipeline processes uploaded documents by extracting textual content, converting it into vector embeddings, and storing them in a Vector Database (VDB) for efficient similarity-based retrieval. Finally, the database (DB) layer stores chat messages, generated summaries, and document metadata to ensure long-term access to discussion history and referenced knowledge.

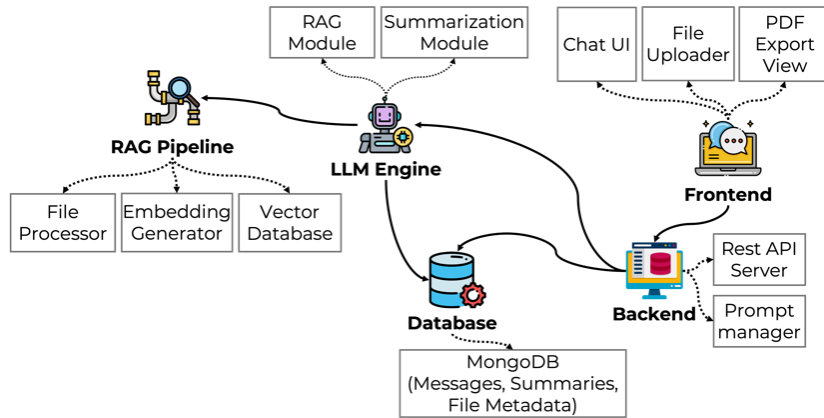


Fig. 1. Architecture of the system.

The system follows two primary workflows: conversation summarization and document-based knowledge retrieval, as illustrated in Fig. 2 where both workflows begin with user input through the chat interface

In the conversation summarization workflow, when a user requests a summary, the system fetches relevant chat data from the database, processes it, and passes it to the Summarization Module. The module formats the conversation and sends it to the LLM engine, which generates a summary at one of three levels: Basic, Intermediate, or Advanced, based on the detail required. The generated summary is then displayed in the chat interface and can be exported as a PDF report.

In the document-based knowledge retrieval workflow, when a user queries an uploaded document, the system performs a semantic search in the VDB to locate relevant content. The RAG Module retrieves the most relevant document segments and enriches the query before sending it to the LLM engine. The LLM then generates a context-aware response, which is returned to the chat interface for the user.

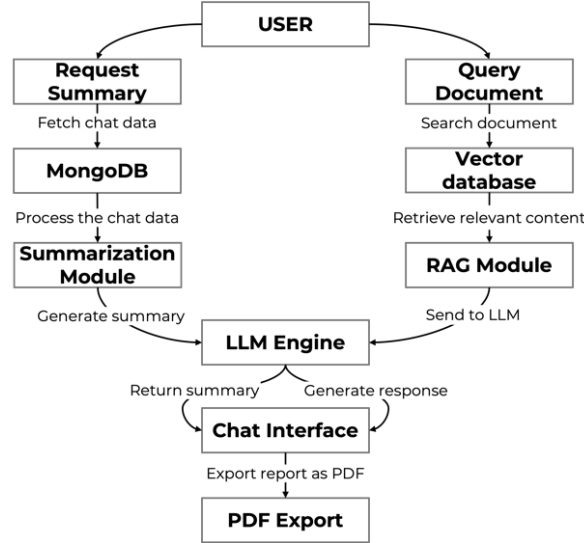


Fig. 2. Workflow of the system.

By integrating conversation summarization and retrieval-augmented document access, the system enhances collaborative discussions by providing structured summary and persistent access to shared knowledge.

III. CHALLENGES AND ISSUES

The chat application has been developed with full support for group conversation and 1:1 messaging, as shown in Fig. 3. While the core chat functionalities are operational, several challenges remain in implementing the system efficiently. A key challenge is response accuracy in document-based retrieval, as errors in semantic search within the VDB can lead to incomplete or misleading answers. Optimizing document chunking and retrieval ranking is crucial to improving reliability.

Another issue is computational efficiency in conversation summarization. Generating structured summaries for long chat histories requires significant processing power. Techniques such as summary caching, incremental summarization, or fine-tuned LLMs will be considered to mitigate this challenge. Additionally, managing long-term chat memory remains challenging. While MongoDB stores past messages, retrieving relevant context dynamically for summarization and RAG queries can be resource intensive. Optimizing database queries and indexing is necessary for efficient knowledge retention. Despite these challenges, this approach offers a solid foundation for an LLM-powered chat system with automated summarization and knowledge retrieval.

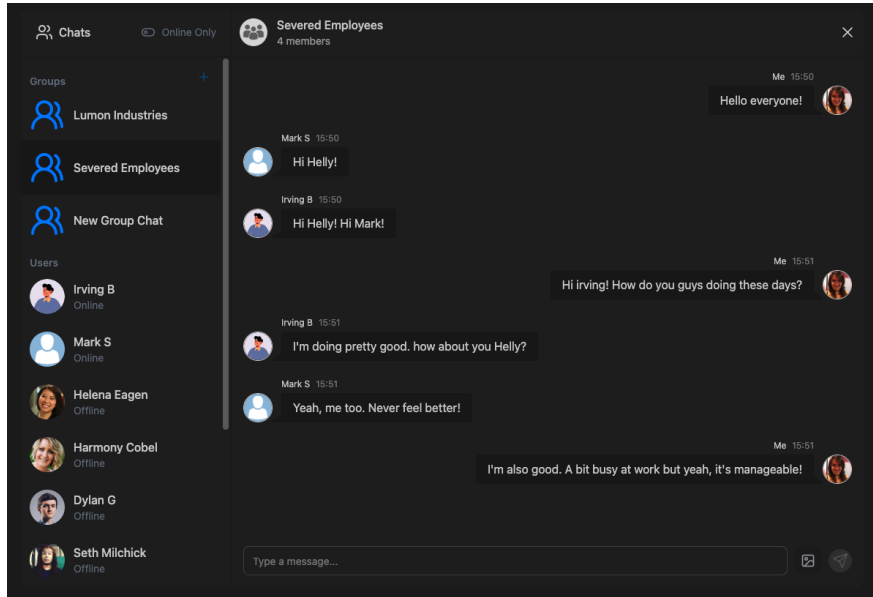


Fig. 3. Screenshot of the chat application.

IV. DISCUSSION AND CONCLUSIONS

This paper introduces an LLM-powered chat application designed for group-based collaboration with integrating automated conversation summarization and document-based knowledge retrieval. Using RAG framework, the system enables users to efficiently summarize discussions and query uploaded documents through natural language interactions. The proposed architecture combines frontend messaging, backend processing, LLM integration, a RAG pipeline, and database storage to provide structured conversation insights and persistent access to shared knowledge. While the system successfully supports 1:1 and group conversations, challenges remain in optimizing retrieval accuracy, computational efficiency, and long-term context management.

Aside from implementing the feature, future work will focus on improving semantic search ranking, reducing summarization latency through optimized caching techniques, and enhancing knowledge retention by refining query indexing strategies.

REFERENCES

- [1] Y. Cheng, J. Chen, Q. Huang, Z. Xing, X. Xu, and Q. Lu, "Prompt Sapper: A LLM-empowered production tool for building AI chains," *ACM Transactions on Software Engineering and Methodology*, vol. 33, no. 5, pp. 1–24, 2024.
- [2] T. T. Tin, S. Y. Xuan, W. M. Ee, L. K. Tiung, and A. Aitizaz, "Interactive ChatBot for PDF content conversation using an LLM language model," *International Journal of Advanced Computer Science and Applications*, vol. 15, no. 9, pp. 1033-1039, 2024.
- [3] L. Cao, "DiagGPT: An LLM-based and multi-agent dialogue system with automatic topic management for flexible task-Oriented Dialogue," *arXiv [cs.CL]*, 2023.

Sentiment Analysis of Major Wildfires in South Korea and the United States Using Text Mining Techniques

Aria Bisma Wahyutama¹, Kim Deok Hyun², and Mintae Hwang^{1*}

¹Department of Information and Communication Engineering, Changwon National University,
Changwon 51140, Republic of Korea

²Department of Management Information Systems, Gyeongsang National University,
Changwon 52849, Republic of Korea

[Email : wahyutama@gs.cwnu.ac.kr, kdh294@gnu.ac.kr, mthwang@cwnu.ac.kr]

Abstract

This study investigates the underlying causes and public discourse surrounding recent large-scale wildfires, including the 2025 Southern California Wildfire and the 2025 Simultaneous Nationwide Wildfires in South Korea, which have escalated beyond human control. While the immediate causes of such disasters are largely attributed to human negligence, global warming is increasingly being recognized as the fundamental driving factor. Utilizing text mining methodologies, this research conducts a comparative analysis of major wildfire incidents in both countries. Keywords, sentiment patterns, and trends are extracted from diverse web-based data sources, including search queries, unstructured textual content, and news mention frequencies. Through sentiment classification of associated positive and negative terms, the study aims to reveal public perception and thematic patterns related to the occurrence of extreme wildfires. The findings offer insight into the evolving narrative around climate-induced disasters and underscore the importance of data-driven environmental analysis.

Index Terms: Climate Change, Sentiment Analysis, Text Mining, Wildfire Disasters

I. INTRODUCTION

In recent years, wildfire disasters have intensified globally, causing severe environmental, economic, and human losses. The 2025 Southern California Wildfire and simultaneous wildfires in Andong and Sancheong, South Korea, exemplify the destructive force of climate-driven events. Marked by rapid spread and difficult containment, these incidents underscore the urgency of addressing wildfires amid climate change. Despite geographic and cultural differences, both countries faced similar challenges in managing large-scale outbreaks.

This growing threat has fueled public discourse on wildfire causes and impacts. As media and online platforms capture evolving sentiment, a large volume of unstructured text has emerged. Yet the subjective nature of expressions, ranging from fear to frustration, makes it challenging to extract clear insights using traditional methods. Without

* Corresponding Author

systematic analysis, policymakers risk missing perspectives critical for shaping adaptive disaster strategies [1].

This study uses text-mining techniques to conduct a comparative sentiment analysis of wildfire-related discourse in South Korea and the United States to address the problem above. Analyzing sentiment polarity, keyword trends, and thematic patterns from online news and search data aims to uncover how the public emotionally responds to wildfire events. The goal is to provide data-driven insights that support more effective, context-sensitive disaster communication strategies [2].

II. RESEARCH FRAMEWORK AND METHODOLOGY

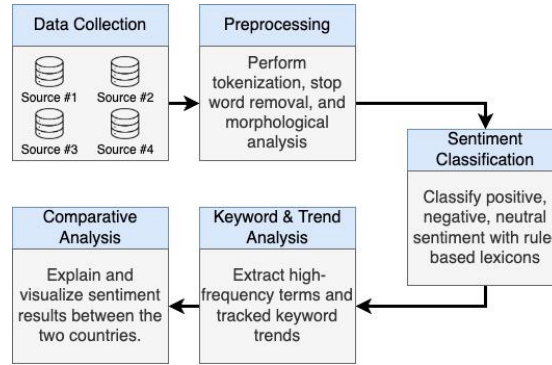


Fig. 1. Research framework and methodology of the study

As illustrated in Fig. 1, this study follows a structured workflow to analyze public sentiment regarding major wildfire events in the U.S. and South Korea during the respective periods when the disasters occurred. A total of over 900 South Korean and U.S. news articles were collected from various sources and reputable news outlets such as CNN, The New York Times, Reuters, YTN, Chosun Ilbo, and Yonhap News Agency covering the timeframes in which the wildfires occurred. South Korean news data were sourced using wildfire-related keywords such as '산불' (Wildfire), '산청' (Sancheong), '안동' (Andong), '화재' (Fire), and '대형 산불' (Large-scale wildfire). In contrast, U.S. data were retrieved using keywords like 'California', 'Forest Fire', 'Mountain Fire', and 'Wildfire'.

The full text of each article was then extracted. Korean texts were cleaned and tokenized using the Komoran analyzer from KoNLPy, while English texts were preprocessed using NLTK, including stopword removal and word tokenization. Sentiment analysis was then conducted using rule-based lexicons: KoSAC for Korean [3] and VADER for English, with each article categorized as positive, negative, or neutral.

Finally, keyword frequency and Term Frequency-Inverse Document Frequency (TF-IDF) analysis were performed to identify prominent wildfire-related terms in each dataset. Sentiment trends were also tracked over time to examine shifts in public perception. This cross-country approach offers insights into how wildfire disasters are framed differently in South Korean and the U.S. media.

III. SENTIMENT ANALYSIS RESULTS

The sentiment analysis reveals distinct patterns across the two countries, offering insight into how each society responds to wildfire disasters. In the U.S. dataset, most articles carried a negative tone (57.8%), with 36.5% positive and only 5.7% neutral. Conversely, South Korean coverage leaned heavily positive (75.1%), with 16.4% neutral and just 8.5% negative. Fig. 2 illustrates the sentiment trends in both countries during their respective wildfire events. These differences likely stem from media tone or narrative framing and from deeper structural, environmental, and logistical factors that influenced how the events unfolded and were reported.

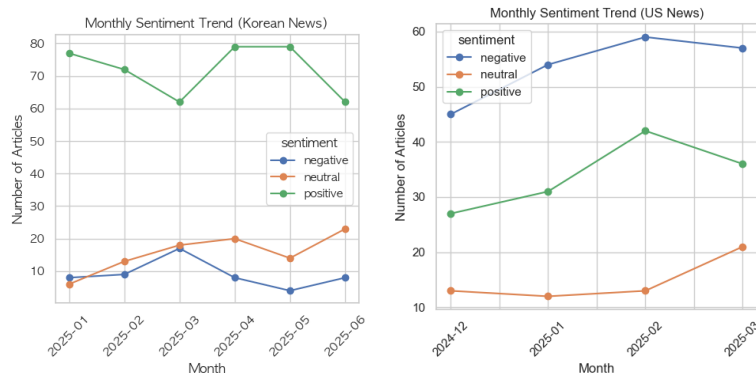


Fig. 2. Sentiment Trend Graphs of South Korean and US News During the Wildfire

These differences reflect variations in national reporting styles, disaster narratives, and levels of institutional trust. The predominance of negative sentiment in U.S. news focuses on destruction, casualties, emergency response, and policy critique related to the event. TF-IDF keywords such as “fire,” “county,” “crews,” and “officials” emphasize institutional actions and damage. In contrast, positive articles highlight terms like “health,” “air,” and “program,” reflecting narratives of recovery and public support.

South Korean news, even when reporting negatively, often features terms like “화재,” “산불,” and “대책” (measures), showing emphasis on government coordination. Positive articles frequently include “위한” (for the sake of), “대형” (large-scale), and “안동,” pointing to localized efforts and successful response. Keyword frequency further shows that “wildfire” and “California” keywords dominate U.S. articles, while “산불,” “화재,” “안동,” and “산청” appear most often in Korean articles. Notably, terms like “mountain fire” and “forest fire” were rare in U.S. articles, likely due to differing terminology.

Underlying these linguistic patterns are structural factors. California’s vast land size (423,970 km²) hampers fast response, especially in remote areas, often delaying help from neighboring counties. In contrast, South Korea’s smaller area (1,522 km² for 안동 and 794.7 km² for 산청) allows quicker mobilization. Higher population density in South Korea also necessitates efficient emergency systems and civilian voluntary response, enabling faster intervention and reassuring coverage.

Technological and environmental conditions further shape sentiment. South Korea’s integrated ICT fire detection and rainfall in early April supported fire suppression.

Moreover, access to nearby water sources, such as the Nakdong River, provided a geographical advantage for firefighting operations. Even in less river-accessible areas like Sancheong, the presence of local reservoirs and tributaries contributed to operational effectiveness. These conditions were frequently highlighted in the media, reinforcing a tone of control, preparedness, and recovery. Meanwhile, California faced drought, Santa Ana winds, and limited water access in inland zones, contributing to worsening fires and more critical media narratives. The contrast in environmental conditions further influenced how the media framed institutional competence and public reassurance.

These findings offer a cross-cultural perspective on crisis communication, showing how linguistic choices and sentiment framing differ under similar disaster contexts. In essence, sentiment analysis is not merely a measure of emotional polarity but a window into how societies process and communicate crises.

IV. CONCLUSIONS AND FUTURE STUDIES

This study conducted a comparative sentiment analysis of wildfire-related news in South Korea and the U.S. during major 2025 wildfire events. Using language-specific preprocessing, rule-based sentiment classification, and keyword analysis revealed that U.S. news coverage was largely negative, focusing on destruction, emergency response, and institutional critique. In contrast, South Korean news leaned strongly positive, highlighting government measures and recovery. These findings reflect how national media frame disaster narratives through cultural and linguistic lenses.

As wildfires intensify due to 이상 기후 (abnormal climate) and the ongoing 기후위기 (climate crisis), governments must strengthen not only disaster response but also public communication. Media sentiment can shape trust and compliance during emergencies. Thus, real-time sentiment monitoring should inform policies that promote transparency, readiness, and collective resilience.

Advanced AI models such as BERT can improve sentiment classification across languages for future studies. Integrating real-time social media and news data into interactive dashboards may also enhance early response and public understanding. Expanding this approach to other climate-related disasters could offer broader insights into public perception and risk communication on a global scale.

REFERENCES

- [1] Z. Han, M. Shen, H. Liu, and Y. Peng, "Topical and emotional expressions regarding extreme weather disasters on social media: a comparison of posts from official media and the public," *Humanities and Social Sciences Communications*, vol. 9, no. 421, 2022
- [2] Y. E. García *et al.*, "Wildfires and social media discourse: exploring mental health and emotional wellbeing through Twitter," *Frontiers in Public Health*, vol. 12, p. 1349609, 2024.
- [3] Seoul National University, Korean Sentiment Analysis Corpus [Internet], Available: <http://ling.snu.ac.kr/kosac/>.

Empowering Education with AI: Generating Indonesian High School Math Exams Using ChatGPT 4.0

Muhamad Jamaludin¹, Handoko Supeno¹, Aria Bisma Wahyutama²,
and Mintae Hwang^{2*}

¹Department of Informatics Engineering, Pasundan University, Bandung, 40153, Indonesia

²Department of Information and Communication Engineering, Changwon National University,
Changwon, 51140, Republic of Korea

[Email : 213040057.Jamaludin@mail.unpas.ac.id, handoko@unpas.ac.id,
wahyutama@qs.cwnu.ac.kr, mthwang@cwnu.ac.kr]

Abstract

The Fifth Industrial Revolution emphasizes human-AI collaboration, personalization, and intelligent systems that enhance various sectors, including education. One such advancement is GPT-4o, developed by OpenAI, which enables automated content generation through a large language model. This research aims to develop a web-based application that leverages GPT-4o technology to automate the generation of high school mathematics questions. This application assists teachers in efficiently preparing questions for exams and daily exercises. It supports both multiple-choice and essay formats, as well as automatically generate illustrative images, while maintaining alignment with the national education curriculum. The implementation of this technology in Indonesian high schools demonstrates that the application accelerates question creation, provides a wider of questions, and enhances the overall quality of assessments activities.

Index Terms: Exams, Generative AI, Mathematics Education, Web-Based Application

I. INTRODUCTION

Recent technological developments have automated many manual processes and introduced AI systems capable of mimicking human reasoning [1]. Today, AI plays a vital role in education by streamlining tasks such as test creation and resource management. For instance, OpenAI's GPT-4o can quickly generate contextually relevant educational materials, thereby reducing educators' workloads [2].

As a vast archipelagic nation, Indonesia faces significant disparities in school infrastructure across its regions. In remote areas, where teachers often lack sufficient support, manually preparing exam questions is both time-consuming and inconsistent, as it depends heavily on each teacher's individual capabilities and the constraints they face. This contributes to ongoing educational inequality nationwide. The advancement of AI technology offers a promising solution to these challenges. As noted in [3], AI can

* Corresponding Author

enhance the development of assessment materials. Similarly, research in [4] demonstrates that automated tools can generate curriculum-aligned questions while reducing the repetitive workload typically placed on teachers.

AI-driven educational tools are increasingly recognized for their potential to enhance personalized learning, enable adaptive assessments, and support intelligent tutoring systems [5]. UNESCO has also emphasized the importance of AI in education, advocating for policies that ensure equitable access and ethical implementation of these technologies in classrooms [6]. Moreover, iterative prototyping approaches have proven effective in developing systems that meet the practical needs of educational stakeholders [7]. Recent studies further highlight that AI can help address major educational challenges, from improving teaching methodologies to accelerating progress toward global education goals [8].

The key contributions of this research are as follows:

1. Development of a system that automates the generation of high school mathematics question using GPT-4o.
2. Enhancing teachers' efficiency in test preparation by minimizing repetitive manual tasks.

II. SYSTEM MODEL AND METHODS

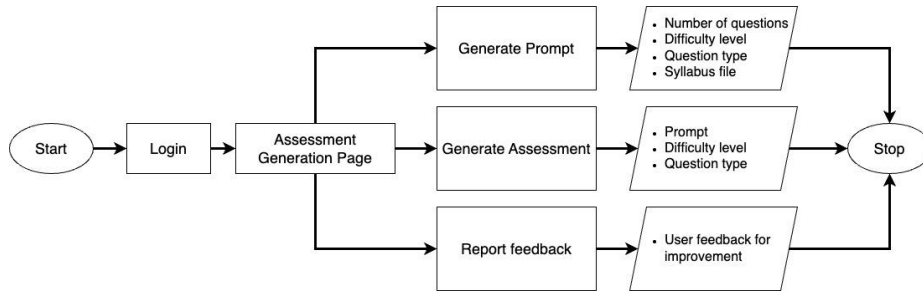


Fig. 1. User Workflow for Assessment Generation System

The development of the system follows an iterative prototyping model, which includes several key stages. The process begins with communication, where active engagement with mathematics teachers helps identify their needs and define the system's objectives. Based on this input, a quick planning phase outlines the core features, priorities, and rough system structure. This is followed by a quick design modeling stage, where initial sketches and use case diagrams are created to visualize user interactions and system flow. In the prototype construction phase, a working version is built using technologies like Next.js and the Azure OpenAI API, incorporating prompt engineering to ensure appropriate question generation. Finally, the prototype is deployed to high school teachers and students for real-world use, and feedback is gathered to guide further refinements in the next iteration.

The outcome of this development is a web-based platform that allows users to automatically generate exam questions, as illustrated in Fig. 1.

III. RESULTS

In the web application, teachers can perform several tasks to generate exam questions efficiently. The process begins by creating a question list, where teachers provide specific inputs: an AI instruction prompt, the desired number of questions, the Bloom's Taxonomy difficulty level (Random, C1–C6), the question type (Random, Essay, or Multiple Choice), and a PDF version of the syllabus. Once all required fields are filled, the teacher clicks the "Generate" button, prompting the AI to automatically produce a list of exam-ready questions. This process is illustrated in Fig. 2.

Automatic Math Problem Generator
Automatic high school level math problem generation using AI

Prompt:

Cognitive Level: C1 (Remember) Question Type: Essay Generate

Simple Assessment: Create one high school level math assessment about [...]

Assessment with Image: Create one high school level math assessment about [...] including an image and its information.

Exam-Style Assessment: Create a practice exam

Prompt:

Cognitive Level: C1 (Remember) Question Type: Essay Generate

Title:

Description:

Preview:

Diberikan sistem persamaan linear dengan tiga variabel sebagai berikut, dan gambarnya adalah representasi dari sistem persamaan tersebut dalam bentuk grafik 3D:

1. $x + y + z = 6$
2. $2x - y + z = 3$
3. $x - 2y + 3z = 4$

Selesaikan sistem persamaan di atas dan tentukan nilai dari masing-masing variabel: x , y , dan z . Sebagai tambahan, gambarkan bentuk grafik dari sistem persamaan ini dalam ruang 3D.

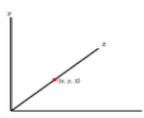


Fig. 2. The question generation interface for teacher (Left) and an example of a generated exam question (Right)



Fig. 3. A teacher interacting with the question generation system (Left) and Students taking a paper-based exam using the generated questions (Right)

After reviewing the generated questions, teachers can select items, provide feedback via a star-rating system and optional comments, and exit using the 'Logout' button. They then choose the exam delivery method—most selected paper-based exams in this study. Post-exam, surveys were distributed to both teachers and students. Likert scale results indicated that the questions aligned well with the curriculum and were generally easy to understand. The implementation process at an Indonesian high school is illustrated in Fig. 3

IV. DISCUSSION AND CONCLUSIONS

The implementation of GPT-4o for automated mathematics question generation has significantly improved educational efficiency by reducing teacher workload and enhancing the quality of assessments in alignment with curriculum standards. Evaluation results indicate that 85% of participating teachers found the system effective in both curriculum alignment and the selection of appropriate difficulty levels. Future development should focus on refining AI-generated visual elements, particularly mathematical diagrams, enabling seamless integration with Learning Management Systems (LMS), and expanding training datasets to improve both accuracy and pedagogical relevance.

ACKNOWLEDGMENTS

The authors thank the Skyshi Digital Indonesia, Universitas Pasundan, and Changwon National University for funding this research.

REFERENCES

- [1] Devianto, Y. & Dwiasnati, S. (2020). Kerangka Kerja Sistem Kecerdasan Buatan dalam Meningkatkan Kompetensi Sumber Daya Manusia Indonesia. *InComTech*, 10(1), 19–24.
- [2] Gumelar, G. (2023). Catatan Editor: Menavigasi Tantangan dan Menciptakan Peluang, Peran Vital Ilmu Psikologi di Era Kecerdasan Buatan. *JPPP*, 12(1), 1–4.
- [3] Chan, K. S. & Zary, N. (2019). Applications and Challenges of Implementing Artificial Intelligence in Medical Education: Integrative Review. *JMIR Medical Education*.
- [4] Prasetyo, V. R., Benarkah, N. & Chrisintha, V. J. (2021). Implementasi Natural Language Processing Dalam Pembuatan Chatbot Pada Program Information Technology Universitas Surabaya. *Jurnal TEKNIKA*, 10(2), 114–121.
- [5] Cunningham-Nelson, S., Boles, W., Trouton, L. & Margerison, E. (2019). A review of chatbots in education: Practical steps forward. In the 30th Annual Conference for the Australasian Association for Engineering Education (AAEE 2019), 299–306.
- [6] Hopcan, S., Polat, E. & Ozturk, M. E. (2022). Artificial intelligence in special education: a systematic review. Manuscript submitted for publication.
- [7] Putri, N., Prabowo, N. A., & Widyanto, R. A. (2019). Implementasi Metode Prototyping pada Perancangan Aplikasi Electronic Ticket (E-Ticket) berbasis Android. *Jurnal Komtika*, 3(2), 62–68.
- [8] Nugroho, A. S. (2023). Pengembangan Media Pembelajaran Matematika Interaktif Berbasis Pendekatan Kontekstual. Presented in June 2023.

Grading Single Oyster based on Their Features Using AI Technologies

Thi-Ngot Pham^{1,2} ‡, Daesung Lee³ ‡, Hee-Jung Lee^{4,*}, Jun-Ho Huh^{5,6,7,**}

¹Postdoctoral Researcher of Department of Data Informatics, National Korea Maritime and Ocean University, Busan 49112, Republic of Korea

²Postdoctoral Researcher of Department of Interdisciplinary Major of Ocean Renewable Energy Engineering, National Korea Maritime and Ocean University, Busan, Republic of Korea

³Full Professor (Tenure Track) of Division of Maritime AI and Cyber Security, National Korea Maritime and Ocean University, Republic of Korea

⁴Researcher of Southeast Sea Fisheries Research Institute, National Institute of Fisheries Science (NIFS), Tongyeong 39769, Republic of Korea

⁵Full Professor (Tenured) of Department of Data Science, National Korea Maritime and Ocean University, Republic of Korea

⁶Full Professor (Tenured) of Department of Interdisciplinary Major of Ocean Renewable Energy Engineering, National Korea Maritime and Ocean University, Republic of Korea

⁷Sabbatical Year at School of Computer Science and Engineering, Yeungnam University, Republic of Korea

‡ These authors contributed equally to this work.

* Corresponding author: sobht@korea.kr ** Corresponding author: 72network@kmou.ac.kr

Abstract

Commercial shellfish farming operations experience substantial benefits through the deployment of automated grading systems for single oyster specimens, resulting in reduced labor costs and enhanced operational efficiency. This research utilizes advanced machine learning methodologies to detect, segment, and classify oysters according to their external physical characteristics. The conference proceeding combines sophisticated deep learning object detection frameworks, particularly YOLOv9-segmentation for oyster identification, with machine learning classification techniques, specifically Support Vector Machine (SVM), to perform detailed feature analysis of 2D RGB image data. The oyster classification results provide quality assessments that support subsequent quality control procedures and performance evaluation processes.

Index Terms: Grading Oyster; Object Detection; Instance Segmentation; Classification; YOLO; SVM; Single Oyster.

I. INTRODUCTION

Evaluating oyster quality, especially shell characteristics, serves as a key element in quality assessment and market pricing within shellfish farming operations [1]. The evaluation process regularly examines length, width, shape, surface texture, density, elongation, and plumpness measurements. These physical characteristics directly affect product quality standards and consumer pricing frameworks. Traditional oyster grading relies on manual inspection performed by experienced technicians, but this approach

* Corresponding author ** Co Corresponding author

proves time-consuming, labor-intensive, and subject to inspector variation. To address these operational challenges, we propose an automated system using artificial intelligence and computer vision technologies [2,3]. This conference proceeding presents a method for automatically classifying single oysters as acceptable (Pass) or unacceptable (Fail) by combining CNN (YOLOv9-segmentation) and SVM algorithms. YOLOv9 functions as a sophisticated deep learning detection and segmentation framework that executes accurate object identification and boundary delineation in digital imagery through bounding box and mask generation processes. The SVM algorithm serves as an established machine learning classification methodology that utilizes previously extracted oyster characteristic data to conduct quality evaluations. This combined approach significantly reduces manual processing requirements while creating an effective and scalable system for oyster quality assessment applications.

II. SYSTEM MODEL AND METHODS

This conference proceeding introduces a system for grading single oysters based on their external characteristics using artificial intelligence methods. We suggest using two primary algorithms for oyster grading purposes. The first model uses a deep learning instance segmentation algorithm to locate and identify oysters. The second model employs a machine learning approach, Support Vector Machine (SVM), to examine oyster features from 2D RGB photographs. This approach is illustrated in Figure 1 through three main stages.

Phase 1 involves loading models and setting up directories. The YOLOv9 segmentation model and SVM classifier are prepared for use. After the models load into computer memory, the system creates the required folders to save processed results. A CSV file gets prepared to store organized data including image names, extracted features, and grading outcomes.

Phase 2 handles importing test images. The system examines each image in the designated input folder. The system checks each input file for proper format. When the image uses .jpg or .png format, this acceptable format image moves forward to the grading process. Otherwise, the system rejects it.

Phase 3 executes the oyster grading workflow. First, the valid image file goes through the trained YOLOv9-segmentation model to perform object detection and instance segmentation. The result shows detected oysters, with each found oyster displaying a bounding box and polygon outline. Then, the system extracts geometric measurements from each oyster's polygon outline. Next, these extracted measurements go to the SVM classifier. The SVM model received training on a dataset of labeled oyster examples. The classifier produces a predicted quality label (such as Good-Grade, Bad-Grade), indicating the quality or condition of each detected oyster. We establish two grading categories: Good grade oysters have teardrop or sector shapes, while Bad grade oysters have other shapes.

Following classification, the system shows the grading results on the original image with predicted classification and confidence scores as shown in Figure 2. This graded image gets saved in the output folder. Simultaneously, the prediction results and related features are recorded in a CSV file and saved in JSON format for easy use in other applications.

III. RESULTS

To assess the performance of our proposed automated classification framework, we conducted comprehensive testing using varied single oyster specimens. We began by training our artificial intelligence models using curated datasets. The YOLOv9-segmentation model received training on 500 mature oyster images. The SVM model was trained using 78 mature oyster images. Feature extraction procedures were executed from segmentation results and underwent proper calibration. We then integrated these trained models into our unified classification system. We evaluated the framework using oyster samples sourced from online databases with standardized white backgrounds. The classification outcomes for test specimens (not included in training data) are displayed in Figure 2. The identified oysters were classified into two categories: good-grade (Pass) and bad-grade (Fail) with confidence scores above 70%. These findings confirm the practical effectiveness of our proposed methodology.

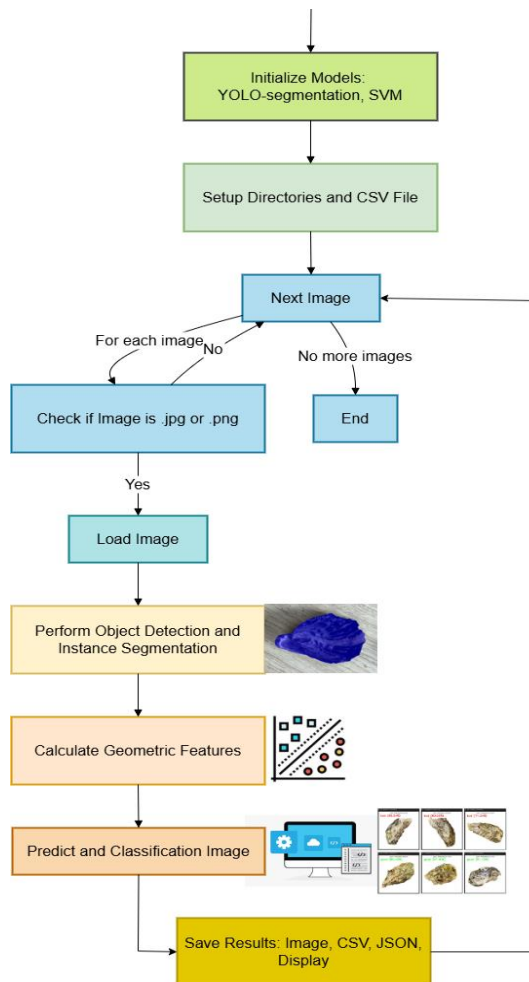


Figure 1. Proposed framework of grading oyster using AI technology.

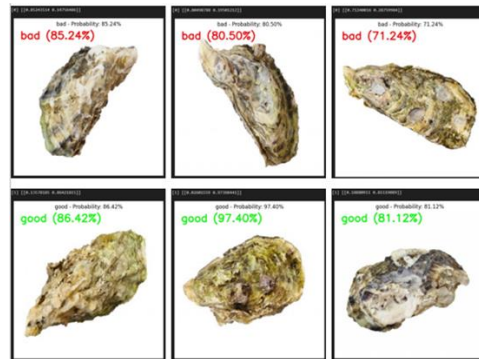


Figure 2. Examples of automated oyster classification results.

IV. CONCLUSIONS

This conference proceeding creates a practical system for automatically sorting oysters that brings together modern computer detection methods with traditional sorting approaches that people can easily understand. The system works well at checking oyster quality by using YOLOv9 technology to identify oysters and SVM methods to classify them properly. The research is still being developed and needs more oyster samples to make the computer models work better, plus improved ways to analyze oyster features so the whole system performs at its best.

ACKNOWLEDGMENTS

This research was supported by the National Institute of Fisheries Science, Ministry of Ocean and Fisheries, Korea(R2025027).

REFERENCES

- [1] Ministry of Oceans and Fisheries, Republic of Korea. Oyster Industry Report. <https://www.mof.go.kr/doc/en/selectDoc.do?docSeq=59946&bbsSeq=90&menuSeq=485>, last accessed 2025/05/25.
- [2] Dah-Jye Lee, Xiaoqian Xu, Robert M. Lane, and Pengcheng Zhan. "Shape analysis for an automatic oyster grading system." In Two- and Three-Dimensional Vision Systems for In-spection, Control, and Metrology II, Proc. SPIE 5606, (16 December 2004).
- [3] Oyster Open-source dataset: <https://universe.roboflow.com/curvy-oysters/oysters-cjt7n>, last accessed 2025/05/25
- [4] Oyster Open-source dataset: <https://universe.roboflow.com/631classwork/oyster-shell-classifier>, last accessed 2025/02/25
- [5] Wang, Chien-Yao, Yeh, I-Hau, and Liao, Hong-Yuan Mark. "YOLOv9: Learning What You Want to Learn Using Programmable Gradient Information." European Conference on Computer Vision (ECCV). Cham: Springer Nature Switzerland, 2024
- [6] <https://scikit-learn.org/stable/modules/svm.html>
- [7] S. Ghosh, A. Dasgupta and A. Swetapadma, "A Study on Support Vector Machine based Linear and Non-Linear Pattern Classification," 2019 International Conference on Intelligent Sustainable Systems (ICISS), Palladam, India, 2019, pp. 24-28, doi: 10.1109/ISS1.2019.8908018.

Session AI-C

AI and Intelligent Information System

Session BI-A

Biomedical Imaging and Engineering

Session IT-C

IT Convergence Technology

Session MD-B

Multimedia and Digital Convergence

Session CS-A

Communication System and Application

Session US-A

Ubiquitous Sensor Network

Session SC-A

Semiconductor and Communication Services

AI-Based Travel Schedule Management Mobile Application Using Microservices Architecture and Cross-Platform Framework

Geun-Min Na¹

¹Department of AI Convergence, Ulsan University, Ulsan 44610, South Korea
/Corresponding author: rmsals8@naver.com/

Abstract

This paper presents an intelligent travel schedule management mobile application integrating AI-based recommendation systems with comprehensive microservices architecture. The system utilizes Flutter framework for cross-platform compatibility and incorporates GPT-4 for AI-powered travel destination recommendations. The application features hybrid FastAPI-Spring Boot backend architecture, five-API integration for location search, and automated schedule conflict resolution. Experimental results with 50 university students over four weeks demonstrate 97% location search accuracy, 89% natural language processing accuracy, and 92% user satisfaction rate, significantly outperforming existing solutions.

Index Terms: Artificial Intelligence, Cross-Platform Development, Microservices Architecture, Travel Planning

I. INTRODUCTION

Modern travel planning suffers from limited personalization capabilities, poor service integration, and lack of intelligent optimization [1]. Current solutions rely on single-API data sources, resulting in incomplete location information, particularly for region-specific searches.

This research develops a comprehensive intelligent travel management system combining cutting-edge AI, enterprise-grade microservices architecture, and cross-platform development. The solution leverages GPT-4 for natural language processing, implements novel five-API integration for enhanced location accuracy, and utilizes hybrid FastAPI-Spring Boot architecture.

Main contributions: (1) seven-tier microservices architecture extending beyond traditional three-tier systems, (2) hybrid framework integration combining FastAPI for AI processing with Spring Boot for business logic, (3) five-API integration achieving 97% location accuracy, and (4) enterprise-grade implementation with 99.7% service availability.

II. SYSTEM MODEL AND METHODS

A. Seven-Tier Microservices Architecture

The system implements seven distinct microservices: (1) **Authentication Service** with OAuth2 social login and JWT security, (2) **User Management Service** for complete lifecycle management, (3) **Subscription Service** with FREE/PREMIUM tiers, (4) **Visit History Service** for preference learning, (5) **Location Search Service** with five-API integration (T-map + Foursquare + Kakao + Google + OpenAI), (6) **Schedule Optimization Service** with GPT-4 natural language extraction, and (7) **Navigation Service** for real-time GPS tracking.

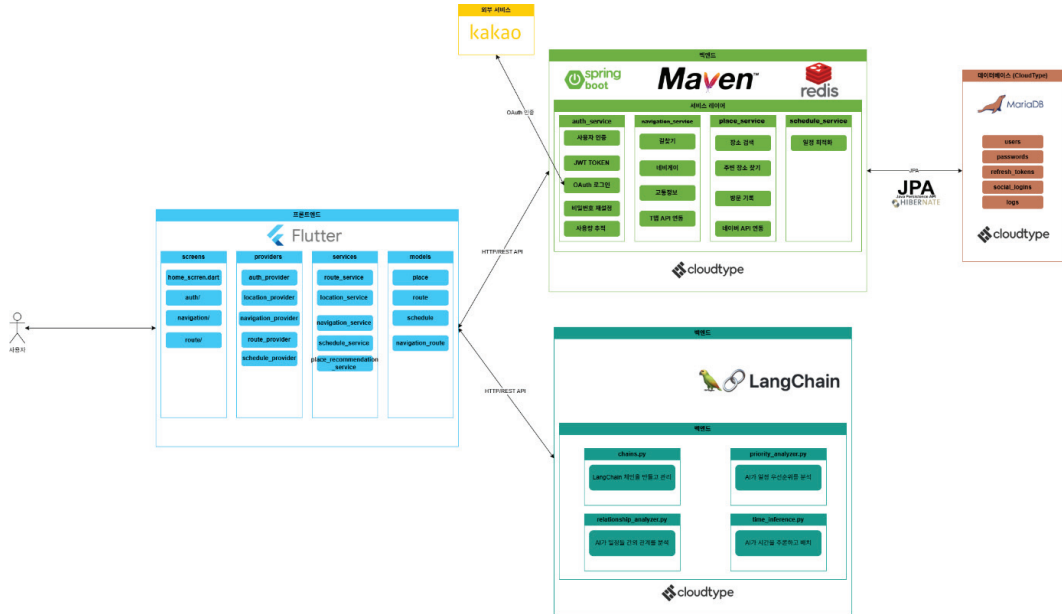


Fig. 1. System architecture with seven-tier microservices, Flutter frontend, FastAPI-Spring Boot hybrid backend, and five-API integration.

B. AI Processing and Optimization

Five-API Integration Algorithm: The system employs sequential API calls with validation - Foursquare (global accuracy) → Kakao (Korea specialization) → Google Places (fallback) → T-map (transportation) → GPT-4 (contextual analysis). This approach achieves 97% location accuracy compared to 73% with single-API usage.

Schedule Optimization: GPT-4 processes natural language input to extract travel schedules with Korean context understanding. The system implements fixed-flexible schedule integration with $O(n \log n)$ complexity, analyzing time slots and applying real-time traffic optimization.

C. Cross-Platform Implementation

Flutter framework enables unified iOS/Android deployment with Material Design UI, real-time synchronization, and native integration. Backend combines FastAPI for AI processing with Spring Boot for business logic, Redis caching, and PostgreSQL persistence.

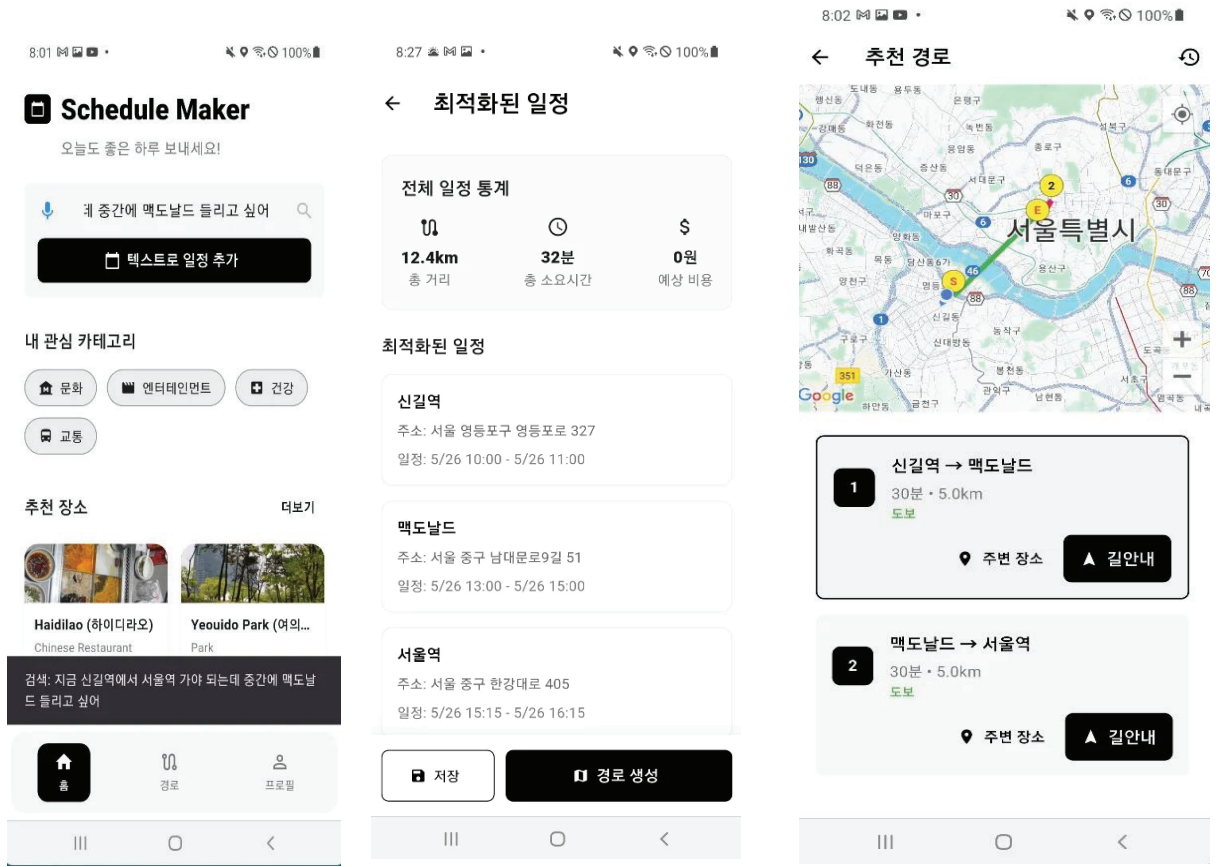


Fig. 2. Mobile application screenshots: (a) Schedule Maker dashboard with voice input, (b) optimized schedule results, (c) five-API location search, (d) real-time map navigation.

III. RESULTS

A. Performance Analysis

System Performance: Stable processing up to 1,000 concurrent users, 100,000 daily requests capability, 95th percentile response time within 3 seconds, and 512MB average memory utilization. **Microservice Performance:** Authentication (200 logins/sec), Location Search (2-second five-API processing), Schedule Optimization ($O(n \log n)$ complexity). **User Testing:** 50 university students, 4-week evaluation, 847 schedules created, 92% satisfaction rate.

B. Technical Innovation Verification

AI Accuracy: Natural language extraction (89%), location search (97%), time conflict resolution (100%), personalized recommendations (94% satisfaction). **System Stability:** 99.7% service availability, 100% data consistency, zero security incidents, 30-second average failure recovery.

Table I. Performance Comparison

Metric	Google Trips	TripAdvisor	Proposed System
Location Accuracy	73%	78%	97%

Metric	Google Trips	TripAdvisor	Proposed System
AI Integration	None	Limited	GPT-4 Full
Concurrent Users	100	500	1,000
Service Availability	98%	96%	99.7%

IV. DISCUSSION AND CONCLUSIONS

A. Technical Contributions

This research provides: (1) **Seven-Tier Architecture** enabling unprecedented scalability, (2) **Hybrid Framework Integration** maximizing FastAPI and Spring Boot strengths, (3) **Five-API Integration** achieving 24% accuracy improvement over single-API implementations, (4) **Real-time AI Optimization** fusing GPT-4 with traffic information, and (5) **Complete Lifecycle Management** with comprehensive audit trails.

B. Superiority and Impact

Differentiation: Google Trips (simple listing) → AI-based optimization; TripAdvisor (static reviews) → real-time personalization; Yanolja (accommodation focus) → complete travel integration. **Academic Impact:** Practical microservices implementation, AI-traditional algorithm integration methodology, multi-API data quality validation, and performance-accuracy balance discovery.

C. Commercial Readiness and Future Work

The system demonstrates enterprise-grade security, 1,000+ user scalability, complete monitoring systems, and 99.7% availability. **Future directions:** Multi-country API integration, AR/VR fusion, blockchain travel history, and advanced pattern prediction using machine learning.

This research successfully demonstrates AI-powered recommendation systems with enterprise-grade architecture effectiveness. GPT-4, five-API orchestration, and hybrid microservices provide robust foundation for next-generation travel applications, validated by positive user feedback and commercial-grade metrics.

ACKNOWLEDGMENTS

This research was supported by the MSIT(Ministry of Science, ICT), Korea, under the National Program for Excellence in SW, supervised by the IITP(Institute of Information & communications Technology Planning & Evaluation) in 2025(2024-0-00025)

REFERENCES

- [1] S. Chen, K. Liu, and M. Zhang, "Microservices Architecture for Mobile Applications," *Int. J. Software Engineering*, vol. 15, no. 3, pp. 45-62, 2023.
- [2] A. Johnson and R. Smith, "Cross-Platform Mobile Development with Flutter," in *Proc. Int. Conf. Mobile Computing*, pp. 123-130, 2023.
- [3] L. Wang, T. Kim, and J. Lee, "AI-Powered Recommendation Systems in Travel Applications," *IEEE Trans. Intelligent Transportation Systems*, vol. 24, no. 8, pp. 2156-2167, 2023.
- [4] M. Brown, "GPT-4 Integration in Mobile Applications," *ACM Computing Surveys*, vol. 56, no. 2, pp. 1-28, 2024.
- [5] OpenAI Team, "GPT-4 Technical Report," *arXiv preprint arXiv:2303.08774*, 2023.
- [6] Google Developers, "Flutter Documentation," [Online]. Available: <https://flutter.dev/docs>

AlCheNet: A Pathology-informed Graph Representation of Channel-wise EEG for Alzheimer’s Disease Classification

Dawoon Kwon¹, and Daehwan Kim^{1*}

Department of IT Convergence, University of Ulsan, Ulsan 44610, Korea
[Email : dobylyve01@gmail.com, daehwankim@ulsan.ac.kr]

Abstract

Electroencephalogram (EEG) is widely used for clinical diagnosis and brain-computer interface (BCI) applications. In Alzheimer’s disease (AD), disrupted functional connectivity between brain regions is a well-established biomarker. However, conventional EEG analysis methods often overlook inter-channel dependencies and are susceptible to noise due to the inherently low signal-to-noise ratio (SNR) of EEG signals, which limits robust feature extraction.

To address these limitations, we propose *AlCheNet*, a hybrid graph-based neural network that learns frequency-specific features and inter-channel dependencies in EEG data. The model first applies one-dimensional convolutions to extract frequency-specific features, then constructs a graph where each node corresponds to an EEG channel. A graph-based neural network is then employed to learn inter-channel relationships and denoise the signal by enforcing inter-channel smoothness. To avoid over-smoothing and preserve disease-relevant regional patterns, we used pathology-informed graph to guide adjacency matrix construction. We evaluated *AlCheNet* on a clinical EEG dataset comprising patients with AD, frontotemporal dementia, and healthy controls. *AlCheNet*, built upon the ChebNet architecture, outperformed a comparative ChebNet-based model by 1.82% in accuracy and 4.39% in F1-score, demonstrating superior representation learning capability.

Index Terms: Electroencephalogram, Functional connectivity, Graph neural network, Alzheimer’s disease

I. INTRODUCTION

Electroencephalogram (EEG) is widely used for clinical diagnosis and brain-computer interface (BCI) applications. In Alzheimer’s disease (AD), disrupted functional connectivity between brain regions is a well-established biomarker. However, conventional EEG analysis methods often overlook inter-channel dependencies and are susceptible to noise due to the inherently low signal-to-noise ratio (SNR) of EEG signals, which limits robust feature extraction.

Graph-based neural networks have recently emerged as a promising approach[1,2] for EEG analysis, where each channel is encoded as a node and inter-channel relationships are defined as edges, often using signal similarity measures such as Dynamic Time Warping (DTW). However, this approach overlooks two critical issues: the over-smoothing problem

common in spectral- and spatial-based graph convolution neural networks (GCN), and the functional connectivity alterations observed in patients with neurological disorders.

Over-smoothing, common in fully connected graphs, causes node representations to converge and lose their discriminative characteristics. This is particularly problematic for identifying subtle regional differences in EEG signals, as seen in Alzheimer’s disease. To address these limitations, we propose AlCheNet, a hybrid EEG graph-based neural network that incorporates disease-specific functional connectivity patterns to enhance representation learning and improve classification performance.

II. AlCheNet

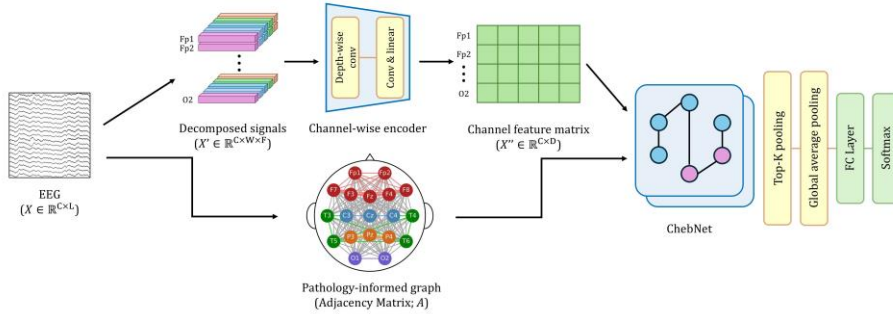


Fig. 1. Overview of AlCheNet.

We propose a novel architecture (Fig. 1), AlCheNet, which integrates graph convolution with channel-wise encoding and pathology-informed graph construction to capture complex dependencies in EEG signals.

A. Graph Convolution Networks

Graph-based learning methods have shown great potential in analyzing EEG data, particularly due to their ability to model spatial dependencies between channels. In EEG analysis, researchers have explored two primary approaches: spatial-based methods that focus on channel-wise connectivity, and spectral-based methods that emphasize frequency-domain graph construction. Since EEG signals naturally reflect interactions among different brain regions, graph neural networks (GNNs) are especially effective in capturing inter-channel relationships. In this study, we evaluated four representative GNN architectures: GCN[3], GraphSAGE[4], GAT[5], and ChebNet[6]. These models are applied to the same feature matrices and adjacency matrices to identify the most suitable architecture for EEG-based analysis of Alzheimer’s disease.

B. Channel-wise Encoder

To derive meaningful representations for each EEG channel as graph nodes, we first considered the original EEG signals $X \in \mathbb{R}^{C \times L}$, where C is the number of channels and L is the length of the sequence. We then transformed the raw signals into the frequency domain. Based on the understanding that brain activity is reflected in frequency-specific patterns, we decomposed the signals into five standard frequency bands (Delta, Theta, Alpha, Beta, and

Gamma) and computed the Power Spectral Density (PSD) for each. This process yielded a frequency-based representation $X' \in \mathbb{R}^{C \times W \times F}$, where W is the number of frequency bands and F the number of frequency bins. To extract features from each channel independently, we applied depth-wise convolution followed by a linear transformation, resulting in a feature matrix $X'' \in \mathbb{R}^{C \times D}$, where D is the dimensionality of each node representation (set to 256 in this study), which serves as the input to the graph neural network.

C. Pathology-informed Graph

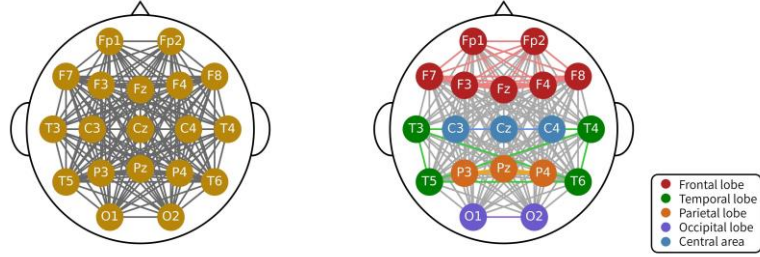


Fig. 2. Fully connected graph representing a conventional EEG-based approach (Left), and a pathology-informed graph in which edge weights are reduced except for connections between lobes. Colored edges indicate preserved weights, while gray edges represent weakened connections. (Right)

Fully connected graphs based on similarity measures such as Dynamic Time Warping (DTW) are widely used in EEG-based GNNs, but they often suffer from over-smoothing and fail to capture inter-lobar brain activity differences characteristic of Alzheimer's disease (AD)[7, 8]. Over-smoothing occurs when excessive connectivity causes node features to become indistinguishable, diminishing the model's ability to represent meaningful neural patterns. While one might consider retaining only inter-lobar edges to mitigate over-smoothing, this results in an overly sparse graph that lacks sufficient structural information, ultimately hindering the extraction of meaningful representations.

To address this, we propose a pathology-informed graph that retains original edge weights within the same lobe while down-scaling inter-lobar edge weights by a scalar factor (0.5). This approach balances the trade-off between over-smoothing and excessive sparsity, preserving critical lobe-specific features while reducing redundancy, thereby enabling our model, AlCheNet, to better reflect AD-related EEG dynamics.

III. EXPERIMENTS & DISCUSSION

A. Dataset

The dataset used in this study consists of EEG resting-state recordings (eyes closed) collected from a total of 88 subjects, including 36 diagnosed with Alzheimer's disease (AD group), 23 with frontotemporal dementia (FTD group), and 29 cognitively normal individuals (CN group). For evaluation, four subjects from each group were designated as the test set. The EEG signals were segmented into 10-second segments to construct the train, validation, and test sets. As part of the pre-processing, a band-pass filter between 1 and 40 Hz was applied to retain the relevant frequency components of the EEG signals.

B. Experimental Settings

All experiments were conducted using the same training configuration: the AdamW optimizer, 100 epochs, learning rate of $1e-4$, batch size of 16, and the cross-entropy loss function. All GNN models used in the experiments were implemented with 6 layers. To evaluate the impact of our proposed pathology-informed graph, we first assessed the performance of each baseline GNN model and then applied the pathology-informed graph to the best-performing architecture to examine its effect on classification performance.

C. Results & Discussion

Among the evaluated models, ChebNet achieved the highest classification performance, with the best results observed when the K-hop parameter was set to 3. Furthermore, the application of the pathology-informed graph led to consistent performance improvements, with an increase of 1.82% in accuracy and 4.39% in F1-score compared to ChebNet-based model. These findings suggest that EEG signals from Alzheimer's patients exhibit distinguishable inter-lobar differences, and incorporating such pathology-specific information contributes to learning more meaningful and discriminative representations.

Table 1. Overall performance comparison of all models. *CW-Encoder* refers to the Channel-wise encoder, and *K* denotes the K-hop parameter. *AlCheNet* represents the combination of CW-Encoder, ChebNet (K=3), and the proposed pathology-informed graph.

Category	Architecture	Accuracy	F1-Score	Precision	Recall
Baseline	CW-Encoder	48.82	46.13	46.09	46.77
Spatial convolution	CW-Encoder + GraphSAGE[3]	46.13	44.96	45.44	44.61
	CW-Encoder + GAT[4]	47.53	44.13	44.38	45.55
Spectral convolution	CW-Encoder + GCN[5]	55.70	52.63	52.29	53.41
	CW-Encoder + ChebNet[6] (K=2)	58.71	57.24	57.99	57.35
	CW-Encoder + ChebNet (K=3)	60.65	55.90	62.48	58.01
	CW-Encoder + ChebNet (K=4)	58.71	55.67	57.95	56.71
	AlCheNet (Ours)	62.47	60.29	64.17	61.09

IV. CONCLUSION

This study proposes a pathology-informed graph to address the often-overlooked issue of over-smoothing in graph-based EEG analysis, enabling the extraction of more meaningful representations. Experimental results demonstrated that the proposed approach led to improved classification performance. Furthermore, considering that EEG patterns vary depending on the type of neurological disorder, future work will explore adaptive methods for constructing adjacency matrices that reflect the patient's brain state dynamically.

ACKNOWLEDGMENTS

This research was supported by the MSIT(Ministry of Science, ICT), Korea, under the National Program for Excellence in SW, supervised by the IITP(Institute of Information &

REFERENCES

- [1] A. Demir, T. Koike-Akino, Y. Wang, M. Haruna, and D. Erdogmus, "EEG-GNN: Graph neural networks for classification of electroencephalogram (EEG) signals," in *Proc. 43rd Annu. Int. Conf. IEEE Eng. Med. Biol. Soc. (EMBC)*, Nov. 2021, pp. 1061–1067.
- [2] T. K. K. Ho and N. Armanfard, "Self-supervised learning for anomalous channel detection in EEG graphs: Application to seizure analysis," in *Proc. AAAI Conf. Artif. Intell.*, vol. 37, no. 7, pp. 7866–7874, Jun. 2023.
- [3] W. Hamilton, Z. Ying, and J. Leskovec, "Inductive representation learning on large graphs," in *Advances in Neural Information Processing Systems*, vol. 30, 2017.
- [4] P. Velickovic, G. Cucurull, A. Casanova, A. Romero, P. Lio, and Y. Bengio, "Graph attention networks," *Stat*, vol. 1050, no. 20, pp. 10–48550, 2017.
- [5] T. N. Kipf and M. Welling, "Semi-supervised classification with graph convolutional networks," *arXiv preprint arXiv:1609.02907*, 2016.
- [6] M. Defferrard, X. Bresson, and P. Vandergheynst, "Convolutional neural networks on graphs with fast localized spectral filtering," in *Advances in Neural Information Processing Systems*, vol. 29, 2016.
- [7] D. Rodinskaia, C. Radinski, and J. Labuhn, "EEG coherence as a marker of functional connectivity disruption in Alzheimer's disease," *Aging and Health Research*, vol. 2, no. 3, p. 100098, 2022.
- [8] K. Baik, J. H. Jung, S. H. Jeong, S. J. Chung, H. S. Yoo, P. H. Lee, Y. H. Sohn, S. W. Kang, and B. S. Ye, "Implication of EEG theta/alpha and theta/beta ratio in Alzheimer's and Lewy body disease," *Scientific Reports*, vol. 12, no. 1, p. 18706, 2022.

Deep Learning-Based Modeling of Ammonia Decomposition for Hydrogen Production

Gi Jin Kim^{a,1}, Hee Jae Kim^{b,2}, Yun Ju Yu^{a,2}, Seok Hwan Jin^{a,2}, Seong bin Ga^{a,*}

^a Department of chemical engineering, University of Ulsan, Ulsan 44610, South Korea

^b Department of Biomedical engineering, University of Ulsan, Ulsan 44610, South Korea

[Email: sgasqa@ulsan.ac.kr]

Abstract

Ammonia (NH₃) serves as a promising hydrogen carrier due to its ease of storage and transport. This study proposes a deep learning approach to predict the behavior of an ammonia decomposition reactor. A Deep Neural Network (DNN) is trained using simulation data generated over varying temperature, pressure, and volume conditions. Among tested activation functions, ReLU shows the best performance (MSE = 0.0074, R² = 0.997). The model accurately captures conversion and flow rates and generalizes well beyond the training range. These results demonstrate the potential of data-driven models for efficient and flexible reactor analysis.

Index Terms: Ammonia Decomposition, Deep Learning, AI, DNN

I. INTRODUCTION

Hydrogen has emerged as a promising clean energy source due to its high energy density and zero carbon dioxide (CO₂) emissions during combustion. However, its storage and transportation remain technically challenging, particularly because liquefied hydrogen requires cryogenic conditions below −253°C. As an alternative, liquefied ammonia (NH₃) has attracted significant attention as a hydrogen carrier. Ammonia can be liquefied under milder temperature and pressure conditions and benefits from a well-established global supply and storage infrastructure. Moreover, hydrogen can be extracted from ammonia via a catalytic decomposition reaction, making it a practical medium for hydrogen transport.

Conventional analysis of ammonia decomposition reactors has largely depended on experimental studies, which are often time-consuming, costly, and limited in their ability to explore wide operating conditions. To address these limitations, recent research has turned to data-driven approaches, such as deep learning, for more efficient and flexible prediction of reactor behavior under diverse process parameters.

II. SYSTEM MODEL AND METHODS

A. Target system: Ammonia Decomposition

The target process investigated in this study is the catalytic decomposition of ammonia (NH₃) to produce hydrogen (H₂) and nitrogen (N₂). A simplified process flow diagram (PFD) is shown in Fig. 1. The feed stream, consisting of ammonia, is first pressurized using a compressor and preheated via a heat exchanger to reach the desired operating pressure and temperature. The reaction occurs in a plug flow reactor (PBR), where

ammonia undergoes thermal decomposition. The reversible reaction governing ammonia decomposition is expressed as follows:



This endothermic reaction requires heat input, favoring higher temperatures for enhanced conversion. The reaction kinetics for the ammonia decomposition process follow the expression:

$$r_{\text{forward}} = k_{\text{forward}} \cdot P_{\text{NH}_3}^2, \quad r_{\text{reverse}} = k_{\text{reverse}} \cdot P_{\text{N}_2} \cdot P_{\text{H}_2}^3 \quad (2)$$

B. Flow chart

The overall research workflow is illustrated in Fig. 2. Process simulation data were generated based on the ammonia decomposition reactor model. Each data point includes three input variables—temperature (K), pressure (bar), and reactor volume (m^3)—and corresponding output values such as NH_3 conversion (X_{NH_3}), and the outlet molar flow rates of H_2 , and N_2 . A total of 3,375 data points are generated through simulation, of which 80% are used for training and 20% for validation. Deep learning models are trained on this dataset: a Deep Neural Network (DNN). Model accuracy is evaluated using the Mean Square Error (MSE) and the coefficient of determination (R^2), comparing predicted values against the simulated ground-truth data. This approach allows for the quantification of predictive capability and the comparative assessment of model performance.

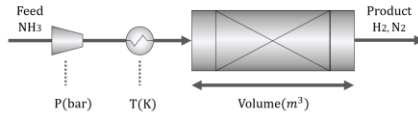


Fig 1 Target system of Ammonia decomposition process

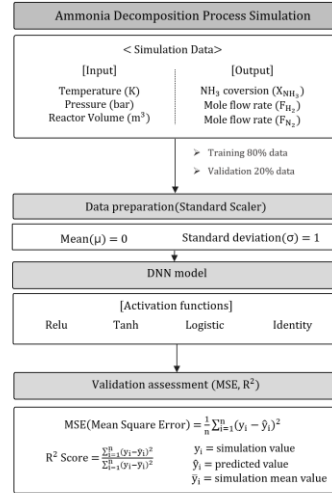


Fig 2 Flowchart of target system

III. Model Training and Evaluation

AI model is produced based on data extracted through an ammonia decomposition reactor modeled by Python. Deep Neural Network (DNN) is a class of artificial neural networks that consist of multiple hidden layers between input and output layers. Each neuron in a layer applies a nonlinear activation function to a weighted sum of its inputs, enabling the model to capture complex nonlinear relationships in the data. Fig 3 shows DNN algorithm of target system.

A. Activation Function Selection

To investigate the effect of different activation functions on model performance, four commonly used functions are tested: Relu, Tanh, Logistic (Sigmoid), and Identity.

The network architecture is fixed with hidden layers of sizes [64, 32,16], and training is performed for 3000 iterations using a fixed random seed to ensure reproducibility. Model performance is evaluated on the validation set using Mean Squared Error (MSE) and the coefficient of determination (R^2 score).

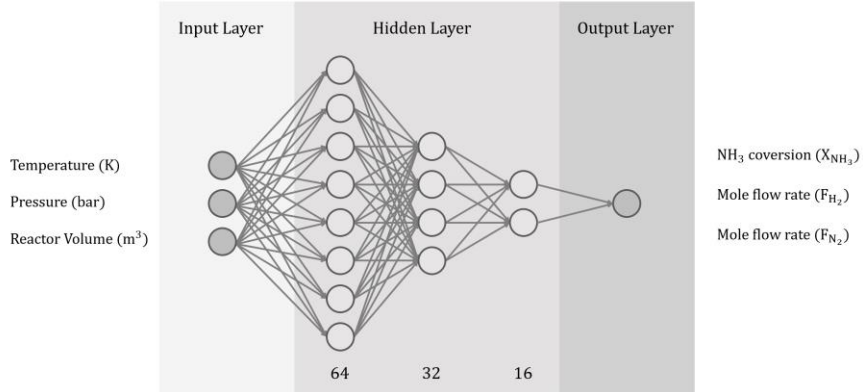


Fig 3 DNN algorithm of target system: Three Input(T,P,V), Hidden layer[64,32,16], Output layer(X_{NH_3} , F_{H_2} , F_{N_2})

Table 1 Comparison of Activation Functions Based on MSE and R^2 Score for NH_3 Conversion

Activation Function	MSE	R^2 Score
Relu	0.0074	0.997
Tanh	2.81	-6.26e-05
Logistic(Sigmoid)	2.81	-1.69e-05
Identity	18.4	-5.55e-05

Among the tested functions, Relu exhibits the best performance, achieving the lowest MSE and highest R^2 score. In contrast, the other activation functions result in significantly higher prediction errors and near-zero or negative R^2 values, indicating poor generalization. The negative R^2 scores suggest that the models performed worse than a simple mean-based prediction.

B. Model Training and Validation

The dataset, which includes simulation outputs such as NH_3 conversion and product flow rates, is split into a training set (80%) and a validation set (20%). A multilayer perceptron (MLP) model is implemented using the MLPRegressor module from scikit-learn, with Relu as the activation function. After training, the model's predictions on the validation set are compared against the corresponding simulation data. Fig. 4 shows the comparison between predicted and actual values for output. Predicted values from the trained model closely match the simulation results for both NH_3 conversion and product flow rates. The scatter plots demonstrate that the model not only captures the overall trends but also provides high accuracy across a wide range of input conditions.

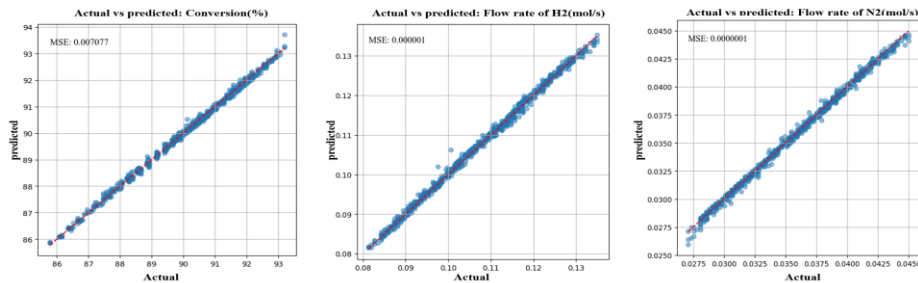


Fig 4 Comparison between actual and predicted values for each data: Conversion, Flow rate

Fig 5 illustrates the extrapolation performance of the DNN model beyond the training and validation ranges. The model demonstrates a strong predictive capability across wider temperature and pressure conditions, closely following the underlying physical trends.

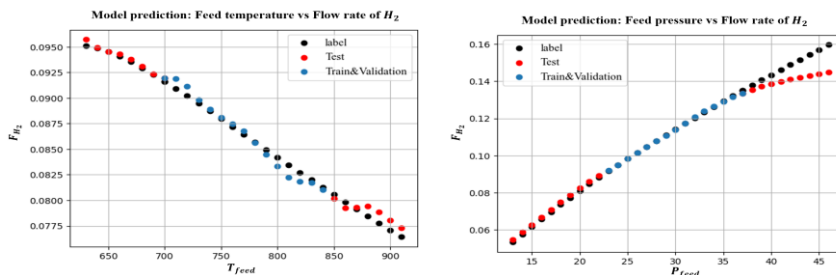


Fig 5 Extrapolation results for feed temperature and pressure vs. H₂ flow rate using the DNN model

IV. CONCLUSIONS

This study demonstrates the feasibility and accuracy of applying deep learning, particularly a DNN model with Relu activation, to predict the performance of an ammonia decomposition reactor. Using simulation data generated from a Python-based process model, the trained network achieves strong agreement with ground-truth values, with an R^2 score of 0.997 for ammonia conversion. Furthermore, the model maintains robust extrapolation capability beyond the original data range, effectively capturing the system's nonlinear behavior across varying conditions. These results suggest that data-driven modeling can serve as a powerful tool for rapid and flexible reactor analysis, reducing reliance on costly and time-intensive experiments. This approach may be further extended to optimize reactor design and operation in real-time industrial settings.

ACKNOWLEDGMENTS

This research was supported by the MSIT(Ministry of Science, ICT), Korea, under the National Program for Excellence in SW, supervised by the IITP(Institute of Information & communications Technology Planning & Evaluation) in 2025(2024-0-00025)

REFERENCES

- [1] Lee, Jaewon, et al. "Carbon-free green hydrogen production process with induction heating-based ammonia decomposition reactor." *Chemical Engineering Journal* 457 (2023): 141203.
- [2] Li, Guanpeng, et al. "Understanding error propagation in deep learning neural network (DNN) accelerators and applications." *Proc. of SC*, 2017.

Neural ODEs with Hierarchical Embeddings for Financial Forecasting

Yeong-Seo Jo

Department of IT Convergence, Ulsan University, 44616, Republic of Korea

[Email : hbys1405@naver.com]

Abstract

This study presents a comprehensive framework that combines Neural ODEs, hierarchical embeddings, and multi-modal data fusion to maximize Sharpe ratio in financial time series forecasting. The framework addresses practical challenges including irregular trading schedules, cross-asset correlations, and prediction lag through systematic integration of continuous-time modeling with GICS-based hierarchical structures.

Index Terms: Neural ODEs, financial time series forecasting, hierarchical embeddings, continuous-time modeling, Sharpe ratio optimization, temporal derivative consistency, dynamic time warping, multi-modal data fusion

I. INTRODUCTION

Financial markets exhibit high volatility with investors frequently underperforming benchmarks. While random walk and efficient market hypotheses emphasize prediction difficulty, observed inefficiencies like daily autocorrelation and sector momentum present opportunities for sophisticated analysis.

Existing LSTM/Transformer models inadequately address irregular time intervals, hierarchical structures, and prediction signal lag, causing temporal alignment errors that reduce investment performance [1]. This research proposes: ① bidirectional continuous-time ODE framework for irregular time series, ② GICS-based hierarchical embeddings for cross-asset correlations, and ③ dual optimization of value prediction and temporal derivative consistency.

II. BACKGROUNDS

A. Time Series Prediction Model Evolution

RNN/LSTM architectures demonstrate strength in learning long-term dependencies but are vulnerable to irregularities in financial data such as market closures and global market asynchronization due to discrete time step assumptions, with gradient vanishing problems occurring in deep network structures. Transformer-based models can capture global dependencies but have limitations in real-time volatility response due to computational complexity and temporal axis distortion issues. To overcome these challenges, Neural ODE/CDE approaches have improved irregular observation

data processing and inflection point alignment accuracy through continuous time domain modeling [2][3], but face trade-offs with increased computational costs from iterative ODE solving.

B. Evaluation Metric System Innovation

Traditional metrics such as MSE and MAE focus on value error measurement while overlooking temporal delays in prediction signals. Dynamic Time Warping (DTW) quantifies time series shape similarity through optimal alignment path exploration, while Temporal Displacement Index (TDI) numerically captures the degree of temporal axis deviation to evaluate prediction timing accuracy [4]. This research maximizes signal alignment performance in volatility periods by integrating DTW and TDI into the loss function, and verifies practical applicability through correlation analysis with investment performance metrics such as Sharpe ratio and maximum drawdown.

$$DTW(X, Y) = \min_{\pi} \sum_{(i,j) \in \pi} d(x_i, y_j) \quad TDI = \frac{1}{n} \sum_{i=1}^n |\tau_i - t_i|$$

III. PROPOSED METHOD

The proposed framework integrates continuous-time Neural ODEs with hierarchical embeddings to address financial time series forecasting challenges. The architecture processes multi-modal data while capturing hierarchical market structures and optimizing for practical trading performance.

Data integration combines 60 features across technical indicators (EMA, MACD, RSI, with backtesting-optimized parameters), financial statements (20 key ratios converted to daily frequency using exponential smoothing), and macroeconomic data. Missing values are imputed using industry averages, with tanh scaling applied for [-1, 1] normalization [5].

The continuous-time processing component handles irregular intervals through explicit time incorporation (Δt) between observations. Hermite cubic spline interpolation generates smooth trajectories, while weighted central difference methods calculate temporal derivatives for the Neural ODE formulation [6]:

$$\frac{dy}{dt} = w_{next} \cdot \frac{y_{i+1} - y_i}{\Delta t_{i+1}} - w_{prev} \cdot \frac{y_i - y_{i-1}}{\Delta t_i}$$

Hierarchical correlation modeling employs a three-tier GICS-based embedding structure (sector→industry→stock) with multi-head attention for dynamic importance weighting. This captures time-varying correlation structures across different market regimes while modeling cross-asset dependencies effectively.

The bidirectional Neural ODE architecture processes forward and backward temporal sequences using RK4 integration for numerical stability [7]. The dual-objective loss function addresses prediction lag by combining MSE with temporal derivative consistency:

$$L_{total} = \alpha \cdot L_{value} + (1 - \alpha) \cdot L_{derivative}$$

Model training employed temporal splitting and hyperparameter grid search with early stopping, learning rate scheduling, and gradient clipping. Model selection used weighted combination of returns and DTW/TDI metrics.

IV. EXPERIMENTS

A. Experimental Setup

Time series data from 12 NYSE stocks (4 each from technology, finance, and consumer goods sectors) over 5 years (2020-2025) were analyzed. Data was split into training (60%)-validation (20%)-testing (20%), with backtesting performed reflecting transaction fees (0.015%) and taxes (0.0008%). Risk-free returns were applied using 3-month US Treasury yields collected through FRED API.

Trading signals were generated by searching threshold values in the 0-0.05 range that maximize Sharpe ratio on the validation dataset, then generating buy/sell signals on the test dataset.

For comparison experiments, vanilla LSTM and CONTIME-RNN were set as baselines, with performance compared by changing only model architecture while maintaining identical data processing and evaluation pipelines.

B. Experimental Results

The proposed CONTIME framework demonstrates superior performance across all evaluated metrics. The model achieves the highest annual return of 23.60%, representing a 17% improvement over vanilla LSTM (20.18%). Regarding temporal alignment metrics, the proposed method shows significant improvements with DTW score reduced by 38% (8.52 vs 13.81) and TDI improved by 34% (10.23 vs 15.60) compared to LSTM baseline. These improvements indicate substantially reduced prediction lag, addressing a critical limitation in financial time series forecasting.

Table 1. Performance Comparison Across Different Models

Models	Return	DTW	TDI	Precision
Buy and Hold	12.21%	-	-	-
LSTM	20.18%	13.81	15.60	52.94%
Contime-RNN	21.35%	8.12	9.62	59.74%
Contime-LSTM	23.60%	8.52	10.23	63.41%

V. CONCLUSIONS

The proposed CONTIME framework addresses prediction lag problems in financial time series through bidirectional neural ODEs and GICS-based hierarchical embeddings. Hermite cubic spline interpolation and continuous-time ODE integration

process irregular interval data, while multi-modal data integration enables comprehensive analysis of technical, fundamental, and macroeconomic indicators. Experimental results achieved 17% improvement in annual return and 34% reduction in TDI compared to LSTM, effectively mitigating signal lag through dual loss function and threshold optimization. The model has been confirmed to maximize practical performance in real investment environments.

ACKNOWLEDGMENTS

This research was supported by the MSIT(Ministry of Science, ICT), Korea, under the National Program for Excellence in SW, supervised by the IITP(Institute of Information & communications Technology Planning & Evaluation) in 2025(2024-0-00025)

REFERENCES

- [1] S. Y. Jhin, S. Kim, and N. Park, "Addressing prediction delays in time series forecasting: A continuous GRU approach with derivative regularization," in *Proceedings of the 30th ACM SIGKDD Conference on Knowledge Discovery and Data Mining*, 2024.
- [2] R. T. Q. Chen, Y. Rubanova, J. Bettencourt, and D. K. Duvenaud, "Neural ordinary differential equations," in *Advances in Neural Information Processing Systems*, vol. 31, 2018.
- [3] P. Kidger, J. Morrill, J. Foster, and T. Lyons, "Neural controlled differential equations for irregular time series," *Advances in Neural Information Processing Systems*, vol. 33, pp. 6696–6707, 2020.
- [4] M. Müller, "Dynamic time warping," in *Information Retrieval for Music and Motion*, 2007, pp. 69–84.
- [5] S. C. Nayak, B. B. Misra, and H. S. Behera, "Impact of data normalization on stock index forecasting," *International Journal of Computer Information Systems and Industrial Management Applications*, vol. 6, pp. 13, 2014
- [6] C. de Boor, K. Höllig, and M. Sabin, "High accuracy geometric Hermite interpolation," *Computer Aided Geometric Design*, vol. 4, pp. 269–278, 1987.
- [7] J. C. Butcher, "Coefficients for the study of Runge-Kutta integration processes," *Journal of the Australian Mathematical Society*, vol. 3, no. 2, pp. 185–201, 1963.

A Case Study on Facility Management Based on Digital Twin Technology

YuDoo Kim^{1*}

¹Department of AI Software, Dongyang Mirae University, 08221, Korea
[Email : kimyudoo@dongyang.ac.kr]

Abstract

In recent years, digital twin technology has emerged as a transformative approach to infrastructure and facility management, enabling real-time monitoring, predictive maintenance, and data-driven decision-making. This paper presents a case study on the application of digital twin technology in the management of building and civil infrastructure facilities. The study focuses on how digital twins were constructed using real-time sensor data, 3D modeling, and IoT integration, and how these models were utilized to monitor structural health, predict failures, and optimize maintenance strategies. The results demonstrate improvements in operational efficiency, cost savings, and risk reduction. This research provides practical insights into the challenges and benefits of deploying digital twin systems in real-world facility management contexts and proposes a framework for broader implementation.

Index Terms: Digital Twin, Facility Management, Predictive Maintenance, Structural Health Monitoring, Smart Infrastructure

I. INTRODUCTION

In recent years, the increasing complexity of infrastructure systems and the demand for more efficient and sustainable facility operations have accelerated the adoption of emerging digital technologies in the built environment. Among these technologies, digital twin (DT) systems have garnered significant attention as a transformative solution for real-time infrastructure monitoring, predictive maintenance, and informed decision-making.

A digital twin is a dynamic digital replica of a physical asset, system, or process, continuously updated through data collected from sensors, Internet of Things (IoT) devices, and Building Information Modeling (BIM). By synchronizing the physical and virtual environments, digital twins allow facility managers to visualize operational states, simulate performance scenarios, and implement proactive maintenance strategies. As urban environments grow more connected and data-intensive, the integration of digital twins into facility management (FM) workflows represents a key enabler of smart, resilient, and cost-effective operations[1].

Despite the growing body of literature on digital twin concepts, their application in real-world facility management remains a developing field, particularly in terms of practical implementation, measurable outcomes, and integration challenges. This research aims to address this gap by presenting a case study in which digital twin technology is applied to

building and civil infrastructure facilities. The study investigates how digital twins were constructed using 3D modeling, real-time sensor data, and IoT integration, and how these models enhanced structural monitoring, failure prediction, and maintenance optimization.

This paper is structured as follows: Section 2 provides a conceptual and technological analysis of digital twin systems, including their architecture, components, and applications in facility management. Section 3 presents a detailed case study demonstrating the implementation of a digital twin in an infrastructure facility, highlighting the workflow, data flow, and results. Section 4 concludes the paper by summarizing the findings and proposing a framework for broader adoption of digital twin technology in infrastructure and facility management.

II. Analysis of Digital Twin

A digital twin for facility management typically comprises the following core layers:

- **Physical Asset Layer:** The actual infrastructure (e.g., buildings, HVAC systems, utilities) that is being modeled and monitored.
- **Sensing Layer:** Includes embedded IoT devices and sensors that capture real-time data such as temperature, vibration, energy consumption, occupancy, or humidity.
- **Data Integration Layer:** Aggregates and processes data from heterogeneous sources (e.g., sensors, enterprise systems, BIM models).
- **Digital Model Layer:** A 3D or semantic model created using CAD, BIM, or GIS technologies to visualize the asset in real-time.
- **Analytics and Intelligence Layer:** Applies machine learning, statistical models, or rule-based systems to predict failures, identify anomalies, and optimize performance.
- **Interface and Control Layer:** Provides users with dashboards, alerts, and simulation interfaces to support decision-making and automate responses.
-

This layered architecture enables interoperability and scalability, allowing facility managers to expand capabilities over time as systems and data evolve[2].

III. Case Study: Facility Management with Digital Twin

The case study was conducted at a mid-sized university campus facility composed of multiple buildings, including academic halls, laboratories, and administrative offices. The objective of the project was to implement a digital twin platform to support smart facility management practices, particularly focusing on predictive maintenance, energy efficiency, and structural health monitoring.

The digital twin system was developed by integrating Building Information Modeling (BIM), real-time sensor data, and IoT infrastructure into a cloud-based platform. The facility management team sought to transition from a traditional schedule-based maintenance model to a data-driven, condition-based strategy enabled by the digital twin[3-6].

Table 1. Summary Table: Digital Twin Implementation for Facility Management

Step	Component	Key Activities	Purpose
Step1	3D Model Generation	- Collected and refined existing BIM models - Used Revit and Navisworks for accurate spatial modeling	Provide structural visualization and foundational model for the digital twin
Step2	IoT Sensor Deployment	- Installed temperature/humidity sensors - Vibration sensors for equipment - Occupancy/motion detectors - Energy consumption meters	Enable real-time data acquisition from the physical environment
Step3	Data Integration Platform	- Constructed cloud data lake - Synchronized with Azure Digital Twins	Aggregate and stream heterogeneous data in near real-time
Step4	Analytics & Predictive Models	- Applied machine learning for failure prediction - Anomaly detection - Energy usage pattern analysis	Optimize maintenance and energy performance with predictive insights
Step5	User Interface & Dashboard	- Developed dashboards with Power BI - Implemented 3D visual environment (Unity) - Displayed alerts, KPIs, and real-time insights	Support facility managers in decision-making and rapid response

Despite the successful outcomes, several challenges were encountered during the implementation:

- **Data Quality Issues:** Sensor drift and occasional communication loss required frequent calibration and redundancy.
- **Integration Complexity:** Ensuring interoperability between legacy BMS systems and modern cloud platforms required customized APIs.
- **Change Management:** Facility staff required training to interpret data and make operational decisions based on the dashboard insights.
- **Initial Costs:** High upfront investment in sensors and infrastructure was a barrier, although offset by long-term savings.
-

These challenges emphasize the importance of phased implementation, stakeholder engagement, and robust data governance in digital twin projects.

IV. CONCLUSIONS

This study demonstrated the practical implementation of digital twin technology in facility management through a real-world case involving building and civil infrastructure. By integrating IoT sensors, real-time data processing, and 3D modeling, a functional digital twin system was developed to support predictive maintenance, energy optimization, and structural monitoring.

The results highlight significant improvements in operational efficiency, cost savings, and risk mitigation. Despite challenges such as system integration and data quality, the benefits of adopting a digital twin approach were evident and measurable.

This research contributes to the growing body of knowledge on smart facility management and offers a practical framework for the broader adoption of digital twin systems. Future work may explore the integration of AI-driven automation, digital twin scalability across multi-building complexes, and long-term impact assessment on facility sustainability.

ACKNOWLEDGMENTS

This research was supported by the internal academic research grant program of Dongyang Mirae University.

REFERENCES

- [1] Akinshipe, O., & Aigbavboa, C. (2022). The Future of Facility Management: A Case for Digital Twin. Retrieved from <https://www.researchgate.net/>.
- [2] Abdelalim, A. M., Sherif, A., & Salem, M. (2025). Optimizing Facilities Management Through Artificial Intelligence and Digital Twin Technology in Mega-Facilities. *Sustainability*, 17(5), 1826.
- [3] Pavón, R. M., Alberti, M. G., & Álvarez, A. A. (2025). BIM-based Digital Twin Development for University Campus Management. *Expert Systems with Applications*.
- [4] Harode, A., Dongre, P., & Thabet, W. (2023). A Tool-Based System Architecture for a Digital Twin: A Case Study in a Healthcare Facility. Virginia Tech. <https://vtechworks.lib.vt.edu/>.
- [5] Locatelli, M., Pellegrini, L., & Seghezzi, E. (2021). Towards an Occupancy-Oriented Digital Twin for Facility Management. *Applied Sciences*, 11(7), 3108.
- [6] Madubuike, O. C., & Anumba, C. J. (2023). Digital Twin–Based Health Care Facilities Management. *Journal of Computing in Civil Engineering*. 37(2), 1-11.

From Subjective Perception to Semantic Embedding: Learning the Latent Structure of Urban Visual Experience

Sang-Il Yoon¹, Kwang-Woo Nam², and Chang-Woo Lee^{3*}

¹Department of Computer Information Engineering Kunsan University, Gunsan 54150,
Republic of Korea

[Email : tkddlf1026@naver.com; kwnam@kunsan.ac.kr; leecw@kunsan.ac.kr;

Abstract

Many cities around the world are promoting the development of eco-friendly and pedestrian-oriented walking environments. While both objective measurements and subjective surveys are used to assess walkability, quantifying subjective perceptions and utilizing them for meaningful spatial analysis remains a significant challenge. This study investigates whether semantic features can be derived from subjective evaluations by utilizing a dataset of 21,138 crowdsourcing walkability scores based on street view images from Jeonju, South Korea. Our experimental results demonstrate that subjective assessments of street views can be naturally embedded into a latent space using convolutional neural networks (CNNs). Furthermore, we validate that these embeddings capture semantic characteristics through vector arithmetic operations. These findings suggest that subjective human judgments inherently contain objective information, and that deep learning techniques can effectively quantify and analyze these subjective evaluations.

Index Terms: Convolutional Neural Networks, Semantic Embedding, Vector Arithmetic Operations, Walkability.

I. INTRODUCTION

Walking is the one of the most fundamental, sustainable, and health-promoting mode of transportation. Numerous studies have found significant correlations between pedestrian activity and various urban issues, including traffic congestion, environmental and social costs, and even increases in housing prices [1, 2].

As the importance of pedestrian environments gains global recognition, diverse urban data are being leveraged to evaluate walkability, and pedestrian-friendly projects are actively being implemented in urban planning and policy-making processes [3].

Walkability is inherently a subjective concept, with evaluations often varying based on individual perceptions, making it difficult to measure objectively and quantitatively [4]. In a previous study, citizens' subjective perceptions of walkability in Jeonju were collected through a crowdsourcing platform, and 21,138 evaluation scores were obtained using the TrueSkill algorithm [5]. Nevertheless, since these scores are based on subjective

* Corresponding author

assessments, there are limitations in interpreting the data as an objective and quantitative representation of walkability.

To address this challenge, the present study utilizes convolutional neural networks (CNNs) to investigate whether crowd-sourced walkability evaluations reflect consistent semantic patterns and encode latent objective features. Specifically, we extract visual features from street view images using a CNN and map them into a low-dimensional embedding space and assess the semantic coherence of these embeddings through vector arithmetic operations.

II. SYSTEM MODEL AND METHODS

A. Dataset

This study utilizes a dataset consisting of 21,138 street view images and their corresponding walkability scores, collected in Jeonju, South Korea. The dataset was created through crowdsourcing and processed using the TrueSkill algorithm to assign walkability scores to each image [5]. The dataset was randomly divided into training (14,796 samples, 70%), validation (3,171 samples, 15%), and test (3,171 samples, 15%) subsets for model training and evaluation.

B. Triplet Network Architecture

To semantically structure subjective walkability evaluations in an embedding space, we employed a Triplet Network. The model takes triplets of images—an anchor, a positive, and a negative—as input and learns to adjust the distances among their embedding vectors accordingly. A positive sample is randomly selected from images whose walkability scores differ from that of the anchor by less than ± 10 , while a negative sample is randomly drawn from those with score differences greater than ± 10 . The network is trained using a triplet loss function with a margin of 1.

As the feature extractor, we adopted a ResNet-50 backbone pretrained on ImageNet. The extracted features are passed through a fully connected layer to generate a 16-dimensional embedding vector. The training objective is to ensure that images with similar walkability scores are embedded closer together in the vector space.

III. EXPERIMENT

A. Embedding Space Visualization

To assess whether the learned embeddings effectively capture semantic similarity, we applied dimensionality reduction techniques—Principal Component Analysis (PCA) and t-Distributed Stochastic Neighbor Embedding (t-SNE)—to project the 16-dimensional vectors into a 2D space for visualization. The resulting plots revealed that images with similar walkability scores formed dense clusters, indicating that the embeddings reflect perceptual consistency as shown in Fig. 1.

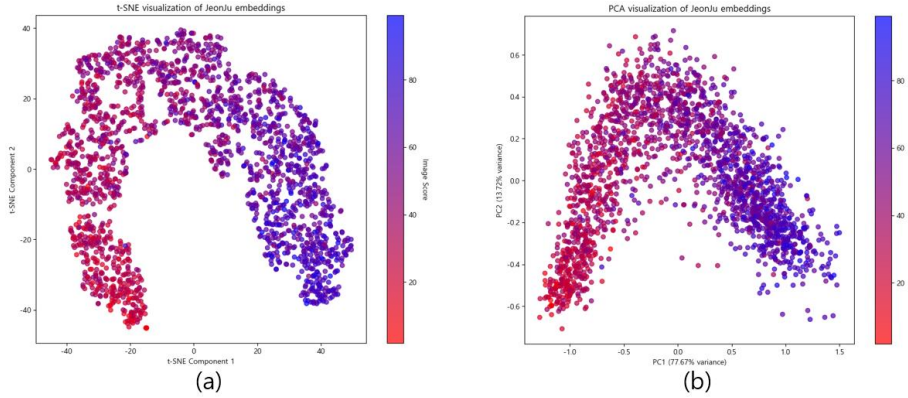


Fig. 1 2D visualization of the learned 16-dimensional embeddings using (a) PCA and (b) t-SNE.

B. Vector Arithmetic in Embedding Space

To further evaluate the semantic structure of the embedding space, we conducted vector arithmetic operations, including both addition and subtraction of embedding vectors. This approach is inspired by similar practices in word embeddings, where analogical reasoning (e.g., *king - man + woman* $\approx$ *queen*) reveals underlying semantic relationships.

As shown in Figs. 2 and 3, our results show that the learned embeddings support meaningful composition and decomposition of visual features through simple arithmetic operations. These findings suggest that the embedding space encodes not only perceptual similarity but also disentangled semantic components, such as tree presence, building density, and openness.

The ability to manipulate scene attributes in this manner opens possibilities for semantic-level image editing and interpretable representation learning in urban environments. Furthermore, it implies that subjective evaluations of walkability are closely tied to identifiable visual elements, which can be recombined or isolated within the latent space.

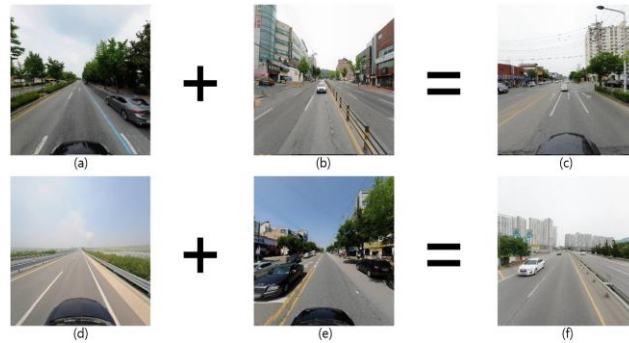


Fig. 2 Vector arithmetic operations in the learned embedding space. (a) + (b) $\approx$ (c): Combining a tree-lined road and a dense commercial street yields a scene that integrates both greenery and urban features. (d) + (e) $\approx$ (f): Adding a sparse riverside road to a vibrant residential area results in a scene reflecting both openness and residential density. In both examples, the resulting vector was closest to the embedding of (c) and (f), respectively.

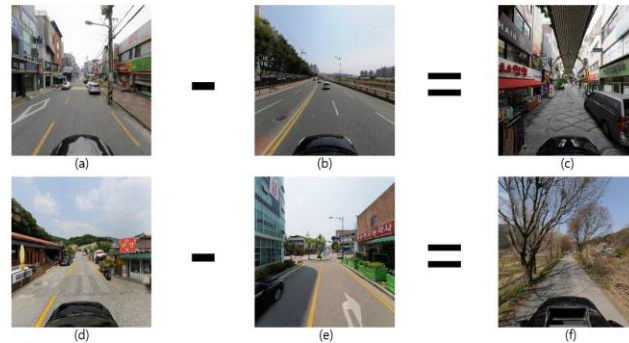


Fig. 3 Vector subtraction operations in the learned embedding space. (a) – (b) $\approx$ (c): Subtracting a wide-open road from a commercial street results in an even denser and more pedestrian-oriented urban scene. (d) – (e) $\approx$ (f): Removing residential features from a mixed-use neighborhood reveals the underlying natural and undeveloped landscape. These examples demonstrate how vector subtraction can isolate and emphasize specific semantic elements within urban environments.

IV. CONCLUSIONS

This study investigated whether crowdsourced walkability evaluations from street view images in Jeonju, South Korea, can be embedded into a semantically meaningful vector space using a Triplet Network. Visualizations (PCA, t-SNE) showed that images with similar walkability scores clustered naturally, revealing that the embeddings captured latent subjective perceptions. Vector arithmetic demonstrated that these embeddings reflect visual characteristics associated with walkability perceptions, linking them to aspects of urban form. Future work will explore alternative embedding methods and integrate temporal variations to model dynamic changes in urban environments.

ACKNOWLEDGMENTS

This work is supported by the Korea Agency for Infrastructure Technology Advancement(KAIA) grant funded by the Ministry of Land, Infrastructure and Transport (Grant RS-2022-00143336).

REFERENCES

- [1] J. Jacobs, *Dark Age Ahead*, New York, NY: Vintage, 2007.
- [2] F. Abastante et al., "Supporting resilient urban planning through walkability assessment," *Sustainability*, vol. 12, no. 19, article 8131, 2020.
- [3] A. Kustar et al., "Mexico Made Safe Mobility a Human Right — Here's How Its Streets Can Become Safer," World Resources Institute, May 8, 2024. [Online]. Available: www.wri.org/insights/safer-streets-pedestrian-mobility-mexico.
- [4] H. Wang and Y. Yang, "Neighbourhood walkability: A review and bibliometric analysis," *Cities*, vol. 93, pp. 43-61, 2019.
- [5] J. Hwang, K. Nam, and C. Lee, "Spatial Image-Based Walkability Evaluation Using Regression Model," *Applied Sciences*, vol. 14, no. 10, article 4079, 2024.

Deepfake Detection Method Based on Transformer Model

Jong-Bae Park¹, Young-Bok Cho^{2*}

¹Department of Radiology, Seoul Care Hospital, 21083, Republic of Korea

²Department of Computer Education, Gyeongbuk National University, 36830, Republic of Korea

[Email : pjcbdr@naver.com , ybcho@gknu.ac.kr]

Abstract

The rapid advancement of artificial intelligence has opened new domains for generative AI, enabling human-like synthesis of media such as images, audio, and text. While this technology holds potential in fields like healthcare, education, and the arts, it has also become a tool for malicious applications such as deepfakes. Deepfake technology leverages GANs, Autoencoders, and Transformers to generate highly realistic synthetic content, posing risks of privacy invasion, identity theft, fake news, and digital crimes. This paper proposes and evaluates a Transformer-based detection method that focuses on identifying malicious deepfake images.

Index Terms: Deepfake, Transformer, Vision Transformer, Image Forensics, Deep Learning

I. INTRODUCTION

The exponential growth of AI technology has brought generative AI to the forefront, allowing synthesis of media at a human-like level. While this has beneficial applications, it also introduces severe risks when exploited for harmful purposes. Deepfakes, created using models such as GANs, Autoencoders, and Transformers, have reached a level where it is nearly impossible to distinguish between real and fake content. This has given rise to various societal problems, including privacy violations and identity fraud. This study proposes a deepfake image detection model that uses only visual frame data. A Transformer-based architecture is adopted for its strength in capturing fine-grained visual inconsistencies like boundary artifacts, unnatural textures, and facial asymmetries

II. SYSTEM MODEL AND METHODS

A. System Overview

The proposed detection system uses video frames to determine whether an image has been manipulated. It consists of four main stages:

* Corresponding author



B. Video Preprocessing

To improve the accuracy of video-based learning, a series of preprocessing steps are applied to the input video, including frame extraction, face alignment and cropping, resizing and normalization, and, if necessary, data augmentation. Frame extraction is performed at a fixed interval of 5 frames per second (5fps) from the input video to ensure temporal consistency during training. Face alignment and cropping are carried out using the MTCNN face detection algorithm, which aligns and crops only the facial region to focus analysis on the subject. Resizing and normalization involve adjusting each frame to 224×224 pixels and scaling pixel values to the [0, 1] range. Data augmentation techniques such as horizontal flipping, brightness adjustment, and random cropping are applied to enhance the model's generalization performance.

C. Transformer-based Model Architecture and Training Configuration

The core of the proposed system is a deep learning structure based on the Vision Transformer (ViT), which, unlike CNNs, processes input image patches as sequential tokens, enabling the model to learn long-range dependencies. In the proposed model, each patch is linearly embedded into a high-dimensional vector through a patch-splitting and tokenization process. During the patch-splitting and tokenization phase, image patches are converted into embeddings, and positional encodings are added to preserve the spatial relationships between patches. These tokenized embeddings are then fed into a Transformer encoder, which consists of multiple self-attention blocks and feedforward networks. This allows the model to effectively learn inter-token relationships and detect artifacts such as boundary distortions, unnatural skin textures, and facial asymmetries.

Finally, a classification head uses the [CLS] token to perform binary classification through a Sigmoid activation function, determining whether the input image is real or a deepfake.

To optimize model performance, this study employs the Binary Cross Entropy (BCE) loss function, which is suitable for probability-based loss calculation. For optimization, the AdamW algorithm is used, which incorporates built-in L2 regularization to reduce overfitting and improve generalization. As for the hyperparameters, training was conducted for 20–30 epochs with a batch size ranging from 16 to 32. Transfer learning was applied using pretrained weights from ImageNet-21k to initialize the Vision Transformer model.

III. RESULTS

A. Experimental Environment

To validate the performance of the proposed deepfake image detection model, experiments were conducted in a PyTorch-based environment running on Ubuntu

20.04 with Python 3.10. The hardware setup included an NVIDIA A100 GPU (40GB VRAM) and 128GB of system RAM.

B. Dataset

The model was fine-tuned using the Celeb-DF v2 dataset, which consists of 590 real videos and 5,639 high-quality deepfake videos. The dataset was split into 70% for training, 15% for validation, and 15% for testing

C. Evaluation Metrics

[Table 1] presents the performance evaluation results for binary classification for each model.

Table 1. Performance Comparison by Model

Model	Accuracy	Precision	Recall	F1 Score	AUC
Xception(CNN)	92.4%	91.5	91.1	91.3	0.935
EfficientNet-LSTM	94.1%	93.2	94.0	93.6	0.954
Proposed(ViT)	96.8%	96.3	96.5	96.4	0.976

According to the experimental results shown in Table 1, the proposed model based on the Vision Transformer (ViT) demonstrated an accuracy improvement of 2.7% to 4.4% compared to the traditional CNN-based Xception and EfficientNet-LSTM models. In terms of F1 Score—which reflects both precision and recall—the proposed model outperformed the others by a margin of 0.022 to 0.041

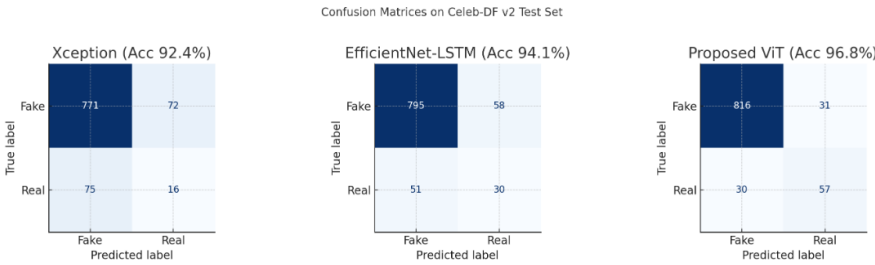


Fig.1. Confusion Matrix by Model

The confusion matrices show the classification results of three models on the Celeb-DF v2 test set. The proposed ViT model outperforms Xception and EfficientNet-LSTM by achieving the fewest false predictions and the highest number of correctly detected deepfakes[Fig.1].

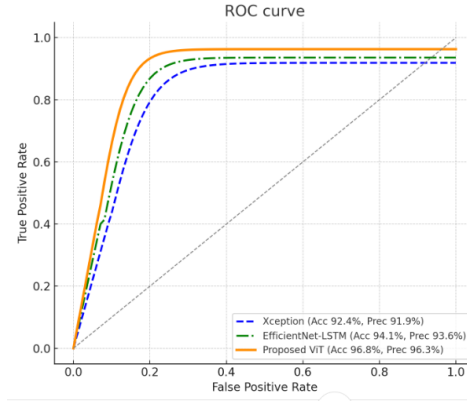


Fig.2. Comparison between CNN based Xception, EfficientNet-LSTM and Proposed ViT

The self-attention-based architecture enables the model to learn global features such as facial texture, boundaries, and lighting variations, making it highly effective in detecting subtle manipulation patterns. In particular, the model demonstrated strong generalization capability by maintaining over 95% accuracy even on realistic datasets such as Celeb-DF v2 [Fig.2].

IV. DISCUSSION AND CONCLUSIONS

The proposed ViT-based model outperformed existing approaches such as Xception and LSTM-CNN combinations in terms of both accuracy and AUC, thereby validating its potential to enhance deepfake detection performance. These findings suggest that the model can serve as a technical foundation for legal and forensic applications in cases of privacy violations caused by deepfakes. Future research should explore expanding this approach into a multimodal Transformer architecture that integrates not only visual information but also audio and textual modalities for more robust and comprehensive deepfake detection.

REFERENCES

- [1] E. H. Choi, H. S. Hwang, and C. S. Kim, "Electron spectroscopy studies on magneto-optical media and plastic substrate interfaces," *International Journal of Information and Communication Engineering*, vol. 9, no. 4, pp. 358-362, 2011.
- [2] J. G. Proakis, *Digital Communications*, 4th ed., New York, NY: McGraw-Hill, 1993.
- [3] J. L. Hennessy and D. A. Patterson, "Instruction-level parallelism and its exploitation," in *Computer Architecture: A Quantitative Approach*, 4th ed., San Francisco, CA: Morgan Kaufmann Pub., pp. 66-153, 2007.
- [4] A. Hashmi, H. Berry, O. Temam, and M. Lipasti, "Automatic abstraction and fault tolerance in cortical microarchitectures," in *Proceeding of the 38th Annual International Symposium on Computer Architecture*, New York, NY, pp. 1-10, 2011.

Hybrid Conservative Q-Learning for Robust Alzheimer’s Disease Stage Classification

Chanyi Lee¹, Ji Won Kim¹, Minseo Choe¹, and Sang Ho Oh^{1*}

¹Department of Computer Engineering and Artificial Intelligence, Pukyong National University, Busan, 48513, Republic of Korea
[Email : shoh0320@pknu.ac.kr]

Abstract

Early and accurate diagnosis of Alzheimer’s disease remains a significant challenge. Conventional supervised learning models often lack adaptability and robustness in uncertain or imbalanced medical contexts. To address this, we propose a Hybrid Conservative Q-Learning (Hybrid CQL) framework that integrates offline reinforcement learning with supervised learning guidance. By jointly optimizing conservative Q-value estimation and cross-entropy loss, and incorporating an entropy-aware reward structure, the model achieves clinically aligned and interpretable decision-making. Experimental results on the ADNI dataset demonstrate that Hybrid CQL outperforms existing baselines in cumulative reward and exhibits stable performance, particularly for cognitively normal and dementia classifications, highlighting its potential in offline clinical decision support.

Index Terms: Alzheimer’s Disease Classification, Conservative Q-Learning, Hybrid Reinforcement Learning, Static Clinical Dataset

I. INTRODUCTION

Alzheimer’s disease poses significant diagnostic challenges due to clinical ambiguity and data imbalance [1]. Traditional supervised learning methods, though effective in processing neuroimaging and biomarker data, struggle with uncertainty, limited adaptability, and suboptimal performance in imbalanced clinical datasets [2]. Offline reinforcement learning (RL) offers a promising alternative by enabling policy learning from static datasets without online interaction [3]. Conservative Q-Learning (CQL) has shown robustness in this context, but its conservative design and misalignment with expert-labeled data may hinder performance in complex clinical cases [4].

To address these limitations, we propose a Hybrid Conservative Q-Learning (Hybrid CQL) framework that integrates supervised learning signals into the offline RL paradigm. Our approach introduces a reward function that reflects both class-specific clinical significance and policy confidence, and unifies conservative Q-learning with cross-entropy loss for stable and interpretable policy learning. Experimental results on the Alzheimer’s Disease Neuroimaging Initiative (ADNI) dataset [5] demonstrate that Hybrid CQL achieves superior cumulative rewards and robust stage-wise classification, particularly for cognitively normal and dementia categories, while maintaining reliable performance under uncertainty.

The remainder of this paper is organized as follows. Section 2 presents the system model and methodological framework. Section 3 presents the experimental results. In Section 4,

* Corresponding author

we offer a comprehensive discussion of the findings. Section 5 concludes the paper by addressing the study’s limitations and outlining potential future research.

II. SYSTEM MODEL AND METHODS

A. Reinforcement Learning Framework

In our framework, the Markov Decision Process (MDP) is defined over a static clinical dataset, where each instance represents a patient profile with multimodal diagnostic features. The MDP is formally defined as follows:

- The state encodes a patient’s clinical indicators, including imaging biomarkers, neurocognitive assessments, and fluid-based markers.
- The action represents the predicted diagnosis: 0 for cognitively normal (CN), 1 for mild cognitive impairment (MCI), and 2 for dementia.
- The reward function emphasizes the clinical importance of diagnostic precision, especially for MCI, and incorporates both structured clinical weights and agent confidence as shown in (1).

$$R(s, a, y) = \begin{cases} w_y + \beta + p_a, & \text{if } a = y \\ -0.5 + \beta, & \text{if } a \neq y \end{cases} \quad (1)$$

Where, w_y is a class-dependent clinical weight (CN = 1.0, MCI = 1.5, Dementia = 1.2), $\beta \sim \text{Beta}(2,5)$ introduces exploration noise to encourage robust policy updates, p_a is the softmax probability of the selected action a , reflecting model confidence.

This reward shaping mechanism integrates diagnostic significance with agent certainty, promoting semantically coherent and clinically aligned behavior.

B. Hybrid CQL with Supervised Guidance

In our work, we extend the CQL framework by incorporating a supervised learning signal to guide the agent’s Q-value learning. This hybrid strategy leverages the robustness of conservative RL while directly aligning value estimates with known diagnostic labels, enhancing both training stability and classification accuracy in a label-constrained clinical setting.

To train the agent, we define a hybrid loss that combines conservative Q-learning with supervised learning objectives. The total loss is formulated as (2).

$$\mathcal{L}_{\text{total}} = \alpha_{\text{RL}} (\mathcal{L}_{\text{Bellman}} + \lambda_{\text{CQL}} \cdot \mathcal{L}_{\text{CQL}}) + \alpha_{\text{sup}} \cdot \mathcal{L}_{\text{sup}} \quad (2)$$

Where, α_{RL} and α_{sup} control the contribution of RL and supervised terms, empirically set to 0.4 and 0.6 respectively. $\lambda_{\text{CQL}} = 0.01$ is the conservative penalty weight.

The Bellman loss, CQL loss, supervised loss are formulated as (3-5).

$$\mathcal{L}_{\text{Bellman}} = \mathbb{E}_{(s,a,r,s')} \left[\left(Q(s, a) - (r + \gamma \max_{a'} Q(s', a')) \right)^2 \right] \quad (3)$$

$$\mathcal{L}_{\text{CQL}} = \mathbb{E}_s \left[\log \sum_{a'} \exp(Q(s, a')) - Q(s, a) \right] \quad (4)$$

Equation (4) encourages conservatism by penalizing overestimated Q-values for out-of-distribution actions.

$$\mathcal{L}_{\text{sup}} = \text{CrossEntropy}(Q(s, \cdot), y) \quad (5)$$

Equation (5) aligns the Q-values with true diagnostic labels to reinforce clinically grounded policy behavior.

III. RESULTS

A. Reward-Based Evaluation

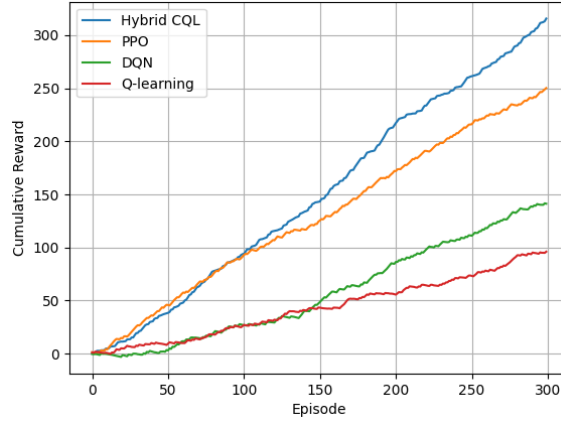


Fig 1. Comparison of Cumulative Reward

The reward-based evaluation demonstrates that the proposed Hybrid CQL model consistently achieves the highest cumulative reward among all compared RL baselines, including Q-Learning, DQN, and PPO. As illustrated in Figure 1, Hybrid CQL exhibits both faster convergence and greater reward stability over 300 training episodes. This performance advantage arises from its integration of conservative Q-value estimation with supervised learning signals, enabling more effective policy learning while mitigating overestimation bias and instability. In contrast, Q-Learning underperforms due to the absence of supervised guidance, while DQN and PPO yield moderate but less stable reward trajectories. The smooth and accelerated reward progression observed in Hybrid CQL underscores its policy-level robustness and learning efficiency, particularly within static clinical environments such as the ADNI dataset, where reliability and safety are paramount. These results affirm the model’s capability to align reward optimization with conservative, clinically meaningful decision-making.

B. Class-wise Diagnostic Performance

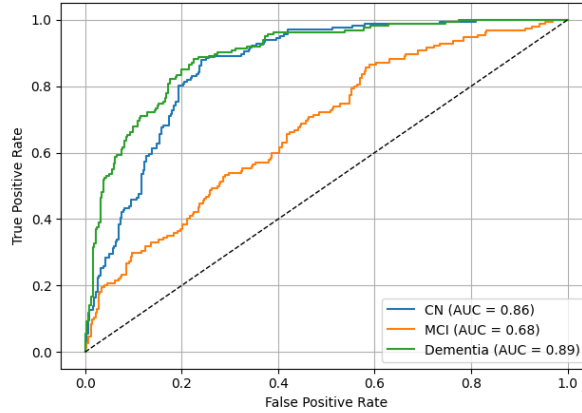


Fig 2. Class-wise ROC Curves of Hybrid CQL

As shown in Figure 2, Class-wise ROC analysis demonstrates that the proposed Hybrid CQL model maintains strong discriminative capability for CN (AUC = 0.86) and dementia (AUC = 0.89), while achieving moderate performance for MCI (AUC = 0.68). The relatively lower AUC for MCI reflects the inherent clinical ambiguity of this intermediate stage, which overlaps in feature space with both CN and dementia. Importantly, the Hybrid CQL model is explicitly designed to prioritize robustness and conservative decision-making under uncertainty, which may lead to cautious predictions rather than overconfident misclassifications in diagnostically subtle cases. This behavior is aligned with the objective of model that maximizes long-term reward while maintaining reliability in offline, static clinical environments, particularly where diagnostic boundaries are not sharply defined.

IV. DISCUSSIONS & CONCLUSIONS

This study proposes a Hybrid CQL framework that effectively integrates offline RL with supervised learning signals to support robust and interpretable classification modeling using the ADNI dataset. Empirical results demonstrate that Hybrid CQL consistently outperforms baseline RL algorithms in cumulative reward optimization, convergence speed, and learning stability. Class-wise analysis confirms strong discriminative performance for CN and dementia classes, while highlighting moderate sensitivity for MCI, a clinically ambiguous category. This outcome reflects the inherent difficulty of the MCI classification task due to clinical overlap with other classes, and the model’s design prioritizes cautious decision boundaries in ambiguous states. Instead of overfitting to diagnostically ambiguous inputs, Hybrid CQL maintains policy reliability and conservative learning, which is critical in healthcare applications. Future work will explore adaptive reward scaling, uncertainty-aware training, and curriculum-based strategies to further enhance sensitivity in diagnostically subtle categories such as MCI, while maintaining the model’s overall robustness and clinical interpretability.

ACKNOWLEDGMENTS

This research was supported by the MSIT (Ministry of Science, ICT), Korea, under the National Program for Excellence in SW), supervised by the IITP (Institute of Information & Communications Technology Planning & Evaluation) in 2025 (2024-0-00018).

REFERENCES

- [1] A. Nazir, A. Assad, A. Hussain, and M. Singh, “Alzheimer’s disease diagnosis using deep learning techniques: datasets, challenges, research gaps and future directions,” *International Journal of System Assurance Engineering and Management*, pp. 1–35, 2024.
- [2] B. H. Aubaidan, R. A. Kadir, M. T. Lajb, M. Anwar, K. N. Qureshi, B. A. Taha, and K. Ghafour, “A review of intelligent data analysis: Machine learning approaches for addressing class imbalance in healthcare—challenges and perspectives,” *Intelligent Data Analysis*, Article ID 1088467X241305509, 2024.
- [3] R. F. Prudencio, M. R. Maximo, and E. L. Colombini, “A survey on offline reinforcement learning: Taxonomy, review, and open problems,” *IEEE Transactions on Neural Networks and Learning Systems*, 2023.
- [4] A. Kumar, A. Zhou, G. Tucker, and S. Levine, “Conservative Q-learning for offline reinforcement learning,” in *Advances in Neural Information Processing Systems*, vol. 33, pp. 1179–1191, 2020.
- [5] S. G. Mueller, M. W. Weiner, L. J. Thal, R. C. Petersen, C. Jack, W. Jagust, et al., “The Alzheimer’s Disease Neuroimaging Initiative,” *Neuroimaging Clinics*, vol. 15, no. 4, pp. 869–877, 2005.

Accuracy of dynamic mode decomposition reconstruction for nonlinear hyperbolic system with respect to number of snapshot

Gwanghyun Jo¹

¹D. of Mathematical Data Science, Hanyang University ERICA, Ansan-si 15588, Republic of Korea

[gwanghyun@hanyang.ac.kr]

Abstract

Dynamic mode decomposition (DMD) is one of the efficient methods to predict long time behavior of the solution of partial differential equations (PDEs). However, the reconstruction error for the last snapshot of the observed solution affects the accuracy of the prediction of solution. Therefore, in this study we numerically assess the accuracy of the reconstruction of DMD solutions versus the number of the snapshot solutions. Our consideration is restricted to nonlinear hyperbolic PDEs.

Index Terms: Dynamic mode decomposition, partial differential equation, reconstruction, hyperbolic.

I. INTRODUCTION

Dynamic mode decomposition (DMD) has been successfully applied to long-time prediction of solutions of partial differential equations (PDEs) [1-3]. The accuracy of solutions depends on the reconstruction of the last snapshot in the DMD mode space. Therefore, studying the accuracy of the reconstruction solution with respect to number of snapshots is important issues. In this study, we study the reconstruction of the last snapshot of the observed data.

II. METHODS

Dynamic mode decomposition finds the eigenvector of the snapshot data. It is natural that the number of snapshots is critical to the accuracy of the reconstruction via the DMD modes. We are studying the accuracy of the reconstruction error versus the number of the snapshots. Our consideration is restricted to nonlinear hyperbolic type PDEs.

III. RESULTS

We numerically test the accuracy of the nonlinear hyperbolic type PDEs. As expected as the number of snapshot increases, the reconstruction errors decrease.

IV. DISCUSSION AND CONCLUSIONS

In this study, we numerically study the reconstruction error of DMD for the nonlinear hyperbolic type PDEs. As the number of snapshot increase, the error tends to decrease. In the future work, we wish to find theoretical reasons.

REFERENCES

- [1] J. H. Tu, "Dynamic mode decomposition: Theory and applications," *Diss. Princeton University*, 2013.
- [2] P. J. Schmid "Dynamic mode decomposition and its variants," *Annual review of fluid mechanics*, vol. 54, no. 1, pp. 225-254, 2022.
- [3] J. L. Proctor, L. B. Steven, and J. N. Kutz, "Dynamic mode decomposition with control," *SIAM Journal on Applied Dynamical Systems*, vol. 15, no. 1, pp. 152-161, 2016.

Comparative Analysis of HSV Preprocessing Combinations for Enhancing YOLOv8 Object Detection in Low-Light Conditions

Hyeon-Jin Woo¹, Kyoung-Yee Kim^{2*}

¹Department of AI Software, Inje University, 50834, Korea

^{2*}Department of AI Software, Inje University, 50834, Korea

[Email : guswls264664@gmail.com¹, kimky@inje.ac.kr^{2*}]

Abstract

This study was conducted to improve the object detection performance of cameras that suffer from performance degradation in low-light environments. Specifically, to enhance camera object detection performance in low-light environments, images using five combinations of adjusted HSV color space channel values were applied, and the YOLOv8 model was trained using these images. The performance of each combination was quantitatively evaluated using mAP50, mAP50-95, Precision, and Recall. The experimental results showed that saturation improved mAP50 by 1.12% to 85.02% compared to before preprocessing.

I. INTRODUCTION

With the advancement of autonomous driving technology, it is now being utilized in outdoor environments such as urban areas where there are many people or static objects. [1] In such situations, the stability of autonomous driving is directly linked to the accuracy of object detection technology and has emerged as a critical issue. Object detection technology utilizes various sensors such as LiDAR and cameras, but among these, cameras experience significantly reduced performance in low-light outdoor environments, such as at night or in backlit conditions, due to the poor quality of the video images they capture. [2] To address this issue, LiDAR, a sensor commonly used as an alternative, is an expensive sensor that requires specialized knowledge to handle, making its practical application challenging.

To reduce this burden, this study aims to enhance the stability of camera-based object detection while reducing reliance on LiDAR. To improve object detection performance in low-light conditions, five HSV channel value combinations are selected, and images incorporating these combinations are used to train an object detection model using YOLOv8[3]. Additionally, quantitative evaluations are conducted using performance metrics to propose an appropriate HSV channel value combination.

II. SYSTEM MODEL AND METHODS

* Kyoung-Yee Kim kimky@inje.ac.kr

The experimental environment for this study was Linux Ubuntu 22.04 and Python 3.12. The preprocessing technique combined a preprocessing method that adjusted the values of each channel in the HSV color space with the YOLOv8 object detection model. This research (paper) used datasets from 'The Open AI Dataset Project (AI-Hub, S. Korea)'. All data information can be accessed through 'AI-Hub (www.aihub.or.kr)'. The dataset was reconstructed from the original data and used in this study. The object detection performance of the YOLOv8 model was compared and analyzed using the original images and five preprocessed images with adjusted HSV channel values, totaling six images.

The five preprocessed images, excluding the original, are “bright”, “saturation”, “gamma”, “balanced”, and “extreme”, and the adjusted values for each channel are as follows.

Table 1. Adjustment values for each preprocessing HSV channel value.

	Hue	Saturation	Value	Gamma
bright	0.0	1.2	1.5	0.8
saturation	0.0	1.5	1.3	0.9
gamma	0.0	1.1	1.4	0.7
balanced	0.0	1.3	1.6	0.75
extreme	0.0	1.4	1.8	0.6

None of the five preprocessing methods adjusted Hue, so there was no change in color. “Bright” showed a brightness-focused correction effect that improved the brightness of dark areas, while “saturation” emphasized both brightness and saturation, producing the most vivid image of the five. “Gamma” brightened dark areas while slightly emphasizing brightness and saturation. “Balanced” showed moderate saturation enhancement and an overall bright image, while “extreme” produced the least noticeable effect among the five preprocessing methods, despite having the highest values.



Fig. 1. Each pre-processing result image.

The evaluation metrics were verified using mAP (Mean Average Precision), Precision, and Recall.

III. RESULTS

The results showed that “saturation” achieved the highest performance with mAP50 of 0.8502, while “original (original image)” achieved the second highest performance with mAP50 of 0.8390. On the other hand, despite the highest value change, “extreme” showed the lowest performance with a mAP50 of 0.7546. The mAP50 values for the remaining “brightness”, “gamma”, and “balanced” were 0.8276, 0.8028, and 0.8305, respectively.

Table 2. YOLOv8 detection performance by preprocessing.

	mAP50	mAP50-95	Precision	Recall
saturation	0.8502	0.6837	0.8695	0.7773
original	0.8390	0.6759	0.8553	0.7723
balanced	0.8305	0.6450	0.8760	0.7585
bright	0.8276	0.6439	0.8570	0.7501
gamma	0.8028	0.6157	0.8189	0.7517
extreme	0.5467	0.7546	0.7937	0.6911

IV. Conclusion

To improve camera object detection performance in low-light environments, we attempted preprocessing by varying the HSV channel values and applied each preprocessing to the YOLOv8 object detection model to compare performance. As a result, “saturation” showed a performance improvement of 0.112 in mAP50 compared to before preprocessing, reaching 0.8502.

On the other hand, “extreme”, which had the highest HSV value, showed the lowest performance, indicating that HSV values are not proportional to performance.

Based on these results, it was confirmed that if HSV channel values are not properly adjusted, performance will be lower than when training with the original image without preprocessing. It is recommended not to lower the gamma value. It was confirmed that appropriately adjusting the saturation and value can partially offset the degradation of object detection performance in low-light environments.

The results of this study demonstrate that preprocessing with appropriate HSV channel value adjustments in low-light environments can improve object detection performance. This approach is expected to reduce the burden on high-cost sensors such as LiDAR and enhance the stability of object detection using cameras.

ACKNOWLEDGMENTS

This research was supported by the MSIT (Ministry of Science ICT), Korea, under the National Program for Excellence in SW, supervised by the IITP (Institute of Information & Communications Technology Planning & Evaluation) in 2022 (2022-0-01091, 1711175863)

REFERENCES

- [1] Wijayathunga L, Rassau A, Chai D, "Challenges and Solutions for Autonomous Ground Robot Scene Understanding and Navigation in Unstructured Outdoor Environments: A Review", *Applied Sciences*, 13(17):9877, 2023.
- [2] C. Son, D. Kim and H. Ko, "A illumination-aware learnable image filtering model for object detection in low-light conditions." *Journal of IKEEE*, vol. 28, no. 4, pp. 503-510,2024.
- [3] N. Kim, J. Park, S. Kim, H. Park, S. Chung, "The Risk Detection Technique for Visually Impaired Pedestrians using YOLOv8 on GloSea6", *Journal of Korea Institute of Information, Electronics, and Communication Technology*, vol. 17, no. 6, pp. 451-461,2024.

Optimized Training Pipeline for Automated Road Sign Detection Using YOLOv8x

Jae-Won Jung¹, Kyoung-Yee Kim^{2*}

¹Department of AI Software, Inje University, City 50834, Korea

^{2*}Department of AI Software, Inje University, City 50834, Korea
[Email : ai20214598@gmail.com¹, kimky@inje.ac.kr^{2*}]

Abstract

Accurate road mark detection technology is essential for the safe operation and efficient route planning of autonomous vehicles. This study proposes an automated training system based on the YOLOv8x architecture to achieve this goal. We converted driving video data labeled for AI Hub Road mark recognition into YOLO format and implemented a feature that automatically adjusts batch size according to GPU memory capacity. Additionally, we have established a training pipeline that automates the entire process from dataset validation to model optimization. The results achieved excellent detection performance with mAP@50 of 71.74 and mAP@50-95 of 57.23, along with an overall accuracy of 90.49% and a recall rate of 63.87%.

I. INTRODUCTION

Road markings are important visual elements that provide traffic rules and route guidance information and accurately recognizing them directly affects the safe operation of autonomous vehicles. In particular, the ability to accurately detect road markings in real time in complex urban environments and under various road conditions is essential for improving the safety of autonomous driving technology [1].

Such road mark recognition technology primarily uses object detection, and the existing object detection model learning process mostly requires manual parameter improvement or additional labeling re-annotation depending on the research environment and the type of learning model. In this process, additional data re-annotation time may be required, or setting incorrect parameters during learning may result in inefficient use of hardware resources.

These issues also arise in model training processes that require safe and accurate object detection technologies, such as road mark object detection for autonomous driving.

This study applies a labeling re-annotation automation process and proposes a hardware environment-adaptive model training pipeline that can apply YOLOv8x to various GPUs [2,3]. This approach not only reduces research and development time and costs by shortening the time required for labeling preprocessing and optimizing hardware resources but also enhances the practicality and safety of road mark detection systems in real-world autonomous driving environments by increasing the possibility of deploying training code across various hardware environments.

* Kyoung-ye Kim kimky@inje.ac.kr

II. SYSTEM MODEL AND METHODS

This study proposes a road mark detection system utilizing YOLOv8x architecture. This research used datasets from “Road Marking Recognition”. All data information can be accessed through ‘AI-Hub(www.aihub.or.kr)’, which was restructured and divided into 30,000 images for training (22,000 images), validation (4,000 images), and testing (4,000 images). The dataset includes 16 types of road markings, such as Direction Arrow, Speed Limit, and Crosswalk.

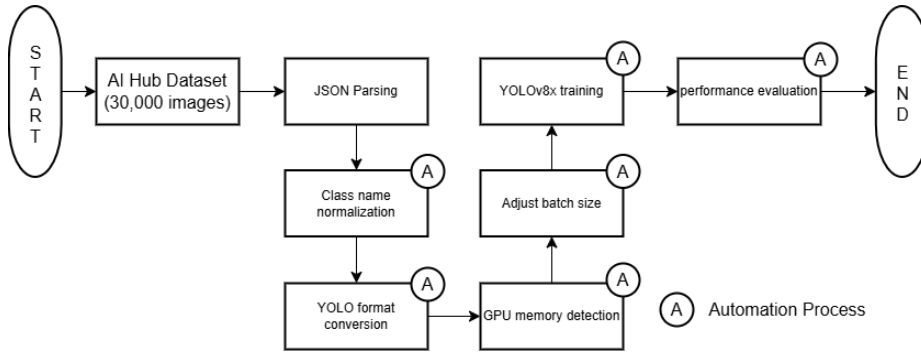


Fig. 1. Automated training pipeline

In the labeling preprocessing stage, the data is converted through a labeling preprocessing process that includes class name normalization and object coordinate conversion of JSON annotations for YOLOv8x learning.



Fig. 2. Ground Truth Labeling Example

During the training process, the batch size was dynamically adjusted based on the available GPU memory: if the memory was less than 6GB, the batch size was set to 4; if it was between 6GB and 12GB, the batch size was set to 8; and if it was 12GB or more, the batch size was set to 16. This implementation ensures that the batch size is automatically adjusted according to the actual available GPU memory.

Training was conducted using the YOLOv8x model in an Ubuntu 22.04 environment. The training parameters included an image size of 640, 100 epochs, and Automatic Mixed

Precision (AMP) set to True. An end-to-end pipeline was built to automate the entire process from model training validation to optimization.

Evaluation metrics include precision, recall, mAP@50, and mAP@50-95 for quantitative assessment.

III. RESULTS

The results achieved excellent detection performance with mAP@50 of 71.74 and mAP@50-95 of 57.23. The overall precision was 90.49, and the recall rate was 63.87.

In the model evaluation results, the confusion matrix analysis showed that Direction Arrow was accurately classified in 94% of the 6,142 cases, while Speed Limit achieved 92% accuracy out of 4,109 cases. However, some classes exhibited a tendency to be misclassified as background, which is likely due to small objects or object imbalance, indicating the need for data improvement.

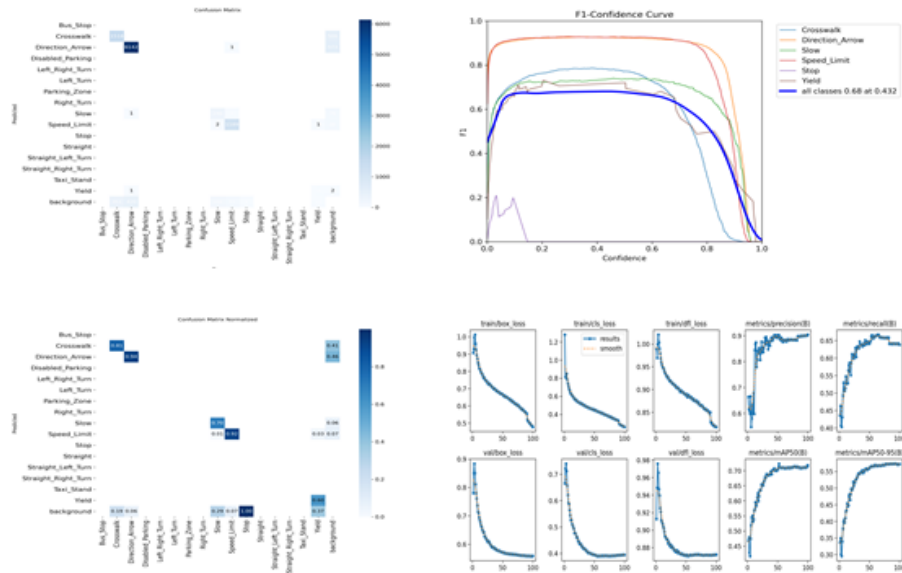


Fig. 3. Comprehensive analysis of performance evaluation results

IV. Conclusion

This study implemented a road mark detection system for safe driving of autonomous vehicles. The implemented system automates processes such as labeling preprocessing, simplifying the data refinement process for model training. As a result, it achieved mAP@50 of 71.74, mAP@50-95 of 57.23, overall accuracy of 90.49, and recall of 63.87, confirming its practical applicability.

However, the current dataset is limited to complete data that does not include adverse weather conditions and damaged road markings, and there are limitations in terms of practicality and safety due to class imbalance caused by the limited dataset and the fact that only a portion of the actual road markings in real road environments were applied.

In future research, we plan to perform learning using not only complete data but also incomplete data to ensure practicality and stability by conducting learning under various weather conditions, road mark damage, and obscured situations.

ACKNOWLEDGMENTS

This research was supported by the MSIT (Ministry of Science ICT), Korea, under the National Program for Excellence in SW, supervised by the IITP (Institute of Information & Communications Technology Planning & Evaluation) in 2022 (2022-0-01091, 1711175863)

REFERENCES

- [1] Wijayathunga, Liyana, Alexander Rassau, and Douglas Chai. "Challenges and solutions for autonomous ground robot scene understanding and navigation in unstructured outdoor environments: A review." *Applied Sciences* 13.17 2023
- [2] Kim, Joon-Yong. "A comparative study on the characteristics of each version of object detection model YOLO." *Proceedings of the Korean Society of Computer Information Conference*. Korean Society of Computer Information, 2023.
- [3] Choi, Woo-Cheol, et al. "Comparative Analysis of Dynamic Traffic Object Detection Performance Using a Deep Learning Model Based on YOLOv8." *Journal of the Korean ITS Society* 23.6 2024

Deep Learning-Based Low-Altitude Flying Object Recognition Using Low-Cost Floating Cameras

Geon Kim¹, Hee-Cheol Kim²

¹Department of Computer Engineering, Inje University, Gimhae 50834, Republic of Korea

²College of AI Convergence/Institute of Digital Anti-Aging Healthcare/u-HARC, Inje University, Gimhae 50834, Republic of Korea

[Email : voicex6358@naver.com]

Abstract

In this study, we conducted an experiment to detect low-altitude flying objects using a low-cost floating camera and a deep learning object recognition model. To overcome the resolution limitations and high cost of radar-based detection systems, a low-cost floating camera mounted on a helium balloon is used to detect low-altitude flying objects. An experimental system utilizing an Arduino-based embedded system, an optical camera module with a resolution of 1280×800, and Wi-Fi communication was built to detect balloon objects from real-time (30 frames per second) video data. The experimental results showed that YOLOv5 detected objects with a confidence range of 30% to 80%, confirming that deep learning-based real-time object detection is possible even in a low-cost hardware environment.

Index Terms: Deep Learning, Floating Camera, Low-Altitude Object Detection, YOLO

I. INTRODUCTION

Recent advances in radar equipment technology have led to the sophistication of detection technology for various objects using radar in fields such as observation and surveillance. General radar, which uses the principle of reflection of electromagnetic waves to detect targets, has a relatively lower resolution than optical devices, making it difficult to detect small objects, and even if an object is detected, it is difficult to recognize the specific object.[1] In addition, systems using radar are expensive. To address these issues, camera-based technologies are essential in situations where detection or precise identification of small flying objects is required.[2] The use of camera-based technologies is essential.

Previous studies that utilize drones or fixed cameras, which are relatively inexpensive compared to radar, as object detection tools have actually been able to detect flying objects.[3] However, drones and fixed cameras are still limited by the fact that they are not cheap enough.[4] In addition, in the case of drones, it is difficult to fly for more than an hour when assuming the use of a general drone,[5] and in the case of fixed cameras, it is not possible to cover a large area due to the limitations of the installation location.

Therefore, this study attempts to recognize objects through the combination of a floating camera system using helium balloons and a deep learning object recognition model, and aims to verify that such a model can actually recognize objects through low-cost hardware through simple experiments, and to present a new methodology to solve the problems of existing research.

II. SYSTEM MODEL AND METHODS

The experimental system model (Fig. 1), which includes an optical camera module (resolution: 1280×800px, processor: ESP32) mounted on an Arduino-based compact embedded system and a YOLOv5s model trained with the Buloons.v22i.coco segmentation dataset, is utilized to perform real-time object detection at 30 frames per second. The floating camera hardware architecture (Fig. 2) integrates 802.11b/g/n Wi-Fi to transmit detection results, which are subsequently verified and analyzed using OpenCV-based verification tools. In particular, outdoor testing was conducted in accordance with South Korea's Aviation Safety Act, which restricts the flight of drones and unmanned aerial vehicles above 150 meters.

III. RESULTS

In the proposed experimental system, we utilized an optical camera module mounted on an Arduino-based compact embedded system and applied YOLOv5 to detect balloons from real-time video data. In the course of the experiment, YOLOv5 showed a confidence score of between 30% and 80% for the detected objects based on the data received from the optical camera module. However, it was also clear that small and distant objects were not detected due to the limitations of the optical camera used. (Fig 3)

In addition, it was found that the problems of flight time, limited installation location, limited coverage of large areas, and high equipment and operation costs of existing drone and fixed camera-based detection methods can be solved with a floating camera platform.

IV. DISCUSSION AND CONCLUSIONS

This study empirically verifies the feasibility of object detection in real-time video data in a real hardware environment by combining a low-cost floating camera with the YOLOv5 object detection model. The experimental results show that the confidence scores of the detected objects range from 30% to 80%, indicating that deep learning-based real-time object detection is feasible even in a low-cost floating camera environment.

This study attempts to differentiate itself by addressing the problems of existing studies by simply attaching an Arduino-based embedded system with a camera to a helium balloon to solve the flight time problems of existing drone-based low-orbit object recognition and the monitoring constraints of fixed camera-based studies, which are still expensive.

If the proposed data utilization methodology is applied to the radar system industry, it will help to overcome the limitations of low-resolution output of conventional radars at a low cost by improving the accuracy of optical equipment-based detection techniques and linking them to radar systems.[7] Furthermore, if the methodology is well utilized, it has the potential to be expanded and utilized in various fields such as environmental monitoring, disaster response, and urban planning.

However, there are limitations to this study, including the inability to detect small or distant objects using a low-resolution optical camera and the difficulty of maintaining a stable altitude with a helium balloon. In the future, further research is needed to improve the detection performance by improving the resolution of additional optical cameras and to find a more suitable way to maintain a stable altitude.

APPENDIX

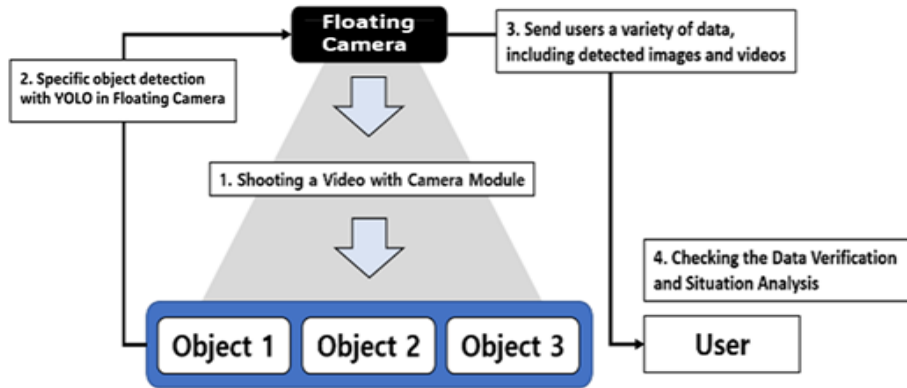


Fig 1. Experiment system model design

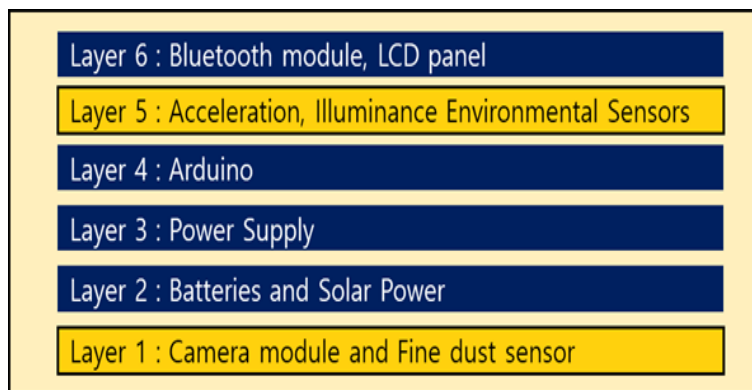


Fig 2 Floating camera hardware structure



Fig 3 Specific object detection

REFERENCES

- [1] Khawaja, Wahab, et al. "A survey on detection, classification, and tracking of UAVs using radar and communications systems." *IEEE Communications Surveys & Tutorials* (2025).
- [2] Bang, Geonho, et al. "Radardistill: Boosting radar-based object detection performance via knowledge distillation from lidar features." *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*. 2024.
- [3] Seidaliyeva, Ulzhalgas, et al. "Real-time and accurate drone detection in a video with a static background." *Sensors* 20.14 (2020): 3856.
- [4] Glüge, Stefan, et al. "Robust Low-Cost Drone Detection and Classification Using Convolutional Neural Networks in Low SNR Environments." *IEEE Journal of Radio Frequency Identification* (2024).
- [5] *Service Robots and Drones Report – 2016*. IHS Markit, Technology Group, 2016.
- [6] Albahri, Ameen Hamza, and Amin Hammad. "A novel method for calculating camera coverage in buildings using BIM." *Journal of Information Technology in Construction (ITcon)* 22.2 (2017): 16-33.
- [7] Bang, Geonho, et al. "Radardistill: Boosting radar-based object detection performance via knowledge distillation from lidar features." *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*. 2024.
- [8] Reyana, A., and P. Vijayalakshmi. "Multisensor Information Fusion for Condition Based Environment Monitoring." *Intelligent Automation & Soft Computing* 36.1 (2023).

Personalized Gesture Recognition Method with Marker-Based User Identification

Hyun-Jun Park¹, Hee-Cheol Kim^{2*}

¹Department of Computer Engineering, Inje University, Gimhae 50834, Republic of Korea

²College of AI Convergence/Institute of Digital Anti-Aging Healthcare/u-HARC, Inje University, Gimhae 50834, Republic of Korea

[Email : guswnsakvk@oasis.inje.ac.kr, heeki@inje.ac.kr]

Abstract

We propose a method for recognizing gestures from a specific user in multi-user environments. Previous studies require expensive equipment or demand users to become familiar with specialized devices, which poses limitations. We utilize OpenCV to detect a specific marker and then employ MediaPipe's real-time hand tracking to recognize only the hand with the marker. This approach effectively reduces malfunctions in multi-user environments, enabling highly reliable gesture recognition. In simultaneous tests with hands both with and without the marker, only the hand with the marker was accurately recognized.

Index Terms: Specific users, gesture recognition, real-time, marker utilization system

I. INTRODUCTION

With the advancement of technology, robots and machines are increasingly collaborating with humans in various industrial and everyday environments. However, most robotic and machine systems operate passively and have limitations in responding quickly when problems arise. To address this issue, technologies that recognize user gestures have attracted significant attention.

Previous studies have often utilized expensive equipment such as Leap Motion or LIDAR sensors [1] to achieve high accuracy, or have required users to become familiar with specialized devices [2], which presents certain limitations. Furthermore, in multi-user environments, these systems may recognize gestures from unintended users, leading to malfunctions.

In this paper, we propose a method that utilizes OpenCV and MediaPipe to recognize only the hand with a specific marker attached. The system can be implemented using a low-cost camera, and by leveraging the relative positional information between the marker and the hand, it accurately recognizes only the specific user's hand. This effectively reduces malfunctions in multi-user environments.

* Corresponding author

II. SYSTEM MODEL AND METHODS

This section describes the overall system model, including the proposed gesture recognition method and the system architecture for marker-based user identification.

A. Method

In this study, we implemented the system using OpenCV and MediaPipe instead of object detection models such as YOLO. Object detection models recognize all hands in the frame, which makes it difficult to distinguish only the hand with a specific marker. Moreover, for a personalized system, object detection models require large datasets to accommodate new markers or gestures, making expansion challenging. In contrast, using OpenCV and MediaPipe allows for relatively easy addition or modification of markers and gestures.

In the proposed method, OpenCV is used to detect the position of the marker, and MediaPipe[3] extracts real-time hand joint landmarks. By utilizing the relative positional information between the marker and the hand, the system can effectively recognize only the specific user's hand.

B. System Architecture

As shown in Figure 1, when a webcam captures the user's hand image, OpenCV[4] detects the position of the specific marker in each frame. The marker's coordinates are then passed to MediaPipe, which extracts the landmark coordinates of all hands in the frame and analyzes the hand gestures. Only the gesture label of the hand closest to the detected marker is output as the system result.

III. RESULTS

As shown in Table 1, the system was tested in various environments. As illustrated in Figure 2, only the hand with the marker was accurately recognized. However, there are limitations in environments with extremely high or low brightness, where the marker may not be detected properly. Additionally, recognition rates decrease when the hand joints are not fully visible in the frame or are partially occluded. This is a limitation of the MediaPipe-based hand joint tracking model, as gesture classification performance drops if the joints are not fully recognized in complex hand shapes or diverse environments.

IV. DISCUSSION AND CONCLUSIONS

In this paper, we proposed a method for recognizing gestures from a specific user by utilizing OpenCV and MediaPipe. This approach combines marker-based user

identification with hand landmark tracking, and experiments demonstrated that only the gestures of the specific user were accurately recognized even in multi-user environments.

The proposed method can be implemented with a low-cost webcam and open-source libraries, making it accessible and cost-effective. By calculating the relative distance between the marker and the hand, the system reduced misrecognition of other users' gestures by more than 90% in multi-user environments.

These characteristics make the method effective for various fields where user-customized gesture-based interfaces are required, such as smart home control, companion robot interaction, and factory automation.

However, there was a limitation in that the recognition rate decreased when the lighting conditions changed or when parts of the hand were occluded in the image. In the future, we plan to focus on enhancing the robustness of the marker by developing marker designs and detection algorithms that can ensure stable recognition even under various lighting (illumination) conditions. In addition, further research will be considered to improve the practicality and reliability of the system, such as implementing user-specific gesture learning functions.

APPENDIX

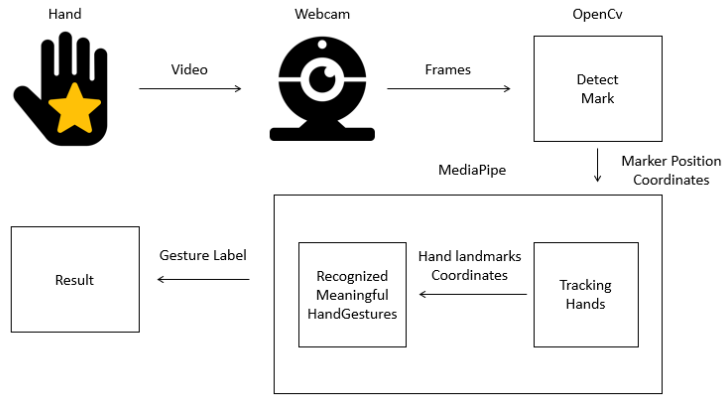


Fig. 1. System Architecture



Fig. 2. Find mark & gesture

Table 1. Recognition rate resultes

Environment	Test count	recognition succes count	Recognition rate (%)
Room	100	95	95
Shade	100	62	62
Playground	100	24	24

REFERENCES

- [1] Y. Zhang, X. Wang, and Y. Liu, "GesturePrint: Enabling User Identification for mmWave-Based Gesture Recognition Systems," in *Proc. 2024 IEEE 44th International Conference on Distributed Computing Systems (ICDCS)*, Jersey City, NJ, USA, Jul. 2024
- [2] J.-W. Lee, K.-J. Kim, and K.-H. Yu, "Implementation of a User-Friendly Drone Control Interface Using Hand Gestures and Vibrotactile Feedback," *Journal of Institute of Control, Robotics and Systems*, vol. 28, no. 4, pp. 349-352, Apr. 2022
- [3] R. Song, Y. Hong, and N. Kwak, "User Interface Using Hand Gesture Recognition Based on MediaPipe Hands Model," *Journal of the Korea Multimedia Society*, vol. 26, no. 2, pp. 103-115, Feb. 2023
- [4] S. Sharma, S. Sharma, and A. Sharma, "Human Computer Interaction Using Marker Based Hand Gesture Recognition," *arXiv preprint arXiv:1606.07247*, 2016.

PCB Surface Defect Detection Using YOLO v12

Seong-Woo Cho¹, Kyoung-Yee Kim^{2*}

¹Department of AI Software, Inje University, 50834, Korea

^{2*}Department of AI Software, Inje University, 50834, Korea
[Email : jsw16977@naver.com¹, kimky@inje.ac.kr^{2*}]

Abstract

In modern industry, quality control of PCBs (Printed Circuit Boards) has emerged as a key issue due to the increasing sophistication of electronic devices. As a result, there is an increasing demand for accurate and fast defect detection technologies. This study investigates the feasibility of applying the YOLOv12 object detection model for PCB surface defect detection. In this study, a dataset including six defect types, including mouse bites, was collected, and the YOLOv12 model was trained based on this dataset. To evaluate the model's performance, we used the following metrics: mAP (mean Average Precision), Precision, Recall, and Inference Time. We compared the performance of the YOLOv12-n version with that of YOLOv12-s, YOLOv12-m, YOLOv5-n, and YOLOv8-n across different versions and models. The results showed that the Precision of YOLOv12-n decreased by 0.36% compared to YOLOv8-n and YOLOv5-n, and the Recall decreased by 0.3% compared to YOLOv8-n and YOLOv5-n. For YOLOv12-m, the Precision increased by 0.78% and 0.12% for Precision, and Recall increased by 0.55% and 0.12%, respectively. When comparing the average inference time for each model, YOLOv5-n measured 15.73 ms, YOLOv8-n measured 15.46 ms, and YOLOv12-n measured 23.09 ms. This indicates that the YOLOv12-n model is not feasible for practical application, and further research into structural changes in the architecture is necessary.

I. INTRODUCTION

In the modern manufacturing industry, artificial intelligence technology is significantly improving the efficiency of quality control, and the accurate and rapid detection of defects on the surface of PCBs (Printed Circuit Boards) has emerged as an essential element in the production process of semiconductors and electronic components. As a result, the importance of vision-based defect recognition technology has grown, with object detection technology, particularly YOLO-based algorithms, gaining attention for their high accuracy and fast processing speeds. In this study, we utilized YOLOv12 to detect defects on PCB surfaces and compared and analyzed performance using metrics such as Precision, Recall, and mAP50. The YOLOv12-n model was compared and analyzed with the YOLOv12-s and YOLOv12-m models, and its performance was also compared with that of earlier versions of YOLO, namely YOLOv5-n and YOLOv8-n, to evaluate the relative performance and industrial applicability of the YOLOv12-n model.

* Kyoung-ye Kim kimky@inje.ac.kr

II. SYSTEM MODEL AND METHODS

In this study, we used the PCB Defect dataset from Kaggle. The PCB Defect dataset consists of a total of 10,668 images with six types of defects (missing hole, mouse bite, open circuit, short, spur, and spurious copper), including 8,534 training images, 1,066 validation images, and 1,068 test images.

Table 1. Data sets Info

	Train	Val	Test	Total
Mouse bite	1,404	179	169	1,752
Spur	1,396	178	166	1,740
Missing hole	1,501	183	168	1,852
Short	1,347	201	184	1,732
Open circuit	1,477	165	190	1,832
Spurious copper	1,409	160	191	1,760
Total	8,534	1,066	1,068	10,668

This study utilized a total of five YOLO versions—YOLOv5-n, YOLOv8-n, YOLOv12-n, YOLOv12-s, and YOLOv12-m—to compare and analyze their performance. For each model, the Epoch was set to 50, Batch Size to 16, and Learning Rate to 0.01.

The learning environment consisted of a Windows 11 operating system, an RTX 3060 12G GPU, and an Intel Core i7-based 64-bit processor. The development environment was based on Python 3.11.9, with PyTorch and CUDA 12.1 used for deep learning.

III. RESULTS

In this study, we compared and analyzed precision, recall, mAP50, and F1-score for YOLOv12. The experimental results are as follows.

YOLOv12-s recorded high values for both precision and recall (0.9728 and 0.9916, respectively) and showed high performance with an F1-score of 0.9821. Notably, YOLOv12-m achieved the best performance in all metrics—precision (0.9740), recall (0.9928), and F1-score (0.9833)—demonstrating the most outstanding overall model efficiency. However, YOLOv12-m showed a slight decrease in the mAP50 score to 0.9761, and the training time was approximately 2.89 times longer than that of YOLOv12-n, indicating that achieving optimal performance comes at a significant cost in terms of training time.

Table 2. Learning verification results

	Precision	Recall	mAP50	Train time
Yolov5-n	0.9813	0.9849	0.9896	3815s
Yolov8-n	0.9820	0.9835	0.9894	3806s
Yolov12-n	0.9842	0.9800	0.9895	5939s
Yolov12-s	0.9805	0.9899	0.9903	8647s
Yolov12-m	0.9912	0.9875	0.6197	17152s

Table 3. Performance comparison test results

	Precision	Recall	mAP50	Inference Time
Yolov5-n	0.9671	0.9904	0.9915	15.73ms
Yolov8-n	0.9699	0.9904	0.9911	15.46ms
Yolov12-n	0.9664	0.9874	0.9909	23.09ms
Yolov12-s	0.9728	0.9916	0.9876	22.23ms
Yolov12-m	0.9740	0.9928	0.9761	24.07ms

IV. Conclusion

YOLOv5-n and YOLOv8-n showed similar levels of performance, with YOLOv8-n demonstrating a slight improvement in inference time (15.46 ms). YOLOv12-n recorded the lowest overall precision and recall, indicating that this model structure has relatively lower performance compared to other versions. Additionally, the YOLOv12-m model demonstrated higher Precision and Recall values compared to YOLOv12-n and YOLOv12-s, which is advantageous for ensuring object recognition accuracy in real-world industrial environments. However, in scenarios where both real-time performance and precision are critical, it is deemed inappropriate to consider the use of YOLOv12 due to its relatively slower Inference Time.

In this study, no additional image preprocessing was performed, and the data was too clean, resulting in limitations in comparing the performance of different versions and models. In future studies, it is necessary to obtain a dataset that includes various types of PCB defects to evaluate the robustness of the model against external variables such as contamination and lighting changes that occur in real-world industrial environments. By confirming the applicability of the YOLOv12 object detection model, this study is expected to contribute to the development of a more accurate and reliable defect detection system for actual manufacturing lines in the future.

ACKNOWLEDGMENTS

This research was supported by the MSIT (Ministry of Science ICT), Korea, under the National Program for Excellence in SW, supervised by the IITP (Institute of Information & Communications Technology Planning & Evaluation) in 2022 (2022-0-01091, 1711175863)

REFERENCES

- [1] TIAN, Yunjie; YE, Qixiang; DOERMANN, David. "YOLOv12: Attention-Centric Real-Time Object Detectors." arXiv preprint arXiv:2502.12524, 2025.
- [2] TIAN, Yunjie; YE, Qixiang; DOERMANN, David. YOLOv12: Attention-Centric Real-Time Object Detectors. 2025. Available at: <https://github.com/sunsmarterjie/yolov12>. License: AGPL-3.0.
- [3] JOCHER, Glenn. Ultralytics YOLOv5. 2020. Version 7.0. Available at: <https://github.com/ultralytics/yolov5>. DOI: 10.5281/zenodo.3908559. License: AGPL-3.0.

[4] JOCHER, Glenn; CHAURASIA, Ayush; QIU, Jing. Ultralytics YOLOv8. 2023. Version 8.0.0.
Available at: <https://github.com/ultralytics/ultralytics>. License: AGPL-3.0.

Effective Optimization of EEG Analysis for Dementia Diagnosis: An Approach to Channel Selection and Feature Extraction

Yujung Sung¹, Jae-Ho Han², Seho Lee^{1,*}

¹Department of AI Convergence, University of Ulsan, Ulsan, 44610, Korea

²Department of Brain and Cognitive Engineering, Korea University, Seoul, 02841, Korea

[Email : ytoofkto@gmail.com, hanjaeho@korea.ac.kr, leeseho*@ulsan.ac.kr]

Abstract

Dementia is a major global health concern that necessitates efficient diagnostic solutions. Therefore, an electroencephalography(EEG)-based framework for effective dementia diagnosis employing feature extraction and optimal channel selection is necessary. Conventional high-complexity algorithms often require advanced hardware, whereas the proposed approach harnesses the computational efficiency of time-domain parameter(TDP) to achieve rapid and accurate classification between healthy individuals and dementia patients. EEG data were obtained from a publicly available dataset comprising 88 subjects, including individuals with Alzheimer's disease and frontotemporal dementia, merged into a unified dementia group, and healthy controls. The Hjorth parameters were implemented to calculate the features from each signal. Subsequently, the selection of important features was executed in two stages: 1) Statistical importance and 2) Importance from results of Model. Initially, the 15 features under consideration were selected through a statistical analysis, and the 6 features were extracted from classifier. As a result, the accuracy for dementia patients with support vector machine was 0.833 based on six features, compared to 0.722, based on 15 features. These findings indicate that the TDP-based channel selection and feature extraction method preserves essential discriminatory information while reducing computational complexity.

Index Terms: Dementia, Electroencephalography, Diagnosis, Time domain parameters, Machine learning

I. INTRODUCTION

Accurate diagnosis of dementia, a significant global health challenge, is crucial [1]. Among available diagnostic approaches, electroencephalography(EEG) offers distinct advantages due to its noninvasive nature, low cost, and ability to monitor and measure real-time changes in brain activity [2]. A number of studies have investigated channel reduction algorithm from EEG to accurate diagnoses[3, 4]. Furthermore, channel reduction algorithm with artificial intelligence were employed for the purpose of reducing the number of channels [5, 6]. For operating algorithm on-device, time-domain parameters(TDP) provided lower computational demands and faster processing speeds, is one of the alternative key[7]. The objective of this study is to optimize channel selection from TDP features on two stages: 1) Statistical importance and 2) feature importance from results of Model.

II. MATERIALS AND METHODS

A. Data Description

A publicly available EEG dataset was employed, containing resting-state, eyes-closed recordings from 88 subjects [8]. The dataset comprised 36 subjects diagnosed with Alzheimer's disease (AD), 23 diagnosed with Frontotemporal Dementia (FTD), and 29 healthy controls (CN). For this study, the AD and FTD groups were combined into a single "dementia" group, and the CN group was labeled as "healthy" to facilitate dementia classification. EEG signals were recorded from 19 scalp electrodes following the international 10–20 system, with a sampling rate was 500 Hz. The EEG recordings underwent standard preprocessing, including band-pass filtering (0.5–45 Hz), re-referencing to mastoid electrodes, and artifact removal. For additional details regarding the preprocessing pipeline, refer to the dataset's associated publication.

B. Time-domain parameters

Time-domain parameters provide a direct quantification of EEG signal characteristics. The Hjorth parameters were employed to capture these features [9]. Activity, which reflects the signal's variance, provides a measure of its power. Mobility, calculated as the square root of the variance of the first derivative of the signal normalized by Activity, serves as an index of the signal's mean frequency. Complexity is defined as the ratio of the standard deviation of the second derivative to that of the first derivative, normalized by Mobility. For each EEG segment obtained via a sliding-window segmentation approach, parameters were computed sequentially on each channel.

C. Channel and Feature Selection

Channel and feature selection was performed in a two-step. First, for each of the Hjorth parameters—activity, mobility, and complexity—statistical tests were conducted to identify parameter-channel combinations exhibiting significant differences between the normal and dementia groups. These parameter-channel pairs were treated as individual features. In the second step, the resulting features ($n=15$) were ranked based on their p -values, and stepwise classification experiments were performed to determine the optimal feature subset.

D. Model

A support vector machine (SVM) with a radial basis function (RBF) kernel was employed as the classifier. The input data consisted of time-domain Hjorth parameters extracted from the selected EEG channels. To ensure that features were on a comparable scale, z -score normalization was applied using a standard scaler. To assess the effect of feature selection, a stepwise modeling approach was implemented, where test accuracy was evaluated while progressively increasing the number of top-ranked features based on statistical significance.

E. Statistical Analysis

To determine the optimal subset of EEG channels for classification, a multi-step channel selection procedure based on statistical significance was implemented. The normality of each channel's distribution was first assessed using the Shapiro–Wilk test. Depending on the normality test, either an independent t -test or the Mann–Whitney U test was applied to compare the normal and dementia groups across the three Hjorth parameters.

III. RESULTS

Table 1. Features ranked by statistical significance and the optimal six features

Rank	Feature	Rank	Feature	Rank	Feature
1	Complexity O2	2	Complexity T5	3	Complexity O1
4	Mobility O2	5	Complexity P3	6	Complexity T6
7	Complexity Pz	8	Mobility O1	9	Activity O2
10	Mobility T5	11	Complexity P4	12	Activity O1
13	Mobility P3	14	Activity T5	15	Mobility T6

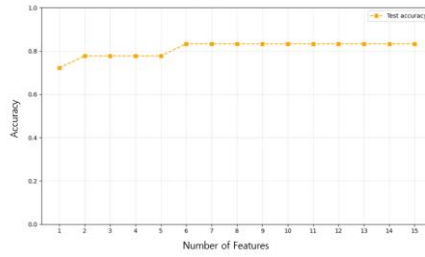


Fig. 1. Classification Accuracy as a Function of the Number of Selected Features.

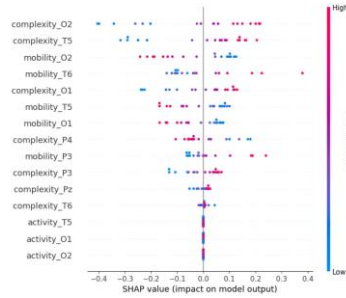


Fig. 2. SHAP Analysis of All 15 Features Derived from the Selected EEG Channels for Dementia Classification.

Features were ranked based on statistical significance, as shown in Table 1. The most significant feature was complexity O2, followed by complexity T5, complexity O1, and mobility O2. Classification performance was evaluated using a stepwise feature selection approach. The test accuracy was 0.722 when using one feature, increased to 0.778 with 2 to 5 features, and reached 0.833 with six features. Notably, the accuracy plateaued at 0.833 beyond six features.

To further interpret the trained model's decision process, SHAP analysis was conducted on the final RBF-SVM model using all 15 selected features. The top contributing features based on mean absolute SHAP values were complexity O2, complexity T5, mobility O2, mobility T6, and complexity O1.

IV. DISCUSSION AND CONCLUSION

By reducing the number of EEG channels and focusing on a minimal informative feature set, the proposed approach offers a streamlined and computationally efficient solution for EEG-

based diagnostics. As a results, the TDP—particularly complexity and mobility—derived from occipital (O1, O2) and temporal (T5, T6) channels were found to provide the most effective discriminative power for dementia classification. This finding is in agreement with earlier studies that identified the occipital and temporal regions as key contributors to dementia classification[10]. The simplicity and efficiency of the proposed methodology make it particularly attractive for the deployment of on-device diagnostic tools in wearable or portable systems where power consumption, processing capacity, and user comfort are critical.

There were several limitations in this study. One limitation is that it did not examine the specific patterns associated with conditions such as Alzheimer's disease and frontotemporal dementia. Future work would execute to investigate the unique EEG characteristics of these disorders. In addition, further studies will explore advanced modeling approaches, including deep learning techniques to improve classification performance.

Acknowledgements

Y.S. and S.L. was supported by the MSIT (Ministry of Science, ICT), Korea, under the National Program for Excellence in SW, supervised by the IITP (Institute of Information & Communications Technology Planning & Evaluation) in 2024-0-00025

J.-H. Han was supported by the MSIT (Ministry of Science, ICT), Korea, under the ITRC (Information Technology Research Center) support program (IITP-2025-RS-2022-00156225) supervised by the IITP (Institute for Information & Communications Technology Planning & Evaluation) and by the National Research Foundation of Korea (NRF) grant (No. RS-2024-00415812)

REFERENCES

- [1] G. B. D. D. Collaborators, "Global, regional, and national burden of Alzheimer's disease and other dementias, 1990-2016: a systematic analysis for the Global Burden of Disease Study 2016," *Lancet Neurol*, vol. 18, no. 1, pp. 88-106, Jan 2019, doi: 10.1016/S1474-4422(18)30403-4.
- [2] N. K. Al-Qazzaz, S. H. B. M. Ali, S. A. Ahmad, K. Chellappan, M. S. Islam, and J. Escudero, "Role of EEG as biomarker in the early detection and classification of dementia," *The Scientific World Journal*, vol. 2014, no. 1, p. 906038, 2014.
- [3] Y. Luo *et al.*, "An EEG channel selection method for motor imagery based on Fisher score and local optimization," *Journal of Neural Engineering*, vol. 21, no. 3, p. 036030, 2024.
- [4] H. Wu, J. Qi, E. Purwanto, X. Zhu, P. Yang, and J. Chen, "Multi-scale feature and multi-channel selection toward parkinson's disease diagnosis with eeg," *Sensors*, vol. 24, no. 14, p. 4634, 2024.
- [5] L. Wang, J. Wang, H. Su, X. Zhang, L. Zhang, and X. Kang, "A zero precision loss framework for EEG channel selection: enhancing efficiency and maintaining interpretability," *Computer Methods in Biomechanics and Biomedical Engineering*, pp. 1-16, 2024.
- [6] J. Shen *et al.*, "An optimal channel selection for EEG-based depression detection via kernel-target alignment," *IEEE Journal of Biomedical and Health Informatics*, vol. 25, no. 7, pp. 2545-2556, 2020.
- [7] P. Geethanjali, Y. K. Mohan, and J. Sen, "Time domain feature extraction and classification of EEG data for brain computer interface," in *2012 9th international conference on fuzzy systems and knowledge discovery*, 2012: IEEE, pp. 1136-1139.
- [8] A. Miltiadous *et al.*, "A dataset of scalp EEG recordings of Alzheimer's disease, frontotemporal dementia and healthy subjects from routine EEG," *Data*, vol. 8, no. 6, p. 95, 2023.
- [9] B. Hjorth, "EEG analysis based on time domain properties," *Electroencephalogr Clin Neurophysiol*, vol. 29, no. 3, pp. 306-10, Sep 1970, doi: 10.1016/0013-4694(70)90143-4.
- [10] C. Simfukwe, S. S. A. An, and Y. C. Youn, "Contribution of Scalp Regions to Machine Learning-Based Classification of Dementia Utilizing Resting-State qEEG Signals," *Neuropsychiatric disease and treatment*, pp. 2375-2389, 2024.

A Differentiation-Based Method for Reliable R-Peak Detection and Signal Quality Assessment with Morphological Constraints of Electrocardiography

SunWoo Park¹, Seho Lee^{1,*}

¹Department of AI Convergence, Ulsan University, Ulsan, 44610, Korea
[Email : [dolkong2802](mailto:dolkong2802@ulsan.ac.kr), leeseho*@ulsan.ac.kr]

Abstract

Accurate detection of R-peaks is essential for reliable ECG interpretation, particularly in applications involving heart rate variability and arrhythmia assessment. However, conventional algorithms are prone to performance degradation in the presence of non-physiological signal distortions. In this study, ECG recordings in the publicly available Siena Scalp EEG Database were analyzed. Signals were segmented into one-minute epochs, with preprocessing performed using a finite impulse response bandpass filter. R-peak candidates were extracted via second-order differentiation and identified through a threshold-based selection process. Implausible detections were excluded based on cardio-physiological characteristics. The proposed method demonstrated strong agreement with a biosppy in segments without observable signal distortion ($r^2 = 0.877$, $p < 0.001$) and a markedly reduced correlation in segments exhibiting significant artifact signal ($r^2 = 0.131$, $p < 0.001$). Root mean square amplitude significantly differed between the two groups. These findings indicate that the proposed approach not only preserves high detection precision under stable conditions but also distinguishes segments likely to contain physiological interference. This enhances its utility in real-world clinical environments, where the minimization of false alarms is critical.

Index Terms: Electrocardiogram; R-peak detection; Second-order differentiation

I. INTRODUCTION

Electrocardiography (ECG) is a widely used technique for monitoring cardiac activity, offering critical insights into heart rhythm, rate, and morphology. Among various ECG features, the R-peak plays a central role in identifying heartbeats and calculating interbeat intervals, which are essential for heart rate variability analysis, arrhythmia detection, and other clinical interpretations[1-3]. Therefore, accurate detection of R-peaks is fundamental to both diagnostic and research applications.

Despite the clinical importance of R-peak detection, conventional algorithms often suffer from reduced accuracy in noisy or artifact-signal recordings. These methods may falsely identify non-physiological peaks or fail to detect peaks, particularly when the signal is contaminated by motion artifacts, muscle activity, or electrode instability. As a result, unreliable R-peak detection can lead to misinterpretation of cardiac conditions and increased false alarms in real-time monitoring systems[4]. While existing approaches have focused primarily on improving detection accuracy, they often lack mechanisms to explicitly evaluate signal quality or distinguish artifact signals.

In this study, we hypothesize that a second-derivative-based approach can achieve high-fidelity R-peak detection in clean ECG segments while simultaneously enabling the identification of artifact-containing signals based on signal characteristics. To examine this hypothesis, a threshold-based detection method employing second-order differentiation was implemented and its performance evaluated against a widely used reference algorithm.

II. MATERIALS AND METHODS

A. Data Description

This study utilized only the electrocardiogram (ECG) recordings from the publicly available *Siena Scalp EEG Database*, which provides synchronized EEG and ECG signals. Among the 14 subjects available in the database, one was excluded due to the absence of ECG recordings, resulting in a final cohort of 13 subjects. All available ECG signals were sampled at 512 Hz and segmented into one-minute epochs. Each segment was manually labeled as either artifact-free or artifact signal of waveform distortions, noise levels, and morphological irregularities in the QRS complexes.

B. Pre-processing

The ECG signals were preprocessed using a finite impulse response (FIR) bandpass filter with cutoff frequencies set to 0.67 Hz and 45 Hz to suppress baseline drift and high-frequency noise. For each subject, a 10-minute segment immediately preceding the onset of the ictal event was selected. These segments were then divided into non-overlapping one-minute epochs, which served as the fundamental units for R-peak detection and subsequent analysis.

C. R-Peak Detection

R-peak detection was performed using a second derivative-based method. For each one-minute ECG epoch, the second derivative of the signal was computed, and the resulting values were sorted. A threshold was applied to identify candidate R-peak locations based on the magnitude of the sorted values.

The candidate peaks were then grouped based on temporal proximity. Groups with inter-peak intervals shorter than 0.60 seconds or longer than 1.2 seconds were excluded from further analysis [5, 6]. The final R-peak annotations were compared to those obtained using the *biosppy* library, a widely adopted algorithm for ECG analysis in both clinical and research settings.

D. Statistical Analysis

To evaluate the performance of the proposed R-peak detection method, interbeat intervals derived from our algorithm were compared with those obtained using the *biosppy*. Pearson's correlation coefficient was computed to assess the agreement between the two methods. The analysis was performed separately for segments. The root mean square amplitude of each one-minute ECG epoch was calculated to quantify signal magnitude. To assess differences in signal characteristics between artifact-free and artifact signal groups, Welch's t-test was performed.

III. RESULTS

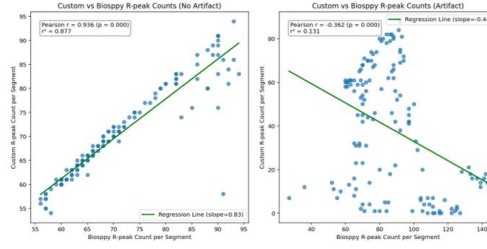


Figure 1. Comparison of R-peak counts per one-minute segment between the custom and Biosppy methods with linear regression analysis. (Left) Artifact-free segments demonstrate a strong positive correlation ($r^2 = 0.877$). (Right) Artifact-containing segments show a weak negative correlation ($r^2 = 0.131$).

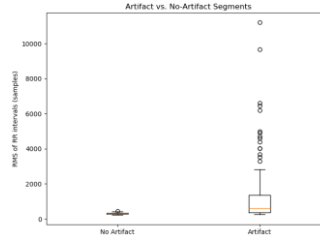


Figure 2. Comparison of RR-interval RMS between artifact-free and artifact segments. ($p < 0.001$)

The proposed second-derivative-based method demonstrated high agreement with the *biosppy* in artifact-free ECG segments, with a Pearson's correlation coefficient of $r^2=0.877$ ($p<0.001$). In contrast, for artifact segments, the agreement was substantially lower, with $r^2=0.131$ ($p<0.001$).

The RMS amplitude of ECG signals differed significantly between groups. The artifact-free segments exhibited a mean RMS value of 318.9 ± 39.4 , while artifact segments showed a much higher and more variable RMS value of 1328.9 ± 1787.7 . The difference in RMS values between the two groups was statistically significant ($p<0.001$).

IV. DISCUSSION AND CONCLUSION

The results demonstrate that the proposed second-derivative-based R-peak detection method performs comparably to the widely used *biosppy*. Moreover, our method shows a distinct advantage in its ability to identify and exclude artifact segments, which are often misinterpreted by conventional algorithms. This dual functionality suggests that our approach not only achieves robust R-peak detection but also offers a practical tool for signal quality assessment.

One possible explanation for this result lies in the nature of signal distortion caused by artifacts. ECG segments with artifact often contain increased high-frequency components [7]. When subjected to second-order differentiation, these components yield prominent peaks that differ significantly from physiological R-peaks. Our thresholding and grouping

strategy distinguish these artificial features, allowing for accurate filtering of non-physiological segments.

This capacity to selectively exclude artifact-heavy regions without falsely detecting R-peaks has practical clinic implications. It enhances the reliability of ECG-based monitoring systems in real-world environments and reduces the risk of false alarms, which are a common source of clinical fatigue. By incorporating physiological plausibility into the detection process, our method may contribute to more dependable and clinician-friendly cardiac monitoring solutions.

Although there are some limitations in this study, the proposed method indirectly detects artifact segments. However, it does not currently estimate R-peaks within those segments. As a result, potentially valuable information in partially contaminated data may be lost. Future work will focus on developing an extended algorithm capable of accurately detecting R-peaks even in the presence of artifacts.

Acknowledgements

This research was supported by the MISP(Ministry of Science, ICT & Future Planning), Korea, under the National Program for Excellence in SW(2024-0-00025) supervised by the IITP(Institute of Information & communications Technology Planing & Evaluation)

REFERENCES

- [1] M. A. Muzammil *et al.*, "Artificial intelligence-enhanced electrocardiography for accurate diagnosis and management of cardiovascular diseases," *Journal of Electrocardiology*, vol. 83, pp. 30-40, 2024.
- [2] L. Zhang, B. Li, and L. Wu, "Heart rate variability in patients with atrial fibrillation of sinus rhythm or atrial fibrillation: chaos or merit?," *Annals of medicine*, vol. 57, no. 1, p. 2478474, 2025.
- [3] J. Saengsuwan, L. Brockmann, C. Schuster-Amft, and K. J. Hunt, "Changes in heart rate variability at rest and during exercise in patients after a stroke: a feasibility study," *BioMedical Engineering OnLine*, vol. 23, no. 1, p. 132, 2024.
- [4] S.-F. Wung, M. Ogoo, J. L. Chow, and A. Muzzy, "False Crisis Alarms in Cardiopulmonary Monitoring:: Identification, Causes, and Clinical Implications," *Critical Care Nursing Clinics*, 2025.
- [5] T. Bombardini, R. Sicari, E. Bianchini, and E. Picano, "Abnormal shortened diastolic time length at increasing heart rates in patients with abnormal exercise-induced increase in pulmonary artery pressure," *Cardiovascular Ultrasound*, vol. 9, pp. 1-9, 2011.
- [6] D. M. Mirvis and A. L. Goldberger, "Electrocardiography," *Heart disease*, vol. 1, pp. 82-128, 2001.
- [7] J. Skoric, Y. D'Mello, and D. V. Plant, "A wavelet-based approach for motion artifact reduction in ambulatory seismocardiography," *IEEE Journal of Translational Engineering in Health and Medicine*, 2024.

Understanding Bias in Stride Length Estimation: Toward a Gating-Based Adaptive Modeling Approach

Hyeran Koo¹, Jong-Yeup Kim^{1,2,3}, Sung Ryul Shim^{1,2}, Yungjin Lee⁴ and Jieun Shin^{1,2*}

¹Konyang Medical data Research Group-KYMER, Konyang University Hospital, Daejeon 35365, Republic of Korea

²Dept. of Biomedical Informatics, Konyang University College of Medicine, Daejeon 35365, Republic of Korea

³Dept. of Otorhinolamgology-Head and Neck Surgery, Konyang University College of Medicine, Daejeon 35365, Republic of Korea

⁴Dept. of Rehabilitation Medicine, Konyang University College of Medicine, Daejeon 35365, Republic of Korea

[*Email : 910037@kyuh.ac.kr]

Abstract

This study investigates the distributional bias present in CNN-based stride length prediction using pressure insole data. A lightweight convolutional neural network (CNN) was trained with leave-one-subject-out (LOSO) cross-validation on 36 time-series features derived from 9 FSR sensors. Despite achieving an overall RMSE of 7.25 cm, model performance varied significantly across stride length ranges. Errors were most pronounced in short and long stride groups, revealing a regression-to-the-mean effect. These findings motivate the adoption of an adaptive architecture such as Gating + Mixture of Experts (MoE), which is proposed for future work to improve robustness in heterogeneous gait data.

Index Terms: stride length estimation, Deep Learning, long-tail distribution

I. INTRODUCTION

Stride length is a fundamental gait parameter for assessing mobility, particularly in elderly or neurologically impaired individuals. Wearable pressure insoles enable in-situ gait measurement, but accurate stride length estimation remains challenging. Prior studies using CNNs have reported moderate errors (4–9 cm RMSE or precision) [1–4], yet often overlook performance inconsistencies across stride length ranges.

This study revisits the stride prediction problem by identifying distributional bias and proposing the need for adaptive modeling. Specifically, we aim to explore the limitations of a single CNN model when applied to heterogeneous gait data.

II. SYSTEM MODEL AND METHODS

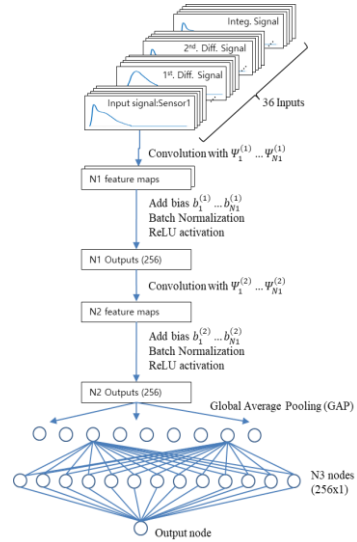
A. Dataset and Features

We used pressure insole data consisting of 9 force-sensitive resistor (FSR) sensors, from which 36 time-series features were extracted. The dataset includes both young and elderly participants, with elderly data augmented using jittering, magnitude scaling, and temporal warping to mitigate class imbalance. All sequences were zero-padded to a fixed length of 256. Target values (stride length) were normalized, while input features were kept in raw scale.

B. CNN Model Architecture

While prior work by Hannink et al. [1] adopted a CNN architecture with max pooling layers, such pooling operations can reduce temporal resolution and hinder the model's ability to capture subtle gait dynamics. In this study, we employed a modified CNN architecture that replaces max pooling with batch normalization (BN) and global average pooling (GAP). This structure improves training stability, preserves temporal resolution, and enhances analytical interpretability.

Although the difference in average performance was marginal (7.25 cm RMSE for BN+GAP vs. 7.43 cm for max pooling), the BN+GAP variant was chosen to align with our analytical objective: characterizing stride-length-dependent error patterns rather than minimizing absolute error.



C. Experimental Settings

The CNN was trained using the Adam optimizer (learning rate = 0.001) for 50 epochs with a batch size of 32. Input sequences were reshaped into $[C \times T]$ format and evaluated using leave-one-subject-out cross-validation. Prediction outputs were denormalized for evaluation. Performance metrics included RMSE, MAE, and bias across subjects and stride length groups.

III. RESULTS

A. Overall Model Performance

The CNN model achieved an average RMSE of 7.25 cm across all participants. However, prediction errors showed a systematic bias: shorter strides were consistently overestimated and longer strides underestimated, demonstrating a classic regression-to-the-mean pattern.

B. Group-wise and Subject-Wise Error analysis

Data were grouped based on stride length into short (<120 cm), medium (120–135 cm), and long (>135 cm). The model's performance varied across these groups:

Short strides: consistently overestimated

Medium strides: lowest errors

Long strides: consistently underestimated with higher variance

Bland–Altman plots (Fig. 1) colored by speed and age group showed no distinct clustering, suggesting that demographic attributes did not strongly influence error distribution. Likewise, prediction error plots versus true stride length (Fig. 2) revealed that error magnitude increased toward both tails of the stride range, regardless of age or speed.

These visual patterns confirm that the model's error is driven more by stride characteristics than demographic variation.

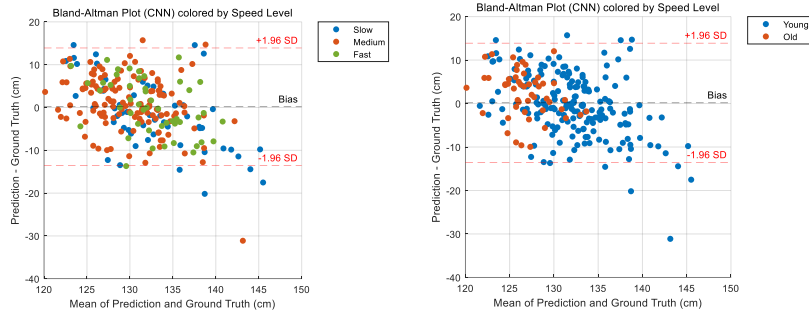


Fig. 1. Bland–Altman plot (CNN), colored by speed level(a) and age group(b)

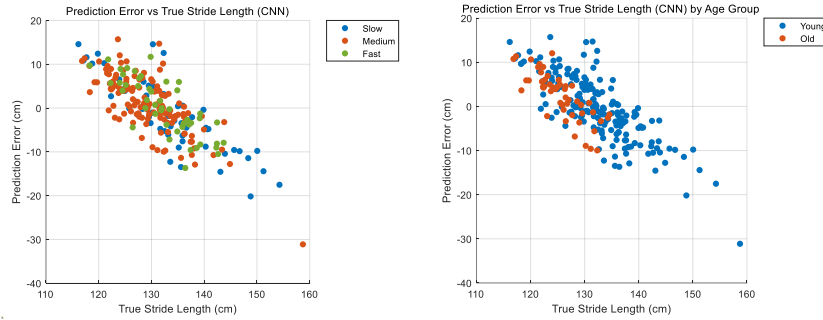


Fig. 2. Prediction error vs. true stride length, by speed level(a) and age group(b)

IV. DISCUSSION AND CONCLUSIONS

Although the CNN model performed reasonably well in terms of overall accuracy, its performance degraded at the extremes of stride length, showing significant underestimation in longer strides and overestimation in shorter ones. This distributional bias suggests a limitation of using a single-model approach for heterogeneous gait data.

Our experimental findings further showed that prediction errors were not significantly affected by walking speed or age. Rather, stride length itself was the dominant factor. These insights validate the need for an adaptive modeling strategy.

To address this limitation, we propose a Gating + Mixture of Experts (MoE) architecture in future work. In this framework, inputs will be classified into stride length-based groups and routed to specialized CNN experts using a gating network informed by temporal and statistical characteristics.

While MoE implementation is left for future study, this analysis provides a strong empirical motivation for its development.

ACKNOWLEDGMENTS

This work was supported by the Technology Innovation Program(or Industrial Strategic Technology Development Program-Digital Healthcare)(RS-2024-00467897, Development of LLM-based NER tool & standardization support tool for utilizing multicenter clinical free text) funded By the Ministry of Trade Industry & Energy(MOTIE, Korea).

This work was supported by the Technology Innovation Program(or Industrial Strategic Technology Development Program-Digital Healthcare)(RS-2025-02306926, Development of Automated Medical Record Generation and Automation Technology Based on Large-Scale AI) funded By the Ministry of Trade Industry & Energy(MOTIE, Korea)

REFERENCES

- [1] J. Hannink, T. Kautz, C. F. Pasluosta, K.-G. Gabmann, J. Klucken, B. M. Eskofier, "Sensor-based Gait Parameter Extraction with Deep Convolutional Neural Networks," *IEEE Journal of Biomedical and Health Informatics*, vol. 21, no. 1, pp. 85-93, 2017.
- [2] A. Zadka, N. Rabin, E. Gazit, A. Mirelman, A. Nieuwboer, L. Rochester, S. Del Din, E. Pelosin, L. Avanzino, B. R. Bloem, U. D. Croce, A. Cereatti, J. M. Hausdorff, "A wearable sensor and machine learning estimate step length in older adults and patients with neurological disorders", *npj Digital medicine*, no. 142, 2024.
- [3] J. Hannink, T. Kautz, C. F. Pasluosta, J. Barth, S. Schüle, K.-G. Gaßmann, "Mobile Stride Length Estimation With Deep Convolutional Neural Networks", *IEEE Journal of Biomedical and Health Informatics*, vol. 22, no. 2, 2018.
- [4] J. R. Verbiest, B. Bonnechère, W. Saeys, P. Van de Walle, S. Truijen, P. Meyns, "Gait Stride Length Estimation Using Embedded Machine Learning", *Sensors*, Vol. 23, No. 16, 2023.

Importance of Condition in Stable Diffusion : AI-generated Image and Semantic map

Si-Yeon Pak¹, Jin-Ho Chung¹, DaeHan Ahn^{1*}

¹Department of AI Convergence, University of Ulsan, 44610, South Korea
[Email: tdus86@mail.ulsan.ac.kr, jinho@ulsan.ac.kr, daehan@ulsan.ac.kr]

Abstract

This study investigates the impact of different conditioning sources in image generation using diffusion-based models. Specifically, we compare AI-generated style images and semantic maps as condition inputs for ControlNet-guided Stable Diffusion. Our experiments examine how the nature of the conditioning affects the visual quality, structure retention, and stylistic richness of the generated images. A unified framework is proposed in which the fixed prompt and initial image are used while only the conditioning input is different. Semantic maps are automatically extracted using SegFormer And AI-generated images are generated by Midjourney. Results show that conditions using AI-generated image tend to produce more vivid and stylistically distinct outputs, whereas semantic maps better preserve spatial layout and object placement. This comparative analysis reveals key trade-offs between structural accuracy and stylistic freedom in conditional diffusion, offering insights for diverse applications such as art generation, scene design, and structure-preserving image synthesis.

Index Terms: AI-generated Image, ControlNet, Semantic Map, Stable Diffusion

I. INTRODUCTION

Diffusion models have emerged as one of the most powerful generative frameworks for producing complex and highly detailed images. These models have continuously evolved to generate higher-quality outputs, with recent advancements emphasizing more precise control over the generation process. Models such as ControlNet [1] have attracted considerable attention for their ability to incorporate diverse external conditions to guide the generative process more effectively. To enable such fine-grained control, various types of conditions—such as AI-generated style images, semantic maps, depth maps, scribbles, and key points—can be employed. As each condition type may result in outputs with distinct visual characteristics and quality, it is crucial to analyze their respective influences on the generation outcomes. In this study, we compare the output characteristics of diffusion models conditioned on two commonly used condition—AI-generated images and semantic maps—and discuss the advantages and limitations of each conditioning approach.

II. SYSTEM MODEL AND METHODS

A. Stable Diffusion

Traditional diffusion models operate directly in pixel space, which leads to extremely high GPU memory consumption. To overcome this limitation, Stable Diffusion utilizes the concept of latent space through a Variational Autoencoder (VAE). This approach significantly reduces computational complexity [2]. As a result, unlike models composed entirely of Transformers, Stable Diffusion performs well even on high-dimensional data, which makes it suitable for use in our study with images of size 1024×1024 pixels.

B. ControlNet Framework

ControlNet extends pre-trained text-to-image models like Stable Diffusion by enabling additional image-based conditions such as poses, edges, or depth maps for more precise control [1]. Traditionally, these inputs reinforce known structural points. However, emerging approaches now use AI-generated images as conditions, introducing synthetic cues that encode stylistic or abstract features beyond the original input. This study compares these two conditioning types to analyze their impact on controllability and output characteristics in diffusion-based image generation.

C. Experimental Setup

We utilize the Stable Diffusion v1-5 model augmented with ControlNet to enable image-to-image generation guided by external conditions. Two types of conditioning inputs are used: (1) AI-generated images that provide stylistic points, and (2) semantic segmentation maps by SegFormer [3] that encode structural information. All experiments share a base image (initial image) and fixed empty prompts to ensure that the only variable is the conditioning input. The model is implemented via the diffusers library and accelerated using NVIDIA CUDA on GPU.

III. RESULTS



Figure 1. Comparison of image generation results across four categories (Animals, Things, Backgrounds, People) using AI-generated images and semantic maps as conditioning inputs.

Figure 1 illustrates the side-by-side results of four categories each consisting of (1) the initial image, (2) AI-generated style image as the conditioning input, (3) the semantic map, and (4) the output using the AI-generated images as the conditioning input, (5) the output using the semantic map as the conditioning input. Across all examples, the influence of the conditioning input is apparent and consistent with our hypotheses.

A. Observations on AI-generated Image Conditioning

The AI-generated Image-based conditioning consistently produces more visually appealing outputs with vivid textures, artistic flair, and enhanced color contrast. In the first two columns of each group, the stylistic guidance leads to novel reinterpretations of the source image, often deviating from the original structure but capturing rich visual semantics. For instance, in animal and portrait cases, facial features and textures are rendered in highly creative ways, sometimes adding painterly or fantasy-like effects.

B. Observations on Semantic Map Conditioning

In contrast, conditioning on semantic maps preserves the global structure and object placement more faithfully. The images generated in the last column of each group exhibit high spatial consistency with their corresponding semantic maps. While stylistic expressiveness is relatively lower, the outputs maintain realistic proportions and recognizable layouts. This is particularly evident in architectural and urban scene cases, where the model adheres closely to building outlines and scene geometry.

V. Trade-off Analysis

Our findings indicate a trade-off between stylistic richness and structural fidelity. AI-generated images are better suited for tasks emphasizing creative aesthetics, while semantic maps are preferable for scenarios requiring structural preservation. The choice of condition should thus align with the intended application, whether it be art synthesis or layout-aware generation. In practice, this means that users must prioritize either visual creativity or spatial consistency based on their goals. Future research could explore hybrid conditioning techniques that balance both aspects for more versatile image generation.

IV. DISCUSSION AND CONCLUSIONS

A. Summary of Findings

This study demonstrates the importance of the conditioning modality in diffusion-based image generation. AI-generated style images offer high aesthetic and expressive flexibility, often leading to more imaginative results. On the other hand, semantic maps serve as reliable guides for structural coherence, preserving spatial layouts with minimal distortion. Each approach reveals unique strengths, and their effectiveness varies depending on the target task.

B. Practical Implications

Our findings can guide practitioners in selecting appropriate conditioning inputs for specific applications. For example, artists and content creators may prefer style-driven generation for its creativity, while urban planners or robotic vision systems

may benefit more from structure-preserving methods. Hybrid conditioning methods could also be explored to combine the best of both worlds.

Table 1. Comparative Summary of Conditioning Inputs

Condition Type	Strengths	Weaknesses	Best Use Cases
AI-generated Image	High aesthetic quality, creativity	Structural deviation, low fidelity	Art, media, stylized generation
Semantic Map	Strong spatial alignment, realism	Less expressive, lower vividness	Scene understanding, CAD, planning

ACKNOWLEDGMENTS

This research was supported by the MSIT(Ministry of Science, ICT), Korea, under the National Program for Excellence in SW, supervised by the IITP(Institute of Information & communications Technology Planning & Evaluation) in 2025(2024-0-00025).

REFERENCES

[1] Zhang, Lvmin, Anyi Rao, and Maneesh Agrawala. "Adding conditional control to text-to-image diffusion models." *Proceedings of the IEEE/CVF international conference on computer vision*. 2023.

[2] Rombach, Robin, et al. "High-resolution image synthesis with latent diffusion models." *Proceedings of the IEEE/CVF conference on computer vision and pattern recognition*. 2022.

[3] Xie, Enze, et al. "SegFormer: Simple and efficient design for semantic segmentation with transformers." *Advances in neural information processing systems* 34. 2021.: 12077-12090.

Scoring Layer for Zero-Day Attack Detection

HoEun Kim¹, Jin-Ho Chung¹, DaeHan Ahn^{1*}

¹Department of AI convergence, University of Ulsan, 44610, Republic of Korea
[Email: likeeun03@mail.ulsan.ac.kr, jinho@ulsan.ac.kr, daehan@ulsan.ac.kr]

Abstract

Detecting zero-day attacks remains a key challenge for Intrusion Detection Systems (IDS), as these attacks involve previously unseen behaviors. While prior studies have mainly focused on improving model architectures and their performance, the role of detection criteria has been relatively underexplored. This study investigates the impact of various anomaly scoring metrics on the performance of AutoEncoder (AE)-based IDS. Experimental results show that while k-Nearest Neighbor (kNN) distance and cosine similarity achieve high performance, their effectiveness varies by attack type. To address this limitation, we introduce a scoring layer that integrates multiple anomaly scoring metrics. Through this approach, the combination of kNN and cosine similarity achieves the best overall performance, while a combination that includes Mahalanobis distance and reconstruction error yields more stable detection across diverse attack types.

Index Terms: Anomaly Detection, Detection criteria, Intrusion Detection System (IDS), Zero-Day Attack.

I. INTRODUCTION

With the advancement of cyber threats, the role of intrusion detection systems (IDS) has become increasingly critical. Traditional IDS, while effective against known attack patterns, struggle to detect zero-day attacks, which refer to novel intrusions involving previously unseen threats. To address this challenge, deep learning-based IDS have gained attention due to their ability to identify unknown patterns and adapt to new types of threats. Among various deep learning approaches, Autoencoder(AE) [1], generative adversarial network (GAN) [2] and Transformer-based models [3] are widely used to capture latent representations of normal data and detect anomalies based on reconstruction errors or latent space deviations.

A key component in such systems is the selection of appropriate anomaly scoring metrics, as it can significantly affect detection performance. However, while prior studies have primarily focused on improving model architectures and enhancing overall detection accuracy, there has been a lack of systematic investigation into the suitability of different scoring metrics for specific models or scenarios—especially in the context of detecting zero-day attacks.

This study aims to address the gap in existing research by empirically analyzing the impact of anomaly scoring metric selection on the performance of AE-based IDS, particularly in the context of zero-day attack detection. We focus on evaluating various detection metrics and analyzing their influence on anomaly detection accuracy. Furthermore, a scoring layer is introduced to combine multiple detection metrics, and practical guidelines are proposed for selecting appropriate metrics based on specific detection scenarios.

II. Methods

A. AutoEncoder-Based Anomaly Detection

An AE model [1] is employed to learn the distribution of benign network traffic. The model is trained only on benign data by minimizing the mean squared error (MSE) between the input and its reconstruction via encoding and decoding. After training phase, anomaly scores are computed for each test sample based on both the reconstruction error and deviations of its latent representation.

B. Scoring Layer

To integrate multiple anomaly detection scores from the AE, we construct a scoring layer based on a multi-layer perceptron (MLP). Each score is log-transformed and standardized before being fed into the MLP that consists of two hidden layers (64 and 32 units) with ReLU activations and outputs a logit for binary classification. The input to the scoring layer can be flexibly configured by selecting different combinations of detection criteria. In this study, we evaluate various combinations and compare their impact on anomaly detection performance.

III. Experiments

A. Experimental dataset

The CIC-IDS2017 dataset [4], one of the most widely used benchmarks in cybersecurity fields for evaluating IDS, was adopted in this study due to its relevance to our problem. The dataset is characterized by realistic network traffic, including benign activity and various attack scenarios captured under real-world conditions. The training dataset consists solely of benign traffic (BENIGN), enabling the AE-based anomaly detection model to learn the distribution of normal patterns and distinguish anomalous behavior during inference. For evaluation, the test dataset includes five attack types: DDoS, DoS, Slowhttptest, FTP-Patator, SSH-Patator, and Web Attack - XSS. To assess the model's generalization capability to previously unseen attacks, a zero-day attack detection scenario was constructed.

B. Detection Criteria

In this study, we evaluate six anomaly detection criteria. (1) Latent k-Nearest Neighbor (kNN) computes the average Euclidean distance between a test sample’s latent representation and its k-nearest neighbors among the training (benign) samples. (2) Mahalanobis Distance (Maha) computes the statistical distance to the mean of the benign latent space, considering the covariance structure. (3) Reconstruction Error (Rec) computes the mean squared error between the input and its reconstruction by the AutoEncoder. (4) Cosine similarity (Cos) assesses the angular similarity between a test sample and benign samples in the latent space. (5) Median-based Score (Med) computes the distance from a test sample to the median vector of the benign latent representations. Finally, (6) Support Vector Data Description (SVDD) constructs a hypersphere that encloses the benign data in the latent space and detects anomalies based on their distance from the center of the sphere.

IV. RESULTS

We compare the AUROC scores across six different anomaly detection criteria (Table 1). Cosine similarity (Cos) achieves the highest AUROC (0.8344), closely followed by k-Nearest Neighbor (kNN) with 0.8334. In contrast, SVDD shows the lowest performance with an AUROC of 0.5466.

Although some individual criteria showed consistently high performance, a detailed analysis by attack type (Table 2) revealed that even high-performance methods such as Cosine and Mahalanobis failed to detect certain attack types effectively. This observation highlights the limitation of relying on a single detection criterion. To address this limitation, we adopt a scoring layer that integrates multiple detection criteria for more robust and generalized anomaly detection.

Due to their low detection performance, Median and SVDD were excluded from this experiment. The scoring layer was trained using different combinations of the following 4 criteria: Latent kNN, Cosine distance, Mahalanobis distance, and Reconstruction Error.

As shown in Table 3, the combination of kNN and Cos achieved the highest average AUROC score of 0.8875, performing particularly well on DDoS and SSH-Patator attacks. In contrast, the combination of Cos, Maha and Rec achieved relatively high performance on attacks that are generally more difficult to detect, such as FTP-Patator and Web Attack-

Table 1. AUROC scores of individual detection criteria.

Detection Criteria	kNN	Cos	Maha	Rec	Med	SVDD
AUROC	0.8334	0.8344	0.7765	0.7482	0.6079	0.5466

Table 2. AUROC scores of individual detection criteria by attack types.

Attack type	kNN	Cos	Maha	Rec
DDoS	0.8391	0.8420	0.7808	0.7454
DoS Slowhttptest	0.9572	0.4972	0.9584	0.9574
FTP-Patator	0.4318	0.5369	0.6203	0.6131
SSH-Patator	0.6896	0.7418	0.4682	0.5086
Web Attack - XSS	0.6342	0.6132	0.7229	0.6821

Table 3. AUROC scores of individual detection criteria by attack type

Attack Type	kNN + Cos	Cos + Maha	kNN + Maha + Rec	Cos + Maha + Rec
DDoS	0.8937	0.8490	0.8136	0.8589
DoS Slowhttptest	0.8401	0.8571	0.9118	0.8892
FTP-Patator	0.5091	0.4643	0.4565	0.5790
SSH-Patator	0.7304	0.5088	0.4937	0.5714
Web Attack - XSS	0.6806	0.6466	0.6730	0.6820
Overall (All Attacks)	0.8875	0.8413	0.8073	0.8529

XSS. This combination also demonstrated stable performance across different attack types. These results suggest that combining multiple detection criteria through a scoring layer enables more robust anomaly detection compared to using a single criterion.

V. CONCLUSIONS

In this study, we propose a scoring layer that combines multiple anomaly detection criteria to overcome the limitations of using a single detection criterion. While single detection criterion performed well on average, they failed to consistently detect certain attack types, revealing a limitation in generalization. To address this issue, our scoring layer integrates complementary information from multiple criteria, resulting in more robust and generalized detection performance. Experimental results demonstrate that the combination of kNN and cosine similarity achieves the highest average AUROC. In contrast, the combination of cosine similarity, Mahalanobis distance, and reconstruction error maintains relatively stable performance on more challenging attack types, such as FTP-Patator and Web Attack-XSS. These findings suggest that combining appropriate detection criteria enhances the anomaly detection effectiveness, especially in zero-day scenarios, compared to relying on a single criterion.

ACKNOWLEDGMENTS

This research was supported by the MSIT(Ministry of Science, ICT), Korea, under the National Program for Excellence in SW, supervised by the IITP(Institute of Information & communications Technology Planning & Evaluation) in 2025(2024-0-00025)

REFERENCES

- [1] Bank, Dor, Noam Koenigstein, and Raja Giryes. "Autoencoders." *Machine learning for data science handbook: data mining and knowledge discovery handbook* (2023): 353-374.
- [2] Goodfellow, Ian, et al. "Generative adversarial networks." *Communications of the ACM* 63.11 (2020): 139-144.
- [3] Vaswani, Ashish, et al. "Attention is all you need." *Advances in neural information processing systems* 30 (2017).
- [4] Canadian Institute for Cybersecurity, "CICIDS2017 Dataset," [Online]. Available: <https://www.unb.ca/cic/datasets/ids-2017.html> [Accessed: Jun. 5, 2025].

Streaming Audio Data Clustering By Topic In Real Time

Jae-Hun Oh¹

¹Department of AI Convergence, Ulsan University, Ulsan 44610, South Korea

Abstract

This paper presents an approach for real-time topic change detection in streaming conversations. By combining CUSUM(Cumulative Sum) change detection with dynamic clustering based on Speech-to-Text(STT) embeddings, the proposed system minimize computational load and reduces time delays compared to traditional methods. Unlike conventional topic detection techniques, which rely on cosine similarity, this approach use Wasserstein Distance (WD) to more effectively capture subtle topic transitions in real-time data.

I. INTRODUCTION

With the development of live streaming platforms, the demand for adaptive systems that respond to real-time conversation content has steadily increased. Early detection of topic shifts in conversations forms the foundation for interventions in various fields such as real-time summarization and content application, significantly enhancing user experience.

Traditional topic transition detection methods are based on offline technologies such as Latent Dirichlet Allocation(LDA)[1] and Dynamic Topic Models(DTM)[2], which analyze topics by receiving large amounts of computational loads and long processing times.

Streaming Embedded Topic Models(StreamETM)[3] aim to solve this problem by gradually updating topic distributions. This research proposes a new approach that minimizes time delays in topic change detection and reduces computational load, enabling real-time operation. The algorithm combines CUSUM(Cumulative sum), a statistical change detection method, with dynamic clustering based on Speech-to-Text(STT) embeddings. This solution provides lower computational burden and minimizes time delays compared to existing modeling methods.

II. SYSTEM MODEL AND METHODS

The structure of the main method is composed as follows : STT, STT preprocessing and embedding, real-time topic change detection, and keyword extraction. It is an algorithm consisting of four stages, and I'll provide an explanation for each stage.

A. stage 1 : STT

The STT(Speech-to-Text) process utilized OpenAi's Whisper large model. The STT is configure to set the minimum conversion unit to 5 seconds and is set to break the text into

sentence units. If a sentence does not end within 5 seconds, the process ensures the text is segmented into sentence units through the STT process.

B. stage 2 : STT preprocessing and embedding

In the preprocessing stage, unnecessary phrases are removed. For example, phrases such as "Thank you", "Thanks", and "See you in the next video" which are unrelated to topic changes, are eliminated. Additionally, repetitive phrases are consolidated into a single instance to reduce redundant embeddings. After this process, embeddings are performed using the Sentence Transformer.

C. stage 3 : Real-time Topic change detection

In the analysis of existing text embedding methods, cosine similarity is commonly used. However, in real-time streaming data, there is often no significant change between sentences, and topic changes are detected through minimal variations. Given this characteristic of the data, it was determined that using cosine similarity could be problematic. Therefore, the approach was redesigned to use Wasserstein Distance(WD) to measure the distance between two embedding vectors, allowing for more effective topic change detection. Additionally, clusters are created for each incoming embedding to group topics more effectively.

D. stage 4 : Keyword Extraction

The criterion for detecting topic changes and starting to form clusters is set to 2-3 sentences being grouped together into a cluster. The keywords of this cluster are then extracted and used to create a representative vector for the cluster

III. RESULTS

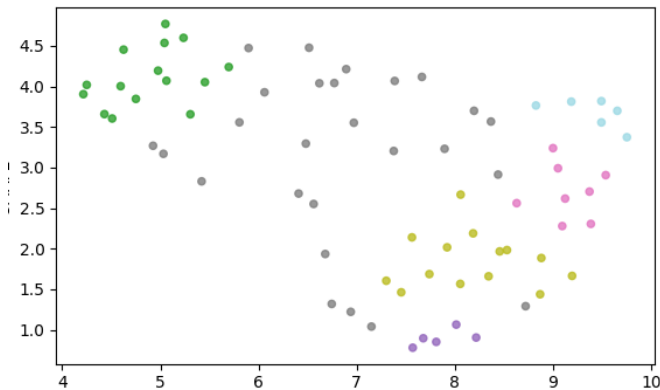


Fig 1. Cluster formation

Figure 1 is a visualization of the embeddings of the given data using UMAP. From the figure, we can see that each time an embedding is introduced, it is projected into the space, and the process of grouping similar clusters together is depicted.

IV. DISCUSSION AND CONCLUSIONS

A. Discussion

The proposed system for real-time topic change detection improves on traditional models by using CUSUM change detection and dynamic clustering based on Speech-to-Text (STT) embeddings. Unlike previous methods that require high computational resources, our approach minimizes time delays and reduces computational load. By utilizing Wasserstein Distance (WD) instead of cosine similarity, the system more accurately detects subtle topic shifts in real-time streaming data.

While the model works well for most conversations, it could be further improved by addressing faster-paced dialogues where transitions occur within a single sentence.

B. Conclusion

This paper presents an efficient method for real-time topic change detection in streaming conversations. By combining CUSUM and WD-based clustering, our approach reduces computational burden and minimizes time delays, making it suitable for real-time applications like content curation and user interaction analysis. The system sets a new standard for adaptive conversation analysis in live-streaming platforms.

ACKNOWLEDGMENTS

This research was supported by the MSIT(Ministry of Science, ICT), Korea, under the National Program for Excellence in SW, supervised by the IITP(Institute of Information & communications Technology Planning & Evaluation) in 2025(2024-0-00025)

REFERENCES

- [1] D. M. Blei, A. Y. Ng, and M. I. Jordan, “Latent Dirichlet Allocation”, *Journal of Machine Learning Research*, vol.3. pp. 993 – 1022, 2003.
- [2] D. M. Blei and J. D. Lafferty, “Dynamic topic models”, in *Proc. 23rd Int. Conf. on Machine Learning(ICML)*, Pittsburgh, PA, USA, pp. 113-120, 2006.
- [3] F. Granese, B. Navet, S. Villata, and C. Bouveyron, “Merging embedded topics with optimal transport for online topic modeling on data streams”, *arXiv preprint, arXiv:2504.07711*, 2025. Available
- [4] A. Petukhova, J. P. Matos-Carvalho, and N. Fachada, “Text clustering with large language model embeddings”, *International Journal of Cognitive Computing in Engineering*, vol. 6, pp. 100 – 108, 2025.

Multilingual LLMs-Based Bias Detection and Correction: Applications in the Korean Media Domain

Han-Eul Park^{1*}, Ga-Yun Kim^{1*}, and Dong-Seop Jin^{2†}

¹Department of AI Convergence, University of Ulsan, Ulsan 44610, South Korea

²Division of AI Convergence, University of Ulsan, Ulsan 44610, South Korea

[Email : kiice@kiice.org]

Abstract

Although biased expressions are often embedded in everyday language, their effects on society are not easily recognized. Such expressions can promote distorted perceptions of social groups or topics and consequently contribute to the formation of discriminatory structures across society. This emphasizes the growing need for technologies capable of detecting and neutralizing biased language. In this study, we propose a framework that utilizes Large Language Models (LLMs) to automatically detect biased expressions and rewrite them in a neutral manner. We demonstrate the practical applicability of the framework with real-world use cases in journalism—a domain where avoiding biased language is imperative.

Index Terms: Bias Detection, Neutralization, Multilingual LLMs, Journalism Ethics.

I. INTRODUCTION

News articles play a pivotal role in shaping public awareness and social discourse. Particularly in sensitive domains such as crime, gender, and immigration, biased language can distort readers’ judgment and exacerbate societal tensions. Such biased expressions also undermine readers’ trust in journalism and may result in reduced news consumption. [1] Therefore, the press should pursue fact-based and neutral reporting, supported by technical tools that assist journalists in detecting and correcting biased language at the drafting stage.

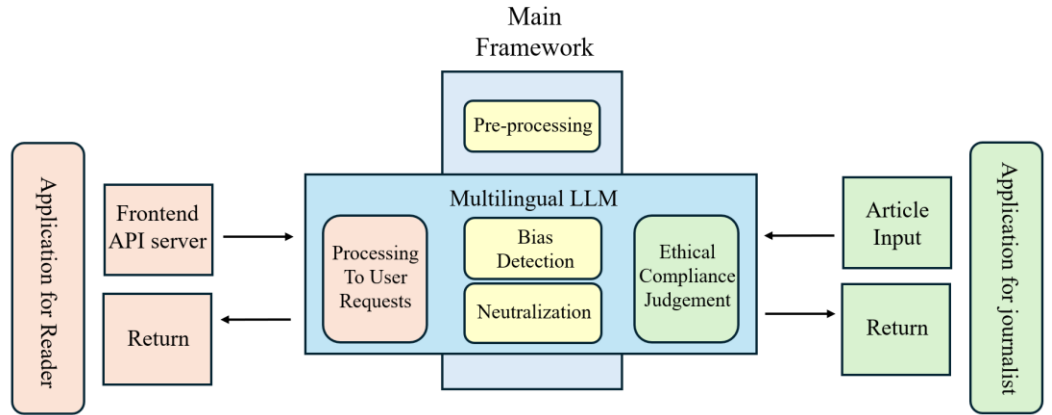
Recent studies have shown that Large Language Models (LLMs) can accurately identify biased expressions in news articles and rephrase them in neutral terms. Kuo et al. [2] defined bias in news as (1) loaded language, (2) selective framing, (3) emotional appeals, and (4) deliberate omission of critical information. Their study compared the bias correction performance of different LLMs under prompt-based instructions. Especially GPT-4o mini demonstrated high performance, aligning with human assessor 92.5% of the time.

Jin et al. [3] highlighted that the existing Bias Benchmark for Question Answering (BBQ) reflects U.S.-centric cultural assumptions and cannot be directly applied to Korean

* These authors contributed equally to this work.

† Corresponding author. Email: dsjin@ulsan.ac.kr

context. For example, perceptions of drug-related crime differ across cultures—Americans



A. Core Framework

The context-aware bias detection and correction pipeline is built using a multilingual LLMs Application Program Interface (API). It is informed by the evaluation methods and results of previous research. The LLMs used in this framework support multiple languages to enable broader applicability. The overall architecture of the proposed bias detection and correction system, including both the core framework and the application design discussed in the following section, is illustrated in Fig. 1. This diagram outlines the sequential flow of article input, LLM-based processing (e.g., preprocessing, bias detection, and neutralization), and output generation for each user group.

1) Preprocessing

The input text is segmented into sentences, and key topics are extracted using the LLM.

2) Bias Evaluation

Each sentence is evaluated based on topic context and classified into three levels of bias severity: high, moderate, or low. For high and moderate levels, the rationale for bias classification is extracted.

3) Bias Correction

The LLM generates neutral rewrites of biased sentences.

4) Output

The system returns suspected biased sentences, reasons for bias detection, bias severity level, and the neutralized rewrite.

B. Application Design

The core framework serves as the foundation for both journalist- and reader-facing applications, with each tailored to meet the specific needs of its users. This section outlines how the shared core processing pipeline is adapted differently in each application.

1) Application for Journalists

This application functions as a pre-publication tool to assess bias and verify compliance with journalistic ethics. It first detects the type and severity of bias in news content and then applies customized prompts based on this classification to generate a neutralized sentence. Next, the system conducts an ethical compliance check referencing guidelines issued by the Korea Journalists Association and other reputable organizations and produces a suggested revision if a potential violation is identified. Finally, the output includes the detected bias type, the neutralized sentence, the relevant ethical code clause, and the proposed revision, thereby supporting journalists in maintaining editorial integrity.

2) Application for Readers

The reader-facing application improves information accessibility through an LLM-based chatbot designed to understand the broader context of social issues. It offers interactive features such as article summarization, factual inquiry, bias explanation and rewriting, and multi-angled interpretation. These capabilities help readers grasp key points and engage with content in a more objective and critical manner, extending the scope of LLMs beyond bias detection and neutralization.

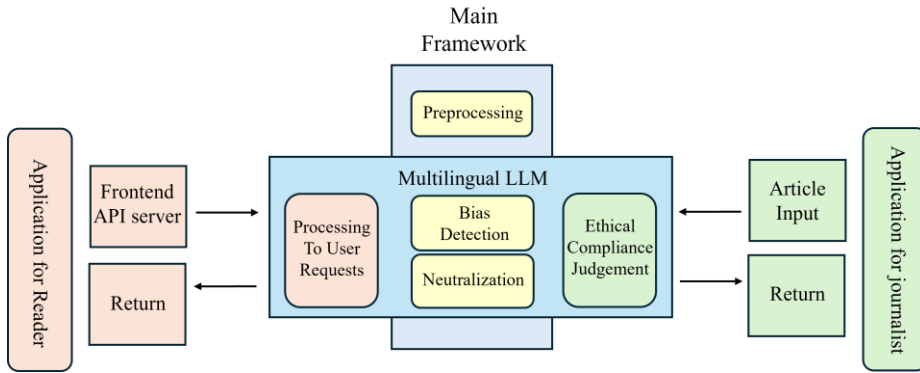


Fig. 1. Core framework workflow and its application across journalist and reader user groups.

III. CONCLUSIONS AND FUTURE WORK

A. Conclusion

This study proposed a multilingual LLMs-based framework for detecting and correcting biased language. By leveraging LLMs capable of understanding diverse sociolinguistic contexts, the proposed framework is applicable across various cultural and national settings. In the journalism domain, its applications include pre-publication editorial support and interactive information access tools, such as chatbots. In conclusion, our framework is adaptable to any domain that requires technologies for detecting and mitigating bias.

B. Future Work

We plan to extend the framework to underrepresented languages for cross-linguistic evaluation, enabling assessment of cultural adaptability beyond the American and Korean contexts. In parallel, we aim to improve the precision of bias correction by developing bias-type-specific strategies with tailored prompts for political, social, and gender-related content.

ACKNOWLEDGMENTS

This research was supported by the MSIT(Ministry of Science, ICT), Korea, under the National Program for Excellence in SW, supervised by the IITP(Institute of Information & communications Technology Planning & Evaluation) in 2025(2024-0-00025).

REFERENCES

- [1] D. Y. Oh, “Time series study on effects of consumers' evaluation of reporter, newspaper news problems and reliability on newspaper news usage,” *Korean J. Commun. Inf.*, vol. 83, no. 3, pp. 96–129, 2017. doi: 10.46407/kjci.2017.06.83.96.
- [2] C. W. Kuo, K. Chu, N. AlDahoul, H. Ibrahim, T. Rahwan, and Y. Zaki, “Neutralizing the narrative: AI-powered debiasing of online news articles,” *arXiv preprint arXiv:2504.03520*, 2025.
- [3] J. Jin, J. Kim, N. Lee, H. Yoo, A. Oh, and H. Lee, “KoBBQ: Korean bias benchmark for question answering,” *Trans. Assoc. Comput. Linguist.*, vol. 12, pp. 507–524, 2024.

Multi-View Learning for Welding Defect Detection in Radiographic Images

Minho Park¹, Daeseop Kim², and Sudong Lee^{1*}

¹Department of Industrial Engineering, University of Ulsan, Ulsan, 44610 Korea

²Department of IT Convergence, University of Ulsan, Ulsan, 44610 Korea

[Email : sudonglee@ulsan.ac.kr]

Abstract

Welding defects pose a risk of structural failure, necessitating reliable techniques for automated early detection. However, detecting subtle Welding defects in grayscale X-ray images is difficult due to low contrast, high level of noise, and indistinct boundaries. We propose a multi view input approach to address the limitations of single-channel images. The grayscale image is expanded to three channels, with distinct preprocessing techniques. frequency-domain filtering, multi-scale edge extraction, and normalization. We used EfficientNet-B0 as the backbone and fine-tuned only selected layers for efficient training. Compared to the single-channel model, the proposed multi view input model achieved a higher F1-score, improving by approximately 6%. The experimental results show that the proposed technique is effective for defect detection in grayscale welding images.

Index Terms: Radiographic Images, Anomaly Detection, Multi-View Learning, Transfer Learning

I. INTRODUCTION

Early detection of Welding defects is critical for industrial safety. However, grayscale X-ray images exhibit low contrast and noise, making automatic detection difficult. Varying material properties further obscure defect visibility. These challenges limit traditional methods, highlighting the need for more robust approaches.

Previous studies have attempted preprocessing techniques to enhance important features in grayscale images [1, 2]. For example, frequency components and Laplacian edge features have been used to highlight defects more effectively. Integrating these features into CNN models, rather than using raw grayscale images alone, has shown improved performance.

In this paper, we propose a method to improve the accuracy of defect detection in grayscale Welding images by using multi view feature fusion input and an EfficientNet-based transfer learning model[3]. From the original grayscale image, we extract features from both the frequency domain and the spatial domain to construct a 3-channel composite image. We validate the effectiveness of the proposed approach through experiments on an actual Welding X-ray image dataset, achieving approximately a 6-percentage point increase in F1-score compared to using a single-channel input.

* Corresponding author

II. METHODS

A. Multi-View Input Image Construction

From a single grayscale Welding image, we extracted three different types of feature maps that are effective for defect detection and combined them to form a 3-channel image. Figure 1 presents the process of constructing the proposed 3-channel input image. The composition of each channel is detailed as follows

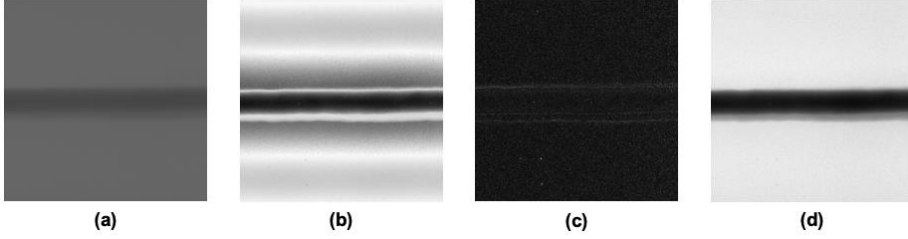


Fig. 1. a) Original and b) FFT c) Laplacian d) Normalized

Channel 1 – High-frequency component (frequency domain):

A Fast Fourier Transform (FFT) is applied to the original grayscale image to convert it into the frequency domain. Low-frequency components, including the DC term and very low-frequency bands, are removed from the center of the spectrum. An inverse FFT is after performed after reconstructing the spatial domain representation. This process suppresses global illumination variations and background trends, while enhancing high-frequency features such as fine textures and edges commonly associated with Welding defects. The resulting Channel 1 is intended to emphasize subtle indications of anomalies, such as micro-cracks or porosities, by capturing fine-grained textural patterns.

Channel 2 – Multi-scale edges (spatial feature)

Laplacian filtering is applied at multiple spatial resolutions to extract edge information from the grayscale image. Gaussian blurring with low and high sigma values generates fine and coarse structures, to which Laplacian operators are applied to create multi-scale edge maps. These maps are combined and normalized into an 8-bit image forming Channel 2, which provides boundary details at multiple scales to aid defect detection.

Channel 3 - Normalized original image

This channel represents a linearly normalized version of the original grayscale image. Pixel intensities are scaled to the range $[0, 1]$ to ensure consistency with the other input channels. Channel 3 retains the absolute intensity distribution and global structural context of the original image, providing a reference for interpreting the frequency- and edge-based features in the other two channels. Its inclusion supports the model in learning both enhanced features and the original background characteristics.

By combining the three processed channels, a three-channel composite image is formed, compatible with standard RGB-based CNN architectures. This design enables compatibility with standard convolutional neural network architectures pre-trained on natural images, such as those trained on ImageNet, without requiring any modifications to

the input layer. The resulting multi view input, which integrates frequency-domain and spatial-domain features, provides a more informative and diverse representation compared to the original single-channel grayscale input.

B. EfficientNet-B0-Based Model and Fine-Tuning

An EfficientNet-B0 architecture pre-trained on the ImageNet dataset was adopted for the binary classification of Welding defects using the constructed three-channel input. To preserve general low-level representations learned during pretraining, the early convolutional layers were kept frozen. Fine-tuning was performed only on the higher-level layers and the final classification head. This transfer learning approach helped effective model training under limited data conditions while mitigating the risk of overfitting.

C. Data Augmentation and Training Setup

A Welding X-ray dataset comprising 700 normal and 100 defective images was collected from an industrial environment, exhibiting a class imbalance with only 12.5% defective samples. To mitigate this, data augmentation was applied to the defective images using random rotations, flips, brightness adjustments, and Gaussian noise. This resulted in an approximately balanced training set. Twenty percent of the total dataset was held out for testing. The model was trained using the AdamW optimizer with a learning rate of $1e-4$ for 50 epochs. Transfer learning was employed by freezing the early layers of the EfficientNet-B0 backbone and fine-tuning the higher layers to improve generalization under limited data conditions.

III. RESULTS

The proposed multi-view input model was compared to a single-channel baseline using the same EfficientNet-B0 architecture and training setup. The baseline received a replicated grayscale image as input, ensuring identical model capacity. Performance was evaluated using metrics such as Accuracy, Precision, Recall, F1-score, AUC, and AUPRC, with emphasis on defect-class detection. Comparative results are summarized in Table 1.

Table 1: Performance Comparison by Input Type

Input Type	Accuracy	Precision	Recall	F1-score	AUC	AUPRC
Grayscale	0.94	0.85	0.88	0.86	0.95	0.84
Multi-View(ours)	0.97	0.91	0.93	0.92	0.98	0.90

As shown in Table 1, the proposed multi-view input model consistently outperformed the single-channel baseline across all evaluation metrics. The F1-score for defect detection improved from 0.86 to 0.92, with corresponding gains in Precision and Recall of approximately 6–7 percentage points. These results suggest improved detection performance, particularly for the minority defective class, without compromising false positive rates.

The multi-view model also achieved higher AUC (0.98 vs. 0.95) and AUPRC (0.90 vs. 0.84), indicating enhanced overall classification ability and better precision-recall trade-

off under class imbalance. These improvements are attributed to the model's ability to capture both frequency- and spatial-domain features via the three-channel input.

Qualitative analysis further revealed that the single-channel model struggled with subtle or low-contrast defects, especially those with unclear boundaries. In contrast, the multi-view input emphasized textural and edge features, leading to improved sensitivity to small or ambiguous anomalies. This confirms the effectiveness of the proposed input representation in enhancing defect localization and classification.

IV. DISCUSSION AND CONCLUSIONS

This study proposed a deep learning-based approach for Welding defect detection in grayscale images by integrating frequency-domain and spatial-domain features into a multi-channel input, combined with an EfficientNet-B0-based model. Experimental results demonstrated that the proposed method achieved notable improvements over conventional single-channel models, as reflected by higher F1-score, AUC, AUPRC, and other performance metrics. These improvements can be attributed to the complementary integration of high-frequency components and edge-based features, which enhanced the model's ability to distinguish defects from the background. Additionally, the use of EfficientNet-B0 with transfer learning facilitated high detection accuracy under limited data conditions.

In summary, the proposed multi view input approach demonstrated strong performance in enhancing Welding defect detection on grayscale X-ray images. The method is considered to be generalizable and may also be applicable to other industrial imaging tasks characterized by low contrast, high noise, and subtle defect patterns.

ACKNOWLEDGMENTS

This research was supported by the MSIT(Ministry of Science, ICT), Korea, under the National Program for Excellence in SW, supervised by the IITP(Institute of Information & communications Technology Planning & Evaluation) in 2025(2024-0-00025)

REFERENCES

- [1] M. I. Rajab, T. A. El-Benawy, M. W. Al-Hazmi, "Application of frequency domain processing to X-ray radiographic images of welding defects," *J. X-Ray Sci. Technol.*, vol. 15, no. 2, pp. 147–156, 2007.
- [2] Z. Xiao, K. Y. Song, M. M. Gupta, "Development of a CNN edge detection model of noised X-ray images for enhanced performance of NDT," *Measurement*, vol. 174, 109012, 2021.
- [3] S. Su, S. J. Pan, I. W. Tsang and Q. Yang, "Multi-view classification with convolutional neural networks," *PLOS ONE*, vol. 15, no. 12, pp. 1–15, 2020.

A Network Optimization Approach Using Congestion Control to Solve Course Registration System Delays

Sangwon Park*

¹Department of Convergence IT Engineering, Ulsan University, 44610, Korea
[Email : ps0020785@gmail.com]

Abstract

The university course registration system often faces problems like connection delays or errors because many students try to access the system at the same time. This happens mainly for two reasons. One is when the student's device has an unstable Wi-Fi connection, and the other is when the server's network is unable to handle too many requests at once, leading to overload. This paper focuses on the second reason, network congestion on the server side.

When the server receives too many requests at once, packet loss or a slow connection happens. To solve this, TCP uses several congestion control algorithms. The main ones include TCP CUBIC and ECN. These methods detect when the network is congested and adjust the speed of data transfer to reduce the problem. This paper analyzes how these congestion control methods work in high-traffic situations like course registration and what improvements can be made. Through this, we confirm the potential of congestion control algorithms to reduce connection delays in the course registration system.

Index Terms: Congestion Control, TCP, CUBIC, ECN

I INTRODUCTION

University course registration is a typical high-traffic situation that almost every student experiences. Even though universities spend a lot of money on high-performance servers to ensure the system's stability, connection delays and errors still happen frequently when registration begins. This leads us to question why this issue keeps happening, and this paper aims to analyze the causes.

The connection delay in the registration system can be divided into two main causes. First, when the user's device has an unstable wireless network, and second, when the server network cannot handle the excessive amount of traffic, causing a bottleneck. This research focuses on the server-side bottleneck, which occurs when the system is overwhelmed by the incoming traffic. In particular, it looks at how the server reacts when traffic suddenly spikes and how the network manages this overload.

To solve these network congestion problems, congestion control algorithms have been developed. TCP uses different algorithms to detect congestion and adjust the data transfer

rate to reduce bottlenecks. However, it is not clear if these algorithms work well in high-traffic situations like course registration.

Therefore, this paper analyzes, based on existing literature, how much TCP-based congestion control algorithms can reduce bottlenecks in the course registration system, exploring both their potential and limitations.

II. Related Work & Theoretical Background

Congestion control is a key mechanism in computer networks that prevents or reduces congestion when traffic becomes excessive. TCP, a connection-oriented protocol, has built-in congestion control features, and over the years, various congestion control algorithms have been developed and improved.

One of the basic algorithms, TCP Reno, detects packet loss as a sign of congestion and adjusts the transmission speed accordingly. Reno uses techniques like slow start, congestion avoidance, and fast retransmit to handle congestion. Floyd and Jacobson [1] showed in their simulation that combining RED and Reno can effectively reduce packet loss and delays through early detection of congestion.

Later, TCP CUBIC was proposed to overcome the limitations of traditional TCP in high-speed network environments. CUBIC adjusts the congestion window(cwnd) using a cubic function over time, allowing for faster recovery and higher throughput. In a study by Rhee et al [2], experiments showed that CUBIC performs better than TCP Reno in situations where congestion occurs, providing higher stability and throughput.

On the other hand, ECN is a method that allows routers to notify the sender about congestion without packet loss. The RFC 3168 standard, defined by Ramakrishnan et al [3], demonstrates that ECN can detect congestion early and reduce packet loss and delays by adjusting the transmission rate without waiting for packet loss. By marking packets with congestion signals, the sender can proactively adjust its transmission rate.

These modern congestion control techniques, especially CUBIC and ECN, are highly important in time-sensitive traffic environments like university course registration systems, where thousands of users access the system at the same time. In situations like this, quick recovery, high throughput, and lossless control are essential to maintaining connection stability and service quality.

III. DISCUSSION

To solve the traffic overload problem in course registration systems, improving only the TCP congestion control algorithms is not enough. This is because there are multiple factors involved in large-scale traffic environments.

First, the fundamental limitation of algorithms. Most TCP congestion control methods only respond to congestion at the network level and have no pre-emptive response to concentration of requests at the application level (e.g. spikes in registration clicks). TCP Reno, CUBIC, and ECN all detect congestion caused by issues like excess bandwidth use or queue overflow as packets travel through the network and adjust the transmission speed. However, application-level bottlenecks or the server's processing limitations, such as thousands of users accessing the server simultaneously, cannot be handled by TCP congestion control alone.

Second, the delay in congestion signals. Most TCP congestion controls only start responding after congestion signals(like packet loss or ECN marks) are detected.

When traffic suddenly spikes, the network queue becomes overloaded, and congestion signals are detected only after packet loss has occurred. As a result, users experience delays and errors before the congestion control kicks in.

To overcome these issues, fundamental improvements in congestion control are needed. Currently, most systems rely on real-time response to traffic overloads. However, real-time responses inevitably involve delays, so there is a need for predictive congestion control systems. For example, using AI-based prediction algorithms, traffic during course registration can be predicted in advance, and server resources and network bandwidth can be automatically expanded or distributed before traffic overload occurs. This method allows for optimization of resources according to the predicted traffic, overcoming the limitations of real-time responses.

ACKNOWLEDGMENTS

This research was supported by the MSIT(Ministry of Science, ICT), Korea, under the National Program for Excellence in SW, supervised by the IITP(Institute of Information & communications Technology Planning & Evaluation) in 2025(2024-0-00025)

REFERENCES

- [1] Floyd, S., & Jacobson, V. (1993). Random Early Detection Gateways for Congestion Avoidance. *IEEE/ACM Transactions on Networking*, 1(4), pp.397–413.
- [2] Rhee, I., Xu, L., Ha, S., & Zimmermann, A. (2008). CUBIC: A New TCP-Friendly High-Speed TCP Variant. *ACM SIGOPS Operating Systems Review*, 42(5), pp.64–74.
- [3] Ramakrishnan, K., Floyd, S., & Black, D. (2001). The Addition of Explicit Congestion Notification (ECN) to IP

Development and EMG-Based Comparative Analysis of a Behavior-Inducing VR Digital Therapeutic System for Forward Head Posture Rehabilitation

Jonghyuk Shin¹, Hyunwoo Jung¹, Yelim Jung¹ and Minsu Jo¹

¹Department of IT Convergence, Ulsan University, Ulsan 44610, Republic of Korea
[Email : sinjeff@naver.com / wldtpdn@naver.com / yelim402@naver.com / minsuu2@gmail.com]

Abstract

This study proposes a behavior-inducing virtual reality (VR)-based digital therapeutic system to support the rehabilitation of Forward Head Posture (FHP), a common musculoskeletal disorder. Traditional rehabilitation typically involves in-person physical therapy, which often poses challenges for patients due to physical accessibility limitations, making regular participation difficult. In contrast, 2D video-based digital therapy offers excellent accessibility and convenience but has limited effectiveness due to low immersion and inadequate posture correction.

To address these issues, this study precisely implemented the Wall Angel Exercise performed by a professional physical therapist using AI-based motion capture technology and incorporated it into a VR content environment. An interactive system was designed that provides real-time feedback and behavioral induction based on the user's electromyography (EMG) signals.

To evaluate the feasibility of the system, a preliminary case study was conducted with a single male in his 20s diagnosed with FHP. The participant performed the same Wall Angel Exercise under three conditions: clinical rehabilitation, 2D video-based therapy, and the proposed VR system. The EMG responses of the serratus anterior muscle were compared and analyzed. Results showed low muscle activation in the initial VR condition, but EMG responses similar to those of clinical treatment were observed after the behavioral induction mechanism was activated. Although limited to a single-subject pilot study, the results suggest that the proposed VR system has potential as a next-generation digital therapeutic tool for musculoskeletal rehabilitation.

Index Terms: Virtual Reality, Digital Therapeutics, Electromyography (EMG), Forward Head Posture, Musculoskeletal Rehabilitation

I. INTRODUCTION

Forward Head Posture (FHP) is a representative musculoskeletal disorder caused by poor posture habits in modern life [1]. Strengthening and correcting specific muscles can alleviate symptoms. Rehabilitation exercises targeting the serratus anterior are known to be effective for shoulder stabilization and posture recovery. Clinical settings

typically use movements like the Wall Angel Exercise, guided face-to-face by a physical therapist [2]. However, repeated clinic visits are required, and patients with geographical or physical constraints face participation challenges.

As an alternative, 2D video-based therapy has been widely used. However, its one-way information delivery limits user immersion and posture accuracy, reducing therapeutic effects [3]. To address these limitations, we propose an interactive digital therapeutic system using AI-based motion capture to replicate a physical therapist's Wall Angel Exercise in a VR environment. The system includes a posture learning phase before starting the exercise and provides real-time EMG-based feedback and behavioral induction to encourage active participation. The behavioral induction mechanism triggers visual stimuli (e.g., paparazzi on horseback events) if EMG activation fails to meet a threshold, thereby promoting muscle activation.

II. SYSTEM MODEL AND METHODS



Fig. 2. AI-based motion capture synchronization: The real-world motion performed by a physical therapist for guide data generation (left) and the corresponding virtual avatar motion (right). The system uses this synchronized data to provide accurate posture guidance in the VR environment.

A. AI Motion Capture-Based Posture Learning System

As shown in Figure 1, the Wall Angel Exercise demonstrated by a physical therapist was precisely captured using AI-based motion capture technology [4]. The 3D motion data was applied to a virtual therapist avatar within the VR environment. Before beginning the main exercise, users underwent a "posture learning" phase to imitate the avatar. This enabled more precise and effective posture correction compared to conventional 2D video-based methods.

B. Real-Time EMG Feedback System

To quantitatively evaluate exercise performance, surface EMG sensors were attached to the serratus anterior to collect real-time data [5]. Before content initiation,

Maximal Voluntary Contraction (MVC) was measured to establish baseline EMG levels. During the exercise, real-time EMG readings were compared to this reference, and immediate visual or situational feedback was provided when values fell short. This real-time biofeedback system not only guides posture but also delivers customized responses, differentiating it from conventional digital therapies.

C. Behavior-Inducing Scenario Design

Behavioral induction plays a key role in VR-based therapy by naturally enhancing user engagement through targeted stimuli during exercises [6]. In this content, users play the role of a bodyguard protecting a celebrity. The Wall Angel Exercise is gamified as an action to block paparazzi from taking photos. If EMG levels fall below the threshold, a new event is triggered where the paparazzi ride a horse to shoot from a higher position. This reactive scenario encourages users to concentrate and exert greater effort, effectively increasing serratus anterior activation

III. RESULTS

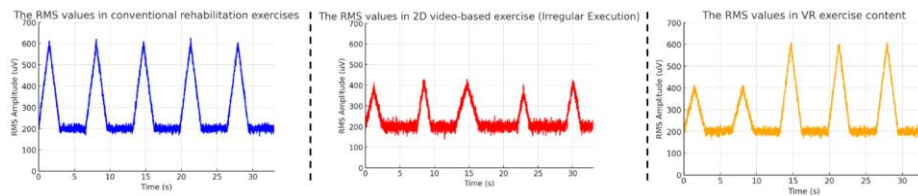


Fig. 2. Comparison of RMS values of the serratus anterior across three exercise conditions: (left) conventional rehabilitation, (center) 2D video-based exercise with irregular execution, and (right) VR-based exercise with behavior-inducing feedback.

As shown in Figure 2, clear differences in EMG responses were observed across the three conditions. In the clinical condition, consistently high RMS values ($\sim 600 \mu V$) and stable waveform patterns were recorded, indicating effective muscle activation guided by professional feedback.

In contrast, the 2D video condition yielded lower average RMS ($\sim 400 \mu V$), with inconsistent waveform intervals and amplitudes. The participant often failed to mimic the video accurately or maintain timing, reflecting limitations in immersion and posture precision.

Interestingly, in the VR condition, the first 1–2 repetitions showed similar low activation levels to the 2D condition. However, from the third repetition—coinciding with the behavioral induction event (paparazzi on horseback)—RMS values rose sharply to levels comparable to the clinical condition ($\sim 600 \mu V$). Waveforms also stabilized, indicating improved consistency and muscle activation.

These findings suggest that although VR initially resembles video-based therapy, the addition of personalized feedback and behavior-inducing scenarios enables muscle activation levels comparable to clinical treatment. Especially for muscles like the serratus anterior that require precise stimulation, interactive VR content may provide superior therapeutic outcomes.

IV. CONCLUSIONS

This study proposed a behavior-inducing VR-based digital therapeutic system for the rehabilitation of Forward Head Posture (FHP) and conducted a single-subject pilot experiment to assess its feasibility. The system integrates AI motion capture of physical therapist demonstrations, 3D avatar-guided posture learning, real-time EMG feedback, and gamified behavioral scenarios to enhance user engagement and exercise intensity.

Results revealed that while the VR condition initially showed low EMG responses similar to 2D video-based therapy, the activation mechanism increased muscle engagement to clinical levels after behavioral cues were introduced. These outcomes suggest that VR systems can compensate for the limitations of video therapy by providing real-time feedback and immersive interaction.

Although this was a preliminary experiment with a single subject, the system showed potential to elicit clinically meaningful muscle activation. Future studies should include a broader demographic and perform statistical validation to establish the therapeutic efficacy and scalability of the system. Further development of personalized behavioral strategies and adaptive feedback algorithms could help expand the clinical applicability of digital therapeutic technologies.

ACKNOWLEDGMENTS

This research was supported by the MSIT(Ministry of Science, ICT), Korea, under the National Program for Excellence in SW, supervised by the IITP(Institute of Information & communications Technology Planning & Evaluation) in 2025(2024-0-00025)

REFERENCES

- [1] E. H. Choi, H. S. Hwang, and C. S. Kim, "Electron spectroscopy studies on magneto-optical media and plastic substrate interfaces," *International Journal of Information and Communication Engineering*, vol. 9, no. 4, pp. 358-362, 2011.
- [2] J. G. Proakis, *Digital Communications*, 4th ed., New York, NY: McGraw-Hill, 1993.
- [3] J. L. Hennessy and D. A. Patterson, "Instruction-level parallelism and its exploitation," in *Computer Architecture: A Quantitative Approach*, 4th ed., San Francisco, CA: Morgan Kaufmann Pub., pp. 66-153, 2007.
- [4] A. Hashmi, H. Berry, O. Temam, and M. Lipasti, "Automatic abstraction and fault tolerance in cortical microarchitectures," in *Proceeding of the 38th Annual International Symposium on Computer Architecture*, New York: NY, pp. 1-10, 2011.
- [5] B. Alavi, "Distance measurement error modeling for time-of-arrival based indoor geolocation," Ph.D. dissertation, Worcester Polytechnic Institute, Worcester: MA, 2006.
- [6] Y. Z. Ben, D. K. John, and Anthony, "Tapestry: An infrastructure for fault-tolerant wide-area location and routing," University of California, Berkeley: CA, Technical Report CSD-01-1141, 2001.

Position-Based Object Mapping Algorithm for Screen Narration to Assist Visually Impaired Users

Park Seong Jun¹, Boo Seungwon²

¹Department of IT convergence, University of Ulsan, 44616, Republic of Korea

²Department of AI, University of Ulsan, 44616, Republic of Korea

[Email: Boo6758@naver.com, psj010602@gmail.com]

Abstract

We propose a hybrid object identification and mapping framework that integrates relative spatial cues with face and body embeddings to support screen narration for visually impaired users. Our approach uses YOLO for object detection, MTCNN for face detection, FaceNet512 for facial embeddings, and PCB-P6 for full-body embeddings when faces are absent. By calculating pairwise relative distances and fusing spatial context with appearance embeddings, our method maintains accurate identification even when individuals wear similar clothing or show only back views. The key is the integration of relative location modeling into traditional embedding-based object identification.

Index Terms: embedding-based object identification, relative spatial modeling, screen narration

I. INTRODUCTION

Accessible visual narration for visually impaired users demands accurate object identification coupled with spatial context modeling. Current approaches predominantly rely on facial embeddings, resulting in failure when faces are occluded or unavailable, particularly in scenarios involving multiple individuals wearing similar clothing such as workplace environments, sports teams, or institutional settings [6], [7].

Our position-aware approach addresses these limitations by incorporating relative spatial relationships as discriminative features. We present a framework that combines: (1) YOLO-based object detection [12], (2) MTCNN branching for face detection-based classification [13], (3) FaceNet512 embeddings for face-present instances [14], (4) PCB-P6 full-body embeddings for face-absent instances [11], and (5) pairwise relative distance measurements for spatial layout modeling [1], [2], [15].

This position-aware embedding fusion enables reliable object mapping regardless of facial information availability.

II. Related Work and System Background

A. Visual Screen Narration Challenges

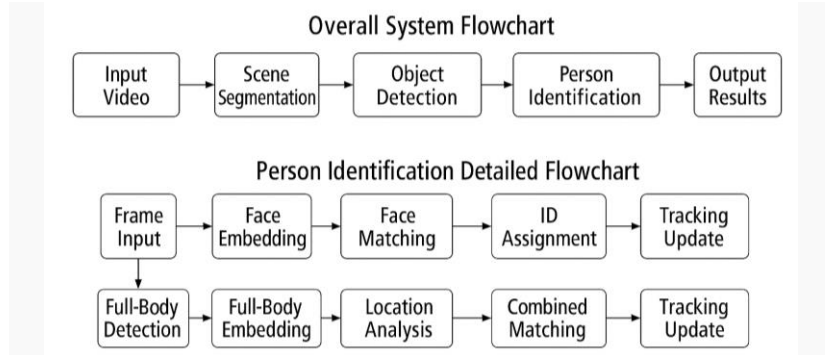
Visually impaired users require accurate, context-aware audio descriptions for dynamic video content where objects continuously enter, exit, or become occluded. Existing frameworks heavily depend on face-based identification, failing when faces are occluded or oriented away from the camera. Moreover, these approaches rarely exploit spatial context for comprehensive scene understanding [5], [7].

B. Spatial Reasoning for Object Re-Identification

Recent methodologies model detected objects as graph nodes with edge weights determined by relative distances. Graph Convolutional Networks (GCNs) and Relational Networks (RNs) learn pairwise relationships by combining appearance embeddings with normalized distance features [1], [2], [15]. Monocular depth estimation techniques provide scale-invariant depth maps that serve as geometric priors for enhanced robustness [3], [4].

III.METHODOLOGY

Framework Overview



Our hybrid framework fuses appearance embeddings with pairwise spatial relationships to maintain consistent identity associations across video frames. The pipeline begins with YOLO-generated bounding box proposals [12], followed by MTCNN-based face presence determination [13]. Subsequently, we extract either FaceNet embeddings [14] or PCB-P6 embeddings [11] depending on face availability. MiDaS depth estimation [4] enables relative distance calculation, culminating in relational reasoning-based fusion [1], [2], [15].

A. Object Detection and Embedding Extraction

- **Object Detection:** YOLO generates bounding boxes with associated class confidence scores and labels [12]
- **Face Detection Branching:** MTCNN determines face presence using $\text{IoU} > 0.5$ overlap threshold [13]

- **Feature Embedding:** Face-present instances utilize FaceNet512 (512-dimensional vectors), while face-absent instances employ PCB-P6 (2048-dimensional vectors) [14], [11]

B. Spatial Distance Modeling

We identify static reference objects appearing in consecutive frames through bounding box center proximity analysis. MiDaS depth estimation [4] facilitates normalized relative distance computation. For face-absent cases, these distance weights are concatenated with PCB-P6 embeddings, yielding "spatially-augmented embeddings."

C. Identity Mapping

Intra-frame matching employs cosine similarity between normalized embeddings. Inter-frame association maintains a global identity gallery, matching detections based on similarity thresholds or instantiating new identities when necessary.

IV. Conclusion

We present a novel hybrid object identification framework that integrates relative spatial cues with facial and body embeddings for enhanced visual narration. Our method demonstrates robust identification capabilities when facial information is unavailable, effectively distinguishing individuals in challenging scenarios involving similar clothing. The systematic integration of appearance embeddings with spatial context yields improved identity mapping accuracy across video sequences, constituting a significant advancement in accessible visual narration systems for visually impaired users.



Acknowledgments

This research was supported by the MSIT(Ministry of Science, ICT), Korea, under the National Program for Excellence in SW, supervised by the IITP(Institute of Information & communications Technology Planning & Evaluation) in 2025(2024-0-00025)

REFERENCES

1. T. N. Kipf and M. Welling, "Semi-Supervised Classification with Graph Convolutional Networks," in *Proc. International Conference on Learning Representations (ICLR)*, 2017.
2. A. Santoro, D. Raposo, D. G. T. Barrett, M. Malinowski, R. Pascanu, P. Battaglia, and T. Lillicrap, "A Simple Neural Network Module for Relational Reasoning," in *NeurIPS*, vol. 30, pp. 4967–4976, 2017.
3. D. Eigen, C. Puhrsch, and R. Fergus, "Depth Map Prediction from a Single Image using a Multi-Scale Deep Network," in *NeurIPS*, vol. 27, pp. 2366–2374, 2014.
4. R. Ranftl, K. Lasinger, D. Hafner, K. Schindler, V. Koltun, "Towards Robust Monocular Depth Estimation: Mixing Datasets for Zero-shot Cross-dataset Transfer," *IEEE Transactions on Pattern Analysis and Machine Intelligence (TPAMI)*, 2022.
5. Y. Li, X. Liu, and X. Gao, "Diverse Part Discovery: Occluded Person Re-identification with Part-Aware Transformer," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, 2021.
6. S. He, Z. Zhang, X. Wang, and J. Qi, "TransReID: Transformer-Based Object Re-Identification," in *Proc. IEEE Int. Conf. Comput. Vis. (ICCV)*, 2021.
7. M. Kiran, S. Muralidhara, and G. R. Murthy, "Holistic Guidance for Occluded Person Re-Identification," *BMVC* 2021.
8. Z. Wang, X. Zhou, and L. Wang, "Feature Erasing and Diffusion Network for Occluded Person Re-Identification," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, 2022.
9. Z. Miao, S. Wu, Y. Yang, Y. Lin, and J. Luo, "Pose-Guided Feature Alignment for Occluded Person Re-Identification," *ICCV* 2019.
10. K. Zhou, Y. Yang, A. Cavallaro, and T. Xiang, "Omni-Scale Feature Learning for Person Re-Identification," *CVPR* 2019.
11. Y. Sun, L. Zheng, Y. Yang, Q. Tian, and S. Wang, "Beyond Part Models: Person Retrieval with Refined Part Pooling (PCB)," in *Proc. European Conf. Comput. Vis. (ECCV)*, 2018.
12. J. Redmon, S. Divvala, R. Girshick, and A. Farhadi, "You Only Look Once: Unified, Real-Time Object Detection," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, 2016, pp. 779–788.
13. K. Zhang, Z. Zhang, Z. Li, and Y. Qiao, "Joint Face Detection and Alignment using Multi-task Cascaded Convolutional Networks," *IEEE Signal Processing Letters*, vol. 23, no. 10, pp. 1499–1503, Oct. 2016.
14. F. Schroff, D. Kalenichenko, and J. Philbin, "FaceNet: A Unified Embedding for Face Recognition and Clustering," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, 2015, pp. 815–823.
15. D. T. Chang, "Distance-Geometric Graph Convolutional Network (DG-GCN) for Three-Dimensional (3D) Graphs," *arXiv preprint arXiv:2007.03513*, Jul. 6, 2020.

A Study on Passenger Ferry Berthing Trajectory Tracking System Using WAVE Communication Technology

Wonsik Kang^{1*} and Kwang-Il Kim²

¹Department of Marine Industry and Maritime Police, Jeju National University, 63294, Republic of Korea

² Department of Marine Industry and Maritime Police, Jeju National University, 63294, Republic of Korea

[Email : wskang@jejunu.ac.kr]

Abstract

This study proposes a berthing trajectory tracking system for passenger ferries using WAVE (Wireless Access in Vehicular Environments) communication technology. In narrow and congested ports such as Jeju Port, safe and efficient berthing operations are crucial due to limited maneuvering space and external forces such as wind, wave, and current. Traditional AIS-based tracking systems have limitations in update frequency and positional accuracy, particularly during low-speed berthing maneuvers. To overcome these limitations, the proposed system utilizes WAVE communication to collect high-frequency positional and navigational data from ferries approaching the pier. By analyzing the collected data, the system can identify standard berthing routes and detect deviations in real time, thereby enabling precise trajectory monitoring and improved operational safety. This paper presents the structure of the WAVE-based tracking system for passenger ferries and outlines the data acquisition protocol. A case study of Jeju Port is also introduced to propose the development direction of the system. The results suggest that the system can serve as a foundational technology for the future development of route deviation alarm systems and for enhancing the safety of berthing operations for passenger ferries.

Index Terms: Wireless Access in Vehicular Environments (WAVE), Automatic Identification System (AIS), passenger ferry, berthing, unberthing, communication technology

I. INTRODUCTION

In recent years, the number of coastal passenger ferry operations has steadily increased, especially at Jeju Port, which accommodates over ten million passengers annually. As ferry traffic grows, ensuring safe berthing in narrow and congested port environments has become a critical issue. Limited maneuvering space and external factors such as wind, wave, and current significantly increase the complexity of berthing operations and the risk of route deviation or collision.

Conventional tracking systems, such as AIS (Automatic Identification System), provide general navigational data but have limitations in update frequency and spatial accuracy, particularly during low-speed berthing maneuvers. To address this issue, this study

proposes a berthing trajectory tracking system for passenger ferries using WAVE (Wireless Access in Vehicular Environments) communication. This system collects and analyzes high-frequency position data in real time, enabling precise trajectory monitoring and improved operational safety.

II. MODEL AND METHODS

The proposed system utilizes WAVE communication technology to **transmit short-range, high-speed data** from ferries approaching the pier. Compared to AIS, WAVE provides more frequent updates and enhanced spatial resolution, making it suitable for detailed monitoring during the final approach to berth.

A. System Architect

The proposed trajectory tracking system is composed of three core components: an onboard transmitter module, a shore-side receiving unit, and a centralized data analysis server. The onboard transmitter, installed on the passenger ferry, continuously collects navigational data including GPS position, vessel ID, speed, and heading. This data is then transmitted using WAVE communication, which enables low-latency, high-frequency data exchange suitable for short-range maritime environments such as port approach areas. The shore-side receiver is located near the berth and is responsible for receiving the transmitted data packets in real time. Once received, the data is forwarded to the analysis server for processing. At the central data server, the system reconstructs the vessel's trajectory and compares it against predefined standard berthing routes. The server also performs real-time analysis to detect deviations or anomalies in the approach pattern. Through this architecture, the system ensures precise monitoring of the vessel's motion and provides a foundation for future development of real-time alerts or decision-support tools.

B. Data Collection

The system collects real-time data within approximately 1 kilo meters of the berth, focusing on the final approach phase. Each data packet includes a timestamp, GPS coordinates, speed over ground (SOG), course over ground (COG), and vessel identifier.

C. Analysis Method

The collected trajectories are compared against predefined standard berthing routes. The algorithm identifies **route deviation, delayed turning, or abnormal approach angles** based on spatial deviation, turning timing, and directional variance.

III. PASSENGER FERRY BERTHING TRAJECTORY TRACKING SYSTEM

A. Definition of Standard Routes

Historical AIS and GPS trajectory data are analyzed to define typical berthing routes for each pier and vessel type. These standard trajectories are segmented into coordinate sequences for comparison.

B. Route Deviation Detection

If the real-time trajectory deviates from the standard route beyond a predefined threshold, or if turning maneuvers are delayed, the system **generates a warning signal**. The detection algorithm incorporates distance deviation, time delay, and approach angle variance. The system can later be expanded into a **berthing deviation alert system**.

C. Case Study at Jeju Port

Using actual berthing trajectory data from Jeju Port, the system was simulated and evaluated. The results confirmed that the WAVE-based approach provides **superior temporal and spatial resolution** compared to AIS and can detect **route anomalies and irregular speeds** during the approach phase.

IV. DISCUSSION AND CONCLUSIONS

The WAVE-based tracking system proposed in this study enables high-precision monitoring of passenger ferry berthing operations by collecting real-time, high-frequency location data. Unlike AIS, it allows detailed observation of vessel movements at low speed, particularly during berthing in constrained environments like Jeju Port.

This system lays the foundation for future development of route deviation alert mechanisms and can be extended to support automated berthing assistance, evacuation simulation integration, and port traffic safety management. The proposed approach has the potential to become a core technology for enhancing safety in passenger ferry operations.

REFERENCES

- [1] J. Ma, Y. Zhou, and Z. Zhu, "Identification and analysis of ship waiting behavior outside the port based on AIS data," *Scientific Reports*, vol. 13, no. 1, p. 11267, 2023.
- [2] A. H. Ali, A. S. Al-Dulaimi, and A. A. Al-Hubaishi, "A Review on IEEE 802.11p for Intelligent Transportation Systems," *Journal of Sensor and Actuator Networks*, vol. 9, no. 2, p. 22, 2020.
- [3] B. Lin, M. Zheng, X. Chu, M. Zhang, W. Mao, and D. Wu, "A novel method for the evaluation of ship berthing risk using AIS data," *SSRN Electronic Journal*, 2023.

Distributed Microservice Management Framework in Multi-Edge Environments

Jinhyo Park, and Joosang Youn*

Department of Artificial Intelligence, Dong-Eui University, Busan 43740, Korea
[Email : jhpark3679@gmail.com, jsyoun@deu.ac.kr]

Abstract

Edge computing is emerging as a promising alternative to address issues such as data transmission delays, network congestion, and security concerns inherent in cloud-based IoT application services. In particular, the enhanced performance of edge devices has led to the need for microservice-based software architectures that emphasize scalability and reusability. To meet this demand, research is needed on discovering and reusing already deployed microservices in environments with multiple edge nodes to reconfigure application services. This paper proposes a “Distributed Microservice Management Framework in Multi-Edge Environments” that automatically discovers and dynamically composes distributed microservices across a multi-edge environment to construct new application services. The proposed framework is designed to facilitate the configuration of IoT application services by enabling real-time state sharing and interoperability among microservices in multi-edge environments.

Index Terms: Edge Computing, Microservice, Internet of Things

I. INTRODUCTION

The Internet of Things (IoT) connects distributed devices, enabling the development of application services across various domains. These IoT application services utilize cloud and edge computing resources to process and analyze large-scale data [1]. However, currently commercialized IoT applications are individually designed, resulting in limitations regarding the reusability of functionality and code. Although research on microservice-based IoT application composition in edge environments is underway to address these issues, improvements are still needed in areas such as real-time state sharing and interoperability across multiple edge nodes [2–4].

In this paper, we propose a framework for discovering and dynamically composing distributed microservices in edge environments to construct application services. In particular, the proposed framework is designed to enable composition of IoT application services by leveraging real-time state sharing and interoperability among microservices across multiple edge environments.

II. Framework and Network Configuration

A. Microservice-based Application Configuration

* Corresponding author

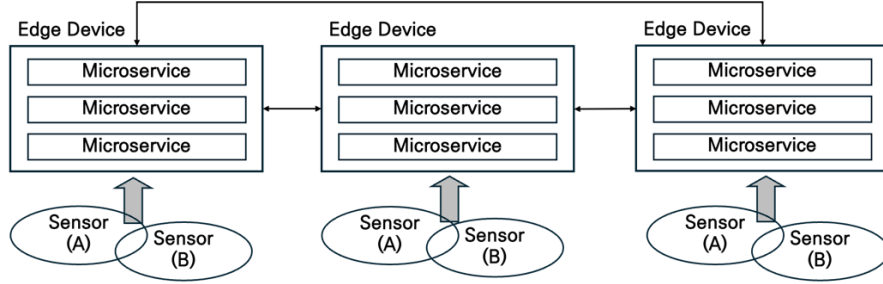


Fig. 1. Distributed microservices network model based on multi-Edge devices

Fig. 1 shows the network architecture for configuring a new application service by selecting and interconnecting microservices distributed across various Edge devices. IoT sensors collect and transmit data to their respective edge devices, and interoperation among microservices is enabled through network connections between these devices. This architecture is supported by the distributed microservice management framework and enables modular function composition, code reusability, flexible interoperability between applications, and service scalability.

B. System Framework

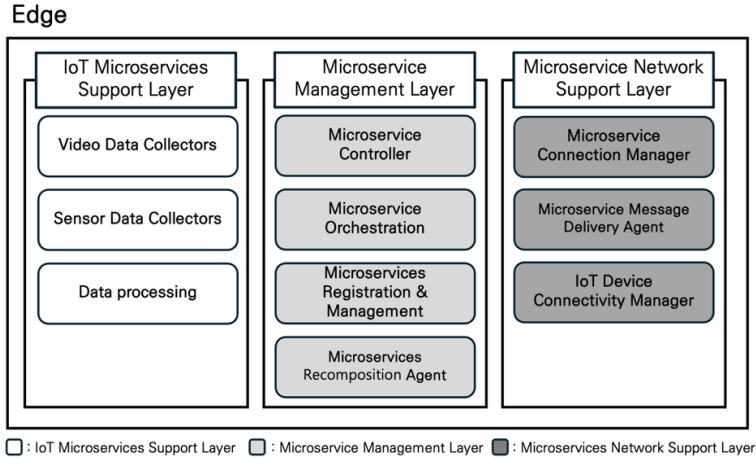


Fig. 2. Distributed microservice management framework for multi-edge environments

Section 2.B defines the distributed microservice framework and its functions for configuring IoT application services. Fig. 2 shows the framework for discovering and composing distributed microservices to create an application within the network environment described in Section 2.A. The framework consists of three layers: IoT Microservice Support Layer, Microservice Management Layer, and Microservice Network Support Layer. Table 1 shows the microservices of each layer and their respective roles.

Table 1. Components of the distributed microservice management framework in multi-edge environments

Microservice	Description
IoT Microservices Support Layer	
Data Collectors	Data collection interface
Data processing	Preprocess and normalize data based on service requirements
Microservice Management Layer	
Controller	Controlling microservice execution
Orchestration	Allocate and deploy resources needed at runtime
Registration & Management	Register, run, monitor health, and manage service quality of microservices
Recomposition Agent	Reconfigure IoT services and set up and tune data flows
Microservice Network Support Layer	
Connect Manager	Manage connection information for message passing between edge devices and microservices
Message Delivery Agent	Configures and routes messages between edge devices and microservices
IoT Device Connectivity Manager	Configures and manages network connectivity between IoT and edge devices

C. Application Service Configuration Network Flow

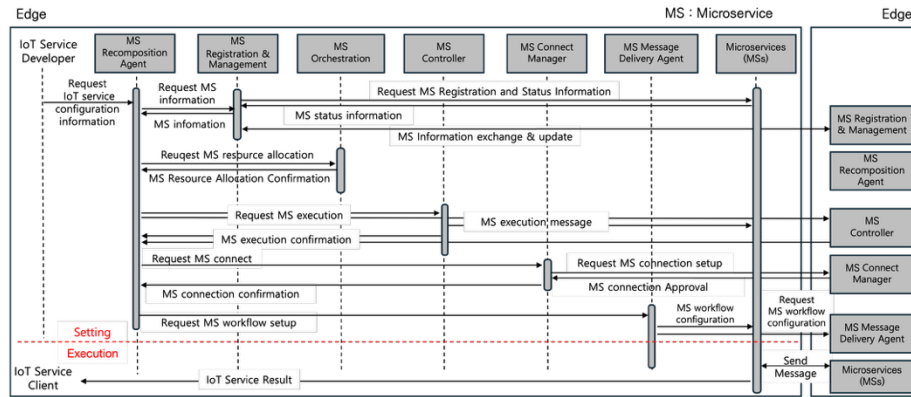


Fig. 3. Framework-Based IoT Service Recomposition Flow

Section 2.C describes the process of configuring a service using the system framework introduced in Section 2.B. When a Service Developer sends a service configuration request to the 'Microservice(MS) Recomposition Agent', the agent requests and receives microservice configuration information from the 'MS Registration & Management'. Subsequently, the 'MS Recomposition Agent' requests resource allocation from the 'MS Orchestration' module, which then determines whether to allocate resources by reviewing currently running or shareable resources. Once resource allocation is complete, the Recomposition Agent requests microservice execution via the 'MS Controller'. The controller then sends an execution command to the corresponding 'Microservices'..

After the service execution, the connection process between microservices begins. The Recomposition Agent requests a connection from the 'MS Connect Manager', which proceeds with the connection setup and approval process with the 'MS Connect Manager' of a neighboring edge. The Recomposition Agent then requests workflow configuration from the 'Message Delivery Agent', and a setup message is delivered to the relevant microservices. When the workflow configuration is complete, data transmission between microservices starts, and the result is returned to the IoT service client.

3. CONCLUSIONS

In this paper, we proposed a distributed microservice management framework for multi-edge environments. The proposed framework was designed to effectively configure IoT application services by leveraging real-time state sharing in a multi-edge environment. In our future work, we plan to define the framework in greater detail and validate its effectiveness through performance comparisons with existing methods.

ACKNOWLEDGMENTS

This research was supported by the Institute of Information & communications Technology Planning & Evaluation(IITP) grant funded by Korea government(MIST) (NO. RS-2024-00346798) and BB21plus funded by Busan Metropolitan City and Busan Techno Park.

REFERENCES

- [1] K. Alanezi and S. Mishra, "Utilizing Microservices Architecture for Enhanced Service Sharing in IoT Edge Environments," in *IEEE Access*, vol. 10, pp. 90034-90044, 2022.
- [2] A. El Akhdar, C. Baidada, A. Kartit, M. Hanine, C. O. García, R. G. Lara, and I. Ashraf, "Exploring the Potential of Microservices in Internet of Things: A Systematic Review of Security and Prospects," *Sensors*, vol. 24, no. 20, Art. no. 6771, 2024. doi: 10.3390/s24206771.
- [3] J. Lu, W. Li, J. Guo, X. Ding, Z. Tang, T. Wang, and W. Jia, "Hybrid Learning for Cold-Start-Aware Microservice Scheduling in Dynamic Edge Environments," *arXiv preprint*, arXiv:2505.22424, 2025. doi: 10.48550/arXiv.2505.22424.
- [4] Y. Chen, C. Wu, F. Zhang, C. Lu, Y. Huang, and H. Lu, "Topology-aware Microservice Architecture in Edge Networks: Deployment Optimization and Implementation," *arXiv preprint*, arXiv:2501.04956, 2025.

Application of Human Activity Recognition Techniques Using Accelerometer and Fast Fourier Transform

Jong-Hoon Youn¹, Do-Eun Park², and Teuk-Seob Song^{2*}

¹Department of Computer Science at the University of Nebraska, Omaha, NE 68182, USA

²Department of Computer Engineering, Mokwon University, Daejeon, 35349, Republic of Korea

[Email : jyoun@unomaha.edu, fffj01@naver.com, teukseob@mokwon.ac.kr]

Abstract

This study proposes a method for effectively recognizing human activities using a single accelerometer embedded in a wearable device. While previous studies have primarily relied on statistical features in the time domain, this study aims to distinguish subtle differences between similar activities through frequency-domain analysis using the Fast Fourier Transform (FFT). The frequency-based features extracted by FFT are applied to a lightweight deep learning model.

Index Terms: Human Activity Recognition, Deep Learning, Fast Fourier Transform, Acceleration

I. INTRODUCTION

Advancements in wearable sensor technology have led to the active use of sensor based data in fields such as healthcare, sports, and industrial safety. In particular, research on automatically recognizing human activities using accelerometers embedded in wearable sensors has been actively conducted. This approach is gaining attention for its ability to monitor and analyze various real-life situations in real time. [1, 2, 3]. Accurate recognition of human behavior allows for the detection of falls and unusual movements. As a result, it is possible to develop systems that can manage the daily lives of elderly individuals or patients with chronic diseases in real time [4, 5].

This study aims to effectively recognize human activities by analyzing acceleration data using Fast Fourier Transform (FFT). Previous studies have mainly classified behaviors using statistical features such as 3-axis acceleration data and the mean and standard deviation of signal vector magnitude (SVM) values. In contrast, using FFT has the advantage of extracting not only time-domain data but also information from the frequency domain. Features such as frequency magnitude and values can be additionally extracted, contributing to a more precise distinction between similar activities.

II. FREQUENCY-DOMAIN ANALYSIS USING FFT

* Corresponding author

Wearable sensors include various components such as accelerometers and gyroscopes. These sensors are used to precisely measure user movements and changes in posture. Among them, the accelerometer measures movement along three axes: x, y, and z. It can detect linear acceleration, gravitational direction, and impact. The output data appears as time-series data representing acceleration over time, which allows for real-time tracking of the user's physical movements.

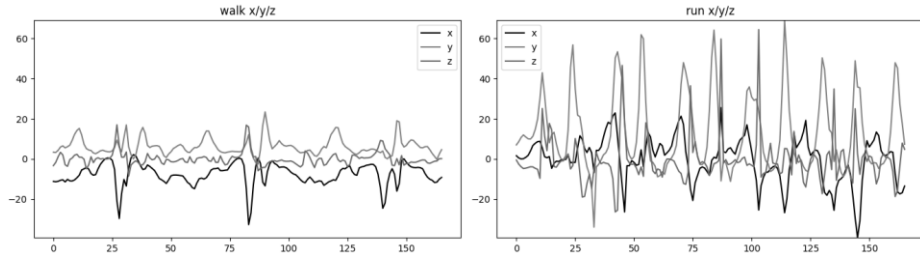


Fig. 1. An example of acceleration signals along the x, y, and z axes for walking and running.

For human activity recognition, it is common to use the Signal Vector Magnitude (SVM), which is calculated as the square root of the sum of the squares of the acceleration signals from the three axes. Figure 2 is the SVM graph of walking and running from Figure 1.

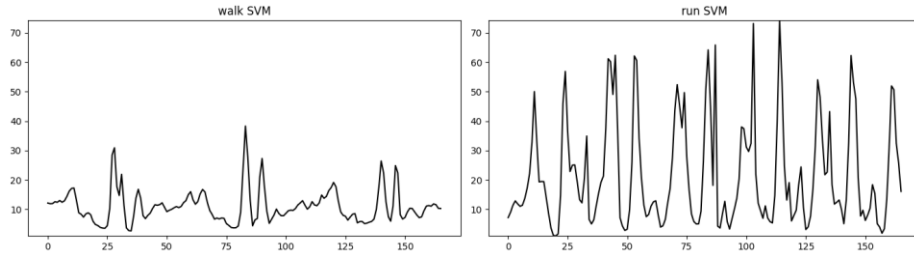


Fig. 2. An example of acceleration signals along the x, y, and z axes for walking and running.

Acceleration patterns vary depending on specific human activities [6]. Movements such as walking, running, and stair climbing exhibit periodic characteristics [7]. The accelerometer serves as an essential component in human activity recognition systems and provides key input features for deep learning-based behavior classification models.

The Fourier Transform is a mathematical technique that converts signals or data from the time domain to the frequency components within a signal. FFT can be used to analyze the periodic patterns of acceleration data related to human activities [8]. It helps more accurately distinguish the characteristics of specific behaviors. For example, “walking” and “running” may appear similar in the time domain, but their differences become clearer when analyzed in the frequency domain using FFT. In this way, FFT reveals periodic

patterns that are not visible in the time domain, quantifies rhythm, and assists the model in better understanding the data.

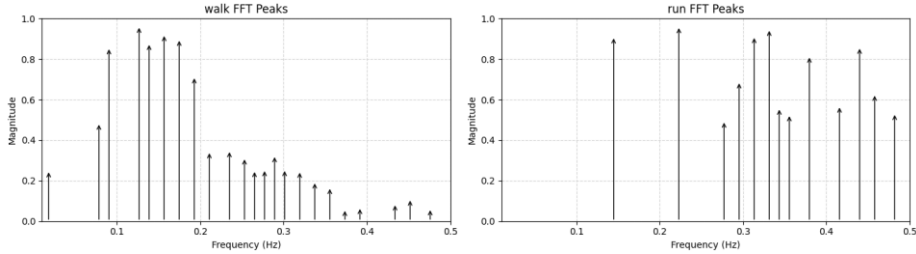


Fig. 3. The result of applying the FFT to the SVM signal, illustrating the frequency components of the acceleration data.

This frequency-domain-based analysis is particularly effective in distinguishing similar time-series patterns. It also enhances the extraction of meaningful features from the data. Representing data in the frequency domain can help deep learning models recognize and generalize important patterns. Based on this approach, this study applies FFT to preprocess human activity data and designs a lightweight deep learning model. The goal of the study is to achieve both high recognition accuracy and real-time processing capability.

III. CONCLUSION

This study presented a new method for accurately recognizing human activities using a single accelerometer. By converting 3-axis acceleration data into the frequency domain, subtle differences between similar activities could be clearly distinguished. This approach demonstrates high applicability in real-time monitoring without the need for complex sensors or high-performance equipment. In particular, it can be utilized in various healthcare fields, such as behavior detection systems for elderly individuals or patients with chronic conditions. Future research will aim to enhance the practicality and generalizability of the model by expanding the range of activity types and conducting experiments in real-world environments.

REFERENCES

- [1] W. Zhai, "Human activity recognition based on acceleration sensor and neural network," *International Conference on Orange Technology*, vol. 8, no. 1, pp. 1–5, 2020.
- [2] Y. Ning, S. Hu, X. Nie, S. Liang, H. Li, and G. Zhao, "Real-time action recognition and fall detection based on smartphone," *IEEE Engineering in Medicine and Biology Society*, vol. 40, no. 1, pp. 4418–4422, 2018.
- [3] P. Jantawong, N. Hnoohom, A. Jitapattanukul, and S. Mekruksavanich, "Time series classification using deep learning for HAR based on smart wearable sensors," *International Computer Science and Engineering Conference*, vol. 26, no. 1, pp. 1–6, 2022.

- [4] S. Ghatak, R. Mitra, W. Ahmed, and K. Chanda, "A simple fall detection scheme for early detection of falls in elderly people," *Evolutionary Algorithms and Soft Computing Techniques*, vol. 1, no. 1, pp. 1–6, 2023.
- [5] L. Shabadi, M. C., L. M., G. P. S., R. K. Deekshith, and L. Narayan, "Real-time elderly fall detection as ambient intelligence in old age homes," *Innovative Technologies in Computer Engineering and Electronics*, vol. 1, no. 1, pp. 1–5, 2025.
- [6] H. F. Nweke, Y. W. Teh, M. A. Al-Garadi, and U. R. Alo, "Deep learning algorithms for human activity recognition using mobile and wearable sensor networks: State of the art and research challenges," *Expert Systems with Applications*, vol. 105, pp. 233–261, 2018.
- [7] S. Ghatak, R. Mitra, W. Ahmed, and K. Chanda, "A simple fall detection scheme for early detection of falls in elderly people," *IEEE Access*, vol. 11, pp. 12345–12352, 2023.
- [8] B. Jakoby and M. J. Vellekoop, "FFT-based analysis of periodic structures in microacoustic devices," *IEEE Transactions on Ultrasonics, Ferroelectrics, and Frequency Control*, vol. 45, no. 5, pp. 1251–1258, 1998.

Acoustic Target Strength of the Giant Jellyfish *Nemopilema nomurai* Using Ex Situ, In Situ, and Model-Based Methods

Euna Yoon and Kwang-Il Kim

College of Ocean Sciences, Jeju National University, Jeju 63243,
Korea

[Email : kki@jejunu.ac.kr]

Abstract

The target strength (TS) of *Nemopilema nomurai* has been estimated relationship bell diameter using comparison method; ex situ, in situ, and model. Ex situ TS measurement, frequency 38 kHz split-beam echosounder synchronized with one camera were used to obtain TS data and behavior of *N. nomurai*. The in situ TS was measured at two frequency 38 and 120 kHz split beam and was sampled Framed Midwater Trawl (FMT) in East China Sea (ESC). The backscattering model to develop TS-to-BD conversions for *N. nomurai* use distorted wave-Born approximation (DWBA) model. Results from TS-to-BD regressions of *N. nomurai* that TS is higher to BD increasing. The regressions of TS_{avg}-to-BD were $TS_{38kHz}=10\log_{10}(\pi\alpha^2)-87.79$ by ex situ and $TS_{38kHz}=10\log_{10}(\pi\alpha^2)-82.72$ and $TS_{120kHz}=10\log_{10}(\pi\alpha^2)-80.18$ by DWBA model. The interceptions of fixed slope difference about 10 dB at 38 kHz and 9 dB at 120 kHz by measurement methods (max, avg; ex situ, model). In situ TS_{avg} is similarly with ex situ TS_{max} and is higher than ex situ TS_{avg} at 38 kHz. These TS-to-BD regressions can be applied to improve acoustic estimates of *N. nomurai* density and biomass.

Index Terms: acoustic, bell diameter, DWBA model, split-beam, echosounder

I. INTRODUCTION

Hydroacoustic techniques are widely used to estimate also biological characteristic such as behavior, abundance, and spatial-temporal distribution of target species in wide waters within quickly times [1, 2]. Acoustic survey was conducted mainly about fishes that it was diversified targets had scatter sound weakly and small size such as krill, copepoda, salp, jellyfish, and whatnot in recent [3, 4].

Jellyfish among target species for measurement has been difficultly expected to detect acoustic not a great deal of difference of impedance between seawater and jellyfish because of composition to water over 95% in body. However, biomass and distribution of jellyfish estimated by [3-5] using acoustic.

To use acoustic techniques, we need to know the target strength (TS) of target species. TS is a measure of the portion of acoustic energy incident on an object that is scattered back towards the sound source and convert jellyfish backscatter into biomass and abundance relevant quantities [4, 6].

Jellyfish is difficult species to measure TS and sampling because if jellyfish has exposed air during experiment and sampling, TS would have reacted errors by highly scattering [7]. Also, jellyfish TS depend on acoustic wave (frequency or wave length), and properties of the species (sound speed contrast and density contrast) physically, and size, shape, swimming angle, pulsation, and symbiont biologically [8, 9].

Here, this study estimated the TS from bell diameter (BD) of *N. nomurai* by the simultaneous collection of acoustic data and video footage of movement within ex situ experiment, comparison single target data and catch data within in situ experiment, and to verify ex situ method simulate according to change of size within model. Because of data based on used to estimate biomass and abundance of target, we were proposed to find TS-to-BD regressions to apply in the field comparison with results above measurement.

II. MATERIELS AND METHODS

1. Ex situ TS measurement

Ex situ experiments were conducted in a 5×5×5 m seawater tank using a 38 kHz split-beam echosounder (Simrad EK500) synchronized with an underwater camera (Fig. 1). Five live *N. nomurai* specimens (BD: 25.0–36.5 cm) were tethered and recorded under controlled conditions. TS values were analyzed using Echoview software with ping filtering for consistent incidence angles ($\pm 10^\circ$). TS distributions were evaluated for maximum ($TS_{\max}$), average (TS_{avg}), and mode values per individual.

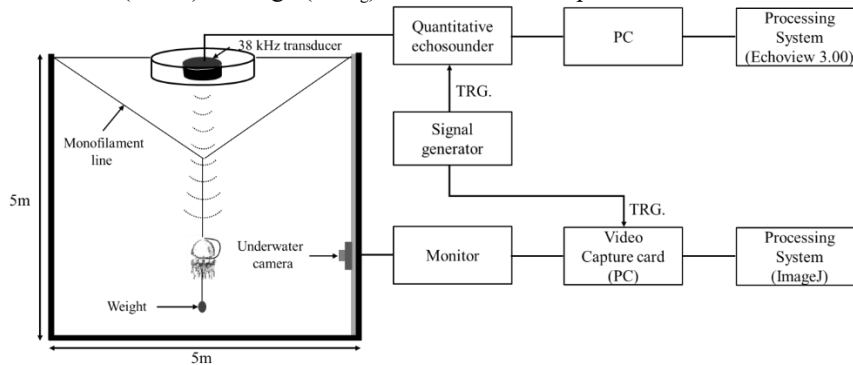


Figure 1. Block diagram of system composition and experiment method to measure target strength of *N. nomurai*.

2. In situ TS measurement

Field measurements were conducted in the East China Sea aboard the R/V Dong Baek. A split-beam EK500 echosounder operated at 38 and 120 kHz. Targets were verified using Framed Midwater Trawls (FMT), which confirmed that jellyfish dominated signals at 12 m depth. TS from matched echoes at both frequencies was analyzed.

3. Acoustic model

The distorted wave-Born approximation (DWBA) model was applied to simulate backscatter from jellyfish modeled as deformed cylinders. Physical parameters (sound speed contrast, $h = 1.0130$; density contrast, $g = 1.0010$) were taken from prior studies. Simulations covered a range of bell diameters and orientation angles (0° and $\pm 10^\circ$).

III. RESULTS

1. Ex situ TS measurement

The target strength (TS) of *Nemopilema nomurai* was evaluated through ex situ tank measurements, in situ field data, and DWBA modeling, revealing consistent relationships between TS and bell diameter (BD). In the ex situ experiment using five live specimens (BD 25.0–36.5 cm) measured at 38 kHz, TS values showed a bimodal distribution, with maximum TS modes ranging from -70 to -53 dB depending on the individual. The variation coefficient (CV) was high (58.2–148.1%), and TS consistently increased with larger BD. Across individuals, the TS range spanned from approximately -75 to -51 dB.

2. In situ TS measurement

In situ measurements at 38 and 120 kHz were conducted in the northern East China Sea, where jellyfish were sampled at 12 m depth using a Framed Midwater Trawl. At two stations, 25–30 jellyfish were captured, with mean BDs of 19.4 cm and 21.1 cm, respectively. Average TS values from matched single targets were -65.4 dB and -65.7 dB at 38 kHz, and -64.2 dB and -62.4 dB at 120 kHz, showing slightly higher TS at higher frequency. The in situ TS distribution also appeared bimodal, reflecting smaller individuals with TS around -69 to -65 dB and larger individuals with TS from -62 to -58 dB.

3. TS-to-BD regressions

Log-linear regressions were developed between TS and BD, assuming a fixed slope of 10 (corresponding to cross-sectional area proportionality, i.e., $TS \propto \log(\pi a^2)$). At 38 kHz, the ex situ regression was $TS = 10 \log_{10}(\pi a^2) - 87.79$ (avg), while DWBA model regressions were -82.72 (avg) and -77.03 (max). At 120 kHz, DWBA regressions were -80.18 (avg) and -71.27 (max). These results showed frequency-dependent TS elevation and a ~ 10 dB range in intercepts between methods.

IV. DISCUSSION AND CONCLUSIONS

The study confirms a positive correlation between BD and TS in *N. nomurai*. In situ TS values matched ex situ $TS_{\max}$ due to possible contributions from symbionts (e.g., medusa shrimp, fish), natural swimming posture (side-view), and gastric content.

DWBA simulations showed higher TS values at 120 kHz than 38 kHz, consistent with frequency response differences. Variability in TS due to swimming angle and pulsation suggests future modeling should incorporate dynamic body forms. Comparisons with previous TS studies (Hirose 2007; Kang et al. 2013) indicate that bubble presence, handling methods, and symbiont load can significantly alter TS results.

This study presents the most comprehensive TS-to-BD regressions for *N. nomurai* to date, based on ex situ, in situ, and model-based approaches. These regression equations can be effectively applied to improve acoustic-based biomass estimation of *N. nomurai* in field surveys. However, further research is required to refine models by incorporating variable morphology, frequency-specific effects, and environmental/biological factors such as symbionts and prey ingestion.

REFERENCES

- [1] J.K. Horne and J.M. Jech, "Multi-frequency estimates of fish abundance: constraints of rather high frequencies," ICES J Mar Sci., 56, 184-199, 1999.
- [2] J. Simmonds and D. MacLennan, "Fisheries acoustic theory and practice," Blackwell Science, Oxford, pp. 176-202, 2005.
- [3] G. Alvarez Colombo, A. Benović, A. Malej, D. Lučić, T. Makovec, V. Onofri, M. Acha, A. Madirolas and H. Mianzan, "Acoustic survey of a jellyfish-dominated ecosystem (Mljet Island, Croatia)," Hydrobiologia, 616, 99-111, 2009.
- [4] T. R. Graham, J. T. Harvey, S. R. Benson, J. S. Renfree and D. A. Demer, "The acoustic identification and enumeration of scyphozoan jellyfish, prey for leatherback sea turtles (*Dermochelys coriacea*), off central California," ICES J Mar Sci, 67, 1739-1748, 2010.
- [5] C. P. Lynam, A. S. Brierley and B. E. Axelsen, "Pinging down the food web: Multifrequency acoustic discrimination of jellyfish and fish," ICES Report CM2004, 1-21, 2004.
- [6] A. D. Robertis and K. Taylor, "In situ target strength measurements of the scyphomedusa *Chrysaora melanaster*," Fish Res, 153, 18-23, 2014.
- [7] M. Hirose, T. Mukai, D.J. Hwang and K. Iida, "Target strength measurements on tethered live jellyfish *Nemopilema nomurai* (in Japanese with English abstract)," Nippon Suisan Gakkaishi, 71, 571-577, 2005.
- [8] E. Mutlu, "Target strength of the common jellyfish (*Aurelia aurita*): a preliminary experimental study with a dual-beam acoustic system," ICES J Mar Sci 53, 309-311, 1996.
- [9] B. C. Monger, S. Chinniah-Chandy, E. Meir, S. Billings, C. H. Greene and P. H. Wiebe, "Sound scattering by the gelatinous zooplankters *Aequorea victoria* and *Pleurobrachia bachei*," Deep-Sea Res II 45, 1255-1271, 1998.

A Pipelined NTT/INTT Processor for Post-Quantum Cryptography ML-KEM

Su-Hyeon Lee^{1*} and Kyung-Wook Shin²

^{1,2}Dept. of Semiconductor System Engineering, Kumoh National Institute of Technology,
39177, Korea

[Email : shlee@kumoh.ac.kr / kwshin@kumoh.ac.kr]

Abstract

This paper presents a high-performance hardware implementation of number theoretic transform (NTT)/inverse NTT (INTT) that is used to speed up the polynomial multiplication required by the module lattice key-encapsulation mechanism (ML-KEM) in post-quantum cryptography (PQC). The processor adopts a radix-4 multi-path delay commutator (MDC) architecture to compute 256-point NTT/INTT in a pipelined fashion. In order to efficiently compute both NTT and INTT using a single MDC structure, the radix-4 butterfly unit and data reordering unit were optimized so as to maximize hardware resource utilization. The processor requires 142 clock cycles to compute a 256-point NTT/INTT, which is equivalent to a latency of 0.31 μ s when operating at its maximum clock frequency of 450 MHz.

Index Terms: NTT, Post-Quantum Cryptography (PQC), ML-KEM, CRYSTALS-Kyber, multi-path delay commutator (MDC)

I. Introduction

Advances in quantum computing technology pose a serious security threat to existing public-key cryptosystems. In particular, the currently popular RSA and elliptic curve cryptography (ECC) have been proven to be vulnerable to Shor's Algorithm when executed on a quantum computer. In response to these threats, NIST led a project to standardize post-quantum cryptography (PQC). As a result, Kyber was adopted as the final standard in key-encapsulation mechanism (KEM) category, formally designated as module lattice-based KEM (ML-KEM) [1] in the official standardization document. The ML-KEM is a type of lattice-based cryptographic system that involves repeated operations between polynomial vectors and matrices. Polynomial multiplication is the most computationally intensive operation, which becomes a bottleneck in the hardware implementation of ML-KEM.

The number theoretic transform (NTT) and inverse NTT (INTT) employed in the ML-KEM are usually used to accelerate polynomial multiplications. It is widely acknowledged that the NTT/INTT can reduce the time complexity of polynomial multiplication from $O(n^2)$ to $O(n \log n)$, and the NIST final standardization document explicitly requires the use of the NTT and INTT for polynomial multiplications. In NTT and INTT computations, how to index the coefficient pairs between each stage is a key factor that directly affects the performance. Most of the existing hardware implementations of the ML-KEM employ a combination of butterfly arithmetic circuits and memory units. While memory-based designs are area-efficient, they often suffer from performance bottlenecks due to repeated memory access and limited parallelism.

This paper presents a pipelined NTT/INTT processor based on a radix-4 multi-path delay commutator (MDC) architecture, which enables continuous data flow without memory stalls. Furthermore, the butterfly unit and the data reordering unit were designed to support both NTT and INTT computations, thereby optimizing hardware resource utilization.

II. NTT/INTT Processor Design for ML-KEM

A. Number Theoretic Transform (NTT)

NTT is an integer-based variant of fast Fourier transform (FFT), defined over a finite field $\mathbb{Z}_q$. The definition of n -point NTT is as Eq. (1)

$$\hat{s}_i = \sum_{j=0}^{n-1} s_j \cdot \omega^{ij} \pmod{q} \quad (1)$$

where ω denotes a primitive n -th root of unity, which satisfies $\omega^n = 1 \pmod{q}$, and $s = (s_0, \dots, s_{n-1})$ represents the polynomial coefficients to be transformed. The ML-KEM performs arithmetic over polynomial quotient ring $\mathbb{Z}_q[X]/(X^n + 1)$, which follows the negative wrapped convolution (NWC) structure. However, for the parameter set $n = 256, q = 3329$ adopted in the ML_KEM, primitive 512-th root of unity in $\mathbb{Z}_{3329}$ does not exist, making the standard NWC-based NTT inapplicable. To solve this problem, the ML-KEM employs Truncated NTT [2]. In this method, the polynomial coefficients are divided into even and odd parts, and each part is processed with an NTT of size $n/2 = 128$. The corresponding operation is shown in Eq. (2)

$$\hat{s}_{2i} = \sum_{j=0}^{127} s_{2j} \cdot \zeta^{(2 \cdot br_7(i)+1) \cdot j} \pmod{q} \quad (2-a)$$

$$\hat{s}_{2i+1} = \sum_{j=0}^{127} s_{2j+1} \cdot \zeta^{(2 \cdot br_7(i)+1) \cdot j} \pmod{q} \quad (2-b)$$

where $\zeta = 17$ is primitive 256th root of unity in $\mathbb{Z}_{3329}$, and $br_7(i)$ denotes the integer represented by bit-reversing the unsigned 7-bit $i \in \{0, 1, \dots, 127\}$. NTT-based polynomial multiplication is computed as Eq. (3)

$$c(X) = INTT(NTT(a(X)) \circ NTT(b(X))) \quad (3)$$

where $a, b, c \in R_q$ and $\circ$ denote coefficient-wise multiplication of the polynomials within the NTT domain. Eq. (3) shows the procedure for performing NTT on each of the input polynomials, then computing point-wise multiplications between the transformed coefficients, and applying INTT on the results to obtain the final polynomial.

B. Radix-4 NTT/INTT processor architecture

The architecture of the pipelined NTT/INTT processor is depicted in Fig. 1, which is based on a radix-4 multi-path delay commutator (MDC) structure. It consists of three stages of radix-4 butterfly units (BU4) and one stage of radix-2 butterfly unit (BU2), with a data reordering unit (RU) between the butterfly units for data reordering. The RU, which comprised of flip-flops and multiplexers, aligns coefficient pairs to be processed in the subsequent stage of butterfly operation. As illustrated in Fig. 1, the NTT operations are processed from left to right, while the INTT operations are processed from right to left.

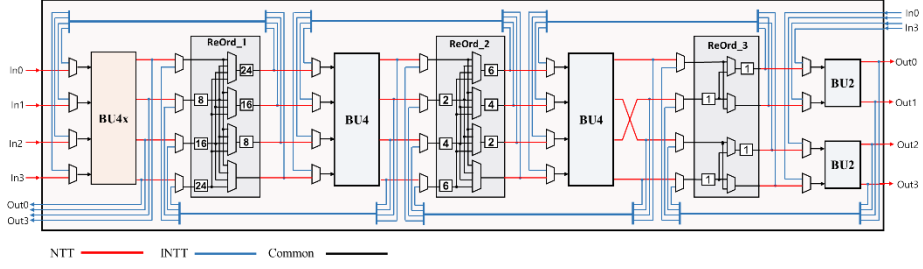


Fig. 1. Pipelined 256-point NTT/INTT processor

As illustrated in Fig. 1, the NTT/INTT processor comprises two types of radix-4 butterfly units, BU4x and BU4. Fig. 2 depicts the internal structure of the BU4x, which consists of four multipliers for twiddle factor multiplications, five reduction blocks for modular reduction of the multiplication results, and four modular adders and subtractors. The K^2_RED block is responsible for computing modular reduction of $C \cdot k^2 \pmod{q}$, where $k = 13$, based on K^2_RED [3]. The BU4x block on the left in Fig. 1 includes a scaling circuit of sINTT that multiplies each coefficient of INTT results by scaling factor of $128^{-1} \pmod{3329}$. This process is implemented through a pre-multiplication of the twiddle factors by $128^{-1} \pmod{3329}$. The BU4 and BU4x blocks have the same internal structure, but the BU4 block does not include the sINTT circuit shown in Fig. 2.

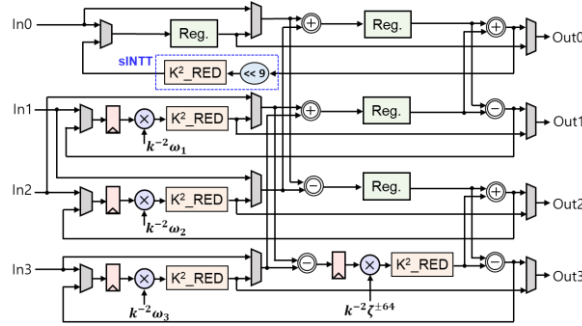


Fig. 2. Butterfly unit BU4x

III. Performance Evaluation

The 256-point NTT/INTT processor was verified by RTL simulation and FPGA implementation. The verifications were carried out by using test vectors that were randomly generated by Python software modeling of NTT/INTT. The FPGA verification was executed using the Zynq UltraScale+ MPSoC platform. It was implemented on the PL region of the ZYNQ Ultrascale+ MPSoC using 4,203 LUTs, 2,518 flip-flops, and 8 DSP blocks. The maximum clock frequency was estimated to be 450 MHz. A single calculation of 256-point NTT requires 142 clock cycles, and subsequent computations of NTTs are continuously processed in 64 clock cycles.

Table 1 shows the performance comparison of NTT/INTT processors for ML-KEM. The hardware complexity of the NTT/INTT processors was evaluated using the equivalent number of slices (ENS) metric, as described in [7]. When compared with the processors in

[4, 5], our processor exhibits approximately two times faster latency and 45% to 50% less ENS. Also, our processor achieves approximately three times faster latency at the expense of approximately 2.2 times higher ENS, as compared to [6].

Table 1. Comparison of NTT/INTT processors for ML-KEM

Design	FPGA device	Hardware resources	Frequency (MHz)	Latency (μ s)	ENS
		LUTs/FFs/BRAMs/DSPs			
This work	Zynq UltraScale+	4,203/2,518/0/8	450	0.31	2,161
[4]	Zynq UltraScale+	3,821/2,970/5/20	322	0.79	4,327
[5]	Artix-7	6,201/3,562/5.5/8	165	0.68	3,896
[6]	Artix-7	429/538/4/0	446	1.03	975

IV. Conclusions

A high-speed pipelined NTT/INTT processor for ML-KEM is presented, which is based on a radix-4 MDC architecture. Our processor efficiently supports both 256-point NTT and INTT computations in a pipelined fashion. The NTT/INTT processor introduced in this paper can be used as a core functional block for high-performance hardware implementation of ML-KEM for PQC.

Acknowledgments

This work was supported by Korea Institute for Advancement of Technology (KIAT) grant funded by the Korea Government (MOTIE) (P0017011, HRD Program).

REFERENCES

- [1] NIST STd. FIPS PUB 203, Module-Lattice-Based Key-Encapsulation Mechanism Standard, National Institute of Standards and Technology, 2024.
- [2] A. Satriawan, I. Syafalni, R. Mareta, I. Anshori, W. Shalannanda and A. Barra, “Conceptual review on number theoretic transform and comprehensive review on its implementations,” *IEEE Access*, vol. 11, pp. 70288-70316, 2023.
- [3] M. Bisheh-Niasar, R. Azarderakhsh and M. Mozaffari-Kermani, “High-speed NTT-based polynomial multiplication accelerator for post-quantum cryptography,” *IEEE 28th symposium on computer arithmetic (ARITH)*, pp. 94-101, 2021.
- [4] D. E. S. Kundi, J. M. B. Mera, P. Y. Strub and M. Hutter, “High-Performance NTT Hardware Accelerator to Support ML-KEM and ML-DSA,” in *Proceedings of the Workshop on Attacks and Solutions in Hardware Security*, pp. 100-105, 2024.
- [5] S. Mandal and D. B. Roy, “KiD: A hardware design framework targeting unified NTT multiplication for CRYSTALS-Kyber and CRYSTALS-Dilithium on FPGA,” *2024 37th International Conference on VLSI Design and 2024 23rd International Conference on Embedded Systems (VLSID)*, pp. 455-460, 2024.
- [6] B. Kieu-Do-Nguyen, N. T. Binh, C. Pham-Quoc, H. P. Nghi, N. -T. Tran, T. -T. Hoang and C. -K. Pham, “Compact and Low-Latency FPGA-Based Number Theoretic Transform Architecture for CRYSTALS Kyber Postquantum Cryptography Scheme,” *Information*, vol. 15, no.7, pp. 2078-2489, 2024.
- [7] T. -H. Nguyen, D. -T. Dam, P. -P. Duong, B. Kieu-Do-Nguyen, C. -K. Pham and T. -T. Hoang, “Efficient Hardware Implementation of the Lightweight CRYSTALS-Kyber,” *IEEE Transactions on Circuits and Systems I: Regular Papers*, vol. 72, no. 2, pp. 610-622, 2024.

A High-Performance Hash Core Supporting Four SHA-2 Algorithms

Song-Hee Lee^{1*}, Ju-Hyung Lee², and Kyung-Wook Shin³

^{1,2,3}Department of Semiconductor System Engineering,
Kumoh National Institute of Technology, 39177, Republic of Korea
[Email : ¹shsong.vlsi@kumoh.ac.kr, ²jhlee.vlsi@kumoh.ac.kr, ³kwshin@kumoh.ac.kr]

Abstract

This paper presents a high-performance hardware implementation of the SHA-2 hash function. Our hash core SHA2_512x supports four hash algorithms including SHA-384, SHA-512, SHA-512/224, SHA-512/256 specified in the NIST standard. The SHA2_512x hash core designed with a 64-bit data path processes two rounds of operations in a clock cycle, and message padding is implemented in hardware to avoid software preprocessing. The hash core was implemented using 6,828 LUTs and 2,232 flip-flops, and the hardware operation was validated using a Zynq UltraScale+ device. It requires 41 clock cycles to process a 1024-bit message block, resulting in a performance of approximately 4.39 mega hashes per second (Mhps) at a clock frequency of 180 MHz.

Index Terms: secure hash algorithm (SHA), cryptographic hash function, hash core, information security, digital signature algorithm (DSA).

I. INTRODUCTION

Cryptographic hash functions are commonly used as a primary component of various security protocols such as digital signature, integrity verification, authentication and key generation. Secure Hash Algorithm (SHA) is the cryptographic hash functions published by the National Institute of Standards and Technology (NIST). The SHA is classified into three families: SHA-1, SHA-2 and SHA-3 [1,2]. The SHA-2 family comprises six hash functions with digests that are 224, 256, 384 or 512 bits; SHA-224, SHA-256, SHA-384, SHA-512, SHA-512/224, and SHA-512/256. SHA-384 is a truncated version of SHA-512 computed with different initial value, and SHA-512/224 and SHA-512/256 are also truncated versions of SHA-512, but the initial values are generated using the method described in FIPS PUB 186-5 [3]. SHA-2 is a widely adopted hash function that is still essential in digital signature protocols such as ECDSA and Ed25519 [3], even after SHA-3 was added as a new standard.

In this paper, we present a high-performance hash core SHA2_512x supporting four SHA-2 hash algorithms including SHA-384, SHA-512, SHA-512/224, and SHA-512/256. The SHA2_512x core was designed with a 64-bit data path and processes two rounds of operations in a clock cycle achieving high throughput. After a brief introduction to the SHA-512 hash function in Section II, the hardware design of our hash processor will be described in Section III. Section IV will discuss the verification and performance evaluation of the processor, then conclusion will be followed in Section V.

II. SHA-512 BASED HASH FUNCTIONS

A. Preprocessing

In the SHA-2 hash function, variable-length message is divided into the blocks of fixed-size by preprocessing. Then, a compression function is applied to each block to obtain a fixed-length digest. In the preprocessing phase, the padding and parsing of the input message and the setting of the initial hash value are carried. The padding adjusts the message length to be a multiple of 1024 bits by concatenating '1' to the least significant bit of the message followed by concatenating additional k bits of '0' and l . The positive integer value k is determined by $l + 1 + k \equiv 896 \pmod{1024}$, where l is the 128-bit representation of the message length. The padded message is parsed into 1024-bit blocks to be processed by compression function with the most significant block first. The initial values to be used in the compression function are defined in the FIPS PUB 186-5 [3].

B. Compression Function

The compression function, applied to each message block, is processed in three steps: a message schedule, 80 rounds of operations and a final word addition. The message schedule involves two procedures: word initialization and expansion. All operations are performed on 64-bit words, and the carry bits exceeding 64 bits are truncated to 64 bits.

The message schedule is defined by Eq. (1), which expands a message block of 16 words, M_t , to 80 words, W_t , to be used in each round operation. The σ_0, σ_1 in Eq. (1) comprise the operations of a circular right shift, a right shift, and an XOR.

$$W_t = \begin{cases} M_t & (0 \leq t \leq 15) \\ \sigma_1(W_{t-2}) + W_{t-7} + \sigma_0(W_{t-15}) + W_{t-16} & (16 \leq t \leq 79) \end{cases} \quad (1)$$

Each round operation requires a 64-bit word W_t , a round constant K_t , and eight working variables ($a \sim h$) as operands. In the round operation for the first message block, each working variable is initialized with the initial hash values. For subsequent message blocks, the working variables are initialized with the output of the compression function of the previous block. The round operation and the working variable update are defined by Eq. (2), where Σ_0 and Σ_1 denote circular right shift and XOR operations, Ch is a choose function, and Maj denotes a majority function, which are defined in [3].

$$\begin{aligned} T_1 &= h + \Sigma_1(e) + Ch(e, f, g) + K_t + W_t \\ T_2 &= \Sigma_0(a) + Maj(a, b, c) \\ a &= T_1 + T_2, b = a, c = b, d = c, e = d + T_1, f = e, g = f, h = g \end{aligned} \quad (2)$$

After 80 rounds of operations are completed, the result of compression function is obtained by adding the working variables and the output of the compression function for the previous message block, word by word. The round operation result on the last block is the message digest.

III. HARDWARE DESIGN

Fig. 1 shows the internal structure of the SHA2_512x core, which consists of a *Message Schedule* block to generate words to be used in round operations, a *Round* block to perform the round operation, a *Digest Update* block to store the initial hash value or the result of the compression function, a *K-Const* block to store the round constant K and a *CTRL FSM* block to control the overall operation. The *Message Schedule* and *Round* blocks were designed with a 64-bit data path. The final word addition was implemented reusing adders in the *Message Schedule* and *Round* blocks to minimize hardware resource utilization.

The *Message Schedule* block is responsible for message padding and message scheduling. A 1024-bit message block is stored in the *Word_Reg*, $W_0 \sim W_{15}$, then the padding process described in Section 2.A is carried out. In the initial 16 rounds among 80 rounds, the words from the message block are directly used in the round operations. The word expansion defined in Eq. (1) is necessary for the rounds from 16 to 79. Fig. 2 shows the *Message Schedule* block, which generates two words to be used in round operations in a single clock cycle. It employs a combination of two stages of carry-save adders (CSAs) and a carry-propagate adder (CPA) to achieve high-speed addition of four operands.

Fig. 3 depicts the round circuit that consists of two identical blocks computing the odd and even rounds, respectively. It processes two rounds of operation within a single clock cycle. At the beginning of the round operations, the *Rnd_Reg* ($a \sim h$) is initialized with H_0 to H_7 , which are extracted from the *Digest_Reg* starting from the most significant 64 bits. The results of round operations ($t_a \sim t_h$) are updated to the *Rnd_Reg* for the next round of operations. The delay due to the dependency of even- and odd-round operations and the path delays requiring the addition of three or more operands is critical. The CSA tree structure is employed to minimize the critical path delay. The critical path delay was reduced by approximately 33% when compared to using CPA.

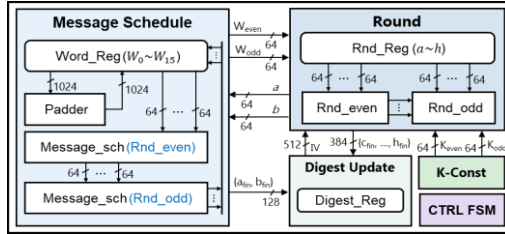


Fig. 1. Architecture of SHA2_512x core

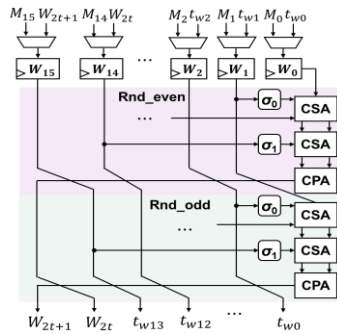


Fig. 2. Message schedule block

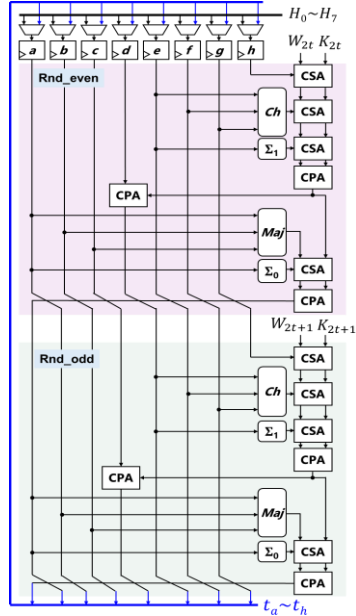


Fig. 3. Round block

IV. PERFORMANCE EVALUATION

The SHA2_512x core was verified using RTL simulation and FPGA implementation. The RTL simulations were carried out using 100,000 test vectors, randomly generated by Python software modeling for messages ranging from 0 to 13 KB. The SHA2_512x core, implemented on the PL region of Zynq UltraScale+ MPSoC device, used 6,828 LUTs and 2,232 flip-flops. It takes 41 clock cycles to process a block of message, and the maximum clock frequency was evaluated to be 180 MHz. Table 1 shows a comparison of SHA2 processors. For a comprehensive comparison, the hash rate and area efficiency metrics in [4] are used. The area efficiency is calculated by considering one slice as equivalent to four LUTs. When compared with the designs in [4-5], our hash core achieves approximately 1.7- and 2.5-times higher hash rate, as well as approximately 1.5- and 4-times better area efficiency.

Design	FPGA device	Freq. [MHz]	Area	#Cycle	Hash rate [Mhps]	Area efficiency [hps/LUTs]
This work	XCZU7EV	180	6,828 LUTs + 2,232 FFs	41	4.39	640
[4]	XCKU5P	106	5,828 LUTs + 4,343 FFs	41	2.59	440
[5]	XC7V855T	145	2,866 Slices	81	1.79	160

V. CONCLUSIONS

This paper presents a high-performance hash core that supports four SHA-2 hash algorithms. The SHA2_512x hash core with a 64-bit data path takes 41 clock cycles to compute the digest of a 1024-bit message block by processing two rounds per clock cycle. It achieves a hash rate of 4.39 Mhps and an area efficiency of 640 hps/LUTs. Our hash core can be used as a cryptographic primitive for embedded security applications and the hardware implementation of digital signature protocols, including Ed25519 and ECDSA.

ACKNOWLEDGMENTS

This work was supported by Korea Institute for Advancement of Technology (KIAT) grant funded by the Korea Government (MOTIE) (P0017011, HRD Program).

REFERENCES

- [1] National Institute of Standards and Technology (NIST), Secure Hash Standard FIPS PUB 180-4, 2015.
- [2] National Institute of Standards and Technology (NIST), SHA-3 Standard: Permutation-Based Hash and Extendable-Output Functions FIPS PUB 202, Aug. 2015.
- [3] National Institute of Standards and Technology (NIST), Digital Signature Standard (DSS) FIPS PUB 186-5, Feb. 2023.
- [4] R. Martino and A. Cilaro, "A flexible framework for exploring, evaluating, and comparing SHA-2 designs," *IEEE Access*, vol. 7, pp. 72443–72456, 2019.
- [5] H. Michail, G. Athanasiou, G. Theodoridis, and C. Goutis, "On the development of high-throughput and area-efficient multi-mode cryptographic hash designs in FPGAs," *Integration*, vol. 47, no. 4, pp. 387–407, 2014.

Session IB-B

IoT and Big Data

Session IT-D

IT Convergence Technology

Session MD-C

Multimedia and Digital Convergence

Session AI-D

AI and Intelligent Information System

Enhancing Construction Site Safety through Smart Wearable IoT for Real-Time Fatigue Prediction

Bubryur Kim^{1,2,3}, Euijung An^{1*}

¹Department of Robot and Smart System Engineering, Kyungpook National University, 80, Daehak-ro, Buk-gu, Daegu, 41566, South Korea

²School of Space Engineering Sciences, Kyungpook National University, 80, Daehak-ro, Buk-gu, Daegu, 41566, South Korea

³Department of Safety Convergence, Kyungpook National University, 80, Daehak-ro, Buk-gu, Daegu, 41566, South Korea

[Email : brkim@knu.ac.kr, ejungan@knu.ac.kr]

Abstract

Despite its importance, the construction industry has shown slower progress in adopting smart safety monitoring systems. This research proposes a real-time framework for fatigue assessment using IoT-integrated wearable devices (PPG and EEG) and sophisticated modeling techniques. The system accurately estimates physical and mental fatigue, achieving low MSEs (0.0008 and 0.0033), and supports early risk detection to improve on-site worker safety.

Index Terms: Complex Model-Based Analysis, Construction Safety, Fatigue Analysis, Internet of Things (IoT), Wearable Sensors

I. INTRODUCTION

The construction sector plays a critical role in economic growth, contributing significantly to global GDP [1]. However, it remains one of the most hazardous industries, with high workplace accident rates leading to severe human and financial consequences, such as decreased productivity, increased costs, and project delays [2, 3]. Additionally, the physically demanding nature of construction work exacerbates health risks, making workers more vulnerable to chronic illnesses, cardiovascular conditions, and musculoskeletal disorders, especially when combined with long shifts and unhealthy lifestyle habits [4]. Studies indicate that mental fatigue is a major cause of hazard detection failures, contributing to nearly half of all fatal construction accidents [3].

Although fatigue poses significant risks, real-time monitoring solutions for fatigue are still insufficiently developed in the construction field. Unlike fields such as healthcare and transportation, where smart safety technologies are widely adopted, construction lacks a standardized system for fatigue assessment [3]. This gap underscores the urgent need for advanced fatigue detection solutions to enhance worker safety and mitigate accident risks.

To address the above issues, this study designs an IoT-enabled, AI-driven fatigue monitoring system that utilizes wearable devices to analyze physiological indicators such as heart rate variability (PPG) and brainwave activity (EEG). By leveraging DeepAR for long-term trend analysis, Temporal Fusion Transformers (TFT) for complex variable

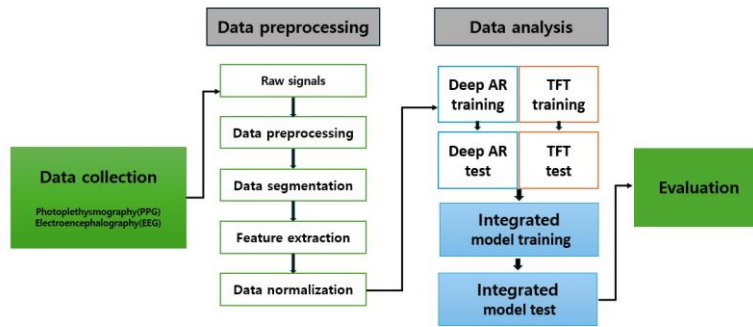
relationships, and XGBoost for enhanced predictive accuracy, this research aims to develop a real-time fatigue detection framework. The proposed system is expected to significantly improve workplace safety and operational efficiency in the construction industry [5].

II. EXPERIMENTS

We conducted real-time fatigue prediction for construction workers using ensemble learning with data from PPG and EEG sensors. The process included wearable-based data collection, preprocessing with Hampel and Savitzky-Golay filters, and normalization using PCA and Z-normalization. DeepAR [6] and TFT were trained with optimized parameters to ensure accurate and robust predictions.

A. Methods

This research introduces a method for gathering information through wearable technologies to observe and forecast construction workers' exhaustion in real-time, utilizing a unified analytical framework illustrated in Fig. 1. We applied a method to combine DeepAR and TFT outputs before applying XGBoost [7] as a meta-learner. DeepAR and TFT were individually trained to capture long-term and multivariate temporal patterns in physiological data. Their predictions for fatigue were then combined using XGBoost as a meta-learner to enhance accuracy. To ensure reliability and minimize evaluation bias, a 5-fold cross-validation approach was employed. Additionally, we conducted a comparative analysis with baseline models, including LSTM, GRU [8], Random Forest, and Support Vector Regression (SVR). These models were chosen for their relevance to time-series fatigue prediction. Performance was evaluated using MSE,



RMSE, and sMAPE to benchmark against baseline methods.

Fig. 1. Flowchart of the Integrated Analysis Model for Fatigue

1) Dataset:

This study collected biometric signals—HR, BVP, HRV, and EEG—from 11 participants (6 males, 5 females). After equipment checks, 20 minutes of baseline data

were gathered, including resting-state EEG and heart-related metrics, followed by a standardized reaction time test to assess cognitive reactivity and mental fatigue.

The dataset was split 8:1:1 for training, testing, and validation. DeepAR with LSTM (sequence length 50) was tuned via grid and random search, with a structured pipeline ensuring reliable preprocessing.

2) Data preprocessing:

Several preprocessing techniques were employed to ready the physiological data for analysis, including the use of Hampel and 3D median filters to address missing values and outliers. Signal smoothing was performed with the Savitzky-Golay and moving average filters, while DC components caused by movement or poor contact were removed through baseline correction.

All signals were normalized for consistency, then segmented into overlapping windows based on fatigue detection sensitivity. Key features from the PPG signal were extracted in both time and frequency domains. Finally, the dataset was standardized using Z-normalization and dimensionality was reduced via principal component analysis (PCA) to improve model efficiency.

B. Analysis Model

This study proposes a hybrid model combining DeepAR, TFT, and XGBoost for real-time fatigue prediction. DeepAR captures long-term trends, while TFT models complex multivariate relationships. Their outputs are combined using XGBoost to enhance accuracy and reduce noise sensitivity. To prevent overfitting, cross-validation and hyperparameter tuning were applied. PCA and feature selection improved model efficiency, and an adaptive weighting mechanism balanced prediction from DeepAR and TFT. Preprocessed physiological data were segmented and used to train DeepAR and TFT. Performance was monitored using standard error metrics and early stopping. Final predictions were refined using XGBoost and validated on a test set, confirming model robustness.

III. RESULTS

The integrated model was evaluated using MSE and RMSE for both physical and mental fatigue prediction. It achieved notably lower errors—MSE of 0.0007 and RMSE of 0.0277 for physical fatigue, MSE of 0.0033 and RMSE of 0.0577 for mental fatigue—compared to individual DeepAR and TFT models. 5-fold cross-validation confirmed the model's robustness and stability. By effectively capturing complex temporal patterns and reducing prediction variance, the integrated approach proved to be a reliable solution for real-time fatigue monitoring in construction settings.

IV. DISCUSSION AND CONCLUSIONS

This study proposes an integrated model for real-time fatigue monitoring of construction workers using PPG and EEG data. By combining DeepAR, TFT, and XGBoost, the model effectively captures long-term trends and complex patterns, achieving high accuracy with low MSE and RMSE values.

To support field deployment, the system incorporates adaptive calibration to handle environmental changes and emphasizes wearable comfort and privacy protection through federated learning. Despite the small sample size, the model shows strong potential for scalable, robust fatigue monitoring in construction environments. Future work will focus on expanding the dataset and validating performance across diverse sites.

ACKNOWLEDGMENTS

This work is supported by the Korea Agency for Infrastructure Technology Advancement (KAIA) grant funded by the Ministry of Land, Infrastructure and Transport (Grant RS-2023-00251002). This work was supported by the National Research Foundation of Korea(NRF) grant funded by the Korea government(MSIT) (No. 2022R1C1C1005409).

REFERENCES

- [1] Y. Yu, W. Umer, X. Yang, and M. Antwi-Afari, "Posture-related data collection methods for construction workers: a review," *Automation in Construction*, vol. 124, p. 103538, 2021. doi: 10.1016/j.autcon.2020.103538.
- [2] H. Jebelli, B. Choi, and S. Lee, "Application of wearable biosensors to construction sites. II: Assessing workers' physical demand," *Journal of Construction Engineering and Management*, vol. 145, p. 04019080, 2019. doi: 10.1061/(ASCE)CO.1943-7862.0001710.
- [3] Z. Mohamed, K. Mostafa, A. Oetomo, T. Hegazy, and P. Morita, "Wearable devices: Cross benefits from healthcare to construction," *Automation in Construction*, vol. 142, p. 104501, 2022. doi: 10.1016/j.autcon.2022.104501.
- [4] W. Lee, K.-Y. Lin, E. Seto, and G. Migliaccio, "Wearable sensors for monitoring on-duty and off-duty worker physiological status and activities in construction," *Automation in Construction*, 2017. doi: 10.1016/j.autcon.2017.06.012.
- [5] Y. Zhang and X. Wang, "Integrating IoT and machine learning for proactive safety management in construction," *Journal of Construction Engineering and Management*, vol. 149, no. 2, p. 04022123, 2023. doi: 10.1061/(ASCE)CO.1943-7862.0002198.
- [6] V. Flunkert, D. Salinas, and J. Gasthaus, "DeepAR: Probabilistic forecasting with autoregressive recurrent networks," *International Journal of Forecasting*, vol. 36, pp. 1181–1191, 2020. doi: 10.1016/j.ijforecast.2019.07.001.
- [7] T. Chen and C. Guestrin, "XGBoost: A scalable tree boosting system," in *Proceedings of the 22nd ACM SIGKDD International Conference on Knowledge Discovery and Data Mining (KDD '16)*, New York, NY, USA: Association for Computing Machinery, 2016, pp. 785–794. doi: 10.1145/2939672.2939785.
- [8] M. Ghafoori, C. Clevenger, M. Abdallah, and K. Rens, "Heart rate modeling and prediction of construction workers based on physical activity using deep learning," *Automation in Construction*, vol. 155, p. 105077, 2023. doi: 10.1016/j.autcon.2023.105077

A Data-Driven Approach to Estimating VTS Operator Staffing Levels Using Vessel Trajectory and Communication Data

Kyung-Rak Kim¹, Kwang-Il Kim^{*2}

^{1,2} College of Ocean Science, Jeju National University, Jeju 63243, Korea

[Email : kk7681@jejunu.ac.kr¹, kki@jejunu.ac.kr^{1,2}]

Abstract

With the rapid growth of global maritime trade, vessel traffic volumes and the workload of Vessel Traffic Service (VTS) operators have increased significantly, raising the risk of maritime accidents due to delays in timely service provision. To ensure safe and efficient vessel traffic management, it is crucial to determine an appropriate staffing level based on traffic volume and near-miss incidents. However, there is currently no standardized quantitative method to calculate the optimal number of VTS workstations a key factor in staffing decisions. This study proposes a data-driven, microscopic methodology for estimating the required number of VTS workstations using automatic identification system (AIS)-based vessel trajectory data and VHF voice communication recordings between VTS operators and ship navigators. The method involves trajectory preprocessing to standardize time intervals and classify operator tasks into three functional modules: information service, navigational assistance, and traffic organization service. The proposed model was applied to Yeosu VTS in Korea and indicated that an additional workstation is required to manage the current operational load. The results demonstrate that the model can effectively determine VTS staffing requirements without relying on long-term statistics, enhancing the ability to provide timely services and reduce the risk of accidents in congested maritime areas.

Index Terms: VTS staffing, workload analysis, AIS data, VHF communication, workstation estimation, vessel traffic management

I. INTRODUCTION

With the increase in global maritime trade, VTS plays a vital role in ensuring navigational safety and environmental protection. VTS operators manage vessel movements using AIS, radar, CCTV, and VHF radio, often under fatigue due to shift work and high workloads. Insufficient staffing can lead to serious safety risks, as shown in past maritime accidents. Despite IALA's staffing guidelines, there is no method for calculating the number of required workstations. This study proposes a vessel traffic flow-based VTS staffing model using AIS and VHF data. The model includes three modules (INS, NAS, TOS) and offers a data-driven method for estimating optimal staffing levels.

II. OPTIMAL VTS OPERATORS STAFFING METHOD

VTS responsibilities are delivered through three core services: INS, NAS, and TOS. The total VTS operator task time is defined as the sum of times required for each service. This study proposes a staffing model that calculates these times using AIS data and VHF voice analysis, supported by a VTS operator questionnaire. The model quantifies sub-item occurrences and durations to estimate the time per module. Required workstations are determined based on hourly T_{task} values. The number of operators per workstation is derived from IALA guidelines using annual and duty hours. Hourly peak workload is compared to current staffing to assess adequacy. Finally, the required daily workstations are computed by aggregating hourly needs.

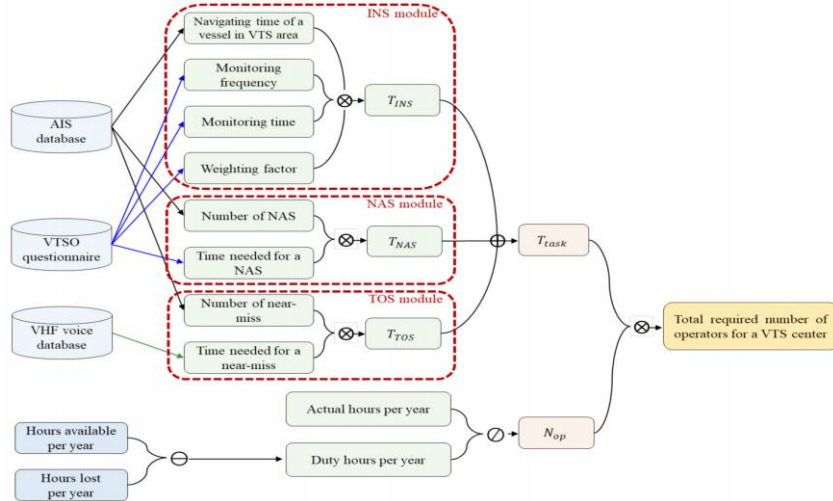


Fig. 1. Optimal VTS staffing model process flow.

III. RESULTS

AIS data from June 2014, the peak traffic month, were analyzed for Yeosu VTS, Korea's busiest harbor with complex routes and 353 piers. The VTS center operates three sectors, each with a dedicated workstation. To estimate TOS time, 1,320 VHF voice clips from 3 days were analyzed, showing an average communication time of 30 seconds per case. This is consistent with prior Busan VTS results.

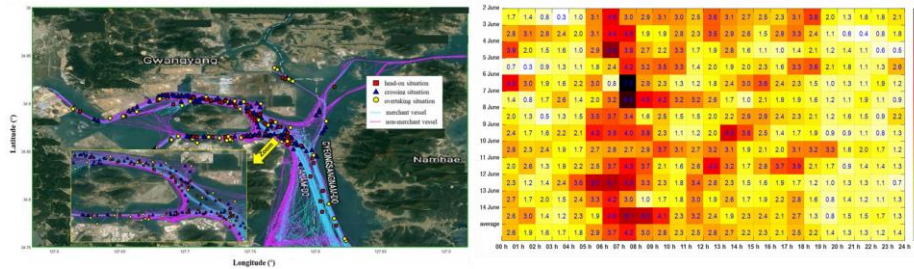


Fig. 2. Scatter plot for the near-miss location between merchant vessels and non-merchant vessels in Yeosu VTS are (left) and required hourly number of workstations for Yeosu VTS(right).

Operators work in two shifts (day/night), rotating workstations every 1.5 hours with a 15-minute overlap. A survey of 20 licensed VTS operators (avg. age 38) provided monitoring frequencies and workload weights. Precautionary zones are monitored every 3 minutes, and TSS every 10 minutes due to lower collision risk. These inputs were used to compute hourly workload per service module. The model outputs the required number of workstations per hour, enabling optimized staffing decisions.

IV. CONCLUSIONS

This study proposed a microscopic method to estimate the optimal number of VTS operators using AIS data, VHF voice analysis, and a questionnaire. Unlike traditional macroscopic models, it allows for real-time, data-driven staffing decisions. The method can help reduce operator fatigue and maritime accidents by ensuring sufficient staffing. Future work should include eye-tracking and dynamic staffing based on traffic patterns and near-miss density.

REFERENCES

- [1] S. D. Gantz and D. R. Philpott, *FISMA and the Risk Management Framework: The New Practice of Federal Cyber Security*, Rockland, MA, USA: Syngress, 2012, pp. 1–551.
- [2] S. A. C. Bukhari, I. Tusseyeva, B.-G. Lee, and Y.-G. Kim, “An intelligent real-time multi-vessel collision risk assessment system from VTS viewpoint based on fuzzy inference system,” *Expert Systems with Applications*, vol. 40, pp. 1220–1230, 2013.
- [3] International Association of Marine Aids to Navigation and Lighthouse Authorities (IALA), *IALA Guideline 1089 Provision of VTS Services (INS, TOS & NAS)*, 1st ed., Saint Germain en Laye, France, 2012, pp. 1–23.

Comprehensive Vulnerability Analysis of Lightweight Authentication Protocols in IoT Healthcare Environments

Haewon Byeon¹

¹Department of Future Technology, Korea University of Technology and Education, Cheonan 31253, Republic of Korea

[Email : bhwuma@naver.com]

Abstract

This paper presents a comprehensive vulnerability analysis of a lightweight authentication protocol for IoT healthcare environments. The study identifies significant risks, including node cloning, insider threats, biometric data insecurity, session management flaws, and scalability challenges. To address these vulnerabilities, enhancements such as Physical Unclonable Functions (PUFs), advanced encryption (e.g., homomorphic encryption), role/attribute-based access control, robust session management, and adaptive cryptography are proposed. The discussion includes future directions toward quantum-resistant cryptography and adaptive AI-based security policies.

Index Terms: IoT Healthcare Security, Lightweight Authentication, Physical Unclonable Functions, Biometric Data Protection, Quantum-Resistant Cryptography

I. INTRODUCTION

The Internet of Medical Things (IoMT) is transforming healthcare by establishing interconnected networks of devices capable of monitoring and transmitting sensitive patient data. This enhanced connectivity offers significant improvements in medical service delivery and patient outcomes. However, it also presents critical security challenges, particularly considering the constrained computational resources of many IoT devices and the transmission of sensitive information across potentially vulnerable networks [1].

To ensure the integrity and privacy of medical data, robust authentication mechanisms are paramount. Traditional single-factor and even two-factor authentication methods have demonstrated limitations in the face of evolving cyber threats [2]. Consequently, lightweight multi-factor authentication protocols, specifically designed for the resource limitations of IoT devices, have become increasingly essential. This paper undertakes an analysis of a representative lightweight authentication protocol employed within the IoMT landscape, with a specific focus on the identification and mitigation of potential security vulnerabilities that could compromise the system.

The aims of this research are twofold: first, to identify and thoroughly analyze security weaknesses inherent in a contemporary lightweight authentication protocol for IoMT [3]; and second, to propose technical enhancements aimed at achieving robust, scalable, and efficient authentication within IoMT environments.

The remainder of this paper is structured as follows: Section II provides a survey of related authentication protocols, highlighting their limitations. Section III details the methodological approaches employed for the vulnerability analysis conducted in this study. Section IV presents the main findings, discussing the identified weaknesses within the analyzed protocol.

II. SYSTEM MODEL AND METHODS

A. Literature Review of IoT Authentication Protocols

Traditional authentication methods relying on passwords or tokens are increasingly proving to be inadequate in the face of evolving security threats. Multi-factor authentication (MFA) and lightweight protocols, which often leverage elliptic curve cryptography (ECC) and cryptographic hash functions, are gaining prominence within the realm of IoT healthcare. These approaches aim to strike a balance between robust security and operational efficiency, particularly crucial for resource-constrained devices. However, the integration of biometric verification mechanisms introduces novel challenges related to user privacy and secure data storage. Furthermore, authentication protocols for medical IoT networks must exhibit flexibility and scalability to accommodate the dynamic and heterogeneous nature of these interconnected systems. The complexity of protocol design is further amplified by the need to adhere to stringent regulatory compliance standards, such as HIPAA and GDPR.

B. Related Works and Existing Vulnerabilities

Existing literature highlights several prevalent attack vectors targeting IoT devices and their authentication mechanisms. These include:

- 1) Node Capture: Physical compromise of IoT devices can lead to the exposure of sensitive cryptographic keys and other confidential information stored within the device.
- 2) Replay Attacks: Malicious actors may intercept legitimate communication messages and subsequently retransmit them to fraudulently gain unauthorized access to the system or its resources.
- 3) Biometric Data Risks: The inherent irreversibility of biometric data underscores the necessity for employing advanced encryption techniques and privacy-preserving transmission methods to safeguard this sensitive information throughout its lifecycle.
- 4) Scalability Challenges: Ensuring consistent security levels as the number of interconnected devices within a medical IoT network expands requires the adoption of standardized, interoperable, and resource-efficient authentication protocols that can effectively manage a growing device ecosystem.

C. Analytical Framework

To rigorously evaluate the security properties of the chosen lightweight authentication protocol [3], a multi-faceted analytical framework was employed, encompassing both formal and practical analysis techniques.

1) Formal Security Analysis

The Burrows-Abadi-Needham (BAN) logic, a well-established framework for formally analyzing security protocols, was utilized. Protocol message flows were meticulously modeled to formally verify the authenticity of communicating entities and the freshness of exchanged messages, thereby ensuring resistance to certain classes of attacks.

2) Cryptanalysis Tools

The Automated Validation of Internet Security Protocols and Applications (AVISPA) tool, a widely recognized platform for security protocol verification, was employed. This tool facilitated the simulation of various known attack scenarios, including replay attacks and man-in-the-middle attacks, to assess the protocol's resilience against these threats in a controlled environment.

3) Mathematical Modelling

Key cryptographic operations underpinning the authentication protocol were subjected to mathematical evaluation. Specifically, the ECC-based session key derivation process and hash-based operations were analyzed. The session key derivation can be represented as:

$$K_{session} = g^{ab} \pmod{p}$$

where g denotes the generator point of the elliptic curve, and a and b represent the private keys of the communicating parties, while p is the modulus of the elliptic curve. This mathematical model allows for a deeper understanding of the cryptographic strength and potential weaknesses of the core security mechanisms.

III. RESULTS

A. Protocol Overview

- 1) Initialization: Upon deployment, each device within the IoMT network undergoes an initialization phase wherein it is provisioned with a unique identifier and associated cryptographic keys. To maintain efficiency for resource-constrained devices, this process employs lightweight cryptographic operations to minimize computational overhead.
- 2) Mutual Authentication: The protocol incorporates a mutual authentication mechanism based on a nonce-based challenge-response exchange. This ensures that both communicating parties, the medical practitioner (MP) and the sensor (S), can confidently verify each other's identities before proceeding with data transmission or service provision.
- 3) Session Key Establishment: To guarantee the confidentiality of subsequent communication, the protocol establishes a session-specific encryption key. This key is derived using principles of Elliptic Curve Cryptography (ECC), a method known for its strong security properties and relative efficiency in resource-limited environments.
- 4) Data Protection: All messages exchanged between the authenticated parties after the session key establishment are encrypted using the derived session key. This measure ensures that sensitive medical data remains protected against unauthorized access and maintains its confidentiality during transmission.

B. Vulnerability Analysis

1) Compromised Gateway Node (GWN) Risk

A critical vulnerability exists within the protocol concerning the potential compromise of a gateway node (GWN). If a GWN is successfully attacked, it could lead to the exposure of all session keys associated with the devices reliant on that particular gateway. This risk arises from the centralized generation or management of session keys by the gateway, as exemplified by the potential key derivation function:

$$KEY = H(MID_{\{MP_i\}} || MID_{\{S_k\}} || RN_{\{S_k\}} || RN_{\{MP_i\}})$$

where H represents a cryptographic hash function, MID denotes the unique identifier of the medical practitioner MP_i and the sensor S_k , and RN signifies the random nonce generated by the sensor S_k and the medical practitioner MP_i . The reliance on a single gateway for key management creates a single point of failure.

Mitigation: To address this vulnerability, it is recommended to implement a distributed approach to session key generation. This could involve leveraging blockchain technology to distribute trust and key management responsibilities across multiple nodes or employing Diffie-Hellman key exchange-based mechanisms to enable independent key agreement between communicating entities, thereby eliminating the reliance on a potentially vulnerable central gateway.

2) PUF Key and Challenge-Response Leakage

The protocol's reliance on Physically Unclonable Functions (PUFs) and the transmission of their responses over public networks, such as a blockchain, introduces a significant security risk. Sharing PUF responses in a publicly accessible manner exposes them to sophisticated modelling attacks. Attackers can potentially analyze a collection of challenge-response pairs to build a model capable of predicting the PUF's response to unseen challenges, effectively compromising the security derived from the PUF.

Mitigation: To mitigate this risk, it is imperative to restrict the use of PUFs to local authentication processes within individual devices. Sensitive PUF responses should never be transmitted or shared on public channels. Alternative methods for verifying the integrity or identity associated with the PUF should be explored that do not involve direct exposure of the raw challenge-response data.

3) Length-Mismatch XOR

The protocol may employ XOR operations for certain cryptographic functions. However, performing XOR operations on data of unequal lengths can inadvertently leak private information. For instance, if $C = A \oplus B$ and the length of A is less than the length of B , the portion of B that extends beyond the length of A will be directly revealed in the ciphertext C .

Mitigation: To prevent information leakage due to length mismatches in XOR operations, it is essential to ensure that all operands involved in such operations are of equal length. This can be achieved by applying a cryptographic hash function to the input data before the XOR operation, guaranteeing a fixed-length output. Alternatively, the use of Hash-based Message Authentication Codes (HMACs) should be considered, as they are specifically designed to provide both data integrity and authenticity without the inherent risks associated with simple XOR encryption.

4) Message Integrity and Replay

The current protocol lacks explicit mechanisms for ensuring message integrity and preventing replay attacks. The absence of Message Authentication Codes (MACs) or digital signatures, coupled with the potential absence of sequence numbers, makes the protocol susceptible to both message tampering and replay attacks, even if timestamps are employed. While timestamps can offer some temporal context, they are not sufficient to guarantee message integrity or to definitively prevent an attacker from retransmitting previously captured valid messages at a later time to achieve unauthorized actions.

Mitigation: To enhance the protocol's security posture, it is crucial to incorporate robust message integrity and anti-replay mechanisms. This can be achieved by including MACs or digital signatures with all transmitted messages, allowing the recipient to verify the sender's authenticity and ensure that the message has not been altered in transit. Additionally, the implementation of sequence numbers for all messages would effectively prevent replay attacks by allowing the receiver to identify and discard any retransmitted messages.

IV. Conclusion,

This paper has identified critical security weaknesses within a representative lightweight authentication protocol for IoMT, specifically concerning gateway node compromise, PUF response leakage, XOR operation vulnerabilities, and the absence of robust message integrity mechanisms. The proposed mitigations, including distributed key generation, localized PUF usage, consistent data length in cryptographic operations, and the implementation of MACs and sequence numbers, offer pathways toward more resilient and trustworthy IoMT systems. Future research should focus on the practical implementation and thorough evaluation of these enhancements to ensure their effectiveness in real-world healthcare environments.

ACKNOWLEDGMENTS

This research Supported by Basic Science Research Program through the National Research Foundation of Korea (NRF) funded by the Ministry of Education (RS-2023-00237287).

REFERENCES

- [1] A. Vyas, S. Abimannan, and R. H. Hwang, "Sensitive Healthcare Data: Privacy and Security Issues and Proposed Solutions," *Emerging Technologies for Healthcare: Internet of Things and Deep Learning Models*, pp. 93-127, 2021.
- [2] J. Kamau and M. Mgala, "A review of Two Factor Authentication Security Challenges in the Cyberspace," *International Journal of Advanced Computer Technology*, vol. 11, no. 5, pp. 1-6, 2022.
- [3] V. O. Nyangaresi and G. K. Yenurkar, "Anonymity preserving lightweight authentication protocol for resource-limited wireless sensor networks," *High-Confidence Computing*, vol. 4, no. 2, p. 100178, 2024.

Enhanced Aquaculture Product Quality Control in Supply Chains: Digital Twin Integration Approach

Ngoc-Bao-Van Le^{1,2}, Daesung Lee^{3,*}, Jun-Ho Huh^{4,5,6,**}

¹Ph.D Candidate of Department of Data Informatics, National Korea Maritime and Ocean University, 49112, Republic of Korea

²Ph.D Candidate of Department of Interdisciplinary Major of Ocean Renewable Energy Engineering (Concurrent Position), National Korea Maritime and Ocean University, 49112, Republic of Korea

³Full Professor (Tenure Track) of Division of Maritime AI and Cyber Security, National Korea Maritime and Ocean University, Republic of Korea

⁴Full Professor (Tenured) of Department of Data Science, National Korea Maritime and Ocean University, Republic of Korea

⁵Full Professor (Tenured) of Department of Interdisciplinary Major of Ocean Renewable Energy Engineering, National Korea Maritime and Ocean University, Republic of Korea

⁶Sabbatical Year at School of Computer Science and Engineering, Yeungnam University, Republic of Korea

* Corresponding author: dslee@kmou.ac.kr ** Corresponding author: 72network@kmou.ac.kr

Abstract

Preserving aquaculture products during transportation and distribution is crucial because their characteristics of sensitivity to temperature and the environment make them more easily perishable. Failure to manage product quality can affect the quality of food and increase great economic losses for farmers. Digital twin technology can help to enhance the aquaculture product supply chain by being capable of high control over physical entities and efficiently managing complicated systems. In this paper, we propose an aquaculture product quality control system integrating digital twin technologies. The proposed system can collect computerized sensors and operation data to simulate them on the digital environment through big data processing, analytics and intelligent decision algorithms. A data architecture to support the crawling, storing, and analysis sensors' data is presented. The paper contributes to promoting the application of digital twin technology in aquaculture quality management. The findings are hoped to serve as a valuable reference for other future works and continuously extend the use of digital twins in supply chain domains.

Index Terms: Aquaculture Product, Quality Control, Monitoring System, Internet of Things, Big Data Analysis, Digital Twin.

I. INTRODUCTION

* Corresponding author ** Co Corresponding author

The aquaculture industry has demonstrated a crucial role because it can provide huge numbers of protein and nutrients sources for humans [1]. Preserving aquaculture products during transportation and distribution is important because of their characteristics of sensitivity to temperature and the environment. Failure to manage product quality can affect the quality of food and increase great economic losses for farmers. Numerous cutting-edge techniques have been researched to enhance quality management. For instance, in food safety management, Artificial Intelligence (AI) is used to track in the food supply chain, anticipate contamination, and identify microorganisms. Smart sensors and machine learning can improve transparency and food quality management while increasing the precision and effectiveness of toxic substance detection.

In the context of Internet of Things (IoT) has had a great development, digital twin has gained attraction from scholars and enterprises. Digital twin technology can help to enhance the aquaculture product supply chain by being capable of high control over physical entities and efficiently managing complicated systems. In this paper, we propose an aquaculture product quality control system integrating digital twin technologies. The proposed system can collect computerized sensors and operation data to simulate them on the digital environment through big data processing, analytics and intelligent decision algorithms.

II. RESEARCH BACKGROUND

Generally, a digital twin can be defined as a virtual representation of a physical system, process, or entity [2]. It works as a computer-generated model that accurately represents a physical product within cyber-physical systems by fusing online measurements with IoT communication capabilities, allowing for real-time control and optimization [3]. In the supply chain domain, digital twin is researched and utilized to monitor food quality in previous works. Zou et al. (2025) integrated digital twin technology and AI to manage quality and reduce food losses in the cold chain [4]. The authors utilized long-short term memory (LSTM) model to estimate food shelf life and forecast temperature. The findings demonstrated that the accuracy of temperature prediction was less than ± 0.3 °C, which led to an error in estimating the shelf life of perishable fruits of less than ± 1.2 days. Compared to conventional techniques, this system decreased food losses by 8.6% to 15.5%. According to the study by Sergio et al. (2023), the digital twin model integrated with 4.0 technology, can optimize the grain supply chain from planting to distribution, enhancing sustainability and farmers' quality of life [5]. However, the application of digital twin in aquaculture sectors is limited. Also, preserving aquaculture products during transportation and distribution is important because of their characteristics of sensitivity to temperature and the environment. It is crucial to promptly check the quality of the seafood and the surrounding environment while it is being transported. In this paper, we propose an aquaculture product quality control system integrating digital twin technologies.

III. RESULTS AND DISCUSSION

The design of digital twin aquaculture product quality control system is presented in Figure 1. IoT devices at the physical layer gather data in real time from actual seafood products in supply chain, including manufacturing, packaging, shipping and retailing. The sensors gather information from the real world. A variety of sensors, including cameras,

accelerometers, temperature, and pressure, are used to gather information on different facets of the system or physical asset. After that, this data is sent straight to the service layers. The system gathers vast amounts of data, which are processed using data analytics and artificial intelligence techniques. To maximize performance, anticipate failures, and make well-informed decisions, these technologies evaluate the data, draw conclusions, and enable predictive and prescriptive analytics. In a digital twin environment, modeling and simulation techniques faithfully depict physical assets or systems. These models allow for analysis and testing in a virtual environment by capturing the physics, interactions, and behavior of the physical asset.

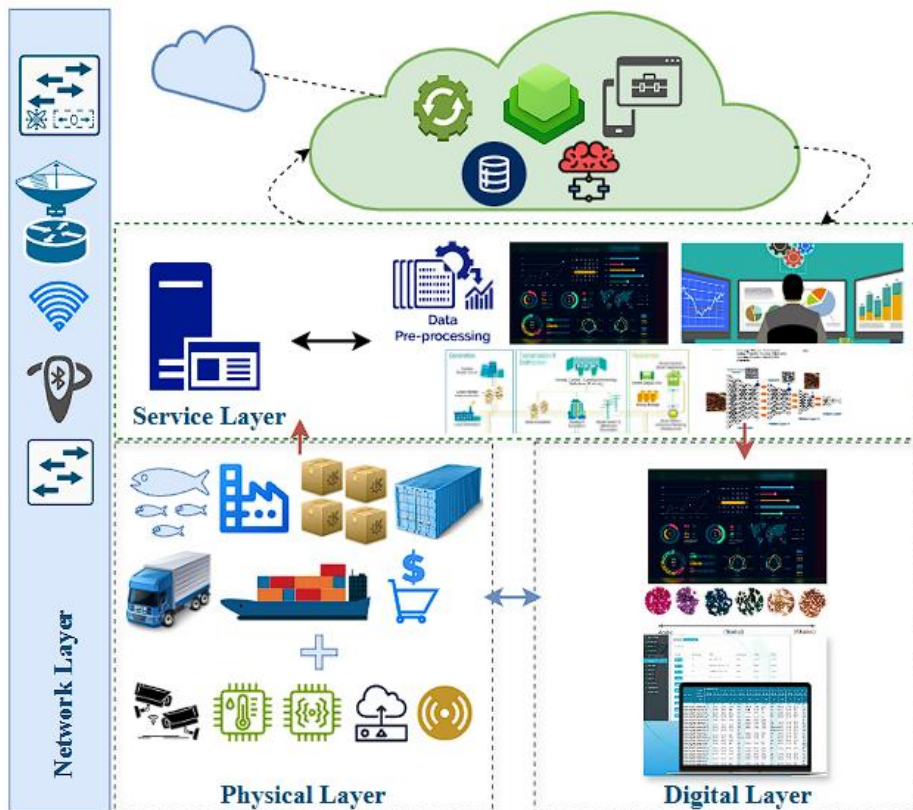


Fig. 1. Design of Digital Twin Aquaculture Product Quality Control System

The system uses cloud computing infrastructure to pre-process, store and analyze massive amounts of data during this process. Cloud platforms offer the scalability, high-performance computing, and storage capabilities required to manage the intricate and resource-intensive operations of a digital twin system.

To guarantee smooth data transfer between the virtual and physical assets at the network layer, digital twins depend on strong connectivity and communication protocols. Access control, encryption, security protocols, and monitoring systems are implemented further to guarantee the security and integrity of digital twin data.

IV. CONCLUSION

Advanced technologies, particularly digital twin technology, can improve the supply chain of aquaculture products. This paper introduces an aquaculture product quality control system integrating digital twin technologies. Our system can collect computerized sensors and simulate them in the digital environment. The paper contributes to promoting the application of digital twin technology in aquaculture quality management. The findings are hoped to serve as a valuable reference for other future works and continuously extend the use of digital twins in supply chain domains. Future research will consist of creating prototypes and field testing to verify their effectiveness in real-world environments and changing it in response to performance and comments.

REFERENCES

- [1] Wang, Cong, Zhen Li, Tan Wang, Xianbao Xu, Xiaoshuan Zhang, and Daoliang Li. "Intelligent fish farm—the future of aquaculture." *Aquaculture International*, pp. 1-31, 2021.
- [2] Erol, Tolga, Arif Furkan Mendi, and Dilara Doğan. "Digital transformation revolution with digital twin technology." In *2020 4th international symposium on multidisciplinary studies and innovative technologies (ISMSIT)*, pp. 1-7. IEEE, 2020.
- [3] Schroeder, Greyce N., Charles Steinmetz, Carlos E. Pereira, and Danubia B. Espindola. "Digital twin data modeling with automationml and a communication methodology for data exchange." *IFAC-PapersOnLine* 49, no. 30, pp. 12-17, 2016.
- [4] Zou, Yifeng, Junzhang Wu, Xiangchao Meng, Xinfang Wang, and Alessandro Manzardo. "Digital twin integration for dynamic quality loss control in fruit supply chains." *Journal of Food Engineering* 397, pp. 112577, 2025.
- [5] Gallego-García, Sergio, Diego Gallego-García, and Manuel García-García. "Sustainability in the agri-food supply chain: a combined digital twin and simulation approach for farmers." *Procedia Computer Science* 217, pp. 1280-1295, 2023.

Research on cloud-based IoT device vulnerability analysis and evaluation system

Minsu Kim^{1*}

¹Department of Information Security Engineering, Joongbu University, 10279, Korea
[Email : mskim@joongbu.ac.kr]

Abstract

In this study, we propose a cloud-based system that automatically diagnoses and evaluates security vulnerabilities in IoT devices. Various items are checked for smart home devices, and the security level is quantitatively analyzed using CVSS v4.0. The diagnosis results are visualized and provided to support repeated checks and countermeasures. The proposed system can contribute to checking the security status in real time and improving the security of smart homes.

Index Terms: IoT Security, Cloud Assessment System, Vulnerability Diagnosis, CVSS v4.0, Smart Home Devices

I. INTRODUCTION

In an IoT environment, smart home devices connect various devices in the home to a network, enabling remote monitoring and control. Typical examples include televisions, refrigerators, robot vacuum cleaners, door locks, and home cams, and the widespread use of such devices has increased security threats such as the leakage of personal and confidential information.

In fact, various security incidents have occurred, such as illegal home cam monitoring, smart door lock hacking, and DDoS attacks, which shows that vulnerabilities in IoT devices are a real threat. This calls for strong security technology and systematic management. Although information protection agencies have presented guidelines focusing on security items in the development stage, real-time vulnerability analysis and countermeasures after the product is released are lacking.

Hackers continue to attack products by exploiting undisclosed vulnerabilities even after their release, so security checks of devices in operation are also important. Therefore, in this study, we propose a cloud-based system that can automatically diagnose and evaluate the security status of IoT devices.

II. Related Study

Recently, the use of IoT devices has increased, and the accompanying security threats have also increased. As such, ensuring the security of IoT devices is no longer an option, but has become an essential security requirement to consider. To address these security

* Corresponding author

threats, various analysis techniques and practice-driven security guidelines have been prepared at the research and policy levels.

First, in a study dealing with security check procedures for IoT devices, a total of five steps of vulnerability analysis procedures were proposed, from hardware analysis to application analysis, focusing on the post-device development stage. The study derived actual vulnerabilities through experiments on routers and IP cameras, and confirmed that the proposed procedures work effectively. This is significant in that a post-launch diagnostic procedure was established, rather than a post-mortem response [1].

In addition, in a study summarizing threat modeling and security measurement methods for IoT smart devices, various analysis methods such as DFD, Attack Tree, and STRIDE models were applied. Specifically, a quantitative security evaluation method was proposed through CVSS-based score evaluation, and an integrated diagnostic method in which analysis and evaluation are linked was presented [2].

The Korea Internet and Security Agency continuously provides security guides for practitioners. The Home Appliance IoT Security Guide presents security functions and countermeasures that take into account the characteristics of each product type, and is structured so that manufacturers and developers can refer to it from the early stages of product design[3]. The IoT Common Security Guide presents security principles and practice items throughout the entire life cycle, from design, installation, operation, and disposal, and provides general-purpose standards that can be applied in a variety of fields[4]. As such, existing research and guidelines address IoT security in various ways depending on the purpose and context, but most of them focus on basic analysis and the establishment of security principles, and the development of substantial methodologies for applying security diagnosis to actual devices is still insufficient.

III. Research Method

A. Diagnostic system overview

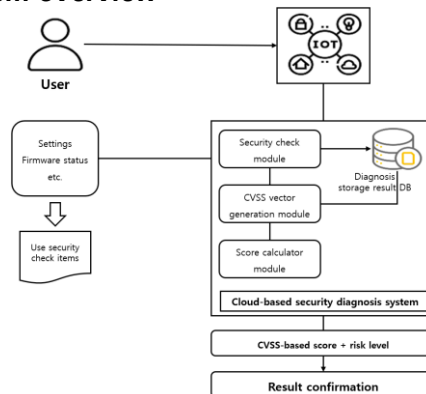


Fig. 1. Cloud-based IoT diagnostic system flow chart

Existing IoT security policies mainly focus on the product development and pre-release stages, and systems for diagnosing and responding to vulnerabilities in real time after

release are relatively insufficient. However, hackers continue to search for new vulnerabilities even after a product is released and attempt attacks that exploit them.

Therefore, in this study, we propose an IoT security diagnostic system in a cloud environment (Figure 1) as a way to compensate for this limitation. The system checks the security settings and network status of smart home devices such as home cameras in an automated manner and provides quantified diagnosis results. This allows users to intuitively understand the security status of their devices and take immediate action when a security threat occurs.

B. Configure the check items

The system configured inspection items based on the Korea Internet & Security Agency's (KISA) "Home Appliance IoT Security Guide" and "IoT Common Security Guide." The configured items are divided into four security areas: (1) account management, (2) service management, (3) network and firmware checks, and (4) data protection. Each item is evaluated as guide compliance (Y/N) and is designed to be associated with CVSS v4.0 evaluation items to calculate a quantitative score.

Each check item includes items directly related to the security of the actual device, such as the presence or absence of CVE-ID, blocking of debug ports, changing initial passwords, application of encryption, and data access control. The diagnosis results are used in CVSS vector configuration and contribute to quantifying the security level.

C. Diagnostic procedures and assessment methods

The flow of the proposed diagnostic system is as follows. First, when a user is authenticated and connects to the system, key data such as homecam settings, network information, and firmware status are collected. After that, an automatic diagnosis is performed according to predefined security check items, and a CVSS vector is generated according to the results. Based on this, a CVSS v4.0 score and risk grade are derived, and the check items are linked to CVSS evaluation elements, and if there are security flaws, they are reflected in the corresponding vector value.

For example, if the initial password has not been changed, it is considered that access is possible with only low privileges or attacks are possible without user operation, and the overall score is increased. The final results are visualized and provided, saved together with recommendations, and used for re-diagnosis and comparative analysis.

D. Diagnosis results and response plan

The diagnosis results consist of the diagnosis status for each item, the applied CVSS vector, the final score, and an official grade, and technical countermeasures are presented along with the main impact factors. In particular, attack scenario-based response strategies are included for vulnerability items, and the check history is stored in the cloud to enable repeated checks and comparative analysis by users and administrators.

This improves the overall security level of the system and ensures the continuous stability and reliability of equipment operation. It also allows the transition of changes in the security status to be understood, contributing to the establishment of long-term security policies.

Table 1. Security Checklist and CVSS Assessment Items Response

Category	Item	Description	Guide compliance	Apply CVSS evaluation item
Software security	Whether external access to the administrator page is restricted	Block access to the administrator page from external networks	Y/N	AV/PR
	Whether CVE-ID exists	Whether there are known vulnerabilities	Y/N	-
Physical security	Blocking debugging ports	Disable or protect debugging ports such as UART and JTAG so that they are not exposed to the outside	Y/N	AV
	Controlling device access	Apply physical locks or seals so that the external case of the device is not easily opened	Y/N	AV/PR
	Protecting critical components	Prevent access to sensitive parts such as flash memory and storage devices	Y/N	VC
	Securing equipment location	Manage the installation location and environment so that the device is not easily exposed to the outside	Y/N	AV
Authentication	Forcing initial password change	Require password change upon initial installation	Y/N	PR
	Account locking function when authentication fails	Block account upon repeated login failures	Y/N	AC
	Password complexity policy exists	8 or more characters, combination of uppercase and lowercase letters, numbers, and special characters	Y/N	UI
	Restricting reuse of previous passwords	Prevent the same password when changing the password	Y/N	UI/PR
Encryption	Restricting guessable passwords	Block use of vulnerable combinations when setting up users	Y/N	PR
	Random number generator security level	Whether an unpredictable random number generator is used	Y/N	AC/VC
	Encrypting stored data	Apply encryption when storing sensitive data such as configuration information and logs	Y/N	VC
	Existing encryption key management policy	Establish policies such as periodic changes to encryption keys and access restrictions	Y/N	VI
Data protection	Reliability of encryption library	Whether an authorized standard library (such as OpenSSL) is used	Y/N	VC
	Encrypting video data	Whether AES256 or higher is applied	Y/N	VC
	Restricting sensitive information log recording	Restrict sensitive information such as video and audio from being stored in logs	Y/N	VC
	Automatic deletion of user data	Whether there is a function to automatically delete user data after a certain period of time	Y/N	SI
Platform security	Data separation according to access rights	Whether accessible data is separated according to user authority	Y/N	PR
	Verifying integrity of firmware	Apply electronic signature and hash value verification	Y/N	SI/PR
	Possibility of update after administrator authorization	User verification before automatic update execution	Y/N	AR
	Creating administrator access audit logs	Whether external logins can be tracked	Y/N	AT
	Auditing security function usage history	Records changes to security settings, login attempts, etc.	Y/N	AT

IV. CONCLUSIONS

In this study, we proposed a cloud-based automated security diagnosis system to solve security issues in IoT devices. The proposed system targets smart home devices, mainly home cameras, and is designed to effectively analyze the security status of devices through check-item-based security diagnosis and quantitative evaluation using CVSS v4.0.

This allows users to intuitively understand the security status and provide technical countermeasures for vulnerabilities, and is expected to improve long-term security levels through repeated diagnosis and history management.

In future research, we will be able to expand the scope of diagnosis to a wider variety of IoT devices and develop the system to increase its scalability and effectiveness by adding real-time warning functions and user-customized security guides.

REFERENCES

- [1] S. G. Choi, "Vulnerability analysis procedure and check items of IoT devices," *Ajou University Graduate School of Information and Communications Master's thesis*, 2020.
- [2] K. T. Lee, "Vulnerability analysis procedure and check items of IoT devices," *Korea University Graduate School of Information Security Doctoral Dissertation*, 2021.
- [3] Korea Internet & Security Agency(KIAS), "Home Appliances IoT Security Guide," 2017.
- [4] Korea Internet & Security Agency(KIAS), "IoT Common Security Guide," 2016.

PD Recognition based on Random Forest Algorithm for Power Transformers

Sung-Wook Kim¹, Gyeong-Yeol Lee², and Gyung-Suk Kil^{3*}

¹Department of Electrical and Electronics Engineering, Silla University, 46958, Korea

²Department of Electrical and Electronics Engineering, National Korea Maritime & Ocean University, 49112, Korea

[Email : number1@silla.ac.kr]

Abstract

Cast resin power transformers have been widely used in power apparatus due to their ease of installation, small size, and low expense of maintenance. In perspective of insulation condition monitoring, identification and classification techniques of internal defect using partial discharge (PD) pulses are getting more important. This study deals with the PD recognition using a Random Forest (RF) algorithm for the PD recognition in the cast resin power transformers. Four different types of PD simulators such as a metal protrusion, a particle fixed on insulator, a delamination, a void defect were prepared, and the PD single pulses were measured by the printed circuit board (PCB) based Rogowski-type PD sensor. Time and frequency domain, and physical shape-based parameters were extracted from each PD pulse according to PD defect types. From the experimental results, the RF algorithm proposed in this paper had over 90% of the accuracy rate of PD defect recognition.

Index Terms: PD recognition, Random forest algorithm, Partial discharge, Power transformers

I. INTRODUCTION

A cast-resin power transformer has been widely used in electrical systems due to its low installation and maintenance costs and self-extinguishing properties [1-2]. However, once insulation defects are generated in the epoxy-insulation, partial discharge (PD) may occur and lead to progress deterioration, resulting in significant electrical accidents [3]. From the perspective of PD recognition, in this paper, the RF classifier model was prepared to identify typical types of PD defects using various PD features extracted from PD single pulses measured by a printed circuit board (PCB)-based Rogowski-type PD sensor. From the results, it was confirmed the RF classifier model could recognize the PD defects over 90%.

II. EXPERIMENTAL SETUP AND METHODS

Four types of PD defect models, shown in Fig. 1, were prepared to simulate typical insulation defects in the cast-resin power transformers. Fig. 2 shows the PD simulation test setup used in this paper. An oil-immersed HV transformer with an output voltage of 50 kV

* Corresponding author

(Model 770-5, Hypotonic) was used to apply high voltages. The PD signals were recorded using a digital storage oscilloscope (MSO 5204B, Tektronix) [4].

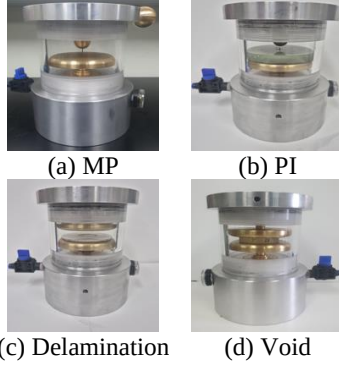


Fig. 1. PD defect models.

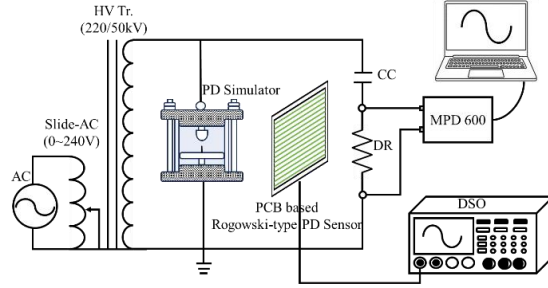


Fig. 2. Experimental setup.

III. RESULTS AND ANALYSIS

Fig. 3 shows typical PD single pulses and their FFT spectra according to four types of PD defect models. Total 100 PD pulses were recorded to systematically extract PD features including rising and falling times, pulse widths, kurtosis, skewness, and maximum frequencies [5]. It was confirmed that the RF classifier model could identify PD defect models with over 90% of PD recognition accuracy with optimal combinations of hyperparameters using the number of estimators, the maximum depth of decision trees, and the minimum number of samples to split.

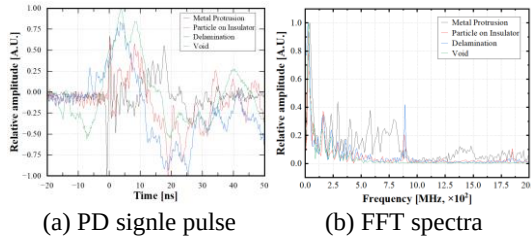


Fig. 3. PD single pulses and their FFT spectra.

Table 1. Confusion matrix.

PD defects	Identification [%]			
	MP	PI	Dela	Void
MP	93	4	0	2
PI	0	93	0	7
Dela	7	0	91	2
Void	0	0	4	96

IV. CONCLUSION

This paper dealt with the RF classifier algorithm model for PD recognition in cast-resin power transformers. The thin and plane structure PCB-based Rogowski PD sensor was used to detect PD single pulses. Total 6 PD features based on time, frequency, and physical shape-domains, were extracted from each PD single pulses. From the experimental results, the recognition accuracy using the RF algorithm could achieve 91.33% in accordance with four types of PD defect models. Future research about other algorithms should be conducted.

ACKNOWLEDGEMENTS

Following are results of a study on the "Busan Regional Innovation System & Education(RISE)" Project, supported by the Ministry of Education and Busan Metropolitan City

REFERENCES

- [1] G.-M. Ma, H.-Y. Zhou, C. Shi, Y.-B. Li, Q. Zhang, C.-R. Li, "Distributed Partial Discharge Detection in a Power Transformer Based on Phase-Shifted FBG," *IEEE Sensors Journal*, vol. 18, no. 7, pp. 2788-2795, Apr. 2018.
- [2] M. Sacha, T. Stefan, F. Kurt, "Detection and location of partial discharges in power transformers using acoustic and electromagnetic signals," *IEEE Transactions on Dielectrics and Electrical Insulation*, vol. 15, no. 6, pp. 1576-1583, Dec. 2008.
- [3] R. Yao, J. Li, M. Hui, L. Bai and Q. Wu, "Feature Selection Based on Random Forest for Partial Discharges Characteristic Set," *IEEE Access*, vol. 8, pp. 159151-159161, Aug. 2020.
- [4] G.-Y. Lee, S.-W. Kim, G.-S. Kil, "Design and Fabrication of Rogowski-type Partial Discharge Sensor for Insulation Diagnosis of Cast-Resin Transformers." *Journal of the Korean Institute of Electrical and Electronic Material Engineers*, vol. 35, no. 6, pp. 594-602, Nov. 2022.
- [5] G.-Y. Lee, G.-S. Kil, "Insulation Defect Diagnosis Using a Random Forest Algorithm with Optimized Feature Selection in a Gas-Insulated Line Breaker." *Electronics*, vol. 14, no. 10, pp. 1940, May 2025.

Dual-LLM Architecture for Aeronautical Information Question Answering with User Feedback Adaptation

Jeongmo Je¹, Byung Wook Kim^{1*}

¹Department of Information and Communication Engineering, Changwon National University,
Changwon 51140, Republic of Korea

[Email : bwkim@changwon.ac.kr]

Abstract

This paper aims to design dual-large language model (LLM) architecture for intelligent question answering on the Aeronautical Information Manual (AIM), integrating retrieval-augmented generation (RAG) with adaptive prompt enhancement driven by user feedback. This system leverages the LangChain framework to vectorize AIM, store it in a vector database, retrieve information relevant to user queries, and use a primary LLM to generate responses based on the retrieved contents. To overcome the limitations of RAG in handling complex and domain-specific documents, the system incorporates user feedback to enhance contextual understanding and improve response accuracy. When a response is rated negatively, users can provide additional feedback, which is processed by an auxiliary LLM to adjust the prompt and enhance the quality of future responses. This approach is expected to improve system performance by delivering more accurate and adaptive responses tailored to user needs in complex domain-specific queries.

Index Terms: Aeronautical Information Manual, Chatbot, Large Language Model, Retrieval-Augmented Generation, etc.

I. INTRODUCTION

In the field of aviation, the Aeronautical Information Manual (AIM) serves as a comprehensive document that consolidates essential information related to operational procedures, air traffic control methods, and the management of navigation aids, intended for pilots, air traffic controllers, and other aviation personnel. It is compiled from official publications such as aviation regulations, the Aeronautical Information Publication (AIP), and Notices to Airmen (NOTAM), and it plays a critical role in enhancing the practical understanding of aviation professionals [1]. However, due to its considerable length, technical complexity, and frequent updates, accessing relevant information efficiently remains challenging. Users are often required to manually search through extensive documents to locate specific guidelines, which is time-consuming and may lead to reduced operational efficiency as well as the risk of overlooking critical information.

Recent advancements in large language models (LLMs) and retrieval-augmented generation (RAG) techniques have opened new possibilities for building intelligent systems capable of understanding and responding to natural language queries using domain-specific knowledge [2]. However, applying such techniques to documents like the AIM, which is a complex and voluminous manual referenced by professionals engaged in

diverse tasks in aviation and characterized by highly specialized terminology, poses significant challenges. In particular, it is impractical to predefine prompt templates that effectively address the diverse and context-dependent informational needs of all users. This limitation highlights the necessity of user-driven approaches that enable the prompt to adapt dynamically based on feedback.

This paper presents a user-friendly intelligent chatbot system based on dual-LLM architecture designed to facilitate efficient access to the AIM. To address the challenge of manually crafting detailed prompts, this study adopts a method that dynamically refines prompts based on user feedback. Users can evaluate the chatbot’s responses, and in cases of negative feedback, the input is analyzed by a dedicated language model responsible for prompt refinement. The results of this analysis are then reflected in subsequent prompts to improve the quality of future responses. This feedback loop allows the system to adapt to user needs and improve accessibility to complex documents like AIM.

II. SYSTEM MODEL AND DESIGN

This system requires several core components: a document embedding module to vectorize the AIM text, a vector database to store the resulting document vectors, a primary LLM to generate responses, an auxiliary LLM to refine prompts based on feedback, and an evaluation module to assess response quality. These components are orchestrated using the LangChain framework, which enables a seamless pipeline from retrieval to generation and evaluation.

Fig. 1 shows the overall architecture of the system, illustrating the interaction among the AIM document, vector database, LLMs, evaluator module, and user feedback loop.

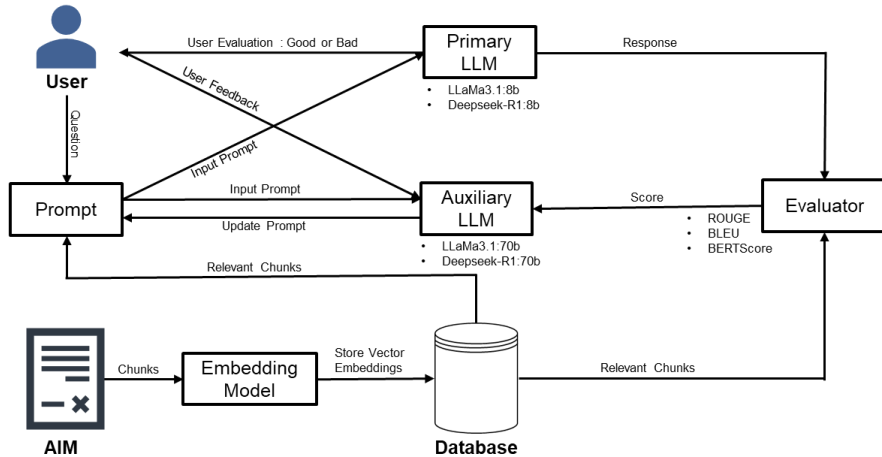


Fig. 1. Overview of System Architecture

Initially, the AIM document is segmented into individual paragraphs, each forming a document chunk, and is then vectorized through a document embedding module. The

resulting embeddings are stored in a vector database. When the user submits a query, the system retrieves the most semantically relevant document chunks from the vector space, to extract the relevant information. Based on the retrieved content, a locally hosted LLM such as LLaMA 3.1 or DeepSeek-R1, generates a natural language response. The quality of the generated response is quantitatively evaluated using standard automatic evaluation metrics such as BLEU, ROUGE, and BERTScore, which provide quantitative measures for assessing the response's relevance and quality. Specifically, these metrics reflect n-gram-based precision (BLEU), recall-oriented summary evaluation (ROUGE), and semantic similarity using pretrained language models (BERTScore) [3–5]. Additionally, users can directly evaluate the responses as either “Good” or “Bad.” In the case of negative feedback, the input is analyzed by an auxiliary LLM dedicated to processing such feedback. The analysis results are then incorporated into the prompts used for similar future queries, enabling the system to gradually improve the quality of its responses over time.

Fig. 2 shows the initial User Interface (UI) design of the AIM chatbot system. On the left image, the basic conversation flow is illustrated, where the user enters a question about the AIM and a primary LLM generates a corresponding response. On the right image, the evaluation interface is shown, where quantitative evaluation metrics (ROUGE, BLEU, BERTScore) are presented for the response, and the user can select either "Good" or "Bad" and provide additional feedback. This feedback is then processed by an auxiliary LLM, which enhances subsequent prompts to improve the quality of future responses.

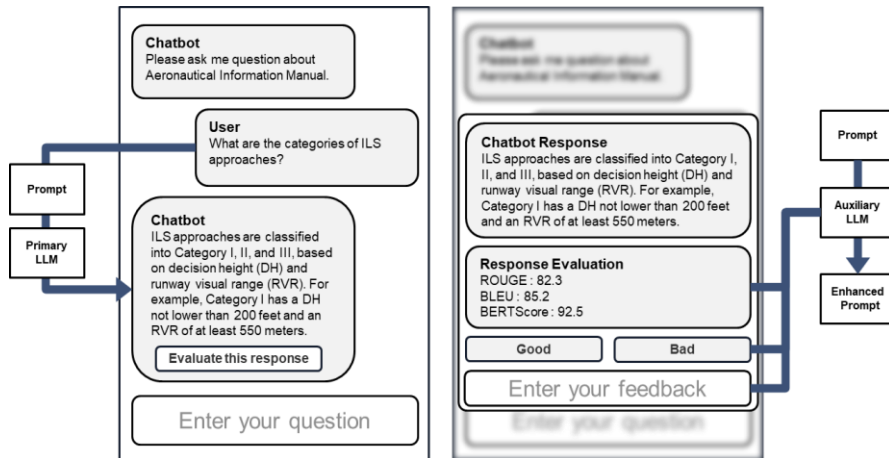


Fig. 2. Initial User Interface (UI) design for AIM chatbot system

III. CHALLENGES AND ISSUES

The AIM encompasses a wide range of content, including aviation regulations, procedures, and operational information. This breadth of content often results in inconsistent formatting and structural variations throughout the document. In particular, information presented in tables, itemized lists, and visual materials is frequently formatted in ways that are not directly compatible with RAG-based systems. Therefore, data

preprocessing procedures such as parsing table structures, extracting visual captions, and reorganizing listed content are essential and are expected to significantly influence the retrieval performance of the RAG system.

Additionally, the dual-LLM architecture adopted in the system enables adaptive prompt refinement but requires substantial computational resources, as it involves running both a primary and an auxiliary LLM in parallel. To address this challenge, the auxiliary LLM could be assigned to a separate server computer to achieve more efficient resource utilization and improved real-time responsiveness.

Furthermore, since the system refines prompts based on user feedback, inconsistent or overly brief input may hinder effective optimization. To address this, users could be guided to select from a predefined set of feedback categories, thereby improving both consistency and utility in the feedback process.

IV. CONCLUSIONS AND FUTURE STUDIES

This paper presents user-friendly dual-LLM architecture designed to enhance the quality of responses in an AIM chatbot system. By integrating an auxiliary LLM for adaptive prompt refinement based on user feedback and evaluation metrics (ROUGE, BLEU, BERTScore), the system effectively mitigates the limitations of standard RAG approaches in processing complex and domain-specific documents. This architecture is expected to enable a user-friendly chatbot system that enhances both practical usability and user satisfaction by allowing aviation professionals in diverse operational roles to tailor response formats to their specific needs.

In future work, we plan to develop preprocessing techniques to convert various data formats within the AIM, including tables, itemized lists, and visual materials, into coherent paragraph-style text. To further improve the consistency and quality of user feedback, we also intend to enhance the user interface by introducing predefined feedback categories. Additionally, we aim to explore the applicability of dual-LLM architecture to other domain-specific documents beyond the aviation field.

REFERENCES

- [1] Ministry of Land, Infrastructure and Transport (MOLIT), Aeronautical Information Manual (AIM), Republic of Korea, 2020.
- [2] P. Lewis, E. Perez, A. Piktus, F. Petroni, V. Karpukhin, N. Goyal, H. Küttler, M. Lewis, W. Yih, T. Rocktäschel, S. Riedel, and D. Kiela, “Retrieval-augmented generation for knowledge-intensive NLP tasks”, in *Advances in Neural Information Processing Systems*, vol. 33, pp. 9459–9474, 2020.
- [3] K. Papineni, S. Roukos, T. Ward, and W.-J. Zhu, “BLEU: a method for automatic evaluation of machine translation”, in *Proc. 40th Annu. Meeting of the Association for Computational Linguistics (ACL)*, Philadelphia, PA, USA, pp. 311–318, 2002.
- [4] C.-Y. Lin, “ROUGE: a package for automatic evaluation of summaries”, in *Proc. ACL Workshop on Text Summarization Branches Out*, pp. 74–81, 2004.
- [5] T. Zhang, V. Kishore, F. Wu, K. Q. Weinberger, and Y. Artzi, “BERTScore: Evaluating text generation with BERT”, in *Proc. Int. Conf. on Learning Representations (ICLR)*, 2020.

PD Location for Cast Resin Power Transformers by Rogowski-type PD sensors

Sung-Wook Kim^{1*}

¹Department of Electrical and Electronics Engineering, Silla University, 46958, Korea
[Email : number1@silla.ac.kr]

Abstract

This study proposes a novel method to locate a partial discharge (PD) source in cast resin power transformers using Rogowski-type PD sensors. A PD is a critical factor in the deterioration of insulation systems and can occur from various defects such as voids, delamination or contaminants introduced during manufacturing process or operational stress in the power transformers. When PD occurs in solid insulation systems, locating PD source is essential for effective maintenance and inspection, as it enables precise actions taken by PD location. In this paper, a method was designed to locate a PD source using multiple Rogowski-type PD sensors using the time difference of arrival (TDOA). The PD electrode systems were fabricated to simulate floating defect. By analyzing the TDOA of PD signals detected by multiple sensors, the system can estimate the exact location of PD source. From the results, this experimental validation can enhance the accuracy of PD location to enable preventive maintenance actions and improve the reliability of power transformer operations.

Index Terms: partial discharge, PD location, time difference of arrival, cast resin power transformers

I. INTRODUCTION

Partial discharge (PD) fault is one of the important factors contributing to the deterioration of insulation systems in high-voltage power equipment. In cast-resin power transformers, PD occurs from internal defects including void, delamination, or contaminants introduced during the manufacturing process. When a PD occurs in solid insulation systems, it is not only important to detect the PD fault accurately, but also critical to identify the exact location of the PD source for effective maintenance and inspection. In this paper, a method was developed to locate the PD source using multiple Rogowski-type PD sensors using the time difference of arrival (TDOA). From the results, it was confirmed that this experimental validation could enhance the accuracy of PD location and achieve a mean localization error of less than 00mm.

II. EXPERIMENTAL SETUP AND METHODS

Fig. 1 shows an experimental setup. It consists of a high-voltage transformer to apply high voltage up to 20 kV, a PD electrode system to simulate the floating defect, and three Rogowski-type PD sensors embedded on the test bench to detect PD fault and estimate the

PD source location at 460mm. Each sensor was installed 350 mm away from the PD source along the x-axis, and positioned along the y-axis at the top position (100 mm), the middle position (500 mm), and the bottom position (900 mm), respectively. For estimation of the PD source location, each PD signal was measured using a digital oscilloscope (MSO 5204B, Tektronix).

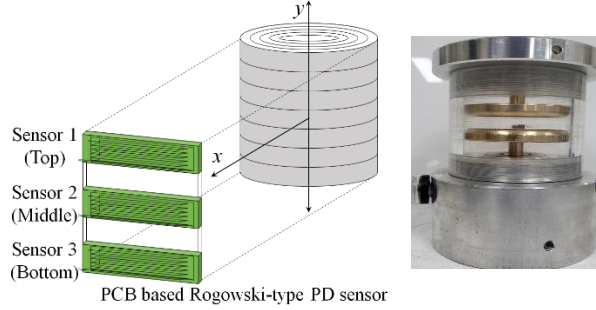


Fig. 1. Experimental setup.

III. RESULTS AND ANALYSIS

Fig 2 shows an example of PD single pulses measured by three Rogowski-type PD sensors. While the waveform shapes were nearly identical, time differences estimated using a cross-correlation algorithm were clearly observed and used to compute the location of the PD source through the TDOA method.

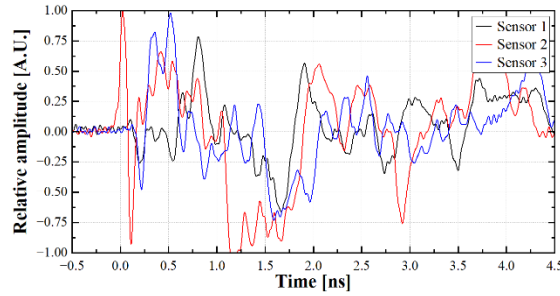


Fig. 2. An example of PD single pulses measured by each sensor

The estimated coordinates of the PD source were compared against the known physical location of the defect in the cylindrical epoxy resin insulation. Through repeated PD experiments, the proposed method consistently achieved a localization error of less than 8 mm. From the experimental results, it was confirmed that the reliability of the TDOA-based localization using Rogowski-type sensors could be validated to locate PD source for effective maintenance and inspection of cast resin power transformers.

IV. CONCLUSION

This study proposed a method to locate partial discharge (PD) sources in cast resin power transformers using multiple Rogowski-type PD sensors and the time difference of arrival (TDOA) technique. An experimental setup was made with a PD electrode system designed to simulate floating defect within a cylindrical epoxy resin insulation. The PD signals measured by the sensors were analyzed using a cross-correlation algorithm to determine the time differences to locate the PD source. Experimental results showed that the proposed method achieved a localization error of less than 10 cm. It was confirmed the reliability of the TDOA-based localization method using Rogowski-type sensors is potential for practical applications in condition monitoring and maintenance of cast resin power transformers.

REFERENCES

- [1] S. Sharifinia, M. Allahbakhshi, A. Akbari, H. R. Mirzaei, "A New Application of Rogowski Coil Sensor for Partial Discharge Localization in Power Transformers," *IEEE Sensors Journal*, vol. 21, no. 9, pp. 10743–10750, Feb. 2021.
- [2] F. Wu, L. Luo, T. Jia, A. Sun, G. Sheng, and X. Jiang, "RSSI-Power-Based Direction of Arrival Estimation of Partial Discharges in Substations," *Energies*, vol. 12, no. 18, pp. 3450, Sep. 2019.
- [3] N. Xue, J. Yang, D. Shen, P. Xu, K. Yang, Z. Zhuo, L. Zhang, and J. Zhang, "The Location of Partial Discharge Sources Inside Power Transformers Based on TDOA Database with UHF Sensors," *IEEE Access*, vol. 7, pp. 146732–146744, Oct. 2019.
- [4] N. Xue, J. Yang, D. Shen, and P. Xu, "The Location of Partial Discharge Sources inside Power Transformers Based on TDOA Database with UHF Sensors," *IEEE Access*, vol. 7, pp. 146732–146744, Oct. 2019.

Reinforcement Learning for Efficient Scheduling of Test Sequences in Multi-Room Environments

Hyeokjin Shim¹, Byung Wook Kim^{1*}

¹Department of Information and Communications Engineering, Changwon National University,
Changwon 51140, Republic of Korea
[Email : bwkim@chanwon.ac.kr]

Abstract

This paper proposes a reinforcement-learning-based scheduling method to minimize the total completion time of test sequences. Various types of tests can be performed in multiple test rooms, and the duration of each test depends on the equipment specifications. Test schedule records were analyzed, and a flowshop scheduling approach was implemented using Q-learning. Based on factors such as product type and test type, the time span of each test was estimated and used for training. The trained Q-learning model generated an optimized test-to-test room allocation matrix that reduced the total testing span. Validation with random test sets demonstrated the system's effectiveness in minimizing total test time and improving resource utilization.

Index Terms: Reinforcement-learning-based scheduling, Test completion time, Q-learning

I. INTRODUCTION

In test stations in the manufacturing environment, various tests can be performed in multiple test rooms, and the required time varies according to the test specimens and test types. While appropriately allocating these time-varying tests to each test room significantly impacts the operational efficiency of test stations, achieving optimal operations remains challenging due to continued reliance on scheduling personnel's expertise and experience. Conventional manual scheduling is inefficient and has limitations in minimizing the overall test completion time (makespan)[1]. Recent advances in deep learning technology have achieved significant progress across various application domains, with an increasing number of cases applying reinforcement learning models to scheduling problems[2]. Particularly, Q-learning-based models[3] demonstrate superior performance in scheduling problem solving due to their ability to learn optimal policies through trial-and-error interactions with the environment without requiring prior knowledge of system dynamics, and their capability to handle discrete action spaces and sequential decision-making processes effectively[4]. This study proposes a Q-learning-based scheduling system that minimizes total testing time through Q-learning-based flowshop scheduling techniques.

II. SYSTEM MODEL AND METHODS

A. Q-learning-based Scheduling Model

This study aims to optimize the scheduling of test lists using a Q-learning algorithm. Q-learning is a reinforcement learning method that consists of four key elements: state, action, reward, and policy. The agent interacts with the environment and learns a policy that maximizes the cumulative reward through repeated experiences. The following presents the Q-learning algorithm components and their application to the scheduling problem.

1) Q-learning Algorithm

- a) *Q-table: A table that stores the expected reward values for each state-action pair*
- b) *Q-function: A function that estimates the long-term reward of an action.*

2) Scheduling Application

- a) *State: The current job assignment status of each test room and unassigned tests.*
- b) *Action: The action of assigning a specific test to a specific test room.*
- c) *Reward: A higher reward is given when the total test completion time is shorter*
- d) *Policy: A strategy for selecting optimal test assignments in the current state.*
- e) *Q-table: A table storing expected rewards for each test assignment action.*

B. Q-learning Training Process

After data preprocessing, the Q-learning training process begins by matching each selected test instance (defined by product and test type) to its corresponding job ID. The training is conducted over 10 epochs, with each epoch consisting of 10,000 iterations in which the Q-function and V-function are iteratively updated. An ϵ -greedy policy[5] is employed, where a random action is selected with probability ϵ in the early stages of training. As training progresses, the policy gradually shifts toward selecting the optimal action with probability $(1-\epsilon)$. The Q-function is initialized as a square matrix of size (number of jobs $\times$ number of jobs), with all values set to zero. The V-function represents the expected cumulative reward from each state, combining immediate costs and future returns. These two functions are updated according to the following equations:

$$Q[s][a] = Q[s][a] + \alpha \times (\text{cost} + \gamma \times \text{lowest_cost}(Q, s') - Q[s][a]) \quad (1)$$

$$V[s] = V[s] + \alpha \times (\text{cost} + \gamma \times \text{lowest_cost}(Q, s') - V[s]) \quad (2)$$

Here, α is the learning rate, cost refers to the scheduling cost associated with the test plan, and γ is the discount factor. The function $\text{lowest_cost}(Q, s')$ returns the minimum expected cost from the next state s' . After the updates in equations (1) and (2) are applied iteratively, the Q-table converges to a policy that yields the optimal test sequence minimizing the total scheduling cost.

C. Scheduling Matrix Algorithm

Each test is assigned to a specific test room, and the scheduling matrix is used to compute the cumulative workload for each test room. The makespan, which represents the total completion time, is calculated as the maximum workload across all test rooms. To minimize the makespan, a Q-learning algorithm is employed. As illustrated in Fig. 1, the overall scheduling optimization process is composed of both the scheduling matrix computation and the Q-learning loop. Beginning with a list of tests, a new test sequence is generated in each episode using an ϵ -greedy policy. Based on this sequence, the scheduling matrix is updated to calculate cumulative workloads and the resulting makespan. A reward is computed according to the makespan, and the Q-function and value function are updated accordingly. The ϵ value is gradually reduced to encourage exploitation in later episodes. This process is repeated over multiple episodes, and the optimal test sequence is finally obtained from the Q-values that yield the shortest makespan.

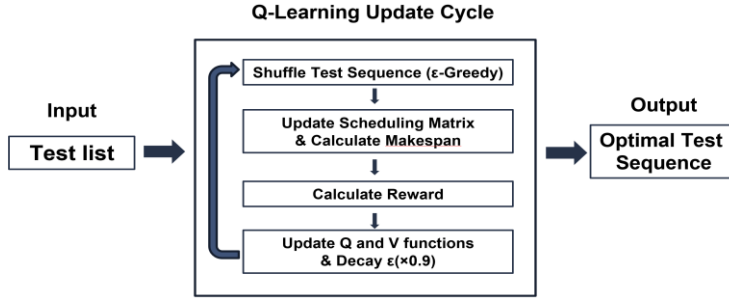


Fig. 1 Q-learning-based scheduling matrix updates

III. RESULTS

A. Performance Evaluation

To validate the performance of the proposed Q-learning-based scheduling algorithm, experiments were conducted using scheduling matrix algorithms to deploy test sets of varying scales 25, 50, and 100 tests across 5 test rooms to derive optimal scheduling results. Training was performed over 10 epochs, as shown in Fig. 2.

Q-learning Convergence Analysis by Test Set Size

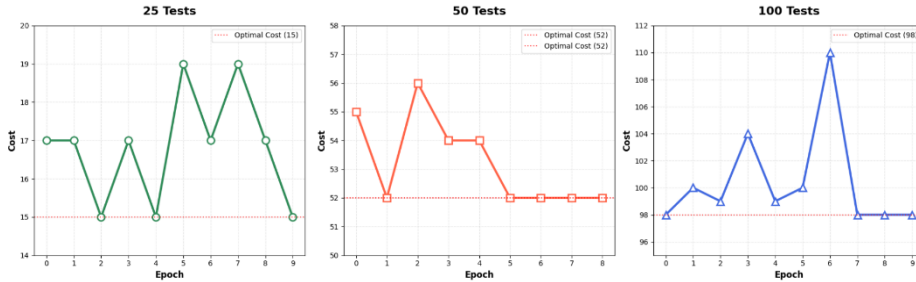


Fig. 2 Convergence analysis of the Q-learning algorithm for different test set sizes. The cost represents the total test time (makespan).

The initial cost represents the makespan (total completion time) obtained from random test scheduling without optimization, while the final cost indicates the optimized makespan achieved after applying the Q-learning algorithm. The experimental results demonstrate significant makespan improvements across different scales: the 25-test scenario achieved a makespan reduction from 17 to 15 time units (11.8% improvement), the 50-test scenario showed a makespan reduction from 55 to 52 time units (5.5% improvement), and the 100-test scenario optimized the makespan from initial 98 to final 98 time units, confirming convergence to the optimal solution. The Q-learning algorithm demonstrated stable convergence across all test scales, with the ϵ -greedy strategy effectively balancing exploration and exploitation, validating the proposed system's effectiveness for test facility scheduling optimization.

IV. Conclusion

This study developed a system for effectively minimizing total test completion time across multiple test rooms through Q-learning-based scheduling. The system generates optimal test-test room matching matrices through Q-learning training based on test data characteristics including product types and test categories, with continuous reduction in total test time confirmed as epochs increased. Validation results on test sets of 25, 50, and 100 tests demonstrated that the proposed system successfully minimizes total test time, achieving improved test facility management efficiency and maximized resource utilization. These research findings contribute to reducing decision-making time and operational costs in industrial settings by transforming conventional manual scheduling approaches into automated intelligent systems, presenting a versatile solution applicable to production scheduling optimization problems across various manufacturing sectors.

REFERENCES

- [1] S. Albers and A. Eckl, "Scheduling with testing on multiple identical parallel machines," *arXiv preprint arXiv:2105.02052*, May 2021.
- [2] C. Zhang, W. Song, Z. Cao, J. Zhang, P. S. Tan, and C. Xu, "Learning to dispatch for job shop scheduling via deep reinforcement learning," in *Advances in Neural Information Processing Systems*, vol. 33, pp. 1621-1632, 2020.
- [3] C. J. C. H. Watkins and P. Dayan, "Q-learning," *Machine Learning*, vol. 8, no. 3-4, pp. 279-292, 1992.
- [4] V. Mnih, K. Kavukcuoglu, D. Silver, A. A. Rusu, J. Veness, M. G. Bellemare, A. Graves, M. Riedmiller, A. K. Fidjeland, G. Ostrovski, S. Petersen, C. Beattie, A. Sadik, I. Antonoglou, H. King, D. Kumaran, D. Wierstra, S. Legg, and D. Hassabis, "Human-level control through deep reinforcement learning," *Nature*, vol. 518, no. 7540, pp. 529-533, 2015.
- [5] R. S. Sutton and A. G. Barto, *Reinforcement Learning: An Introduction*, 2nd ed., Cambridge, MA: MIT Press, 2018.

A Study on Forage Grass Yield Prediction Model Applying LASSO Regression

Moon-Sun Shin^{1*}, Sang-Ho Moon², and Seung-Hak Yang³

¹Department of Computer Engineering, Konkuk University, Chungju 27478, Korea

²Department of Food Science, Konkuk University, Chungju 27478, Korea

³Division of Forage Crop, National Institute of Animal Science, Cheonan 31000, Korea

[Email: msshin@kku.ac.kr, moon0204@kku.ac.kr, y64h@korea.kr]

Abstract

Due to extreme climate change, livestock farms are increasingly suffering losses in forage crop yields. Therefore, it is essential to assess the vulnerability of forage crop yields and changes in suitable cultivation areas under the impact of climate change. In this study, long-term yield data and climate data for northern-type forage grasses were collected, and LASSO regression was applied to analyze and predict yield changes due to climate change. Experiments were also conducted to optimize the model's hyperparameters. The developed LASSO model for forage grass yield prediction can be utilized to forecast the impacts of climate change on forage crop productivity, identify key influencing climatic factors, and serve as a basis for predicting shifts in suitable cultivation zones.

Index Terms: Climate Change, Forage Crop Yield Prediction, Lasso Regression.

I. INTRODUCTION

With the increasing frequency and severity of abnormal climate events worldwide, the production of forage crops in the livestock sector is declining, and cultivation areas are shifting accordingly. However, comprehensive assessments of the climate vulnerability of forage crops remain insufficient. Therefore, it is necessary to establish appropriate indicators and methodologies for evaluating the impacts and vulnerabilities related to changes in suitable cultivation areas and productivity of forage crops under climate change.

To ensure the stable production and supply of high-quality roughage, research is urgently required on the cultivation areas, production status, and climate vulnerability of forage crops. Through such studies, it will be possible to identify critical climatic factors that significantly affect forage crop production and evaluate their impact on productivity. This knowledge can serve as a foundation for securing stable forage supply and preparing countermeasures against damages caused by extreme weather events.

In this paper, we apply Lasso regression to develop a predictive model for forage crop yield that supports vulnerability analysis under climate impacts. Key climatic variables influencing forage crop yield will be identified, and hyperparameter tuning will be conducted to derive an optimal Lasso regression model. The resulting model can be used to analyze shifts in suitable cultivation zones for forage crops due to abnormal temperature patterns and inform future agricultural planning and policy decisions.

II. BUILDING BASIC DATABASE

The foundational dataset was constructed using forage yield data (dry matter yield, kg/ha) obtained from a forage productivity survey and meteorological data from the Automated Synoptic Observing System (ASOS) provided by the Open MET Data Portal. The yield data were organized by year and region based on dry matter yield to build the initial database. From the ASOS data, various climate indicators such as monthly average temperature, minimum temperature, maximum temperature, precipitation, and GDDFJTA (Growing Degree Days from January to April) were collected on a yearly and regional basis. These indicators were compiled together with the forage yield data to establish the foundational database. Figure 1 presents the constructed basic dataset of forage crop yield data and involved climate data.

	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O
1	DMYTOTA	GDDTOTAL	DDAYS	MAXTAUL	MAXTAUG	MAXTMAR	MAXTNV	MTJAN	MTFEB	MTMAR	PREDJUL	PREDNOV	MINTJAN	MINTFEB	MINTMAR
2	13330	3201.3	1	31.1	30.3	9.8	9.6	-0.5	-0.2	3.6	13	14	-4.7	-5	-2.4
3	12120	3201.3	1	31.1	30.3	9.8	9.6	-0.5	-0.2	3.6	13	14	-4.7	-5	-2.4
4	12660	3201.3	1	31.1	30.3	9.8	9.6	-0.5	-0.2	3.6	13	14	-4.7	-5	-2.4
5	9930	3201.3	1	31.1	30.3	9.8	9.6	-0.5	-0.2	3.6	13	14	-4.7	-5	-2.4
6	13180	3201.3	1	31.1	30.3	9.8	9.6	-0.5	-0.2	3.6	13	14	-4.7	-5	-2.4
7	9750	3038.7	4	27.5	29.7	8.4	10.7	-4.6	-2.9	2.9	12	6	-10.2	-7.7	-2.1
8	9340	3038.7	4	27.5	29.7	8.4	10.7	-4.6	-2.9	2.9	12	6	-10.2	-7.7	-2.1
9	6310	3038.7	4	27.5	29.7	8.4	10.7	-4.6	-2.9	2.9	12	6	-10.2	-7.7	-2.1
10	6340	3038.7	4	27.5	29.7	8.4	10.7	-4.6	-2.9	2.9	12	6	-10.2	-7.7	-2.1
11	6790	3038.7	4	27.5	29.7	8.4	10.7	-4.6	-2.9	2.9	12	6	-10.2	-7.7	-2.1
12	6630	3038.7	4	27.5	29.7	8.4	10.7	-4.6	-2.9	2.9	12	6	-10.2	-7.7	-2.1

Fig. 1. Basic Dataset of Forage Crop Yield and Involved Climate Data(N=1068)

III. FEATURE SELECTION

The climatic factors found to influence pasture yield include the accumulated effective temperature, drought days (DDAYS) counted as the number of consecutive 10-day periods without rainfall, maximum temperature during summer, minimum temperature during winter, and the number of precipitation days in July and August.

Table 1. Feature Selection

Category	Feature	Description
Precipitation	PREDJUL	Precipitation Days in Jul.
	PREDJUL	Precipitation Days in Jul.
	PREDJUL	Precipitation Days in Jul.
Temperature	DDAYS	Drying Days
	MAXTJUL	Max Temperature in July
	MAXTAUG	Max Temperature in Aug.
	MINTJAN	Min Temperature in Jan.
	MINTFEB	Min Temperature in Feb.
GDD	GDDFJTA	Effective accumulated temperature from Jan. to Apr.
	GDDTOTAL	Effective accumulated temperature
Dry Matter Yield	DMYTOTAL	Total DMY

These selected climatic variables are presented in Table 1. Due to the nature of pasture forage crops, which are affected by climatic conditions throughout the entire year following the establishment of pasture, their correlation with specific climate variables appears to be less concentrated compared to winter forage crops such as Italian ryegrass. Nevertheless, the results indicate that pasture yield is relatively influenced by the maximum temperature during July and August, as well as precipitation in August. In addition, the drought indicator DDAYS—defined as the number of periods exceeding 10 consecutive days without rainfall—was also found to have a notable impact.

IV. LASSO MODEL OPTIMAIZATION

To analyze the relationship between pasture productivity and climatic variables, a LASSO regression model was applied, with dry matter yield set as the dependent variable and climate factors as the independent variables. In multivariate regression analysis, multicollinearity among the independent variables posed a challenge; therefore, the LASSO model was employed to address this issue. The predictive model was implemented using Python.

```
herb_all = pd.read_csv('/content/drive/MyDrive/herb_all_2024_N1068.csv', sep=',')

X = herb_all[['GDDFJTA', 'GDDTOTAL', 'DDAYS', 'MAXTJUL', 'MAXTAUG', 'MINTJAN', 'MINTFEB', 'PREDJUL', 'PREDAUG']]
Y = herb_all['DMYTOTAL']

X_train, X_test, Y_train, Y_test = train_test_split(X, Y, random_state=1000)

# Lasso Regression
lasso_coef_df = pd.DataFrame()
lasso_reg = Lasso(alpha=1, max_iter=10000)
lasso_reg.fit(X_train, Y_train)
lasso_pred = lasso_reg.predict(X_test)
lasso_r2 = r2_score(Y_test, lasso_pred)

lasso_coef = pd.Series(data=lasso_reg.coef_, index=X.columns)
print('Lasso R2 Score : ', lasso_r2)

print(lasso_coef)
print('Intercept : ', lasso_reg.intercept_)

Lasso R2 Score : 0.5345728424016352
GDDFJTA      14.137054
GDDTOTAL     -3.150526
DDAYS        -390.096832
MAXTJUL       542.077838
MAXTAUG       122.262670
MINTJAN      -220.418687
MINTFEB       216.813390
PREDJUL       -80.278750
PREDAUG      -105.852176
dtype: float64
Intercept : 1753.7907716134923
```

Fig. 2. Lasso Model for Prediction of Forage Crop Production

Figure 2 illustrates the results of applying the LASSO model in Python to estimate pasture yield. The coefficient of determination (R^2) was found to be approximately 0.53457, indicating a moderate explanatory power; however, further improvement in model performance is deemed necessary.

Table 2. Experiments of Optimization of hyper parameter alpha

alpha α	R^2	alpha α	R^2	alpha α	R^2
$\alpha = 1$	0.5345728	$\alpha = 100$	0.534995	$\alpha = 200$	0.5343111
$\alpha = 0.01$	0.53456318	$\alpha = 1000$	0.5275467	$\alpha = 300$	0.5335289
$\alpha = 0.0001$	0.5345630	$\alpha = 10000$	0.2664168	$\alpha = 150$	0.5347986

In addition, to optimize the hyperparameters of the LASSO model, it was necessary to test a range of alpha (α) values. The model presented in Figure 2 was generated using an alpha value of 1. To identify the optimal alpha, multiple experiments were conducted with alpha values ranging from 0.001 to 10000. The results of these experiments are summarized in Table 2. Among the tested values, the highest R^2 score was obtained when alpha was set to 100.

V. CONCLUSIONS

In this study, we collected data on pasture yield along with related climatic variables and developed a yield prediction model using the LASSO (Least Absolute Shrinkage and Selection Operator) regression method to analyze the climate vulnerability of forage crops specifically pasture grass. Through this approach, we identified the key climatic factors affecting forage crop productivity. Furthermore, we conducted hyperparameter optimization experiments to derive the optimal LASSO model. In future work, we plan to validate the predictive performance of the proposed LASSO-based pasture yield model.

ACKNOWLEDGMENTS

This research was funded by the Cooperative Research Program for Agriculture Science and Technology Development (No. RS-2024-00400758), Rural Development Administration, Republic of Korea.

REFERENCES

- [1] Shin, M.; Hwang, S.; Kim, J.; Kim, B.; Jung, J.-S. A Study on Analyses of the Production Data of Feed Crops and Vulnerability to Climate Impacts According to Climate Change in Republic of Korea. *Appl. Sci.* 2023, *13*, 11603. <https://doi.org/10.3390/app132011603>
- [2] H.Jun, S.H.Lee, W.C.Lee, S.W.Lee, S.M.Hwang and M.S.Shin, "Analysis of Italian Ryegrass Production and Vulnerability Effects of Climate Factors", in Proceeding of the KAIS conference, Daejeon: pp. 528-531, 2021.
- [2] M.S.Shin, S.W.Lee, A.R.Jeong, and S.H.Moon, "Big data analysis for productivity change in feed crops due to climate change" in Proceeding of the KAIS conference, Kyongju: pp. 474-477, 2021.
- [3] M.S.Shin, S.W.Lee, S.M.Hwang and S.H.Moon, "Correlation Analysis of Production Changes of Feed Crop Data according to Climate Change", in Proceeding of the 14th Annual International Conference on Future Information and Communication Engineering, Jeju: pp. 149-152, 2022.
- [4] Jeong, S.J., Park, H.S., Ji, H. J., Kim, Y. K., Lee, S. Y. and Lee, B. H. "Predicting Changes in the Suitable Agro-climate Zone of Italian Ryegrass Cultivars with RCP 8.5 Climate Change Scenario" *Journal of the Korean Society of Grassland and Forage Science* 40(4):196-204, 2020.
- [5] Kim, M.J. and Sung, K.I. "Comparison of Causality of Temperature and Precipitation on Italian Ryegrass (*Lolium Multiflorum* Lam.) Yield between Cultivation Fields via Multi-Group Structural Equation Model Analysis in the Republic of Korea." *Agriculture* 9: 254, 2019.

Real-Time Construction Hazard Detection with SRGAN and Transformer Architectures

Bubryur Kim¹, Hyejune Kim^{1*}

¹Department of Robot and Smart System Engineering, Kyungpook National University, 80, Daehak-ro, Buk-gu, Daegu, 41566, South Korea

²School of Space Engineering Sciences, Kyungpook National University, 80, Daehak-ro, Buk-gu, Daegu, 41566, South Korea

³Department of Safety Convergence, Kyungpook National University, 80, Daehak-ro, Buk-gu, Daegu, 41566, South Korea

[Email : brkim@knu.ac.kr, khj9931@knu.ac.kr]

Abstract

Ensuring safety in construction sites is a critical challenge due to the presence of heavy machinery. This study proposes a real-time hazard detection system integrating deep learning techniques to enhance safety monitoring. The framework combines video super-resolution with SRGAN, object detection using RT-DETR, and behavior classification via DINOv2 and DeiT. SRGAN improves low-quality surveillance footage, enabling better detection accuracy. RT-DETR identifies workers and machinery efficiently, and transformer-based classifiers recognize hazardous interactions. Experimental results show a mean Average Precision of 97.3% and 29 FPS processing speed. The system maintains high performance under low light, occlusion, and motion blur. This research demonstrates the feasibility of AI-based safety systems for dynamic construction environments and lays the foundation for future work involving multi-sensor integration and edge computing deployment.

Index Terms: Construction safety, Deep learning, Object detection, Super-resolution

I. INTRODUCTION

Construction sites are inherently hazardous due to the frequent interaction between heavy machinery and human workers. According to the International Labour Organization (ILO), the construction sector ranks among the most dangerous industries worldwide, with frequent incidents resulting from equipment collisions, falls, and the mishandling of materials [1]. Traditional safety monitoring methods, including manual supervision and fixed-position surveillance cameras, have limited capability in detecting dynamic or evolving hazards, especially under poor lighting conditions, occlusions, or motion blur. Although recent advances in deep learning have enabled automated detection of personal protective equipment (PPE), such as helmets and safety harnesses [2, 3], these approaches remain insufficient for identifying unsafe interactions between personnel and machinery, as well as structural anomalies such as misaligned rebar [4]. To overcome the limitations of low-resolution or degraded surveillance imagery, super-resolution techniques including SRGAN, SeD, and RDN have been utilized. These models apply adversarial training and semantic-aware mechanisms to reconstruct fine visual details, with proven effectiveness in

domains such as satellite and aerial imaging [5–8]. In object detection, two-stage models like Faster R-CNN provide high accuracy but require substantial computational resources, which limits their suitability for time-sensitive applications [9]. In contrast, transformer-based detectors such as DETR and Deformable DETR support fast and efficient inference while maintaining a high level of contextual understanding, and have shown superior performance compared to YOLO-type architectures in complex scenes [10–12]. In parallel, image classification models based on transformers, including DINOv2 and DeiT, leverage self-supervised learning to extract robust features even with minimal labeled data [13, 14]. In addition, anomaly detection approaches like MAD-GAN demonstrate the potential to identify hazardous behaviors by modeling irregular activity patterns from video streams [15]. Motivated by these advancements, the present work introduces a unified hazard detection framework designed for real-time application in construction environments. This system incorporates SRGAN for enhancing video quality, RT-DETR for transformer-based object recognition, and DINOv2 and DeiT for behavior classification, providing a reliable and scalable solution for safety monitoring under challenging visual conditions.

II. SYSTEM MODEL AND METHODS

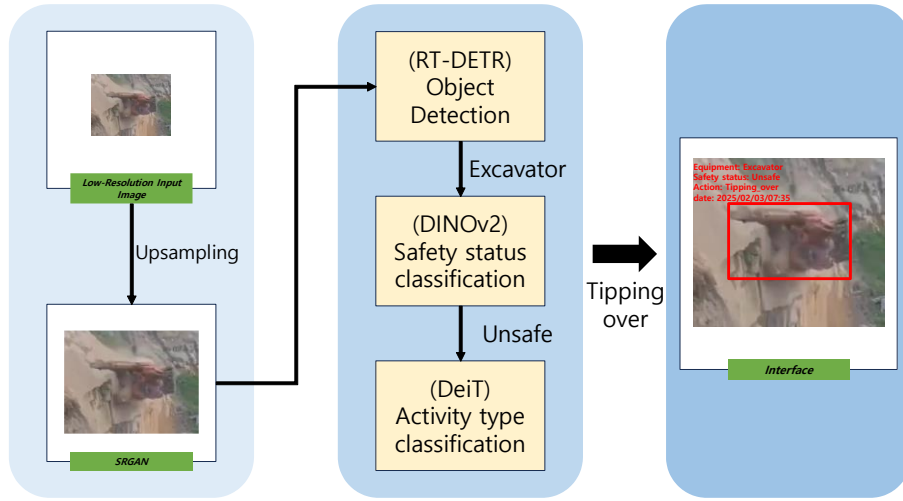


Fig. 1. Workflow of the Proposed Framework

The proposed system is a real-time hazard detection framework for construction environments, consisting of three primary modules: super-resolution, object detection, and behavior classification. As shown in Fig. 1, the system processes surveillance footage often degraded by low resolution, motion blur, or poor lighting. To address these issues, SRGAN is employed to enhance image quality by reconstructing fine structural details. The enhanced frames are analyzed by RT-DETR, a transformer-based detector that performs real-time inference and accurate localization without anchor boxes. RT-DETR identifies dynamic entities such as workers, machinery, and hazardous materials. For analyzing interactions, the system uses transformer-based classifiers like DINOv2 and DeiT, trained via self-supervised learning to extract semantic features and recognize risky

behaviors, such as unsafe proximity between personnel and equipment. Upon detecting a hazardous condition, the system overlays visual alerts on the video stream and sends notifications to safety personnel. All events are logged with timestamps to support post-incident analysis and ongoing safety improvements.

III. RESULTS

The proposed system was tested using construction site video under low lighting, occlusion, and motion blur. Performance was evaluated using mAP, precision, recall, false positive rate (FPR), and inference speed (FPS). As shown in Table 1, RT-DETR outperformed Faster R-CNN and YOLOv5, achieving 97.3% mAP, 96.8% precision, and 97.1% recall. It also met real-time requirements, operating at 29 FPS with 36.3 ms latency. SRGAN improved clarity in low-light scenes, and RT-DETR maintained accuracy under visual degradation. Removing the super-resolution module reduced detection in dark conditions, while excluding the classification module increased false positives, highlighting the importance of each component.

Table 1. Performance of the Proposed System

Model/Configuration	mAP (%)	Precision (%)	Recall (%)	FPS	FPR (%)
Faster R-CNN	85.2	83.7	82.9	15	6.1
YOLOv5	91.2	90.5	90.0	25	4.3
RT-DETR (Proposed)	97.3	96.8	97.1	29	2.5
w/o SRGAN	89.4	–	–	30	5.6
w/o Anomaly Detection Module	94.2	–	–	32	4.1

IV. DISCUSSION AND CONCLUSIONS

A real-time hazard detection framework was developed to enhance construction site safety by combining SRGAN for video super-resolution, RT-DETR for object detection, and transformer-based classifiers such as DINOv2 and DeiT for behavior analysis. The system addresses key challenges in surveillance footage, including low resolution, poor lighting, and occlusion. Evaluation results indicate high performance, achieving a mean Average Precision of 97.3% while maintaining 29 FPS, satisfying real-time operational demands. Ablation studies confirm the complementary roles of each module, particularly under visually degraded and dynamic conditions. Compared to baseline models such as Faster R-CNN and YOLOv5, the proposed approach demonstrates improved accuracy and efficiency. Future research will explore multi-sensor integration, deployment on edge devices, and adaptive learning strategies to ensure sustained performance across diverse construction scenarios.

ACKNOWLEDGMENTS

This work is supported by the Korea Agency for Infrastructure Technology Advancement (KAITA) grant funded by the Ministry of Land, Infrastructure and Transport (Grant RS-2023-00251002). This work was supported by the National Research

Foundation of Korea(NRF) grant funded by the Korea government(MSIT) (No. 2022R1C1C1005409).

REFERENCES

- [1] International Labour Organization, "Construction: A hazardous work," 2023.
- [2] W. Kim, D. Kim, and S. Kim, "Deep Learning-Based Safety Helmet Detection in Construction Sites," *IEEE Access*, vol. 9, pp. 37850–37859, 2021.
- [3] W. Fang, L. Ding, H. Luo, and P. E. D. Love, "Falls from Heights: A Computer Vision-Based Approach for Safety Harness Detection," *Automation in Construction*, vol. 110, p. 103013, 2020.
- [4] C. Zhou, L. Ding, and L. Chen, "Computer Vision-Based Steel Bar Counting and Safety Detection: Deep Learning Algorithms and Applications," *Journal of Computing in Civil Engineering*, vol. 35, no. 3, p. 04021001, 2021.
- [5] C. Ledig et al., "Photo-Realistic Single Image Super-Resolution Using a Generative Adversarial Network," *IEEE Transactions on Pattern Analysis and Machine Intelligence*, vol. 40, no. 2, pp. 4681–4690, 2017.
- [6] B. Li, X. Li, H. Zhu, Y. Jin, R. Feng, and Z. Chen, "SeD: SemanticAware Discriminator for Image Super-Resolution," *IEEE Transactions on Image Processing*, vol. 33, pp. 1234–1245, 2024.
- [7] Y. Zhang, Y. Tian, Y. Kong, B. Zhong, and Y. Fu, "Residual Dense Network for Image Super-Resolution," *IEEE Transactions on Pattern Analysis and Machine Intelligence*, vol. 41, no. 12, pp. 2483–2497, 2018.
- [8] S. Park and J. Kim, "Deep Back-Projection Networks for SuperResolution," *IEEE Transactions on Pattern Analysis and Machine Intelligence*, vol. 42, no. 9, pp. 2161–2174, 2020.
- [9] S. Ren, K. He, R. Girshick, and J. Sun, "Faster R-CNN: Towards realtime object detection with region proposal networks," *IEEE Transactions on Pattern Analysis and Machine Intelligence*, vol. 39, no. 6, pp. 1137–1149, 2017.
- [10] N. Carion, F. Massa, G. Synnaeve, N. Usunier, A. Kirillov, and S. Zagoruyko, "End-to-End Object Detection with Transformers," *IEEE Transactions on Pattern Analysis and Machine Intelligence*, vol. 43, no. 12, pp. 4035–4046, 2020.
- [11] X. Zhu, W. Su, L. Lu, B. Li, X. Wang, and J. Dai, "Deformable DETR: Deformable Transformers for End-to-End Object Detection," in *International Conference on Learning Representations (ICLR)*, 2021.
- [12] Y. Zhao, W. Lv, S. Xu, J. Wei, G. Wang, Q. Dang, Y. Liu, and J. Chen, "DETRs Beat YOLOs on Real-time Object Detection," *IEEE Transactions on Pattern Analysis and Machine Intelligence*, vol. 46, no. 1, pp. 1–14, 2024.
- [13] M. Oquab, T. Darcet, T. Moutakanni, H. Vo, M. Szafraniec, V. Khalidov, P. Fernandez, et al., "DINOv2: Learning Robust Visual Features without Supervision," *IEEE Transactions on Pattern Analysis and Machine Intelligence*, vol. 45, no. 7, pp. 1234–1250, 2023.
- [14] H. Touvron, M. Cord, M. Douze, F. Massa, A. Sablayrolles, and H. Jegou, "Training Data-Efficient Image Transformers & Distillation through Attention," in *Proceedings of the 38th International Conference on Machine Learning (ICML)*, 2021, pp. 10347–10357.
- [15] W. Sultani, C. Chen, and M. Shah, "Real-World Anomaly Detection in Surveillance Videos," *IEEE Transactions on Pattern Analysis and Machine Intelligence*, vol. 40, no. 10, pp. 6479–6488, 2018.

Multi-Level Security Architecture for Cloud Collaboration Tools (SaaS)

Da-Eun Oh¹, Ho-Jun Lee¹, DongHwi Lee^{1*}

Department of information security, Dongshin University, 58245, Republic of Korea
[Email : dauenhi@dsu.ac.kr, lhj011126@dsu.ac.kr, dhclub@dsu.ac.kr]

Abstract

Cloud collaboration tools (SaaS) provide benefits such as centralized data management, accessibility, and flexibility. However, their adoption is limited by security concerns under current network separation policies. This paper proposes a multi-level security (MLS) system that combines WebAuthn authentication, UTM-based access control, and network monitoring to mitigate potential Adversary-in-the-Middle (AiTM) attacks.

Index Terms: Multi-Level Security (MLS), Adversary in the Middle (AiTM), Network Access Control (NAC), Software as a Service (SaaS), Unified Threat Management (UTM)

I. INTRODUCTION

SaaS enhances productivity but faces challenges in network-separated environments due to its reliance on internet connectivity. Uncontrolled SaaS adoption increases data leakage risks. Although multi-factor authentication (MFA) mitigates some threats, AiTM attacks that bypass MFA continue to evolve. To address these issues, a multi-level security system is proposed, integrating UTM-based web filtering, NAC, and WebAuthn authentication to ensure secure SaaS access.

II. Literature Review

A. Major Security Vulnerabilities in SaaS

SaaS allows users to access software provided by suppliers directly over the network without installation, offering high accessibility and ease of use. Although SaaS services are rapidly expanding, security threat cases caused by supply chain management issues and network connections within business networks are increasing.

Table 1. Security Threats in SaaS Environments

Vulnerability	Description
Man-in-the-Middle Attack [1]	Stealing authentication information through DNS landing page spoofing and session hijacking. <i>Example:</i> The hacking group Lapsus\$ spoofed the Microsoft Office 365 login page to bypass MFA targeting over 10,000 users and gained unauthorized access to email accounts.
Access Control Misconfiguration [2]	Incorrectly configured permissions allow unauthorized access to resources or functions, leading to information leakage and system damage. <i>Example:</i> The default Get User permission was excessively broad, making it possible for customer information of multiple institutions to be exposed without authentication, leading to proactive response.

* Corresponding author

Authentication Bypass [3]	Unauthenticated or unauthorized users access accounts by appearing as authenticated users. <i>Example:</i> A Russian hacker group phished OAuth authentication codes from users, bypassed formal authentication procedures, gained unauthorized access to Microsoft 365 accounts, and continuously infiltrated by registering devices.
---------------------------	---

B. Adversary in the Middle Attacks

AiTM is an advanced form of man-in-the-middle attack that can bypass MFA introduced to prevent man-in-the-middle attacks. AiTM attacks employ phishing and proxy techniques to redirect users to fake login pages, rather than merely acting as simple relays.[1] The reason AiTM is dangerous is that it is used together with Business Email Compromise (BEC) to impersonate identities and exploit trust relationships to infiltrate organizations.

C. Multi Level Security

Multi-level security involves applying tiered security controls based on the classification of business systems and data into levels such as Classified(C), Sensitive (S), and Open (O).[5] Applying security technologies such as encryption, access control, and security auditing in multiple levels according to data classification levels, data is protected from both internal and external threats.

E. Unified Threat Management

UTM (Unified Threat Management) is a security solution that integrates various security functions into a single platform for centralized management. It centrally manages security features such as IPS, web filtering, VPN, and DLP, and provides services to detect and respond to security threats occurring both inside and outside the network.[6]

III. SYSTEM MODEL AND METHODS

A. Multi-Level Security Network for SaaS

Data is classified (C/S/O) and network separation is enforced. WebAuthn authentication is integrated for business networks, enhancing SaaS access security. Access to SaaS is restricted to authorized users with RBAC and ABAC policies.

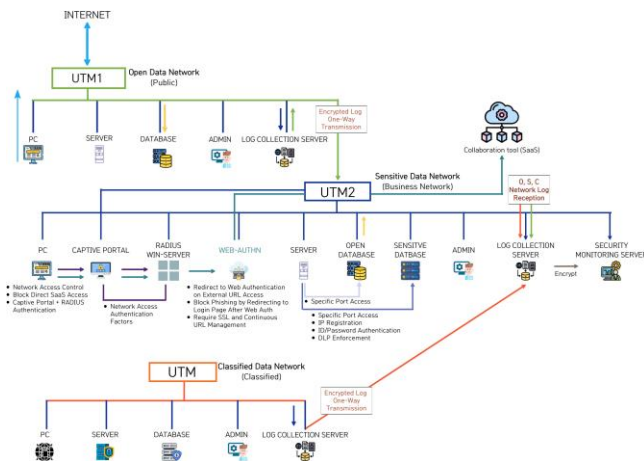


Fig. 1. Multi-Level Security (MLS) Network for SaaS in Enterprise Environments

B. Building Web Authentication for SaaS Access

The SimpleWebAuthn library was used to implement a custom web authentication system, with the server and APIs built using Express and Node.js. User management was handled using MongoDB, and various packages provided by Node.js were employed.

Table 2. WebAuthn Implementation Environment

Category	Description
Backend Runtime	Node.js (supports ESM modules)
Web Framework	Express
Web Authentication Library	@simplewebauthn/server (supports WebAuthn / FIDO2)
Database	MongoDB (mongoose ODM)
Environment Variables Management	dotenv
HTTPS Server	Node.js built-in https module

C. Countermeasures Against Adversary-in-the-Middle Attacks

Certificates are distributed to each device, and network access policies and data access controls are configured using UTM. A whitelist of URLs related to collaboration tools is added through web filtering settings.

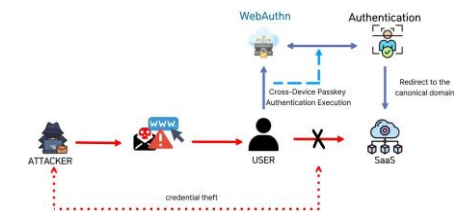


Fig. 2. Secure Redirect via WebAuthn



Fig. 3. Passkey Authentication Flow

Access to collaboration tools (SaaS) is restricted to official domains only. When attempting to access shared URLs received via email or URLs containing specific paths, users are redirected to a web authentication page. After web authentication, users are directed to the SaaS home page specified on the UTM web authentication page, which terminates connections to attacker-controlled servers and blocks malicious entry routes. To implement and manage an in-house web authentication system, continuous management of SSL certificates and URLs related to collaboration tools is required.

D. Security Management of Internet-Connected Internal Networks

Logs contain data such as user account information and sensitive database network connection details, so encryption was applied to prevent potential security threats during the collection process.

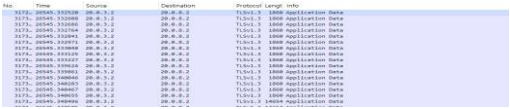


Fig. 4. TLS-Protected Log Transmission Traffic

Log collection servers are deployed on endpoints within each network segment, with secure, encrypted, unidirectional transmission configured to the business network's central

log collection server. Distributed logs are visualized and analyzed in real-time using the ELK stack, enabling centralized monitoring for rapid detection and response. Upon detection of abnormal traffic, the affected system is isolated and network access is blocked through UTM to prevent further attacks.

IV. DISCUSSION AND CONCLUSIONS

A. Discussion

The proposed MLS system integrates WebAuthn authentication with traditional captive portal methods to bolster SaaS security. Data classification-based network separation, internal web authentication, and real-time threat monitoring enhance defense against AiTM attacks.

B. Conclusions

As IT threats grow sophisticated, businesses must balance security with operational costs. This paper demonstrates a lightweight security solution leveraging existing UTM infrastructure and open-source tools to improve security and operational efficiency.

ACKNOWLEDGMENTS

These results were supported by the Innovation Convergence Campus Project Group at Dongshin University, funded in 2025 by the Ministry of Land, Infrastructure and Transport (MOLIT) with support from Jeollanam-do Province and Naju City. (No. 2025ICCS-001)

REFERENCES

- [1] Microsoft, "From cookie theft to BEC: Attackers use AiTM phishing sites as entry point to further financial fraud," Microsoft Security Blog, Jul. 12, 2022. [Online]. Available: <https://www.microsoft.com/en-us/security/blog/2022/07/12/from-cookie-theft-to-bec-attackers-use-aitm-phishing-sites-as-entry-point-to-further-financial-fraud/>
- [2] Varonis, "Abusing misconfigured Salesforce Communities for recon and data theft," Varonis Blog, 2021. [Online]. Available: <https://www.varonis.com/blog/abusing-salesforce-communities>. [Accessed: June. 6, 2025].
- [3] S. Gallagher, "Phishing for Codes: Russian Threat Actors Target Microsoft 365 OAuth Workflows," *Volexity*, Apr. 22, 2025. [Online]. Available: <https://www.volexity.com/blog/2025/04/22/phishing-for-codes-russian-threat-actors-target-microsoft-365-oauth-workflows/>
- [4] M.-M. Han and J.-H. Sun, "The detection system for hosts infected malware through behavior information of NAC post-connect," *The Journal of Korean Association of Computer Education*, vol. 13, no. 6, pp. 91–98, 2010.
- [5] National Intelligence Service, *National Network Security System Guidelines (Draft)*, Seoul, South Korea: National Intelligence Service, Jan. 2025.
- [6] Fortinet, "Unified Threat Management (UTM)," *Fortinet CyberGlossary*. [Online]. Available: <https://www.fortinet.com/kr/resources/cyberglossary/unified-threat-management>. [Accessed: June 4, 2025].

Non-Destructive Stress Detection in Metallic Materials Using the Metal Magnetic Memory Approach

Kyung Kwon Jung ^{1*}, and Sung Chan Pai ²

¹Department of Electrical Engineering, Dongshin University, 58245, Korea

²KEPCO KPS, 58326, Korea

[Email : kkjung@dsu.ac.kr]

Abstract

The Metal Magnetic Memory (MMM) method is an emerging non-destructive testing technique used to assess stress concentration zones and detect early structural damage in metallic materials. In this paper, an experimental system was developed utilizing an array of magnetic sensors, arranged in two rows, to measure magnetic field variations on metal surfaces. The sensor array was integrated with a Raspberry Pi-based data acquisition platform to record the natural remanent magnetic field without the application of an external magnetic source. Tensile tests were performed on metallic samples of thickness (4T) using a mechanical tensile testing machine. During the loading process, the sensor array captured real-time changes in the self-magnetic leakage field. The results demonstrated that the MMM method effectively detected magnetic anomalies correlated with stress concentration and material deformation, and that thickness influenced the magnetic response under tensile loading. These findings highlight the potential of the MMM approach for non-destructive evaluation and real-time structural health monitoring of metallic components.

Index Terms: non-destructive stress detection, metal magnetic memory, magnetic sensors, real-time sensing, tensile tests.

I. INTRODUCTION

Nondestructive testing (NDT) plays a critical role in evaluating the integrity of metallic structures without causing physical damage. Conventional NDT techniques such as ultrasonic testing (UT), radiographic testing (RT), magnetic particle testing (MT), and eddy current testing (ET) are widely used across industries to detect surface and subsurface flaws. However, these methods often require complex equipment, skilled operators, or preconditioning of the test object, and may not effectively capture early-stage stress concentration or micro-defects in materials.

In recent years, the Metal Magnetic Memory (MMM) method has emerged as a promising alternative for early detection of stress-induced damage in metallic components. Unlike traditional magnetic testing methods that require magnetization, MMM utilizes the natural self-magnetic leakage field (SMLF) that forms in ferromagnetic materials under mechanical stress. This passive approach allows for the identification of stress

concentration zones and potential failure points without the need for external magnetic excitation. Several studies have explored the theoretical foundations and practical applications of MMM in fields such as pipeline inspection, welded joint monitoring, and fatigue assessment of structural components [1].

The accuracy and resolution of MMM measurements largely depend on the number and arrangement of magnetic field sensors. Prior research has employed sensor arrays ranging from single-point probes to 16-channel AMR (Anisotropic Magnetoresistance) arrays for capturing magnetic field variations. While these configurations have demonstrated feasibility, they are often limited in spatial resolution or scalability. Furthermore, many existing systems focus on sequential scanning, which may not capture dynamic stress changes in real time[2].

To address these limitations, this study proposes a sensor array-based MMM measurement system that enables higher spatial resolution and more cost-effective implementation. The system consists of 30 LIS3MDL magnetometer sensors arranged in two rows of 15 sensors each, placed in a diagonal zigzag pattern to increase surface coverage and sensitivity.

To validate the proposed system, tensile tests were conducted on metal specimens of 4T using a mechanical tensile testing machine. The sensor array, integrated with a Raspberry Pi-based data acquisition platform, recorded the magnetic field variations in real time as the specimens underwent elastic and plastic deformation. This study evaluates the performance of the proposed array-based MMM system and its applicability to stress detection in metallic materials.

II. EXPERIMENTAL SETUP

The proposed device was designed to continuously collect and record magnetic flux signals in the X, Y, and Z directions during mechanical loading. The system operates without any external magnetic excitation, relying solely on stress-induced magnetic responses inherent to the material.

To evaluate the system's performance, tensile tests were conducted using a universal testing machine under static loading conditions. The test specimens were threaded steel bars (TSBs), as illustrated in Figure 1. These samples were subjected to uniaxial tensile loading until plastic deformation occurred, enabling the capture of SMFL signal variations associated with stress concentration and microstructural changes.

The magnetic signal acquisition module was custom-built, as shown in Figure 2. The sensor array consisted of 30 LIS3MDL three-axis magnetometer sensors arranged in two zigzag rows of 15 sensors each. This layout was designed to maximize spatial resolution and enhance the sensitivity of magnetic field detection across the metal surface.

Magnetic field data from each sensor were collected using a Raspberry Pi-based platform, which served as the central controller and data logger. The Raspberry Pi simultaneously recorded X, Y, and Z magnetic components from all sensors and stored the time-series data in structured files for further analysis.



Fig. 1. Photographs of the threaded steel bar specimens used in the test (4T).

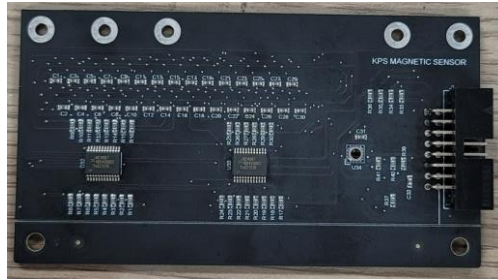


Fig. 2. Sensor array module.

III. RESULTS

The variation of the SMFL above the surface of the points of TSBs for different thickness during the tensile loading process is plotted, as shown in Figure 3. In order to better study the variation law of SMFL during the loading process of TSBs, the loading displacement is put together with the tensile-displacement curve of TSB.

Figure 4(a) shows the magnetic field data of 30-channel sensor array. Figure 4(b) presents the interpolated average X-axis magnetic field values measured across the 30 LIS3MDL sensors. The magnetic field exhibited notable fluctuations in response to increasing displacement. In particular, several regions showed significant changes in the magnetic field gradient, suggesting the presence of stress concentration zones and microstructural changes within the specimen.

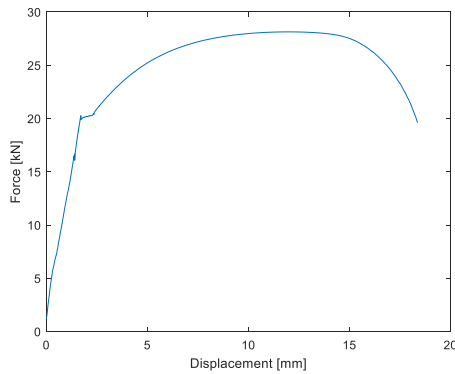


Fig. 3. Tensile-displacement curve

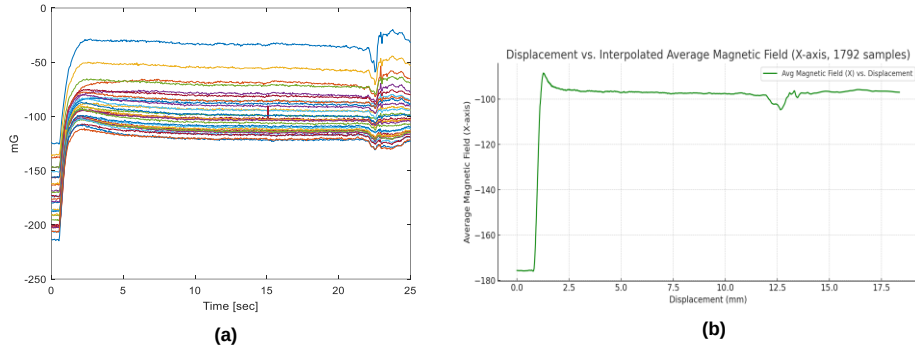


Fig. 4. Sensor data analysis.

IV. DISCUSSION AND CONCLUSIONS

This study demonstrated the effectiveness of the Metal Magnetic Memory (MMM) method for non-destructive stress detection in metallic materials. By employing a 30-sensor array and interpolating magnetic data to match the mechanical sampling rate, meaningful correlations were observed between stress-induced deformation and variations in the self-magnetic leakage field. The proposed approach offers a low-cost and scalable solution for high-resolution structural health monitoring, particularly suitable for early detection of stress concentration zones.

ACKNOWLEDGMENTS

This work is a result of the Dongshin University Innovation Convergence Campus Project, conducted with funding from the Ministry of Land, Infrastructure and Transport in 2025 and support from Jeollanam-do and Naju-si.(2025ICCS-001).

REFERENCES

- [1] Sheng Bao, Meili Fu, Shengnan Hu, Yibin Gu, and Huangjie Lou, "A Review of the Metal Magnetic Memory Technique," in Proceedings of the ASME 2016 35th International Conference on Ocean, Offshore and Arctic Engineering, pp. 1-9, 2016.
- [2] Zhichao Li, Steve Dixon, Peter Cawley, Rollo Jarvis, and Peter B. Nagy, "Study of metal magnetic memory (MMM) technique using permanently installed magnetic sensor arrays," in Proceedings of AIP Conference, Vol. 1806, No. 1, pp. 110011-1-8, 2017.

Effectiveness of Digital Acceptance and Commitment Therapy (ACT) for Cancer Pain: A Meta-Analysis of Randomized Controlled Trials

Seyeon Park^{1*}, Jeongeun Lee¹, Suna Choi^{2*}

¹College of Nursing, Chungnam National University, Daejeon 35015, Republic of Korea

²Department of Anesthesiology and Pain Medicine, Chungnam National University Hospital, Daejeon 35015, Republic of Korea

[Email : park_sy@cnu.ac.kr/ csa323@naver.com]

Abstract

This study aims to synthesize evidence from randomized controlled trials (RCTs) to evaluate the effectiveness of digital Acceptance and Commitment Therapy (ACT) in reducing pain and improving psychological outcomes among cancer patients. A systematic literature search was conducted in PubMed, CENTRAL, Embase, and PsycINFO for RCTs published up to April 2025. Inclusion criteria were RCTs involving adult cancer patients receiving ACT interventions via mobile apps, web-based systems, or telehealth platforms, compared with standard care or waitlist controls. Four RCTs involving a total of 170 participants were included. The meta-analysis showed that digital ACT significantly reduced cancer pain (SMD = -0.66, 95% CI: -1.08 to -0.23), depression (SMD = -2.12, 95% CI: -3.99 to -0.25), and improved acceptance (SMD = 1.02, 95% CI: 0.10 to 1.93) and psychological flexibility (SMD = 0.94, 95% CI: 0.54 to 1.35). However, no significant effect was found for anxiety (SMD = -0.94, 95% CI: -2.47 to 0.59). Risk of bias was assessed using the Cochrane RoB 2 tool, with two studies rated as low risk and two as having some concerns. These findings suggest that digital ACT may be a promising modality for improving cancer pain and psychological adaptation, supporting the integration of psychological interventions into digital health platforms for oncology care. These findings support the use of digital ACT as an effective non-pharmacological intervention for cancer patients, particularly for managing pain, alleviating depressive symptoms, and enhancing psychological resilience in clinical oncology settings.

Index Terms: Cancer pain, digital health, acceptance and commitment therapy, IT convergence, intelligent healthcare, meta-analysis, randomized controlled trials

I. INTRODUCTION

Cancer-related pain significantly affects patients' quality of life and is often exacerbated by psychological factors like anxiety and catastrophizing [1]. Acceptance and Commitment Therapy (ACT), a third-wave CBT, promotes psychological flexibility through acceptance

* Corresponding author

and values-based action and has shown effectiveness in chronic pain management [2]. Advances in information and communication technology (ICT) have enabled ACT delivery via mobile apps, web platforms, and telehealth, offering scalable, accessible solutions [3], [4]. However, studies on digital ACT in cancer care show mixed results. This meta-analysis aims to synthesize findings from RCTs evaluating the effectiveness of ICT-based ACT interventions on pain and psychological outcomes in oncology settings.

II. METHODS

Using PRISMA 2020 guidelines [5], four databases (PubMed, CENTRAL, Embase, PsycINFO) were searched up to April 2025. Eligible studies were RCTs involving adult cancer patients receiving ACT via digital platforms, measuring pain and psychological variables. Data extraction and bias assessment followed Cochrane RoB 2. Meta-analysis was conducted using RevMan 5.4, calculating standardized mean differences (SMDs) with random-effects models and assessing heterogeneity via I^2 .

III. RESULTS

This study systematically reviewed four randomized controlled trials (RCTs) evaluating digital Acceptance and Commitment Therapy (ACT) in cancer patients. Interventions were delivered via mobile apps, web platforms, or telehealth and included patients with breast, lung, and other cancer types. Meta-analysis showed that digital ACT effectively reduced pain and depression, and improved acceptance and psychological flexibility. However, effects on anxiety were limited. Variability in outcomes appeared to depend on intervention intensity, patient condition, and delivery method (see Figure 2a–e).

IV. DISCUSSION AND CONCLUSIONS

Digital ACT showed beneficial effects on cancer pain and psychological flexibility, aligning with previous evidence of its efficacy in remote settings [6]. Krok et al. also supported ACT's role in improving acceptance and quality of life in cancer patients [7]. However, anxiety outcomes suggest limitations in ACT's standalone effectiveness. Given its feasibility and accessibility, digital ACT can serve as a cost-effective tool in tech-enabled oncology care, though more robust studies are needed to confirm long-term and anxiety-related effects.

A. Figures and Tables

1) Figures

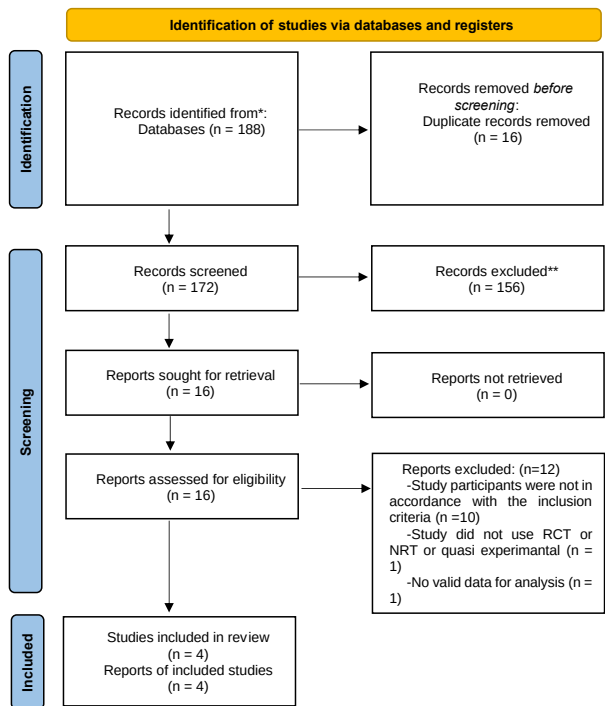
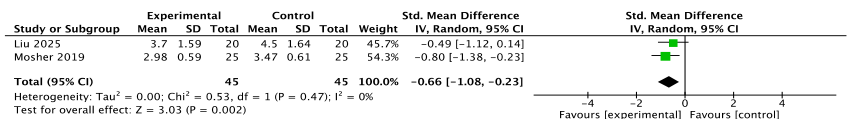
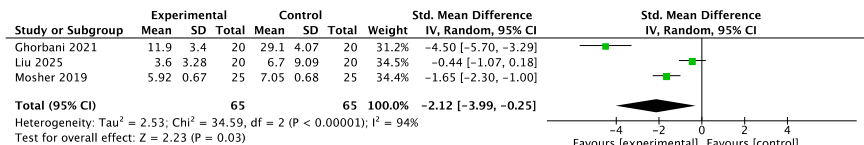


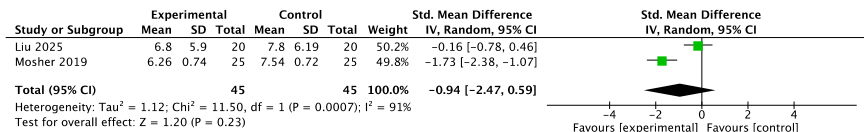
Fig. 1. Study Flow Diagram



(a) pain



(b) depression



(c) anxiety

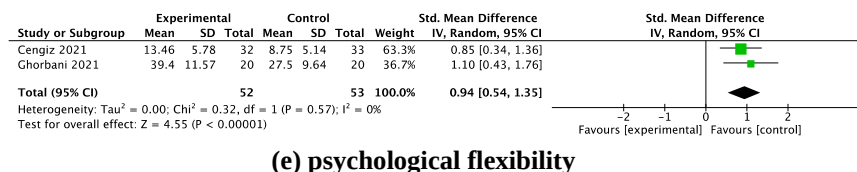
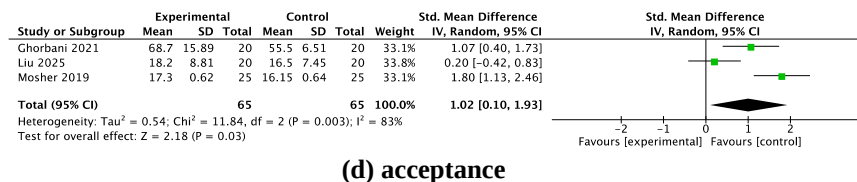


Fig. 2. Effect Sizes of ACT Interventions by Outcome Variable

ACKNOWLEDGMENTS

This research was supported by a grant from the National Research Foundation of Korea (NRF), funded by the Korean government (RS-2024-00360136).

REFERENCES

- [1] M. S. Herbert, N. Afari, L. Liu, P. Heppner, T. Rutledge, K. Williams, S. Erally, K. VanBuskirk, C. Nguyen, M. Bondi, J. H. Atkinson, S. Golshan, and J. L. Wetherell, "Telehealth versus in-person Acceptance and Commitment Therapy for chronic pain: A randomized noninferiority trial," *The Journal of Pain*, vol. 18, no. 2, pp. 200–211, 2017, doi: 10.1016/j.jpain.2016.10.014.
- [2] F. Garcia-Torres, M. García-Carmona, S. J. Rubio, et al., "Feasibility study of a mobile application (ACT-ON) to complement acceptance and commitment therapy (ACT) intervention in cancer patients," *Supportive Care in Cancer*, vol. 31, no. 11, pp. 761–772, 2023, doi: 10.1007/s00520-023-07745-w.
- [3] B. D. Darnall, P. Krishnamurthy, J. Tsuei, and J. D. Minor, "Self-administered skills-based virtual reality intervention for chronic pain: Randomized controlled pilot study," *JMIR Formative Research*, vol. 4, no. 7, p. e17293, Jul. 2020, doi: 10.2196/17293.
- [4] D. G. Rogers, K. Santamaria, E. K. Seng, and A. S. Grinberg, "Behavioral health, telemedicine, and opportunities for improving access," *Current Pain and Headache Reports*, vol. 26, no. 12, pp. 919–926, Dec. 2022, doi: 10.1007/s11916-022-01096-w.
- [5] M. J. Page, J. E. McKenzie, P. M. Bossuyt, I. Boutron, T. C. Hoffmann, C. D. Mulrow *et al.*, "The PRISMA 2020 statement: An updated guideline for reporting systematic reviews," *BMJ*, vol. 372, p. n71, Mar. 2021, doi: 10.1136/bmj.n71.
- [6] L. S. Hughes, J. Clark, J. A. Colclough, E. Dale, and D. McMillan, "Acceptance and Commitment Therapy (ACT) for chronic pain: A systematic review and meta-analyses," *The Clinical Journal of Pain*, vol. 33, no. 6, pp. 552–568, Jun. 2017, doi: 10.1097/AJP.0000000000000425.
- [7] D. Krok, E. Telka, and M. Morón, "Dyadic effects of attachment on illness acceptance in patients with breast cancer and spousal caregivers: Sense of coherence as a mediator," *Journal of Clinical Medicine*, vol. 13, no. 21, p. 6425, 2024, doi: 10.3390/jcm13216425.

A Theoretical Review on Enhancing Communication with Patients Using Artificial Intelligence in Nursing Education

Sung-Yun Ahn^{1*}

¹Department of Nursing, Pai Chai University, 35345, South Korea
/Email : syahn@pcu.ac.kr

Abstract

This study explores theoretical strategies for enhancing nursing students' communication skills with patients by integrating artificial intelligence (AI) technologies into nursing education. While traditional nursing curricula include lectures, clinical practice, and simulation, they often lack effective methods for developing and evaluating interpersonal communication, especially emotional empathy and patient-centered interaction. Based on existing literature and applications of AI in healthcare, this paper examines how tools such as natural language processing, emotion recognition, and chatbot simulations can be utilized to strengthen communication competencies. AI offers repeatable, feedback-driven learning environments that can be tailored to individual learner needs, enabling personalized and immersive educational experiences. The findings underscore the importance of instructional design in simulation-based learning and propose the potential for AI-assisted training to overcome limitations in conventional methods. Additionally, ethical, clinical, and contextual considerations are discussed to support the development of responsible and practically relevant AI-based communication programs. This theoretical review provides a foundational perspective for designing future nursing curricula that integrate AI technologies to advance communication education.

Index Terms: Nursing education, Artificial Intelligence, Communication, Patient-centered care, Natural language processing, Emotion recognition

1. Introduction

A nurse's communication competency is a critical factor directly related to patient safety and treatment outcomes. However, conventional nursing education presents structural limitations that hinder the systematic development of communication skills. Recently, as artificial intelligence (AI) technology has been applied in various educational fields, new approaches are being sought in nursing education as well. This study aims to theoretically examine how AI technology can contribute to enhancing communication competency in nursing education.

2. Theoretical Background

2.1 Nursing Education and Communication

The application context of focusing on image and signal processing with tracking, monitoring, or the classification of activity and health, followed by care coordination and communication, demonstrates that AI can contribute to enhancing effective communication through patient condition analysis and interactive systems [1]. In Kim's

(2022) study examining the effects of simulation-based learning among nursing students, learners' satisfaction and confidence were found to be positively associated with overall nursing competency [2]. This provides empirical evidence that simulation learning can enhance satisfaction and confidence, which in turn improves nursing competency. Although traditional nursing education incorporates lectures, clinical practice, and various forms of simulation, it often lacks effective methods for teaching and assessing communication with patients. In particular, emotional empathy, active listening, and information delivery skills can be improved through repeated practice and structured feedback.

2.2 Overview of AI Technology and Its Application in Healthcare

In Kim's (2022) study on the effects of simulation-based learning among nursing students, the instructor's teaching methods and instructional design were shown to be positively associated with overall nursing competency [2]. This finding supports the importance of designing AI-based instructional strategies. Applying AI technologies in nursing education enables the provision of repeatable simulation environments, real-time feedback, and personalized learning tailored to the student's level. In particular, tools such as conversational chatbots, virtual patient simulations, and voice analysis technologies serve as effective means to replicate real-world communication with patients.

2.3 Potential for AI Application in Nursing Education

In a study by Kim (2022) on the effects of simulation-based learning among nursing students, the instructor's teaching methods and instructional design were found to be positively correlated with overall nursing competency [2]. This finding underscores the importance of designing AI-based instructional approaches. Applying AI technology to nursing education enables the provision of repeatable simulation environments, real-time feedback, and instruction tailored to the learner's individual level. Specifically, conversational chatbots, virtual patient simulations, and voice analysis technologies are effective tools for replicating real-life communication scenarios with patients.

3. Educational Applications of AI Technology

3.1 NLP-Based Simulation

Chatbots based on natural language processing (NLP) technology allow nursing students to engage in conversations with AI acting as virtual patients and receive real-time feedback. Such an environment is effective for analyzing communication patterns and identifying areas for improvement.

3.2 Use of Emotion Recognition Technology

In Kim's (2022) study on the effects of simulation-based learning among nursing students, it was found that learners' satisfaction and confidence need to be considered when implementing simulation to develop nursing competency [2]. Therefore, it is necessary to design educational programs that incorporate AI's feedback and emotion recognition functions. Emotion recognition AI can analyze facial expressions, voice tone, and word choice to assess the emotional state of patients, allowing students to train in responding appropriately. This contributes to the enhancement of emotional empathy.

3.3 Feedback-Based Communication Training

In Kim's (2022) study on simulation-based learning among nursing students, participants' age and the instructional design were identified as predictive factors that explained 21.5% of nursing competency [2]. Therefore, it is necessary to design personalized AI simulations. AI can support learners by analyzing their communication content and providing immediate feedback, enabling them to assess and improve their own communication styles.

4. Conclusion

AI technology presents a promising solution to the practical challenge of enhancing communication skills in nursing education. However, few studies have reported the effects of AI applications on clinical or organizational outcomes, lacking particularly in data gathered outside laboratory conditions [1]. Therefore, AI-based education should evolve into a communication training approach that is applicable in real-world clinical settings. Meanwhile, ethical, legal, and social implications reflect the discourse on technology use in health care but have mostly not been discussed in meaningful and potentially encompassing detail [1]. Despite these limitations, AI has the advantage of providing repetitive and feedback-driven learning environments without the constant presence of a human educator. This conclusion aligns with the finding that integration of AI into real-time decision support and feedback systems represents a new direction for digital healthcare training [3].

5. Suggestions

Challenges include standardization, ethical considerations, and ensuring clinical relevance in AI applications [3]. The application of AI requires addressing issues related to standardization, ethical responsibility, and the practical validity of clinical implementation. In addition, close collaboration between nursing educators and technology developers is essential for the development of AI-based communication programs that are integrated into real clinical settings.

References

- [1] Seibert, K., Domhoff, D., Bruch, D., Schulte-Althoff, M., Fürstenau, D., Biessmann, F., & Wolf-Ostermann, K. (2021). Application scenarios for artificial intelligence in nursing care: Rapid review. *Journal of Medical Internet Research*, 23(11), e26522. <https://www.jmir.org/2021/11/e26522/>
- [2] Kim, S. (2022). Effects of simulation learning on nursing competency in nursing college students. *Journal of Digital Convergence*, 20(1), 481–488. <https://doi.org/10.14400/JDC.2022.20.1.481>
- [3] Gao, S., Wang, X., Xia, Z., Zhang, H., Yu, J., & Yang, F (2025). Artificial intelligence in Dentistry: A narrative review of Diagnostic and Therapeutic Applications. *Med Sci Monit*.8(31),8(31), e946676. doi: 10.12659/MSM.946676

Lightweight Coordinate-Classification Framework for Real-Time 3-D Pose and Rotation Estimation

Min-Gyu Kim¹, Hwan-Do Kim¹, Che-Ri Kim¹, Dae-Hwan Kim*

¹Department of Electrical, Electronic and Computer Engineering, University of Ulsan, Ulsan, 44610 Korea

*Department of IT Convergence, University of Ulsan, Ulsan, 44610 Korea

[Email mingnine@ulsan.ac.kr, uio889@ulsan.ac.kr, cheriontop@ulsan.ac.kr, daehwankim@ulsan.ac.kr]

Abstract

This study presents a lightweight framework that leverages stereo depth to improve the accuracy of three-dimensional joint coordinates for real-time, multi-person pose estimation. By feeding RGB-D streams from a stereo camera, the model simultaneously predicts 3-D joint positions and roll–pitch–yaw (RPY) angles, yielding high-resolution motion features that are essential for virtual-human synthesis. A coordinate-classification module drastically reduces computation while preserving accuracy. With ONNX-based optimization, the system achieves real-time inference on a single GPU and records per-frame position and rotation in JSON format, simplifying downstream animation and analytics. A two-stage strong-to-weak augmentation strategy, together with an attention mechanism that fuses global and local spatial cues, enhances robustness to small or partially occluded subjects. The proposed approach enables more natural and precise motion synthesis for metaverse content, virtual-character animation, and sports biomechanics, underscoring its considerable industrial potential.

Index Terms: Real-time Human Pose Estimation, Precise 3-D Joint Localization, RGB-D Fusion, Lightweight Model, Coordinate-Classification

I. INTRODUCTION

Accurate human-pose estimation underpins human–computer interaction, action recognition, sports analytics, healthcare, and surveillance. The industrial demand for real-time, multi-person, 3-D tracking is further fueled by VTuber production and metaverse applications. Traditional 2-D approaches infer joint locations from color images but lack depth cues, whereas 3-D methods that rely on stereo vision often incur high latency because of heavy heat-map regression.

To meet latency constraints without sacrificing accuracy, we adopt a one-stage detector based on RTMO [1] and enrich it with depth cues from a ZED 2 stereo camera. The resulting framework attains real-time throughput on laptop-grade hardware while maintaining benchmark-level precision.

* Corresponding author

II. SYSTEM MODEL AND METHODS

The inference pipeline consists of four stages, closely following the RTMO design.

A. Feature Extraction and Detection.

The full-resolution RGB frame is passed through a CSPDarknet[2] backbone and hybrid encoder, producing multi-scale feature maps.

B. Dynamic Coordinate Classification(DCC)

The Dynamic Coordinate Classifier (DCC) extends the SimCC paradigm [3]. Taking the probability weighted expectation of the bin centers yields sub-pixel joint coordinates while completely avoiding dense 2-D heat-maps.

C. RGB-D Fusion

The network fuses the stereo depth map with the RGB feature tensor and, using the paired 2-D joint detections plus depth values, triangulates the 3-D coordinates (X, Y, Z) of each joint[4]. it then analytically derives the roll–pitch–yaw (RPY) angles from those coordinates via arctangent operations, following [5].

D. Skip-frame detection

Skip-frame detection followed by a One Euro filter [6] stabilizes the trajectories. The system then writes per-frame position-and-rotation tuples to JSON for downstream animation and analytics.

E. Training

The network is trained on COCO Train2017 and validated on COCO Val2017. Optimization employs AdamW (learning rate 0.001, batch size 32) for 900 epochs; the best checkpoint is selected at epoch 625.

F. Inference Optimization

The trained PyTorch model is exported to ONNX (opset 17). On a single RTX 3080-laptop GPU the system processes each frame in 66.4 ms, achieving 15.07 FPS with RGB-D input enabled.

III. RESULTS

A. Quantitative Evaluation on COCO Val2017

Table 1. Quantitative Evaluation on COCO Val2017.

Metric	Value
AP	0.531
AP50	0.763
AP75	0.577
AR	0.843

B. Real-Time Performance

The average per-frame processing time is 66.4 ms, corresponding to 15.07 FPS, satisfying the ≥ 15 FPS real-time criterion.

C. Qualitative Analysis

Fusion of RGB and depth inputs produces stable skeleton overlays and RPY-angle visualizations across diverse poses and lighting conditions, as illustrated in supplementary material.

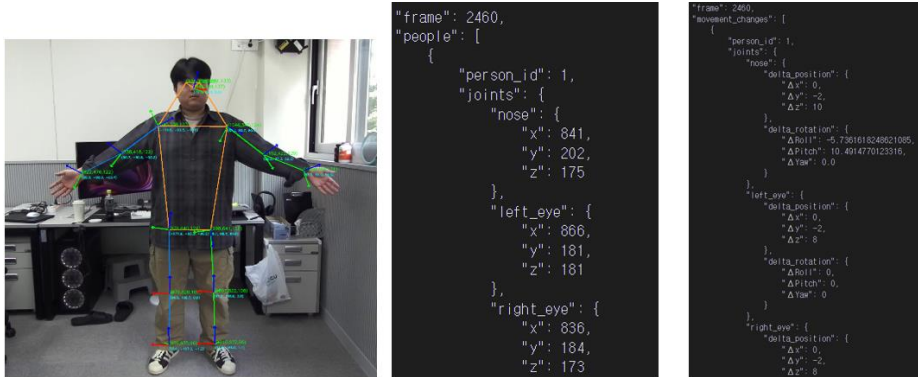


Fig. 1. Pose inference images and JSON files returned by inference

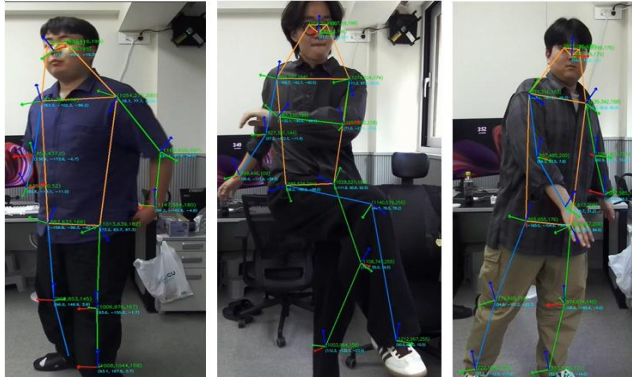


Fig. 2. Example images of inference results from different postures

IV. DISCUSSION AND CONCLUSIONS

The proposed framework depth fusion resolves scale ambiguity and enables RPY estimation; the lightweight attention fusion enhances robustness in crowded scenes. Remaining challenges include reduced accuracy for very small subjects and domain shifts to outdoor lighting.

Future work will explore knowledge distillation, post-training quantization, multi-scale feature pyramids and collection of ground-truth RPY labels for rigorous rotational benchmarking.

ACKNOWLEDGMENTS

REFERENCES

- [1] Lu, Peng, et al. "RTMO: towards high-performance one-stage real-time multi-person pose estimation." *Proceedings of the IEEE/CVF conference on computer vision and pattern recognition*. 2024.
- [2] Mahasin, Marsa, and Irma Amelia Dewi. "Comparison of CSPDarkNet53, CSPResNeXt-50, and EfficientNet-B0 backbones on YOLO v4 as object detector." *International journal of engineering, science and information technology* 2.3 (2022): 64-72.
- [3] Li, Yanjie, et al. "Simcc: A simple coordinate classification perspective for human pose estimation." *European conference on computer vision*. Cham: Springer Nature Switzerland, 2022.
- [4] Izadi, Shahram, et al. "Kinectfusion: real-time 3d reconstruction and interaction using a moving depth camera." *Proceedings of the 24th annual ACM symposium on User interface software and technology*. 2011. [5] B. Alavi, "Distance measurement error modeling for time-of-arrival based indoor geolocation," Ph.D. dissertation, Worcester Polytechnic Institute, Worcester: MA, 2006.
- [5] Diebel, James. "Representing attitude: Euler angles, unit quaternions, and rotation vectors." *Matrix* 58.15-16 (2006): 1-35.
- [6] Casiez, G ry, Nicolas Roussel, and Daniel Vogel. "1  filter: a simple speed-based low-pass filter for noisy input in interactive systems." *Proceedings of the SIGCHI Conference on Human Factors in Computing Systems*. 2012.

Resolution-Aware Conditional Diffusion for Small Human Pose Recovery

Hwan-do Kim¹, Min-gyu Kim¹, Che-ri Kim¹ Dae-hwan Kim¹

¹Department of Electrical, Electronic and Computer Engineering, University of Ulsan, Ulsan, 44610 Korea

[Email : uiop889@ulsan.ac.kr, mingnine@ulsan.ac.kr, cheriontop@ulsan.ac.kr,
daehwankim@ulsan.ac.kr]

Abstract

This study proposes a conditional diffusion model that enables more precise pose estimation for small, low-resolution human objects in images. Due to resolution degradation, existing pose estimation models struggle to accurately predict the keypoints of small objects. To solve this problem, this paper combines various conditional information such as image features, bounding box information, object resolution level, and confidence to construct a conditional vector, which is then used to construct a diffusion model with the structure of Deterministic Denoising Diffusion Implicit Models (DDIM). The proposed framework generates structurally consistent keypoint information through the process of restoring noise-injected heatmaps and refines the condition encoding and loss function design to be more sensitive to small objects. In particular, loss weights based on object size are applied to maximize learning for small figures, and resolution information is effectively incorporated into the inference process. This approach compensates for the limitations of existing pose estimation models and enables robust keypoint estimation in a wide range of resolution environments.

Index Terms: AI, Human Pose, Diffusion, Skeleton

I. INTRODUCTION

In recent years, deep learning-based human pose estimation has played a key role in a variety of computer vision challenges, with applications in sports analytics, surveillance systems, and augmented reality. In particular, 2D heatmap-based approaches have shown high accuracy for clearly visible human bodies in high-resolution images and have been adopted as the standard in many studies.

However, in real-world environments, people are often far away and appear very small in the image, and traditional pose estimation models degrade rapidly in this situation. This is because the smaller the object size, the more information is lost, and resolution-sensitive heatmap-based estimation models have difficulty recognizing small features. In addition, small objects have unclear boundaries with the background, making it difficult to clearly distinguish key points.

To solve these problems, we propose a conditional diffusion-based generative model that enables robust pose estimation even for small human objects at a distance. The

proposed model is designed to extract semantic features through a ResNet based image encoder and utilize them as conditional information for a diffusion-based U-Net to incrementally reconstruct the heatmap. In particular, a fast sampling strategy based on DDIM (Denoising Diffusion Implicit Model) is introduced to generate high-resolution heatmaps to improve inference efficiency [1].

The experiments utilize the CrowdPose dataset, and a cropped subset centered on low-resolution (distant) people is constructed for training and evaluation. We analyze various quantitative metrics such as the accuracy of keypoint estimation for small objects and the structural consistency of the sample and demonstrate the potential for performance improvement through comparison with existing regression-based models.

As a result, this paper demonstrates that the conditional diffusion model is a novel approach that can achieve both robustness and accuracy in pose estimation for small figures and has the potential to be effectively applied to human behavior recognition and smart surveillance systems in real-world environments [2-4].

II. Related work

A. 2D Human Pose Estimation

2D human pose estimation is the task of predicting the major keypoints of the human body from a given image and is typically categorized into top-down and bottom-up methods. Top-down methods first detect people and then estimate the pose for each person, and representative models include HRNet and SimpleBaseline. Bottom-up methods detect all keypoints in the entire image and then cluster them to estimate poses for individual people. These methods are efficient in crowd situations where many people are present. However, both methods have a common limitation: their performance decreases rapidly for small human figures [2]. This is due to loss of resolution, ambiguity in the object representation, and difficulty in determining the interrelationships between keypoints.

B. Diffusion Models in Vision Tasks

Denoising Diffusion Probabilistic Models (DDPMs) and their extension DDIMs were originally proposed as probabilistic generative models for image generation and have recently been applied to various conditional prediction problems such as heatmap prediction, segmentation, and keypoint localization. In particular, the conditional diffusion model has the advantage of generating more precise output by denoising based on conditional information (e.g., text, image features) [5-7]. In the field of human pose estimation, there are still limited applications, and most of the existing studies extract human keypoints in a conventional way and post-process them or utilize them only as auxiliary information [8-9].

III. Methods

In this work, we propose a conditional diffusion-based framework that utilizes resolution-sensitive conditional information to enable accurate pose estimation for small

and distant human objects. The overall model follows the structure of reconstructing heatmaps generated from images, where different forms of conditional information are considered together to achieve both learning stability and granularity.

The input image and the corresponding correct heatmap, bounding box information, object resolution level, and keypoint confidence information are each independently preprocessed, and then combined into a single condition vector. In particular, the features extracted from the image contain multi-scale information based on the ResNet50 backbone and FPN structure, while the bounding box, resolution, and confidence information are projected into a low-dimensional space through separate embedding processes. All these condition factors are finally combined into a single vector, which is used as input along with temporal information in a U-Net based predictor.

The correct heatmap is noisy at randomly selected timesteps via a diffusion process, and the resulting noisy heatmap is given as input to the model. The network predicts the injected noise based on this heatmap, condition information, and time embeddings, and uses the prediction results to restore the original heatmap. The diffusion scheduler used here follows the DDIM structure, which allows for fast convergence and stable learning [5-6]. The loss function includes more than just a simple root mean square error. To improve performance, especially for small objects, the loss function is weighted proportional to the size of the bounding box, and Focal Loss is used to adjust the learning weight for hard-to-learn samples. In addition, a structural term is added to maintain structural consistency in the heatmap, which encourages the preservation of relationships between keypoints and their spatial distribution [6-7]. The overall loss is a weighted average of these four terms, designed so that smaller objects have a stronger impact.

The AdamW optimizer is used to train the model, with an initial learning rate that gradually decreases through a cosine decay schedule after a warm-up period. At regular intervals, the quality of the reconstructed heatmap is visually checked by DDIM-based sampling, and the accuracy is evaluated using the Point-wise Correct Keypoint (PCK) metric. This configuration plays a key role in ensuring that the overall framework can perform structured and reliable pose reconstruction for small human objects.

IV. DISCUSSION AND CONCLUSIONS

In this work, we propose a novel pose estimation framework based on conditional diffusion to perform accurate pose estimation for human objects with low resolution or small representation in images. To overcome the limitation that existing pose estimation models do not fully reflect the structural characteristics of small objects, we design an approach that incorporates and utilizes various forms of conditional information, such as bounding box size, resolution level, and keypoint confidence, as well as visual features extracted by an image encoder.

The proposed model follows a diffusion-based structure that gradually reconstructs the noisy heatmap based on the condition information, and an efficient sampling strategy using DDIM is utilized to ensure practicality in the inference process [6-7]. In addition, we designed a resolution-weighted loss function to be more sensitive to small objects, so that the size and difficulty of objects can be reflected in the learning process.

By applying a generative method utilizing complex conditions to the pose estimation problem, beyond a simple regression-based approach, this study shows the possibility of securing structural consistency and expressiveness even in situations that are difficult to handle with existing methods. The proposed methodology provides a foundation for future contributions to the problem of pose recognition in various resolution environments and complex crowd situations [10].

REFERENCES

- [1] Fan, Junqiao, et al. "Diffusion model is a good pose estimator from 3d rf-vision." European Conference on Computer Vision. Cham: Springer Nature Switzerland, 2024.
- [2] Tan, Dayi, et al. "Diffusionregpose: Enhancing multi-person pose estimation using a diffusion-based end-to-end regression approach." Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition. 2024.
- [3] Ta, Calvin-Khang, et al. "Multi-modal Pose Diffuser: A Multimodal Generative Conditional Pose Prior." arXiv preprint arXiv:2410.14540 (2024).
- [4] Liu, Zhihua, and Yuru Wang. "Uncertainty-guided Diffusion Model for 3D Human Pose Estimation." Neurocomputing (2025): 130306.
- [5] Qiu, Zhongwei, et al. "Learning structure-guided diffusion model for 2d human pose estimation." arXiv preprint arXiv:2306.17074 (2023).
- [6] Hu, Shengxiang, et al. "Diffusion-Refinement Pose Estimation with Hybrid-Representation." IEEE Transactions on Instrumentation and Measurement (2025).
- [7] Feng, Runyang, et al. "Diffpose: Spatiotemporal diffusion model for video-based human pose estimation." Proceedings of the IEEE/CVF International Conference on Computer Vision. 2023.
- [8] Xu, Jinglin, Yijie Guo, and Yuxin Peng. "Finepose: Fine-grained prompt-driven 3d human pose estimation via diffusion models." Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition. 2024.
- [9] Wang, Weiquan, et al. "DiPose: Discrete Diffusion Model for Occluded 3D Human Pose Estimation." CoRR (2024).
- [10] Wan, Xiaoyue, et al. "Dual-Diffusion for Binocular 3D Human Pose Estimation." Advances in Neural Information Processing Systems 37 (2024): 78079-78103.

Development and Linear Growth Verification of an ATmega328 Microcontroller-Based Nanocoating 3D Printed Automation System

Min-Ho Yoon¹, Sun-Kyoung Kang^{2*}, Chungyeon Cho^{3*}, Ik Whang Chang^{1*}

¹Department of Automotive Engineering, College of Engineering
Wonkwang University, Iksan, 54538, Republic of Korea

²Department of Computer Software Engineering, College of Engineering
Wonkwang University, Iksan, 54538, Republic of Korea

³Department of Carbon Convergence Engineering, College of Engineering
Wonkwang University, Iksan, 54538, Republic of Korea

[Email : doctor10@wku.ac.kr, cncho37@wku.ac.kr, ikwhang2@wku.ac.kr]

Abstract

This study details the development of a low-cost, high-efficiency automated Layer-by-Layer (LbL) nanocoating system and the crucial experimental verification of its capability for linear film growth. The system's structural components were entirely fabricated using 3D printing technology, and it operates under the control of an ATmega328 microcontroller-based board, an open-source GRBL shield, A4988 stepper motor drivers, and OpenBuilds GRBL CNC controller software. Using this G-code operated system, nanometer-thick films were fabricated with the representative polyelectrolyte pair, BPEI/PAA. Subsequent measurements experimentally verified a precise linear increase in film thickness with the number of deposited layers, demonstrating that precise and predictable nanomaterial fabrication is achievable without expensive commercial equipment. This automated system, leveraging the design freedom of 3D printing, simplifies the LbL process and maximizes experimental efficiency, offering significant potential for enhancing accessibility and reproducibility in chemical experimental processes worldwide.

Index Terms: Automated LbL coating, ATmega328 microcontroller, 3D Printing, Open-source hardware, G-code, PETG, BPEI/PAA nanofilm, Linear Growth

I. INTRODUCTION

With the advancement of nanotechnology, the fabrication of functional thin films is crucial in various fields. Layer-by-Layer (LbL) deposition technology is extensively researched for its capability to control thickness at the nanometer level. However, traditional manual LbL processes suffer from issues like time consumption and low reproducibility, while commercial automated equipment is often prohibitively expensive. To address these problems, this study aimed to develop a low-cost, high-precision automated LbL coating system, with all structural components fabricated using 3D printing with PETG material for its mechanical strength and eco-friendly characteristics. A

primary objective was to rigorously verify the system's performance, specifically its ability to achieve consistent and linear film growth. The system is based on open-source hardware, including an ATmega328 microcontroller, and G-code based software. The successful development and linear growth verification are intended to expand the accessibility of advanced nanocoating technology and enhance research efficiency.

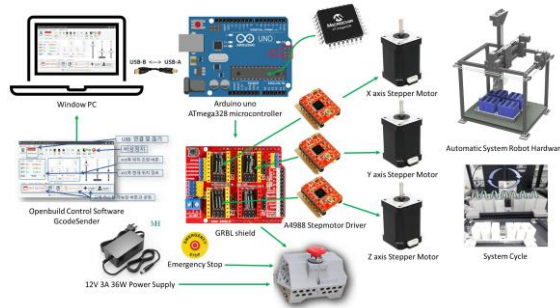


Figure 1. Schematic diagram of the automated coating system

II. SYSTEM DEVELOPMENT AND LINEAR GROWTH PERFORMANCE VERIFICATION

This section details both the development of the automated LbL coating system and the experimental verification of its linear film deposition performance. The developed system achieved precise Z-axis actuation using an ATmega328 microcontroller, a CNC shield, and A4988 drivers. A key aspect of its development is that all its custom mechanical parts were fabricated using 3D printing technology. Specifically, PETG filament was used with an internal wall count of 2 during printing to ensure enhanced mechanical strength and durability. The system operates entirely based on G-code commands, allowing users to easily set parameters like coating sequence, dipping time, and speed via software such as OpenBuilds CONTROL.

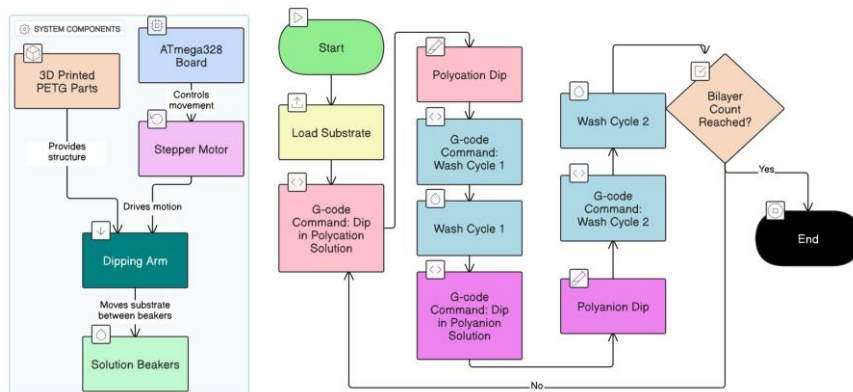


Figure 2. Step-by-step flow chart of an automated layer-by-layer thin film deposition process via G-code commands.

For performance verification, BPEI/PAA polyelectrolytes were coated onto glass substrates with varying numbers of bilayers (5, 10, 15, and 20). Ellipsometry measurement results showed that the film thickness increased linearly with the number of bilayers. This linear growth verification confirms the system's precision. .

III. CONCLUSIONS

This study successfully developed an open-source, G-code based automated LbL coating system, with all structural components 3D printed using PETG material. Crucially, the system's precision and reproducibility were validated through the experimental verification of linear thickness growth of BPEI/PAA nanofilms. The successful development and demonstrated linear growth capability highlight that the utilization of 3D printing makes the system highly customizable and implementable at a very low cost. This significantly supports precision nanocoating research in budget-constrained laboratories and contributes to streamlining chemical experimental processes, showcasing the potential for research innovation and accessibility through the synergy of open-source technology and additive manufacturing.

ACKNOWLEDGMENTS

This work was supported by the Institute of Information & Communications Technology Planning & Evaluation(IITP)-Innovative Human Resource Development for Local Intellectualization program grant funded by the Korea government(MSIT)(IITP-2025-RS-2024-00439292)

REFERENCES

- [1] P. F. Feng, Y. Su, Y. J. Zheng, Y. F. Wang, and M. S. Zheng, "Surface Modification with Particles Coated or Made of Polymer and Their Biomedical Applications," *Polymers (Basel)*, vol. 14, no. 22, p. 4972, 2022.
- [2] Y. M. Lvov, R. R. Price, J. Singh, S. S. R. K. C. K. Yarraguntla, K. M. Zakharova, and P. K. Dobbins, "Nanocoating for biomolecule delivery using layer-by-layer self-assembly," *Expert Opinion on Drug Delivery*, vol. 13, no. 3, pp. 355-368, 2016.
- [3] J. F. Grizante, R. S. R. S. S. Teixeira, C. A. Rodrigues, F. N. Crespilho, and C. J. L. Constantino, "Inkjet Deposition of Layer by Layer Assembled Films," *Journal of Nanoscience and Nanotechnology*, vol. 11, no. 7, pp. 6171-6175, 2011.
- [4] A. Sharma, P. Sharma, and S. Singh, "Design and Fabrication of Arduino Based Flexible Manufacturing Process on the Desk: 3D Printing," *International Journal for Research in Applied Science & Engineering Technology (IJRASET)*, vol. 6, no. 4, pp. 2084-2088, 2018.

Articulated Assistive Manipulator for Visually Impaired Individuals

Viet Trinh^{1,2}

¹ University of Economics and Law, Ho Chi Minh City, Vietnam

² Vietnam National University, Ho Chi Minh City, Vietnam

[Email : tqviet@uel.edu.vn]

Abstract

Current assistive technologies for individuals with visual impairments (VI) largely prioritize self-localization and navigation via verbal instructions, neglecting diverse communication channels critical for comprehensive integration. This paper introduces a novel articulate manipulator designed to bridge this gap by enabling effective communication between VI individuals and others with varying disabilities. Our system translates sign language inputs into Braille characters, addressing the overlooked need for multi-modal interaction. The research leverages advanced principles in robotics, specifically kinematics and computer vision-based gesture recognition. Key technical components include spatial transformations, forward and inverse kinematics, and sophisticated control algorithms for accurate 3D pose localization and precise gesture recognition. This interdisciplinary approach aims to enhance communicative capabilities for VI individuals and foster inclusivity across diverse disability groups, setting a new standard in assistive technology and contributing to the broader discourse on accessibility and disability inclusion.

Index Terms: AI Application, Assistive Technology, Robotics.

I. INTRODUCTION

The past decade has witnessed an unprecedented surge in interest and investment in the field of assistive technology for individuals with visual impairments. This growing focus spans a wide array of stakeholders, from agile start-up ventures pushing the boundaries of innovation to established multinational corporations dedicating significant resources to research and development. The impetus behind this intensified effort is clear: to enhance the independence, mobility, and overall quality of life for millions worldwide living with visual impairments [3]. Consequently, numerous sophisticated solutions have emerged, largely centered on facilitating self-localization and navigation within complex environments. These existing systems predominantly rely on verbal instructions, guiding users through spatial information and directions [1].

However, despite these significant advancements, a critical limitation persists within the current landscape of assistive technologies. The prevailing emphasis on verbal navigation, while undeniably valuable, inadvertently creates a narrow communicative paradigm. This singular focus often overlooks the rich tapestry of alternative communication channels and modes of interaction that are essential for true comprehensive integration and accessibility. Communication is inherently multimodal, involving diverse languages, non-verbal cues, and varying sensory inputs. For individuals with VI, limiting interaction primarily to verbal cues can inadvertently isolate them from broader communicative exchanges, especially with individuals who may communicate differently, such as those relying on sign language [2], [4]-[7].

Recognizing this critical gap, the primary objective of the proposed project is to transcend the conventional boundaries of assistive technology by empowering visually impaired individuals to engage in effective and confident communication with others, particularly those who may have different types of disabilities. This endeavor introduces the development of a novel articulate manipulator specifically designed for VI individuals. This innovative device aims to bridge communicative divides by performing a crucial translation: converting sign language inputs into tangible Braille characters that can be easily interpreted by visually impaired users. This bidirectional communication pathway promises to open new avenues for social interaction and inclusion that are currently underserved.

The research and development underpinning this system are deeply rooted in advanced principles of robotics, drawing heavily from the domains of kinematics and computer vision-based gesture recognition. The technical scope of this ambitious project encompasses several key, interconnected components. This includes mastering spatial transformations to accurately map the physical environment, developing robust manipulator forward and inverse kinematics for precise movement and positioning, and implementing sophisticated control algorithms to ensure

seamless and responsive operation. Collectively, these elements are crucial for enabling accurate 3D pose localization of sign language gestures and their precise recognition, ultimately facilitating the seamless and reliable conversion of dynamic sign language into static, tactile Braille output. Through this deeply interdisciplinary approach, the project not only aspires to profoundly enhance the communicative capabilities of visually impaired individuals but also actively seeks to foster greater inclusivity by promoting meaningful interaction across diverse disability groups. By leveraging cutting-edge technological advancements, this articulated assistive manipulator endeavors to establish a new benchmark in assistive technology, significantly contributing to the broader discourse on accessibility and disability inclusion within contemporary society.

II. METHOD AND RESULTS

Figure 1 generally illustrates the proposed system's pipeline, which can be divided into five main stages: *Input and Image Acquisition*, *Gesture Recognition*, *Coordinate Localization*, *Data Communication*, and *Output Generation*.

- **Input and Image Acquisition:** The system initiates with visual data of hand gestures representing sign language. This is achieved through a camera, which serves as the primary sensor for acquiring real-time images or video streams of the user's hand movements. Accompanying the camera input, the diagram shows examples of American Sign Language (ASL) letters (A, B, C, D), indicating the type of gestures the system aims to interpret.
- **Gesture Recognition:** This is the core processing unit powered by OpenCV (Open Source Computer Vision Library), which involves identifying and mapping key points on the hand. As detailed in the zoomed-in section of Figure 1, this process uses computer vision techniques to detect specific landmarks on the hand, such as fingertips, knuckles, and the base of the palm. The detailed breakdown of the gesture recognition process involves several image processing steps:
 - Initial frames or images are analyzed to sample color information, likely to identify skin tones or specific hand regions.
 - The sampled color data is converted into binary images. This step typically involves thresholding, where pixels within a certain color range (e.g., skin color) are highlighted, separating the hand from the background.
 - A *Median Blur* filter is applied to the binary images to reduce salt-and-pepper noise, preserving edges while smoothing the image.
 - Bilateral Blur: A Bilateral Blur is then applied. This filter is effective at noise reduction while simultaneously preserving sharp edges, which is crucial for maintaining the distinct features of the hand gesture.
 - The culmination of these steps is a refined, *noise-free binary image* that accurately represents the hand gesture, ready for feature extraction and classification.
- **3D Pose and Coordinate Localization:** The detected points from the previous step are then used to form a skeletal or polygonal representation of the hand, providing a 3D pose and coordinate information. This allows the system to understand not just the 2D shape of the hand but also its orientation and position in three-dimensional space, which is critical for distinguishing between similar gestures.

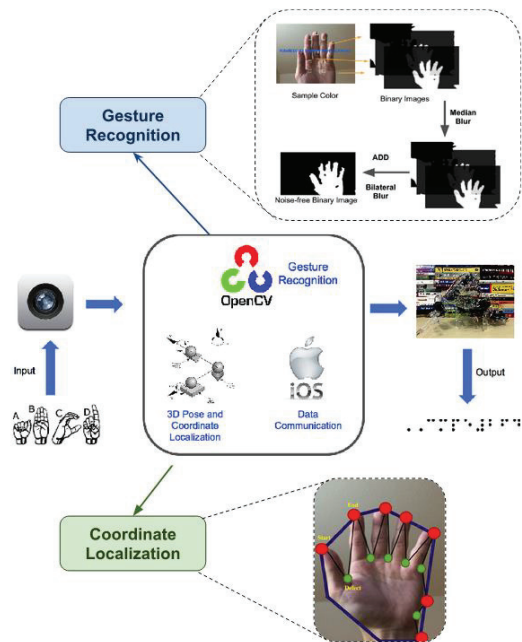


Figure 1. The Articulated Assistive Manipulator for Visually Impaired Individuals' pipeline.

- **Data Communication:** The system also incorporates Data Communication functionalities, explicitly mentioning iOS. This indicates that the processing and control logic, or at least a significant part of it, is designed to run on or communicate with an Apple iOS platform. This suggests the potential for a mobile application interface for user interaction, data display, or direct control of output devices.
- **Output Generation:** Finally, the recognized gesture information is translated into a tangible output. The diagram shows the processed data being sent to a robotic manipulator. This articulated manipulator is designed to physically generate Braille characters based on the recognized sign language. The output is clearly depicted as a series of raised dots, representing Braille. This completes the cycle, providing a tactile interpretation of the visual sign language input for visually impaired individuals. Figure 2 and Figure 3 shows outputs and errors of a couple runs from our system.

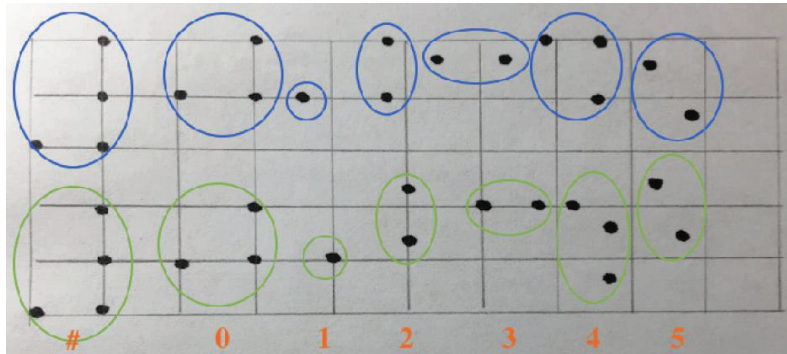


Figure 2. Braille numbers are dotted on a paper. The first and second runs are circled in blue and green, respectively. The orange character explains its corresponding Braille.

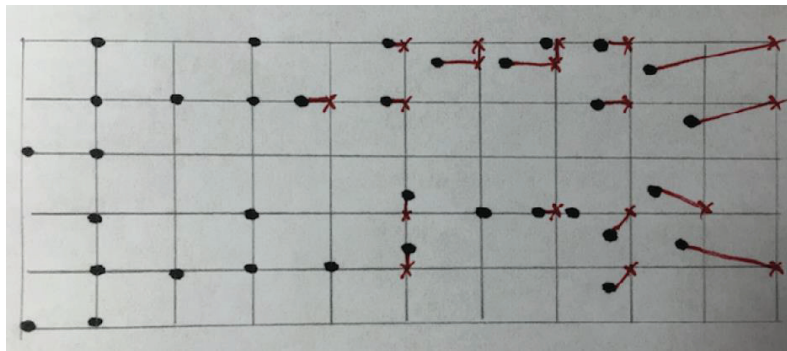


Figure 3. The differences between the correct locations (red crosses) and the system's dotted positions (black points).

III. CONCLUSION

This research introduces a novel articulate manipulator that transforms assistive technology for individuals with visual impairments. By enabling the translation of sign language into tactile Braille, our system addresses the limitations of current verbal-centric solutions, fostering multimodal communication between VI individuals and those with other disabilities. Leveraging robotics, kinematics, and computer vision, we've established the technical foundation for accurate gesture recognition and Braille output. This achievement is a significant stride toward greater inclusivity and social integration. Future work will focus on refining the system's performance and expanding its applications, aiming to set a new standard for accessibility and disability inclusion in assistive technology.

REFERENCES

- [1] Okolo, G. I., Althobaiti, T., & Ramzan, N. (2024). Assistive systems for visually impaired persons: challenges and opportunities for navigation assistance. *Sensors*, 24(11), 3572.
- [2] Macuch Silva, V., Holler, J., Ozyurek, A., & Roberts, S. G. (2020). Multimodality and the origin of a novel communication system in face-to-face interaction. *Royal Society open science*, 7(1), 182056.
- [3] Jiang, H., Wachs, J. P., & Duerstock, B. S. (2013, June). Integrated vision-based robotic arm interface for operators with upper limb mobility impairments. In 2013 IEEE 13th International Conference on Rehabilitation Robotics (ICORR) (pp. 1-6). IEEE.
- [4] Lin, L., Song, Y., Yang, Y., Feng, H., Cheng, Y., & Pan, H. (2015). Computer vision system R&D for EAST Articulated Maintenance Arm robot. *Fusion Engineering and Design*, 100, 254-259.
- [5] Cabre, T. P., Cairol, M. T., Calafell, D. F., Ribes, M. T., & Roca, J. P. (2013). Project-based learning example: controlling an educational robotic arm with computer vision. *IEEE Revista Iberoamericana de Tecnologías del Aprendizaje*, 8(3), 135-142.
- [6] M. M. Samli and I. Dogan, "An artificial neural network for predicting the presence of spermatozoa in the testes of men with nonobstructive azoospermia," *The Journal of urology*, vol. 171, no. 6, pp. 2354–2357, 2004.
- [7] Chaudhury, A., Ward, C., Talasaz, A., Ivanov, A. G., Huner, N. P., Grodzinski, B., ... & Barron, J. L. (2015, June). Computer vision based autonomous robotic system for 3D plant growth measurement. In 2015 12th Conference on Computer and Robot Vision (pp. 290-296). IEEE.
- [8] Szabo, R., & Gontean, A. S. (2013). Full 3D Robotic Arm Control with Stereo Cameras Made in LabVIEW. In *FedCSIS (Position Papers)* (pp. 37-42).

Disciplinary Landscapes of Deep Learning: Cross-Domain Insights via LDA Topic Modeling

Daesoo Choi^{1*}

¹Department of Convergence Engineering, Joongbu University, 10279, Korea
[Email : daesooc100@gmail.com]

Abstract

This study applies Latent Dirichlet Allocation (LDA) to examine research trends in deep learning across the disciplines of engineering, natural sciences, and social sciences during the period from 2020 to 2024. A corpus of 3,000 research paper titles was analyzed to extract latent topics within each field. Four distinct topics were identified per domain, reflecting domain-specific variations in thematic focus and research maturity. Engineering research emphasized automation and control systems, natural sciences focused on medical imaging and data-driven analytics, whereas the social sciences revealed growing integration with ensemble modeling and algorithmic frameworks. The findings provide a cross-disciplinary comparison of deep learning research and suggest future directions, including the adoption of explainable AI, multimodal data integration, and domain-specific methodological adaptations.

Index Terms: Cross-Disciplinary Research, Deep Learning, Latent Dirichlet allocation, Topic Modeling

I. INTRODUCTION

The academic and industrial impact of deep learning is not confined to the traditional boundaries of computer science. In recent years, its applications have expanded significantly into interdisciplinary and domain-specific fields such as biomedical imaging, financial market forecasting, climate modeling, and smart city development. These fields, which often require the integration of heterogeneous data sources and the uncovering of subtle patterns, benefit greatly from the expressive capacity of deep learning architectures.

Despite this widespread diffusion, much of the existing literature on deep learning remains siloed within individual disciplines. Prior survey studies have predominantly focused on reviewing models and applications within specific scientific domains—often emphasizing methodological innovations or performance benchmarks relevant to that particular field. As a result, there remains a lack of comprehensive, cross-disciplinary analysis that reveals how the research agendas, thematic trends, and topical emphases surrounding deep learning differ—or converge—across domains.

To address this gap, the present study adopts a data-driven, quantitative approach to investigate how deep learning has shaped research trajectories in three distinct scientific

* Corresponding author

disciplines: engineering, natural sciences, and social sciences. Utilizing Latent Dirichlet Allocation (LDA), an unsupervised probabilistic topic modeling technique, we extract latent thematic structures from a curated corpus of 3,000 research paper titles spanning these three fields. Unlike manual content analysis or narrative reviews, LDA enables the identification of hidden topic distributions that reflect the underlying semantic patterns within large-scale textual data.

Through this lens, we aim to uncover both shared and domain-specific themes in the way deep learning is conceptualized, applied, and discussed in scholarly discourse.

II. MATEIALS AND METHODS

To investigate the cross-disciplinary landscape of deep learning research, this study conducted a comprehensive topic modeling analysis using Latent Dirichlet Allocation (LDA) across three major scientific domains: engineering, natural sciences, and social sciences. The primary objective was to compare and contrast thematic structures emerging from research discourse within each field.

A. Data Collection and Corpus Construction

The data corpus was compiled from peer-reviewed journal articles published between 2020 and 2024, focusing specifically on works that explicitly engaged with deep learning technologies. A total of 3,000 research paper titles were collected, with 1,000 titles assigned to each disciplinary corpus. Titles were retrieved from reputable academic databases such as IEEE Xplore, ScienceDirect, SpringerLink, Web of Science, and Scopus, using targeted keyword queries that included terms such as “deep learning,” “neural networks,” and “AI applications.” Only articles published in English were included to maintain linguistic consistency across the datasets.

B. Text Preprocessing

Raw title texts were subjected to a standardized preprocessing pipeline to ensure compatibility with LDA modeling requirements. Preprocessing was carried out using a combination of Python libraries, including NLTK and spaCy, chosen for their extensive NLP capabilities and integration with lemmatization tools [1].

C. LDA Model Implementation and Training

For topic extraction, we utilized the Gensim implementation of LDA, which is widely recognized for its computational efficiency and compatibility with large-scale text corpora [2]. To maintain consistency and interpretability across all three disciplinary corpora, the number of topics was initially fixed at four. This decision was informed by prior studies suggesting that lower topic numbers enhance interpretability in short-text contexts such as titles, and was later verified through coherence score optimization.

D. Evaluation Metrics and Model Selection

Model evaluation was conducted using both intrinsic and extrinsic metrics. The primary coherence metric used was the *c_v* score, which assesses the degree of semantic similarity among the top keywords within each topic. This measure is known to align well with human judgments of topic quality [3]. To complement this, the *u_mass* coherence score was also computed to provide a probabilistic estimate of internal consistency. For generalizability assessment, perplexity scores were calculated on held-out validation subsets to estimate how well the model could predict unseen data [4].

E. Visualization and Topic Interpretation

Interpretability of the resulting topic models was addressed using both quantitative and qualitative methods. Quantitatively, we employed LDAvis, an interactive visualization tool that enables users to explore the relationship between topics and their constituent keywords in two-dimensional space using multidimensional scaling (MDS) [5]. This visualization facilitated the assessment of inter-topic distance, topic dominance, and keyword salience, which proved critical in assigning meaningful labels to each topic.

F. Domain-Specific Model Structuring

To minimize interpretational confounding due to topic overlap across domains, we adopted a domain-separated modeling strategy. Rather than combining all titles into a single global model—which could obscure field-specific nuances—we trained independent LDA models for each domain. This approach preserved the internal semantics of each corpus while enabling more precise comparisons across domains.

III. RESULTS

The modeling outcomes reveal that deep learning research manifests with distinctive thematic priorities across the three disciplines, reflecting their epistemological orientations, data practices, and technological maturity.

In the engineering domain, topics were strongly centered around system-level implementation and control architecture. This thematic focus is indicative of a discipline primarily concerned with optimizing real-time responsiveness, fault detection, and reliability in intelligent systems. In the natural sciences, the topic structures were markedly shaped by data-driven experimentation and imaging technologies. Keywords such as MRI, assessment, tomography, and diagnostic prominently emerged, highlighting the application of deep learning in biomedical image analysis and structured data modeling. This aligns closely with ongoing efforts in computational biology, radiology, and environmental sensing, where AI methods are used to enhance diagnostic accuracy, automate complex measurement pipelines, and facilitate data-intensive discovery. In contrast, the social sciences exhibited a more diverse and exploratory set of topic patterns. Thematic clusters included ensemble learning methods, socioeconomic forecasting, behavioral analytics, and policy modeling. Representative keywords such as forecasting,

ensemble, framework, and algorithm point to a growing interest in integrating computational models into the analysis of complex human systems.

Despite these disciplinary distinctions, the topic models also revealed several shared methodological trends. Across all domains, keywords such as model, analysis, approach, and framework were frequently observed. This convergence indicates that while the applications of deep learning differ by field, the underlying computational strategies are becoming increasingly standardized.

IV. DISCUSSION AND CONCLUSIONS

This study set out to examine the evolving landscape of deep learning research through a cross-disciplinary lens, employing Latent Dirichlet Allocation (LDA) topic modeling to analyze 3,000 research paper titles drawn from engineering, natural sciences, and social sciences. Our results highlight the divergent yet occasionally overlapping ways in which deep learning has been conceptualized, implemented, and adapted across disciplinary boundaries.

In summary, this study affirms that deep learning is not merely a technological phenomenon but a transformational force that is reshaping scientific inquiry across domains. It acts as both an enabler of discovery and a catalyst for methodological innovation. By adopting a comparative framework, we offer a novel perspective on how different academic communities are internalizing and operationalizing AI technologies. Ultimately, this work contributes to a broader conversation on the epistemological and institutional implications of machine learning, encouraging further meta-analytical research that bridges disciplinary divides and fosters integrative knowledge production.

ACKNOWLEDGMENTS

This paper was supported by Joongbu University Research & Development Fund.

REFERENCES

- [1] S. Bird, E. Klein, and E. Loper, *Natural Language Processing with Python*, Sebastopol, CA: O'Reilly Media, 2009.
- [2] R. Rehurek and P. Sojka, "Gensim—Statistical Semantics in Python," [Online]. Available: <https://radimrehurek.com/gensim/>, 2010.
- [3] S. Syed and M. Spruit, "Full-text or abstract? Examining topic coherence scores using latent Dirichlet allocation," in *Proc. IEEE Int. Conf. Data Sci. Adv. Anal. (DSAA)*, pp. 165–174, 2017.
- [4] J. Tang, Z. Meng, X. Nguyen, Q. Mei, and M. Zhang, "Understanding the limiting factors of topic modeling via posterior contraction analysis," in *Proc. 31st Int. Conf. Machine Learning (ICML)*, pp. 190–198, 2014.
- [5] C. Sievert and K. Shirley, "LDAvis: A method for visualizing and interpreting topics," in *Proc. ACL Workshop on Interactive Language Learning, Visualization, and Interfaces*, pp. 63–70, 2014.

Analysis of Technology Acceptance Intentions for Prospective Elementary School Teachers in Introduction to Artificial Intelligence Classes

Semin Kim¹, and SungWon Seo^{2*}

¹Department of Computer Education, Jeonju National University of Education, Jeonju 55101, Republic of Korea

²Academic Career Counselor Center, Korea Advanced Institute of Science and Technology, Daejeon 34141, Republic of Korea

[Email : imsil303@hotmail.co.kr, rockcast@kaist.ac.kr]

Abstract

This study analyzed the technology acceptance intention of prospective elementary school teachers attending universities of education in Korea in the introduction to artificial intelligence classes. The research subjects were 44 people, and the technology acceptance intention test tool was used to analyze the results using descriptive statistics. The results of analyzing the sub-factors of the research test tool, such as performance expectancy, effort expectancy, social influence, facilitating conditions, and intention to use, showed that the facilitating conditions had the highest average and the smallest deviation. It is expected that this study will serve as a basis for research on the introduction to artificial intelligence course for prospective elementary school teachers.

Index Terms: Technology Acceptance Intention, Prospective Elementary School Teachers, Introduction to Artificial Intelligence, Artificial Intelligence Education, etc.

I. INTRODUCTION

As the importance of artificial intelligence is emphasized in the Korean education sector, the curriculum for prospective teachers has also changed significantly. In particular, following overseas cases of artificial intelligence education, artificial intelligence education has recently been reflected in the Korean curriculum, so it is necessary to understand the technology acceptance intentions of prospective elementary school teachers [1-4].

Accordingly, this study aims to analyze the technology acceptance intentions of prospective elementary school teachers taking the subject “AI Digital Education,” which is an introduction to artificial intelligence.

* Corresponding author : SungWon Seo

II. THEORETICAL BACKGROUND

A. Definition of Technology Acceptance Intentions

Technology acceptance intention means that when introducing a new technology, the ease and usefulness of the technology affect the intention to accept the technology. Perceived usefulness and perceived ease of use are mainly mentioned, and it can be statistically verified whether they have an effect. Although the structure of the model for technology acceptance intention is simple, the explanatory power of the model itself is high, so research is being conducted in various fields to predict user responses using technology acceptance intention. In this study, we aimed to measure technology acceptance intention when prospective elementary school teachers were trying to learn about artificial intelligence, a new technology.

Perceived usefulness refers to the degree to which users believe that using a new technology will increase their efficiency in work or study, and perceived ease of use refers to the degree to which users perceive that there will be no particular difficulties in the process of using a new technology, and whether additional effort is required from the user a lot or little.

B. Related Researchs of Technology Acceptance Intentions

Hong Hyun-mi's study measured the factors influencing prospective teachers' intention to use the metaverse based on the technology acceptance model and presented theoretical and practical implications[1].

Choi Jin-seon et al.'s study measured the factors influencing the intention to use online education using the technology acceptance model framework and presented related discussions and implications[2].

III. RESEARCH METHOD

This study conducted a survey on 45 prospective teachers attending a local education college in A region, and the data of 44 students with no errors in their responses were finally analyzed. The characteristics of the respondents who responded to this study included 14 male students and 30 female students. The research tool used was the questionnaire presented in the study published by Venkatesh et al. (2012) that was suitable for this study[5].

B. Research Model

In this study, in order to consider the educational characteristics of the research model, the model presented in the study by Eun-seon Choi et al[4]. was referenced and the hypothesis was set that performance expectancy, effort expectancy, social influence, facilitating conditions, and intention to use affect the intention to accept technology. The model used in this study is as shown in Fig. 1.

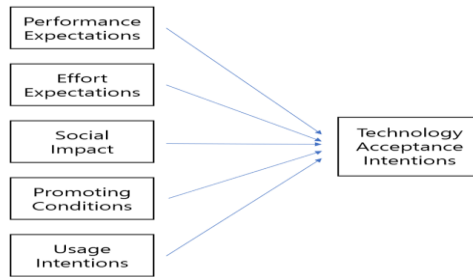


Fig. 1. Research Model of Technology Acceptance Intentions.

IV. RESEARCH RESULTS

The research results are presented in Table 1, showing the results of the five sub-factors of technology acceptance intention. In general, the technology acceptance intention was measured high in the performance expectation and facilitation conditions, while the technology acceptance intention was measured low in the effort expectation. In particular, the facilitation conditions showed the smallest deviation among respondents.

Table 1. Research Results of Technology Acceptance Intentions

Sub-Element	No.	Mean	STDEV.P
Performance Expectations	1	5.841	1.147
	2	5.818	1.302
	3	5.795	1.289
	4	5.886	1.265
Effort Expectations	5	3.750	1.448
	6	4.500	1.270
	7	3.818	1.628
	8	4.523	1.373
Social Impact	9	4.114	1.541
	10	5.182	1.211
	11	5.442	1.106
Promoting Conditions	12	5.295	1.099
	13	5.674	1.115
	14	5.581	1.186
	15	4.955	1.065
Usage Intentions	16	5.409	1.094
	17	4.455	1.514
	18	4.591	1.154

IV. CONCLUSION AND SUGGESTION

This study analyzed the intention to accept technology for the introduction to artificial intelligence class for prospective elementary school teachers. Descriptive statistics were conducted for each sub-factor, and the research results showed that the intention to accept technology for facilitating conditions was the highest. The limitation of the study is that the results were not necessarily generalized because factor analysis through a structural model was not conducted.

Therefore, future research will conduct scientific statistical verification through factor analysis through a structural model. It is expected that this study will serve as a basis for research on the introduction to artificial intelligence course for prospective elementary school teachers.

REFERENCES

- [1] H. M. Hong, "A study on factors affecting intention to use metaverse based on technology acceptance model," *Journal of The Korean Association of Information Education*, vol. 26, no. 6, p. 533-541, 2022.
- [2] J. S. Choi and Y. W. Kim, "A Study on Factors Influencing Intention to Use Online Learning Based on Technology Acceptance Model," *Journal of The Korean Association of Information Education*, vol. 27, no. 5, pp. 567-577, 2023
- [3] E. S. Choi and N. J. Park, "Teachers' Intention to Use Liveworksheets Applying UTAUT Model," *Journal of The Korean Association of Information Education*, vol. 25, no. 2, pp. 413-421, 2021.
- [4] J. B. Song, "Development and Validation of Artificial Intelligence Education on the Environmental Education Based on Unplugged," *Journal of The Korean Association of Information Education*, vol. 25, no. 5, pp. 847-857, 2021.
- [5] V. Venkatech, J. Y. L. Thong and X. Xu, "Consumer Acceptance and Use of Information Technology: Extending the Unified Theory of Acceptance and Use of Technology," *MIS Quarterly*, vol. 36, no. 1, pp. 157-178, 2012.

Proposing an Internal CNN Architecture for Efficient Deepfake Video Identification

Yong-chan Kim¹, Si-gwon Jo¹, Sung-hyun Kim² and Jong-min Kim^{2*}

¹Department of Information Security, Dongshin University, 58245, Republic of Korea

²Department of Computer Science, Dongshin University, 58245, Republic of Korea

[Email : yongchan@dsu.ac.kr / csk102400@dsu.ac.kr / 2110005@dsu.ac.kr / dyuo1004@dsu.ac.kr]

Abstract

With the advancement of artificial intelligence, deepfake technology has developed at a staggering pace. Because it's being used for malicious purposes like spreading fake news, defamation, and financial fraud, it's now recognized as a serious threat across society. To counter these threats, research into deepfake video identification technology is very active, and you could say that CNN-based detection algorithms are the core technology in the field of deepfake analysis. However, even these CNN-based detection algorithms are struggling to keep up with the rapid evolution of deepfake technology and are facing structural limitations. Therefore, in this study, we aim to propose an internal CNN architecture that can help overcome these limitations.

Index Terms: CNN, Deepfake, Artificial Intelligence.

I. INTRODUCTION

Deepfake is a technology that creates fake video and audio by exquisitely synthesizing a person's face or voice, making it very difficult to distinguish from the real thing. This is causing serious social and ethical problems, like spreading fake news, manipulating public opinion, defamation, and invasion of privacy. To counter these threats, a lot of research is going into deepfake detection technology. The key player here is the Convolutional Neural Network, or CNN, which plays a crucial role thanks to its powerful performance in image and video analysis. However, the existing CNN-based detection models are struggling to keep pace with the rapid evolution of deepfake technology, and they're facing a few structural limitations. First, there's the problem of losing subtle forgery patterns. The pooling layers in a CNN or a very deep network structure can end up losing the very evidence we need, like high-resolution information or the subtle texture artifacts that are tell-tale signs of a deepfake. Second, there's the limitation of the static filter structure. Once a CNN's convolutional filters are trained, they're fixed. They apply the same way to every input, which makes it hard for them to flexibly adapt to the diverse and unpredictable micro-variations in new deepfakes. Third is the overfitting problem that comes with deep structures. Deep CNN models have so many parameters that they can easily overfit to the training dataset. This leads to poor generalization performance when they encounter new types of deepfakes or different datasets. And fourth, there's a lack of temporal information. Many CNN-based detection models only analyze a single frame or a limited sequence of frames. This means they can't effectively capture the subtle temporal inconsistencies or unnatural movement patterns that appear across an entire video.

Therefore, in this paper, we propose an architecture that can supplement the existing CNN structure to overcome these limitations and more precisely extract the features of deepfake videos.

II. SYSTEM MODEL AND METHODS

A. CNN(Convolutional Neural Network)

A CNN treats a voice spectrogram as a 2D image, and it's very effective at using convolution operations to extract the local patterns from it—in other words, the acoustic features in specific time-frequency areas. Now, 'Deep Voice' recordings can contain artifacts from the generation process, such as subtle noise, spectral inconsistencies, or artificial patterns in certain frequency bands, and CNNs are particularly strong at detecting these spatial and local artifacts [1, 2]. A typical CNN model is composed of multiple convolutional layers, non-linear activation functions, pooling layers, fully-connected layers, and an output layer.

$$\hat{y} = \sigma(\sum_{i=1}^{128} w_i^{(3)} \cdot ReLU(\sum_{j=1}^N w_{ij}^{(2)} \cdot z_j)) \quad (1)$$

$$z_j = flatten_j(\phi(pool_3(\phi(pool_2(\phi(pool_1(\phi(X * K^{(1)} + b^{(1)})) * K^{(2)} + b^{(2)})) * K^{(3)} + b^{(3)})))))) \quad (2)$$

Conventional CNN architectures face several challenges in high-level classification tasks such as deepfake video detection. These include the loss of subtle forgery patterns, limitations of static filter structures, overfitting due to deep layers, and the inability to capture temporal information. To address these issues, this study aims to structurally enhance the internal architecture of CNNs and design a new model capable of more precisely extracting the distinctive features of deepfake videos.

III. Proposed Architecture Design

While previous formulations focused on learning the global differences of deepfakes using a single criterion (BCE loss) and a simple CNN layer flow, the proposed formulation maintains structurally deeper representations and introduces loss terms that quantify manipulation traces at various levels. This approach is expected to simultaneously improve the model's precision, local sensitivity, consistency, and generalization performance.

$$\hat{y} = 1 / (1 + \exp(-(\sum_{i=1}^{128} w_i^{(3)} \cdot \max(0, \sum_{j=1}^N w_{ij}^{(2)} \cdot flatten(\phi(pool_3(\phi(pool_2(\phi(pool_1(\phi(X * K^{(1)} + b^{(1)})) * K^{(2)} + b^{(2)})) * K^{(3)} + b^{(3)}))))))_j + b_i^{(2)})) + b^{(3)}))) \quad (3)$$

$$L_{total} = -y \cdot \log(\hat{y}) - (1 - y) \cdot \log(1 - \hat{y}) + \alpha \cdot Var(\widehat{y_{patches}}) + \beta \cdot \|F(X) - F(\widehat{X})\|_2^2 + \gamma \cdot Var_{(i,j)}[\phi((X * K)_{i,j})] \quad (4)$$

IV. Research Results

Table 1. Model Validation Results

Classification	Basic CNN (%)	Proposed CNN(%)
Accuracy	30.97	31.20
Precision	40.11	40.28
Recall	57.62	57.99
F1-Score	47.92	47.54

The results confirm that the improved CNN outperforms the baseline CNN across all performance metrics, with particularly notable gains in Recall and F1-score. This indicates that the proposed loss function contributes to enhancing the model’s detection sensitivity.

Although the absolute accuracy remains relatively low, this is largely attributable to limitations in the dataset and the restricted number of training epochs due to the experimental environment. As the number of epochs increases, the performance gap between the two models is expected to widen further.

The observed performance improvements in the enhanced model are closely tied to the structural characteristics of the proposed loss function. The FourierLossLayer, by incorporating frequency domain information of images, accentuates subtle manipulations—such as blurring or compression artifacts—that are hard to detect in the spatial domain. This helps the model learn covert editing patterns more effectively.

Additionally, the PatchVarianceLossLayer encourages the model to minimize the variance of output probabilities, thereby enforcing more decisive classifications in ambiguous regions between classes. This is directly supported by improvements in both Precision and Recall.

Lastly, the ResponseVarianceLossLayer increases the spatial consistency of activation patterns in the feature maps, enabling the CNN to more sensitively detect localized manipulations within images. This directly contributes to improved Recall and, consequently, a higher F1-score.

V. Conclusion

In this study, we designed a novel loss function structure for a CNN-based classification model aimed at deepfake image detection and proposed an approach to enhance the detection performance of existing models. The improved model incorporates three custom loss components that simultaneously consider frequency domain information, output probability stability, and spatial consistency of feature map responses. Experimental results demonstrated modest improvements in precision, recall, and F1-score compared to baseline models.

Notably, the increases in recall and F1-score suggest that the detection model is better at capturing manipulated images without missing them, indicating practical contributions to real-world deepfake mitigation systems. This study is significant in showing that even

within relatively simple CNN architectures, performance can be improved solely through mathematically designed loss functions.

ACKNOWLEDGMENTS

This work is a result of the Dongshin University Innovation Convergence Campus Project, conducted with funding from the Ministry of Land, Infrastructure and Transport in 2025 and support from Jeollanam-do and Naju-si.(2025ICCS-001)

REFERENCES

- [1] Alam, M. F., Feng, Y., Fang, F., & Larcher, A. (2020). A Comparative Study of Feature Representations for Synthetic Speech Detection. *Interspeech 2020*, 2377-2381.
- [2] Lavrentyeva, G., Khvedchenya, D., Vasiliev, E., & Larcher, A. (2019). Stcu r-cnn for synthetic speech detection. *ICASSP 2019*, 6764-6768.

A Comparative Analysis of AI Model Efficiency for Malicious URL Detection

Song-Bi Han¹, Hyeon-Yeong Nam¹, Myeong-Do Cha², Seog-Yeong Yun² and Jong-min Kim^{2*}

¹Department of Information Security, Dongshin University, 58245, Republic of Korea

²Department of Computer Science, Dongshin University, 58245, Republic of Korea

[Email : songbee0318@dsu.ac.kr / banabana@dsu.ac.kr / 2110030@dsu.ac.kr / ysysy9314@dsu.ac.kr / dyuo1004@dsu.ac.kr]

Abstract

URLs have emerged in various forms—such as phishing, malware distribution, and spam propagation—posing serious security threats to users. These attacks can lead to significant losses, including personal information leakage, financial damage, and system paralysis. Moreover, the techniques used are becoming increasingly intelligent and automated. Attackers rapidly generate and mutate malicious URLs to evade detection systems, making traditional blacklist-based or fixed heuristic rule-based detection methods ineffective in responding to newly emerging malicious URLs in real time. Therefore, this paper conducts an in-depth comparative analysis of the performance and efficiency of three representative machine learning models—Random Forest, LightGBM, and SGDClassifier—to evaluate the effectiveness of AI models in detecting malicious URLs in real time. The analysis identifies the most optimal model for malicious URL detection. The results show that the LightGBM model outperforms the others in terms of handling large-scale datasets and efficient training, making it the most suitable model for application in real-time system environments.

Index Terms: Random Forest, LightGBM, SGD classification, Malicious URL.

I. INTRODUCTION

Recently, machine learning and deep learning technologies have gained significant attention in the field of malicious URL detection due to their remarkable ability to recognize complex patterns through large-scale data training. Although various machine learning models can be applied to URL-based threat detection, each model comes with its own characteristics in terms of learning methodology, computational complexity, data requirements, and performance based on hyperparameters. Real-time detection systems demand not only high detection accuracy—particularly low false-negative and false-positive rates—but also fast prediction speed to handle the rapidly growing volume of URL traffic. Additionally, an efficient training process is essential to keep up with constantly evolving malicious URL patterns. Therefore, a model cannot be considered suitable for real-time systems based solely on its detection accuracy; its overall efficiency—including the time and resources required for both training and prediction—must also be taken into account. Accordingly, this paper aims to identify the most efficient AI model for URL-based malicious URL detection by conducting a comparative performance and efficiency analysis of three models: Random Forest, LightGBM, and

SGDClassifier. The analysis evaluates each model in relation to dataset size and number of training iterations, ultimately determining the optimal model for real-time application.

II. SYSTEM MODEL AND METHODS

A. Malicious URL Detection Technology

Malicious URL detection techniques can be broadly categorized into static analysis and dynamic analysis. Static analysis leverages information that can be obtained before accessing a website, such as the URL string itself or the webpage's source code. On the other hand, dynamic analysis involves actually visiting the website to analyze its content or simulate malicious behavior in order to detect threats. Blacklist-based and heuristic-based detection methods typically fall under static analysis, while recent machine learning and deep learning-based approaches utilize features from both static and dynamic analysis, or focus solely on analyzing the URL string itself.

Blacklist-based detection compares URLs against a known list of malicious URLs, making it extremely fast. However, it has a fundamental limitation in that it is ineffective against newly generated malicious URLs. Heuristic-based detection uses predefined rules such as URL length, specific character patterns, or domain characteristics to offer more flexibility, but it is vulnerable to rule evasion attempts and requires constant updates and maintenance to address new attack techniques. Machine learning and deep learning-based detection models overcome these limitations by learning complex patterns from large datasets of benign and malicious URLs, showing high detection performance.

B. Random Forest

This model applies bagging, one of the ensemble learning techniques, to decision trees. It combines the results of multiple decision trees to reach a single outcome. Each decision tree is trained on a different dataset generated through bootstrap sampling, which results in varied outputs every time. This characteristic helps reduce the risk of overfitting. It is also robust to discrete values and missing data, and allows for identifying the importance of features, which is why it was selected as a model for comparison.

C. LightGBM

This model is a type of tree-based Gradient Boosting Machine (GBM) that does not consider tree balancing. It continuously splits the leaf node with the maximum loss value, resulting in deeper and asymmetrical trees. Unlike balanced tree splitting, it focuses on splitting in the direction that reduces the loss value, minimizing prediction error loss. This model is well-suited for handling large-scale data and was selected as a model for comparison.

D. SGDClassifier

Unlike batch gradient descent, which calculates the gradient of the cost function using the entire dataset, this model computes the gradient and updates parameters using one or a few training samples at a time to minimize error. It can quickly approximate solutions to nonlinear optimization problems. It is effective for large-scale datasets, and the noise generated by updates helps escape local optima, which is why it was selected as a model for comparison.

III. Proposed method

A. Dataset

The analysis dataset was based on normal URLs from the global internet access site survey by alex.com [1], and malicious URLs were sourced from the Kaggle malicious URL dataset [2]. Additionally, the dataset was composed of malicious and normal URLs in a 1:1 ratio, with datasets containing 4,000, 10,000, 20,000, 60,000, and 100,000 entries respectively.

B. Evaluation metrics

The experimental process involves observing the changes in evaluation metrics—accuracy, precision, recall, and F1 score—of AI models for each dataset. The F1 score, which evaluates precision and recall, indicates better predictive performance as its value increases. Although all evaluation metrics typically range between 0 and 1, this paper intends to convert them into percentages for use.

$$F1 = 2 \times \frac{\text{Precision} \times \text{Recall}}{\text{Precision} + \text{Recall}} \quad (1)$$

By comparing the rate of increase and the evaluation metrics for each dataset, the model to be used is selected. To identify the optimal dataset, additional datasets of 4,000, 10,000, 20,000, 60,000, 100,000, 120,000, 140,000, 160,000, 180,000, and 200,000 samples are trained. The evaluation metrics are assessed by comprehensively considering all components, but when the differences between metrics are similar or no single result dominates in dataset comparisons, the evaluation favors the result with higher accuracy.

IV. Result Analysis

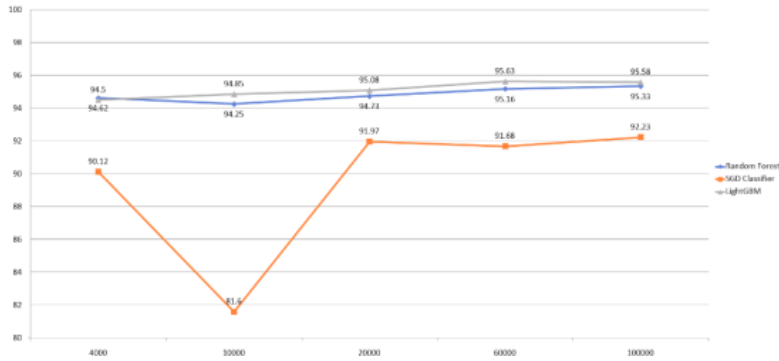


Fig. 1. Changes in AI Accuracy

Fig. 1 shows the change in accuracy, one of the evaluation metrics, when Random Forest, LightGBM, and SGD methods were trained on each dataset. Initially, with 4,000 samples, Random Forest showed strong performance, but starting from 10,000 samples, the evaluation metrics for LightGBM increased, ultimately recording the highest value.

Therefore, LightGBM was selected as the model to use. However, since it continued to show improvement, additional datasets were trained to determine the dataset size that achieves the highest accuracy.

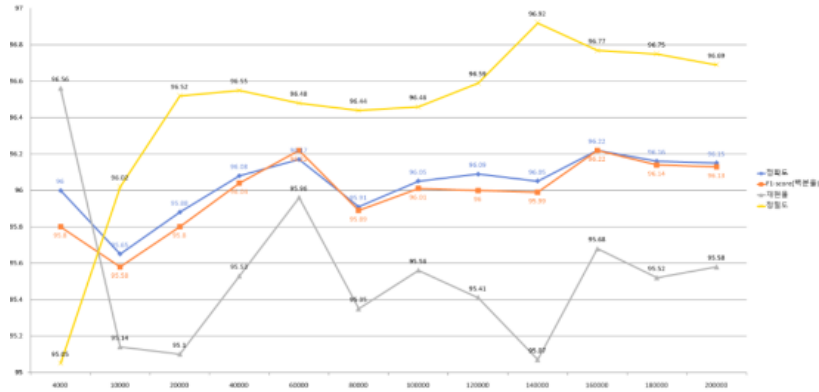


Fig. 2. Evaluation Metrics of LightGBM

Fig. 2 shows a graph of the changes in evaluation metrics according to the size of each dataset. It can be seen that the dataset size producing the maximum value differs for each metric: accuracy, recall, and F1 score peaked at 160,000 samples, while precision reached its highest value at 140,000 samples. Therefore, since most evaluation metrics achieved their best results with the dataset of 160,000 samples, the AI model used for the subsequent system implementation was decided to be LightGBM trained with 160,000 data points.

V. Conclusion

In this paper, we divided the datasets by size and trained Random Forest, LightGBM, and SGD models to compare their evaluation metrics. As a result, LightGBM was selected as the AI model for system implementation, and the optimal dataset size was determined to be 160,000 samples. This outcome enabled us to choose an AI model capable of early detection of continuously emerging malicious URLs and adapting to changes in the dataset. In the future, we plan to further specify the threshold point that optimizes the results between AI and API inspections within the system.

ACKNOWLEDGMENTS

This work is a result of the Dongshin University Innovation Convergence Campus Project, conducted with funding from the Ministry of Land, Infrastructure and Transport in 2025 and support from Jeollanam-do and Naju-si.(2025ICCS-001)

REFERENCES

- [1] <https://github.com/PeterDaveHello/top-1m-domains?tab=readme-ov-file>
- [2] <https://www.kaggle.com/datasets/hassaanmustafavi/phishing-urls-dataset>

A High-Dimensional Encryption Architecture Based on LSB Distributed Embedding and Chaos Shuffling in Unstructured 3D Structures

Tae-Gyu Yang¹, Yang-Hyun Kim², Il-Guk Song² and Jong-min Kim^{2*}

¹Department of Information Security, Dongshin University, 58245, Republic of Korea

²Department of Computer Science, Dongshin University, 58245, Republic of Korea

[Email : xorb4413@gmail.com/ kimyanghyeon12@gmail.com/ilguk9537@gmail.com/dyuo1004@dsu.ac.kr]

Abstract

This paper proposes a novel high-dimensional encryption technique that integrates the structural characteristics of 3D point clouds with chaos theory-based sequences. The proposed method performs encryption by fragmenting and distributing data into the least significant bits (LSBs) of the point cloud, followed by random reordering of the entire point coordinates through chaotic mapping. Decryption relies on the same structure and chaotic seed, and even a slight deviation in seed value or coordinate order renders data recovery impossible. The chaotic seed utilizes highly sensitive real-number initial conditions, theoretically allowing for an infinite key space. Furthermore, the combination of point structure and embedding order achieves extremely high randomness and resistance to prediction. These features meet advanced security requirements such as plaintext concealment, structural obfuscation, and irreversible data deletion. This encryption scheme presents a new paradigm applicable to security-critical industries such as finance, defense, healthcare, and the Internet of Things(IoT).

Index Terms: 3D Encryption, Point Cloud, Chaos Mapping, Structural Security, High-Dimensional Cryptography, Data Concealment, Post-Quantum Cryptography, Information Security

I. INTRODUCTION

Recent advancements in information and communication technologies have led to an explosive growth in data generation, bringing the importance of information security to the forefront more than ever before. While currently used encryption standards such as AES and RSA offer high levels of reliability, they are now facing two major challenges. The first challenge arises from the rapid development of artificial intelligence, which has enabled intelligent attack techniques capable of analyzing encryption patterns and exploiting vulnerabilities in traditional cryptographic systems. The second challenge is the emerging practicality of quantum computing. In particular, Shor's algorithm poses a significant threat to public-key encryption schemes like RSA, as it can solve the underlying factorization problem in polynomial time—potentially rendering today's public key infrastructure obsolete. In light of these evolving threats, a growing consensus is

emerging that merely increasing key lengths is insufficient to fully counter future security risks.

As a fundamental alternative to traditional methods, this paper proposes a novel encryption model that leverages the complexity of three-dimensional (3D) space, moving beyond the conventional two-dimensional plane. The proposed approach integrates the geometric structure of 3D point clouds with the mathematical unpredictability of chaos theory. Data is concealed by being distributed throughout the coordinates of a vast 3D point cloud and then shuffled based on unpredictable sequences generated through chaotic mapping. This introduces extreme randomness and non-linearity into the encryption process, making it fundamentally impossible to infer the position or order of the data. Unlike traditional schemes where the relationship between the key and plaintext is well-defined, the structure that holds the data in this model itself acts as a massive, dynamic key. This paper outlines the necessity of such a shift and introduces a high-dimensional encryption framework that not only transcends the limitations of conventional encryption but also achieves enhanced security and quantum resistance through its structural and mathematical complexity.

II. SYSTEM MODEL AND METHODS

A. Quantum Computers and Encryption Threats

Quantum computers utilize the phenomena of qubit superposition and entanglement to perform levels of parallel computation that are impossible for classical computers. This poses a serious threat to modern cryptography. Peter Shor's "Shor's Algorithm" can factor very large numbers rapidly, which fundamentally breaks public key cryptosystems like RSA and ECC that are based on the difficulty of factorization. In addition, "Grover's Algorithm" reduces the time required to search an unsorted database from N steps to $\sqrt{N}$ steps, effectively halving the effective key length of symmetric encryption algorithms like AES. To address these threats, the National Institute of Standards and Technology (NIST) is leading the standardization of Post-Quantum Cryptography (PQC). The structure-based encryption proposed in this paper can also be considered an alternative approach within the realm of PQC.

B. Chaos Theory

Chaos theory is a field of study that deals with seemingly unpredictable irregularities found in nonlinear dynamic systems. The key characteristic of chaotic systems is their "sensitivity to initial conditions," commonly known as the butterfly effect. This refers to the phenomenon where even an infinitesimal difference in the starting value can amplify over time into completely different outcomes. Even a simple mathematical equation like the Logistic Map, expressed as $x(n+1) = r \cdot x(n) \cdot (1 - x(n))$, can generate extremely complex and pseudo-random sequences. These properties are highly valuable in cryptography. A real-number-based initial seed can offer virtually infinite key space, and the unpredictability of the generated sequence contributes significantly to the security of a cryptographic system [2]. Previous research has already applied chaos sequences in 2D contexts—for example, shuffling pixel values of images or generating keystreams for stream ciphers. This study extends those 2D-based ideas into a 3D unstructured framework, applying chaos theory in a higher-dimensional manner by shuffling the

coordinate order itself, thereby achieving a more advanced level of obfuscation and security.

III. Proposed Method

The encryption method proposed in this paper divides data into fixed-size blocks, then embeds each block into the least significant bits (LSBs) of a 3D point cloud, and finally shuffles the entire sequence of points randomly using chaotic mapping methods such as the Logistic map. In this structure, the embedding coordinates and order themselves function as the decryption key, and the combination of a real-number-based chaotic seed and embedding rules creates an almost infinite key space. Dummy data is randomly inserted into the remaining points, making it difficult to distinguish between real and fake data. During decryption, only by using exactly the same chaotic seed and embedding rules (e.g., block size, number of iterations, coordinate sequence) can the original data order be restored; if the coordinate information differs even slightly, completely different results are produced.

A. Encryption Procedure

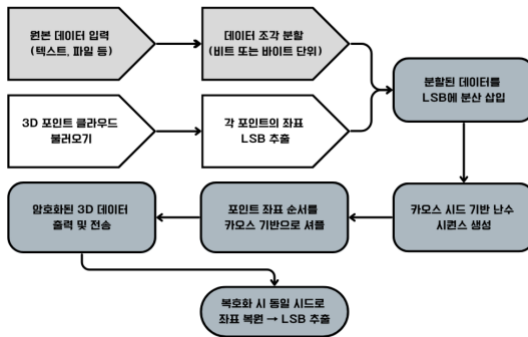
1) Generate/prepare the 3D point cloud

Using GANs, noise-based generative models, or random matrix methods, unique point clouds are generated for each user. In the test environment, a cubic structure of about $3 \times 3 \times 3$ is used, but in real applications, high-resolution models with tens of millions of points or more can be employed, offering a nearly infinite key space.

2) Divide data into blocks and embed into LSBs

The original data to be encrypted is divided into blocks of fixed size (in bits or bytes). Then, each data fragment is sequentially or according to a predefined rule embedded into the least significant bits (LSB) of the (x, y, z) coordinate values of each point constituting the 3D point cloud. For example, one point can hide a total of 3 bits of data—1 bit each in the x, y, and z coordinates.

Fig. 1. Encryption Structure Flowchart



3) Perform coordinate shuffling based on chaotic mapping

By inputting an initial seed value (x_0) and parameter (r), which serve as the secret key, into a chaotic function such as the logistic map, an unpredictable random sequence is generated. This sequence is then sorted to create a permutation of point

indices. According to this permutation, the entire sequence of points in the 3D point cloud is rearranged. This shuffling process completely scrambles the locations where data is embedded, making it impossible for outsiders to determine the order.

4) Insert dummy data

The LSBs of the remaining points, which do not contain actual data, are filled with random values (dummy data) to prevent distinguishing which points hold valid information. This approach is effective in defending against attacks based on statistical analysis.

B. Decryption Procedure

1) Verify key information

The receiver must possess the exact same 3D point cloud structure used for encryption, the identical chaotic seed values (x_0 , r), and the same data embedding rules (such as block size and insertion order).

2) Restore the coordinate order

Using the shared chaotic seed and parameters, the receiver regenerates the same pseudo-random sequence used during encryption, which is then used to restore the original order of the shuffled points.

3) Extract and recombine data

From the point cloud reordered to its original sequence, the receiver sequentially extracts the LSB values of the point coordinates according to the predefined rules. The extracted data fragments are then reassembled in the original order to reconstruct the plaintext data. If even a single piece of key information is slightly different, the point order will be entirely mismatched, resulting in meaningless noise upon extraction. In this way, the proposed method is not merely a conventional encryption algorithm, but a multidimensional security system that uses physical and procedural elements—specifically structure and sequence—as part of the encryption key.

IV. Model Comparison and Analysis

This paper presents a comparative analysis of the proposed model against AES and PQC algorithms(Kyber, SPHINCS+) based on five key security criteria.

Table 1. Model Validation Results

Evaluation Metric	AES-256	Kyber	SPHINCS+	3D Chaotic Encryption
Brute Force Resistance	High	High	High	Excellent
Quantum Resistance	Vulnerable	Excellent	Excellent	Excellent

Structural Obfuscation	Low	Medium	Medium	Excellent
Sequence Randomness	Low	Low	Low	Excellent
AI Guessing Resistance	Medium	Medium	High	Excellent

As a result of the comparison, the proposed 3D chaos-based encryption model offers a new dimension of security that goes beyond the key-length-dependent security of conventional encryption methods by leveraging structural complexity and procedural randomness. In particular, its ability to obscure the very existence of data through structural obfuscation, combined with the unpredictability derived from chaos theory, demonstrates overwhelming superiority over other models in addressing future security threats posed by quantum computing and AI technologies.

V. Conclusion

This paper successfully presents a new high-dimensional encryption paradigm that surpasses traditional planar encryption schemes by combining the structural characteristics of 3D point clouds with the unpredictability of chaos theory. The proposed method redefines the encryption key not as a simple bitstring but as a complex combination of “3D structure,” “chaotic seed,” and “insertion rules,” thereby achieving theoretically infinite key space and strong irreversibility. As a result, it effectively conceals the very existence of the data and maximizes the irregularity of insertion positions and order, offering strong resistance against modern security threats such as pattern-based analysis, quantum algorithm attacks, and AI-driven inference. This approach goes beyond merely encrypting data—it realizes a new security philosophy that hides the data itself and leverages structural complexity for protection. The study confirms the scalability of the proposed method as a next-generation high-trust security technology and holds significant value in proposing an immediately applicable security solution for critical industries handling sensitive information, such as defense, finance, and healthcare.

ACKNOWLEDGMENTS

This work is a result of the Dongshin University Innovation Convergence Campus Project, conducted with funding from the Ministry of Land, Infrastructure and Transport in 2025 and support from Jeollanam-do and Naju-si.(2025ICCS-001)

REFERENCES

- [1] NIST, Post-Quantum Cryptography Project. <https://csrc.nist.gov/projects/post-quantum-cryptography>
- [2] M.S. Baptista, "Cryptography with Chaos", Physics Letters A, 1998.
- [3] Y. Khedmati et al., "2D Hybrid Chaos Map for Image Security", 2018.

Spectral Normalized U-Net for Light Field Occlusion Removal

Mostafa Farouk Senussi¹, Mahmoud Abdalla¹, Mahmoud SalahEldin Kasem¹, Mohamed Mahmoud¹, and Hyun-Soo Kang^{1*}

¹School of Information and Communication Engineering, Chungbuk National University,
Cheongju, 28644, Republic of Korea
/Email: hskang@cbnu.ac.kr/

Abstract

Occlusion artifacts significantly hinder light field (LF) image reconstruction, especially in complex scenes. We propose a spectral normalized U-Net for LF occlusion removal, which begins by stacking LF views and extracting view-dependent features using a local feature encoder. To capture spatial complexity, ResASPP enable multi-scale context aggregation, while channel attention enhances occlusion-related features. Spectral normalization is applied to all convolutional layers to improve training stability and generalization. The encoder-decoder structure with skip connections preserves fine details. Experimental results show our method restores occluded regions more accurately than baselines.

Index Terms: LF Occlusion Removal, Spectral Normalization, Channel Attention, U-Net.

I. INTRODUCTION

Light field (LF) Light field (LF) imaging captures both spatial and angular information of light, enabling post-capture view synthesis and depth-aware processing. However, occlusion artifacts [1] remain a major challenge, especially in scenes with overlapping structures, degrading synthesized views and hindering tasks like object detection and image inpainting [2–4]. These artifacts disrupt scene continuity and lead to inaccuracies in localization and content restoration.

Occlusion removal in LF data is inherently difficult [5], as it requires fusing multi-view information. Traditional inpainting techniques often ignore inter-view dependencies, while early LF-specific methods fail to model the complex interactions between visible and occluded regions. Although deep learning has improved performance via convolutional and attention-based models, challenges in generalization and training stability persist.

To address this, we propose a deep learning model based on a U-Net backbone with spectral normalization. Our method stacks LF views and extracts view-specific features using a local feature encoder enhanced with ResASPP, residual blocks, and channel attention for robust occlusion reasoning. Spectral normalization ensures stable training, while skip connections preserve fine details, enabling high-quality, occlusion-free LF reconstruction. Our approach outperforms existing baselines in both quantitative and qualitative metrics, offering a simple yet effective solution for LF occlusion removal.

* Corresponding author

II. PROPOSED METHOD

We present a deep learning model for LF occlusion removal using a spectral normalized U-Net. As shown in Fig. 1, the architecture comprises a Local Feature Extractor (LFE) and an Occlusion Reconstruction (OR) module. The stacked LF input captures spatial-angular information, which the LFE encodes into multi-scale features. These are processed by the SN-enhanced U-Net for stable and accurate occlusion-free reconstruction.

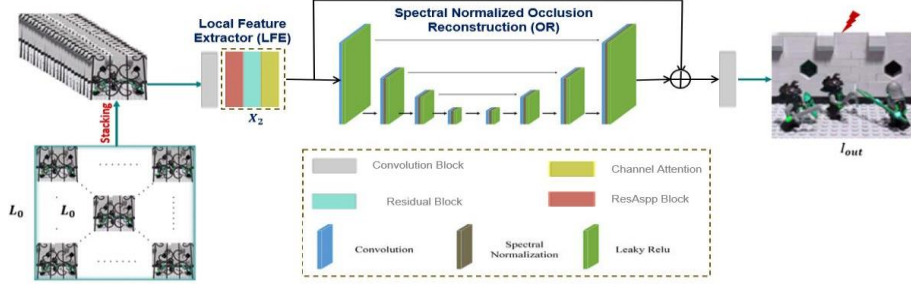


Fig. 1. Overview of the Proposed Spectral Normalized U-Net for Occlusion Removal in LF Images.

A. Local Feature Extractor (LFE)

The LFE starts by stacking all LF views into a 3D tensor L_0 , which is processed by a 3×3 convolutional layer (stride 1, padding 1), capturing spatial and angular correlations. To model multi-scale context, a ResASPP block applies parallel atrous convolutions with dilation rates $\{1, 3, 6, 12\}$, followed by Leaky ReLU activations. Their outputs are concatenated, compressed via a 1×1 convolution, and added through a skip connection.

A residual block further refines, enhancing local structure and feature quality. To emphasize occlusion-relevant features, a channel attention module generates an attention vector using global average pooling followed by two fully connected layers and a sigmoid activation. The attended feature is then passed to the Occlusion Reconstruction module for full-scene recovery.

B. Spectral Normalized Occlusion Reconstruction (OR)

The Occlusion Reconstruction (OR) module is a U-Net architecture enhanced with spectral normalization (SN) in all convolutional layers except the first, improving training stability and generalization. The encoder contains four downsampling stages: the first uses a standard convolution with Leaky ReLU, while the remaining use SN-convolutions. These progressively reduce spatial size while enriching feature depth.

The decoder mirrors this structure with bilinear upsampling and SN-convolutions. Skip connections transfer high-resolution encoder features to corresponding decoder layers, preserving detail and enabling multi-scale context fusion, which is concatenated with encoder features. Finally, a 1×1 convolution generates the center-view occlusion-free output $I_{out} \in \mathbb{R}^{H \times W \times 3}$.

III. EXPERIMENTS AND RESULTS

A. Experimental Setup

We trained on 1,418 DUTLF-V2 [6] images with 5×5 views (256×192), using [7]’s occlusion masks and 21 new dense masks. Training ran 500 epochs with ADAM, batch size 18, and LR 0.001 (halved every 150 epochs), completing in ~ 1 day on an RTX 4090. Evaluation used synthetic (4-Syn, DUTLF-V2), real (CD), and Single/Double Occ datasets from Stanford Lytro and EPFL-10 with disparities 1–4.

B. Experimental Results

1) Quantitative Results

As shown in Table 1, our method achieves the highest PSNR and SSIM across all LF datasets, confirming its effectiveness in occlusion removal. On synthetic sparse datasets, it scores 26.35 dB / 0.849 (4-Syn) and 26.79 dB / 0.857 (9-Syn), outperforming all baselines. For the real sparse CD, it achieves 25.10 dB / 0.863, demonstrating strong generalization. In dense cases, it attains top PSNR on Single Occ (30.05 dB) and Double Occ (28.47 dB), with competitive SSIM, showing robust performance under complex occlusions.

Table 1. Quantitative Analysis of Sparse and Dense LF Datasets Using PSNR/SSIM: Top Result in Red.

LF Type	Name	RFR [8]	LBAM [9]	DeOccNet [7]	Zhang [10]	Ours
Sparse (syn)	4-Syn	19.89 / 0.668	21.11 / 0.677	23.74 / 0.701	14.46 / 0.683	26.35 / 0.849
	9-Syn	20.69 / 0.672	23.04 / 0.725	23.70 / 0.715	22.00 / 0.758	26.79 / 0.857
Sparse (real)	CD	21.13 / 0.646	21.56 / 0.803	22.70 / 0.741	20.19 / 0.832	25.10 / 0.863
Dense (syn)	Single Occ	26.28 / 0.867	27.92 / 0.668	28.67 / 0.914	23.15 / 0.900	30.05 / 0.827
	Double Occ	23.25 / 0.801	24.83 / 0.827	25.85 / 0.847	18.01 / 0.823	28.47 / 0.849

2) Qualitative Results

As shown in Fig. 2, our method outperforms others in reconstructing occluded regions with high structural and textural accuracy across various LF scenes. In the first row (sparse synthetic), single-image inpainting methods like RFR [8], LBAM [9], and DeOccNet [7] exhibit blurring or artifacts, while Zhang et al. [10] fail to maintain geometry. Our model recovers fine details and spatial coherence, especially in repetitive patterns.

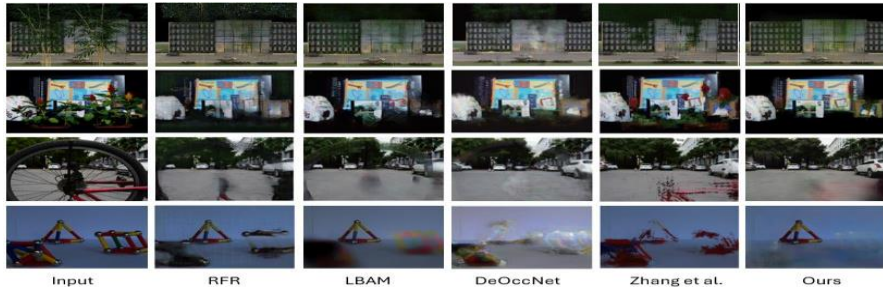


Fig. 2. Visual comparison of our method against existing approaches on sparse and dense LF datasets.

In real-world scenes (rows two and three), competing methods suffer from ghosting and photometric inconsistencies, whereas our model restores sharper, semantically consistent regions. In the dense synthetic scene (fourth row), others show color bleeding and deformation, but our method preserves object boundaries and depth structure, highlighting the effectiveness of our feature extractor and spectral normalized modules.

IV. CONCLUSION

In this paper, we introduced a spectral normalized U-Net architecture for occlusion removal in LF images, combining ResASPP for multi-scale context, residual blocks with channel attention, and spectral normalization for stable training. Our model effectively reconstructs occlusions and outperforms existing baselines.

ACKNOWLEDGMENTS

This work was supported by the National Research Foundation of Korea (NRF) grant funded by the Korean government (Ministry of Science and ICT) (RS-2023-NR076833).

REFERENCES

- [1] M. F. Senussi, M. Abdalla, M. S. Kasem, M. Mahmoud, B. Yagoub and H. -S. Kang, "A Comprehensive Review on Light Field Occlusion Removal: Trends, Challenges, and Future Directions," in *IEEE Access*, vol. 13, pp. 42472-42493, 2025, doi: 10.1109/ACCESS.2025.3548133.
- [2] Kasem, M.S.; Mahmoud, M.; Yagoub, B.; Senussi, M.F.; Abdalla, M.; Kang, H.-S. HTTD: A Hierarchical Transformer for Accurate Table Detection in Document Images. *Mathematics* 2025, 13, 266. <https://doi.org/10.3390/math13020266>.
- [3] Mahmoud, M.; Yagoub, B.; Senussi, M.F.; Abdalla, M.; Kasem, M.S.; Kang, H.-S. Two-Stage Video Violence Detection Framework Using GMFlow and CBAM-Enhanced ResNet3D. *Mathematics* 2025, 13, 1226. <https://doi.org/10.3390/math13081226>.
- [4] Abdalla, M.; Kasem, M.S.; Mahmoud, M.; Yagoub, B.; Senussi, M.F.; Abdallah, A.; Hun Kang, S.; Kang, H.S. ReceiptQA: A Question-Answering Dataset for Receipt Understanding. *Mathematics* 2025, 13, 1760. <https://doi.org/10.3390/math13111760>.
- [5] Senussi, M.F.; Kang, H.-S. Occlusion Removal in Light-Field Images Using CSPDarknet53 and Bidirectional Feature Pyramid Network: A Multi-Scale Fusion-Based Approach. *Appl. Sci.* 2024, 14, 9332. <https://doi.org/10.3390/app14209332>.
- [6] Piao, Y., Rong, Z., Xu, S., Zhang, M., & Lu, H. (2020). DUT-LFSaliency: Versatile dataset and light field-to-RGB saliency detection. arXiv preprint arXiv:2012.15124.
- [7] Wang, Y., Wu, T., Yang, J., Wang, L., An, W., & Guo, Y. (2020). DeOccNet: Learning to see through foreground occlusions in light fields. In *Proceedings of the IEEE/CVF winter conference on applications of computer vision* (pp. 118-127).
- [8] Li, J., Wang, N., Zhang, L., Du, B., & Tao, D. (2020). Recurrent feature reasoning for image inpainting. In *Proceedings of the IEEE/CVF conference on computer vision and pattern recognition* (pp. 7760-7768).
- [9] Xie, C., Liu, S., Li, C., Cheng, M. M., Zuo, W., Liu, X., ... & Ding, E. (2019). Image inpainting with learnable bidirectional attention maps. In *Proceedings of the IEEE/CVF international conference on computer vision* (pp. 8858-8867).
- [10] Zhang, S., Shen, Z., & Lin, Y. (2021, August). Removing Foreground Occlusions in Light Field using Micro-lens Dynamic Filter. In *IJCAI* (pp. 1302-1308).

TOF and Vision Fusion Sensor Night Time Testing and Learning Data Collection

A. Nomin¹, Dai-Hyun Jang², Gwanghyun Jo³, Seong-Yoon Shin^{4*}

¹Ulaanbaatar khot, Khan-Uul duureg 3-r khoroo Gerelt-Urguu khothon 47e-51toot, Mongolia

²LG Uplus, LG Uplus Bldg, 32, Hangang-daero, Seoul, 04389 Republic of Korea

³Department of Mathematical Data Science, Hanyang University ERICA, Ansan-si, 15588, Republic of Korea

⁴School of Computer Science and Engineering, Kunsan University, Kunsan-si, 54150, Republic of Korea

Email: min0904@gmail.com, daijang@lguplu.co.kr, gwanghyun@hanyang.ac.kr, s3397220@kunsan.ac.kr

Abstract

TOF and Vision fusion sensors play a major role as vehicle safety systems. In this paper, we will test the developed fusion sensor during nighttime hours and learn about learning and test data collection using object recognition algorithms.

Index Terms: TOF, Vision Fusion Sensor, Safety System, Data Collection

I. INTRODUCTION

In our previous paper, we developed a TOF and Vision fusion sensor as a vehicle safety support system to prevent fatal accidents in vehicle blind spots [1].

The representative technology applied to vehicles to prevent vehicle collision accidents is ADAS (Advanced Driving Assistant System). However, ADAS technologies currently being supplied and developed are focused on lane departure/forward collision/vehicle gap maintenance/front vehicle departure notifications, etc.

In this paper, we will learn about experiments for vehicle safety at night using this fusion sensor and about object recognition algorithm learning and test data collection.

II. EQUIPPED WITH FUSION SENSOR VEHICLE

A. Vehicle installation

- Laptop-USB extension cable-Fusion sensor 1 set configuration
- Test by attaching and detaching at each location
- Power supply to sensor and laptop using vehicle inverter

B. Fusion sensor test conditions

- ① Day/night distinction
- ② Vehicle stop/Vehicle driving
- ③ Object stop/Object movement
- ④ Sensor installation location (front, side, rear)
- ⑤ Sensor installation angle (0° , 24°)



Fig. 1 Vehicle mounting of fusion sensor prototype

III. NIGHT TIME ZONE VERIFICATION

A. Test condition

- Night / Vehicle stopped / Object moving / Front installation / Installation angle 0°

B. Collected data (Fig. 1)

Time	Seg	Amp	Dist	Flag
874253	4	167.06	4.89	1
874253	2	1.68	14.60	1
874253	2	2.67	23.84	1
874253	0	6.97	16.63	1
874487	7	1.83	9.23	1
874487	7	1.59	18.59	1
874487	6	2.11	17.74	1
874487	5	2.75	18.25	1
874487	4	128.49	5.00	1
874487	3	4.52	5.10	1
874487	2	1.62	14.62	1
874487	2	2.74	23.92	1
874487	0	6.98	16.68	1
874720	7	1.76	9.39	1
874720	7	1.62	18.53	1
874720	6	2.17	17.76	1
874720	5	2.66	18.25	1
874720	4	113.56	5.11	1
874720	3	6.07	5.15	1
874720	2	1.73	14.65	1
874720	2	2.65	23.90	1
874720	0	7.38	16.87	1
874953	7	1.78	9.30	1
874953	7	1.73	18.63	1
874953	6	2.14	17.74	1
874953	5	2.80	18.29	1
874953	4	100.27	5.32	1



Fig. 1. Convergence sensor night object detection experiment

C. Analysis

- Data acquisition while the tester moves freely
- Distance value increases as Amplitude value of front segment 4 decreases

IV. OBJECT LEARNING AND TEST DATA COLLECTION

A. Test conditions

- Daytime / Vehicle driving / Object moving (stationary) / Forward installation / Installation angle 0°

B. Collected data (Fig. 2)

Time	Seg	Amp	Dist	Flag
8301189	3	9.95	16.18	1
8301189	2	15.18	16.40	1
8301189	2	1.37	36.19	1
8301189	1	2.02	20.78	1
8301189	1	1.58	36.89	1
8301189	0	2.31	19.31	1
8301189	0	5.65	37.69	1
8301423	5	11.26	6.42	1
8301423	4	47.59	8.54	1
8301423	3	30.11	8.61	1
8301423	3	10.95	16.07	1
8301423	2	16.98	16.26	1
8301423	1	2.16	20.74	1
8301423	1	7.99	36.76	1
8301423	0	2.02	19.39	1
8301423	0	6.01	37.44	1
8301654	7	1.72	9.12	1
8301654	5	8.18	8.50	1
8301654	4	51.03	8.99	1
8301654	3	24.77	9.01	1
8301654	3	11.16	15.91	1
8301654	2	17.83	16.15	1
8301654	1	2.80	20.57	1
8301654	1	11.36	36.61	1
8301654	0	2.28	19.17	1
8301654	0	5.94	37.49	1
8301888	3	3.07	8.62	1
8301888	5	8.19	9.28	1
8301888	5	1.56	21.15	1
8301888	4	50.81	9.36	1
8301888	3	20.54	9.39	1
8301888	3	11.44	15.75	1



Fig. 2. Convergence sensor weekly object detection experiment

C. Analysis

- Acquisition of city driving test data
- Entire segment changed

V. CONCLUSION

Using the developed TOF and Vision fusion sensor, objects were tested for time, segment, size, and distance during the night time period, and object recognition algorithm learning and test data collection were also performed using the same method.

REFERENCES

- [1] S-Y Shin, Seung-Pyo Cho, Kwang-Seong Shin, Hyun-Chang Lee, "Development of TOF Sensor for Vehicle Safety Support," Proceedings of KIICE Conference 2022, Busan, VOL. 26, No. 1, pp. 670-671, 2022.5

Cartoon Single-Image Super-Resolution

Guangxing Wang¹, Kwang-Seong Shin², Seong Yoon Shin³

¹School of Computer and Big Data Science, Jiujiang University, Jiujiang 332005, China

²Department of Computer Engineering, Sunchon National University, Suncheon 57922, Korea

³School of Computer Science and Engineering, Kunsan National University, Gunsan 54150, Korea

Email: wanggxrs@163.com, waver@scnu.ac.kr, s3397220@kunsan.ac.kr

Abstract

In this paper, we propose an image super-resolution technique based on a generative adversarial network called the Text Localization Generative Adversarial Nets (TLGAN) model. This technique is improved on SRGAN and significantly reduces the computational burden of the model by removing the batch normalization layer. In the TLGAN model, we pre-train the model based on a large-scale dataset, ImageNet, using the transfer learning technique, and then apply the pre-trained model to the Animes dataset, a cartoon image dataset, to achieve image super-resolution.

Index Terms: super-resolution technique, TLGAN, SRGAN, ImageNet

I. INTRODUCTION

Image super-resolution technology refers to a computer software technology that restores high-resolution images from low-resolution images or image sequences [1]. Image super-resolution technology is generally divided into super-resolution restoration and super-resolution reconstruction. Single-image super-resolution is a technology that restores high-resolution images from low-resolution images. Image super-resolution research is largely divided into interpolation-based, reconstruction-based, and learning-based methods. Of course, image super-resolution can also be implemented through hardware technology. The disadvantage is that the cost is too high and the accuracy requirements of the equipment are strict.

II. TLGAN MODEL

A. Activate function

In the original SRGAN model, the generator is mainly composed of convolutional layer, batch normalization (BN) layer, and activation function. The purpose of the BN operation is to regularize the size of the model input unit, avoid gradient disappearance or gradient explosion, and accelerate the model training. However, although the BN layer can normalize the image feature map and solve the problem of inconsistency in the input image feature map's size, compared to solving the classification problem, the BN operation appears to be inadequate in dealing with the image super-resolution problem.

Therefore, in our model, we removed the BN operation and replaced it with a deep residual module.

$ReLU$ is used as the activation function in the GAN network; the difference is that we used the $PReLU$ function as our model activation function. The advantage of this is that in the model training process, the activation function can be adjusted according to the dynamic parameters in the training process so that the image can retain the most extensive feature map, thereby reducing the feature loss during the training process. The mathematical expression of $PReLU$ is shown in formula (1).

$$PReLU(x_i) = \begin{cases} x_i & \text{if } x_i > 0 \\ a_i x_i & \text{if } x_i \leq 0 \end{cases} \quad (1)$$

The above is a detailed introduction of the proposed TLGAN model structure from a mathematical perspective, including the loss function and activation function settings. The TLGAN model structure diagram is shown in Figure 1.

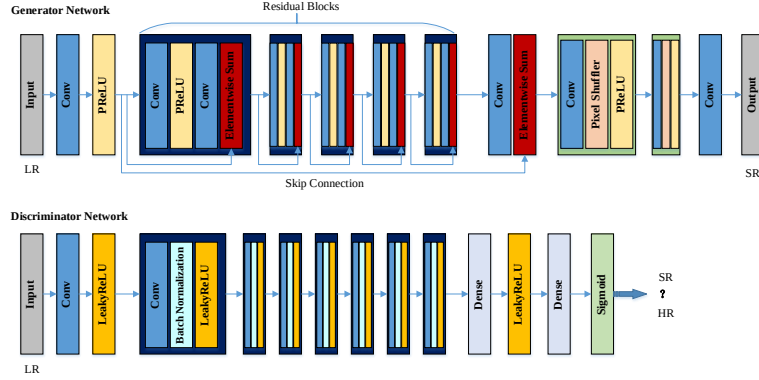


Figure 1. Proposed super-solution model architecture.

B. Image Quality Metric

For the quantitative evaluation of experimental results, we use two commonly used evaluation indicators, peak signal-to-noise ratio (PSNR), and structural similarity index (SSIM). PSNR is an objective indicator for evaluating images, and the definition is as follows.

Given a clean image I and a noise image K with a size of $m \times n$, the definitions of Mean Square Error (MSE) and PSNR are shown in formulas (2) and (3), respectively.

$$MSE = \frac{1}{mn} \sum_{i=0}^{m-1} \sum_{j=0}^{n-1} [I(i, j) - K(i, j)]^2 \quad (2)$$

$$PSNR = 10 \cdot \log_{10} \left(\frac{MAX_I^2}{MSE} \right) \quad (3)$$

In where, MAX_I^2 represents the maximum possible pixel value of the image. If each pixel is represented by 8-bit binary, the maximum value is 255. Generally, if the pixel value of an image is represented by B-bit binary, then $MAX_I^2 = 2^B - 1$.

Of course, this method is to calculate the PSNR of the grayscale image. For color images, we need to calculate the PSNR of the three RGB channels and then average them.

Another evaluation metric is SSIM, which is a measure of the similarity of two images. Among the two images used by SSIM, one is an uncompressed and undistorted image, and the other is a distorted image. The SSIM value is obtained by formula (4).

$$SSIM(x, y) = \frac{(2\mu_x\mu_y + c_1)(2\sigma_{xy} + c_2)}{(\mu_x^2 + \mu_y^2 + c_1)(\sigma_x^2 + \sigma_y^2 + c_2)} \quad (4)$$

Among them, μ_x is the average of x , μ_y is the average of y , σ_x^2 is the variance of x , σ_y^2 is the variance of y , and σ_{xy} is the covariance of x and y . $c_1 = (k_1L)^2$, $c_2 = (k_2L)^2$ are constants used to maintain stability. L is the dynamic range of the pixel value, usually $k_1 = 0.01$, $k_2 = 0.03$. The range of SSIM is 0 to 1. When the two images are the same, the value of SSIM is equal to 1.

III. EXPERIMENT

In the experiment, the size of the low-resolution image input by the proposed model is 64×64, the batch size is set to 8, and the size of the high-resolution image is 256×256, all of which are color images. The model is performed by up-sampling, and the scale scaling factor of the bicubic interpolation method is $\times 4$. The initial learning rate is set to $1e-4$, the optimizer is Adam, and the number of training rounds is 500. The generator uses 16 residual blocks, the convolution kernel size is 3, the step size is 1, and the activation function is *ReLU*. The activation function used by the discriminator is *LeakyReLU*, the momentum is 0.8, and the Sigmoid function is used for classification. Before the image is sent to the model for training, data expansion techniques such as cropping and rotation are used to ensure the sufficient amount of model training data and avoid the phenomenon of non-convergence or over-fitting due to too little data. The data normalization method is used to prevent the occurrence of gradient explosion and gradient disappearance problems. We use PSNR and SSIM to evaluate the prediction results of the model. Finally, the experimental results are saved in the training process and prediction of the model, and the evaluation scores of super-resolution images and the effective display of super-resolution images are compared. The loss rate during model training is shown in Figure 3. Figure 4 is the PSNR evaluation curve of the proposed model during the training process. Figure 5 is the SSIM evaluation curve of the proposed model during the training process. Figure 6 is a comparison of the super-resolution results predicted by different methods in three different data sets. Table 1 is a comparison of the super-resolution prediction scores of the proposed model.

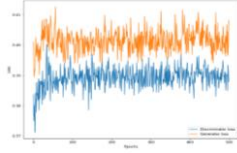


Figure 3. The loss rate curve of the TLGAN model training process.

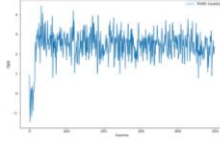


Figure 4. PSNR curve of the TLGAN model training process.

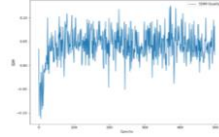


Figure 5. SSIM curve of the TLGAN model training process



Figure 7. Sample evaluation results of the proposed model in different data sets.

Table 1. Test scores of the proposed TLGAN model in different datasets

	Bicubic	SRGAN	TLGAN
Dataset	PSNR(dB)/SSIM	PSNR(dB)/SSIM	PSNR(dB)/SSIM
Anime-characters	21.4474/0.7800	19.8097/0.7289	19.9128/0.7493
Simpsons-characters	19.6024/0.6769	18.8215/0.6155	17.0138/0.5933
AnimeFace	24.6243/0.8471	23.2505/0.8121	19.9527/0.7720

IV. CONCLUSIONS

In this paper, we propose a generative adversarial network (GAN)-based image super-resolution technique called TLGAN model. This technique is improved on SRGAN, and the computational burden of the model is greatly reduced by removing the batch normalization layer. In the TLGAN model, the model is pre-trained on ImageNet, a large-scale dataset, using the transfer learning technique, and then the pre-trained model is applied to Animes, a cartoon image dataset, to achieve image super-resolution.

REFERENCES

- [1] J. Jiang, R. Hu, Z. Wang, and Z. Han, "Noise Robust Face Hallucination via Locality-Constrained Representation," *IEEE Transactions on Multimedia*, vol. 16, no. 5, pp. 1268-1281, Aug. 2014, doi: 10.1109/TMM.2014.2311320.

The Role of Artificial Intelligence in Multimedia Content Creation

Seong-Yoon Shin^{1*}, Kwang-Seong Shin²

¹Department of Computer Science and Engineering, Kunsan University, Kunsan-si, 54150, Republic of Korea

²Department of Computer Engineering, Sunchon National University, Suncheon 57922, Korea

Email: s3397220@kunsan.ac.kr, waver@scnu.ac.kr,

Abstract

This paper delves into the significant role of Artificial Intelligence (AI) in multimedia content creation, examining its influence on trends, challenges, and future directions. AI technologies, such as machine learning, natural language processing, and computer vision, are reshaping the landscape of graphic design, video production, music, sound engineering, and interactive media.

Index Terms: Artificial Intelligence (AI), trends, challenges, future directions

I. INTRODUCTION

Multimedia content creation encompasses the process of designing, producing, and distributing content that integrates multiple forms of media, including text, audio, video, graphics, and interactive elements. This field has evolved significantly with technological advancements, moving from traditional print and broadcast mediums to digital and interactive platforms [1]. Modern multimedia content creation leverages a variety of tools and platforms to engage audiences across different channels effectively. The aim is to deliver a cohesive and immersive experience that can inform, educate, or entertain. With the rise of digital technologies, creators now have the ability to craft more complex and interactive narratives, making multimedia content an integral part of our daily digital consumption. This evolution reflects a shift towards more dynamic and user-centered forms of communication, highlighting the importance of creativity, technical skill, and strategic planning in content creation.

II. TRENDS IN AI-DRIVEN MULTIMEDIA CONTENT CREATION

A. AI in graphic design and video production

Artificial Intelligence (AI) is revolutionizing the fields of graphic design and video production, offering novel tools and techniques that enhance creativity and efficiency. In graphic design, AI algorithms can automate routine tasks such as color correction, resizing

images for different platforms, and even generating entire design layouts based on certain parameters. This not only speeds up the design process but also allows designers to focus on more creative aspects of their work. AI-driven tools are capable of analyzing design trends and suggesting adjustments that can improve the aesthetic appeal of a project.

B. AI in music and sound engineering

In the realm of music and sound engineering, AI is transforming how music is composed, produced, and mixed. AI systems can now compose music in various styles and genres, from classical to pop, by analyzing vast databases of existing music. This capability is not intended to replace human composers or musicians but to serve as a tool for inspiration and collaboration [7]. AI can generate musical pieces that can be further refined by artists, or it can be used to create background music for games, videos, and other multimedia content.

C. AI in interactive media and gaming

AI's impact on interactive media and gaming is profound, enhancing both the development of games and the gaming experience itself. In game development, AI algorithms are used to create realistic environments, generate dynamic narratives, and develop non-player characters (NPCs) that can adapt to player actions in sophisticated ways. This not only makes games more immersive and engaging but also allows for unique storytelling techniques that can respond to the player's decisions .

In terms of gameplay, AI is used to adjust the difficulty level in real-time, ensuring that games remain challenging yet accessible to players of different skill levels. AI-driven analytics are also employed to study player behavior and preferences, which can inform future game design and content updates.

Overall, AI's integration into graphic design, video production, music, sound engineering, interactive media, and gaming is not just about automation or efficiency. It's about expanding the creative possibilities, making high-quality multimedia content more accessible, and providing personalized experiences that were once beyond reach.

Following the discussion of AI-driven trends in multimedia content creation, Table 1 summarizes the key AI technologies and their applications across different multimedia domains.

Table 1. Key AI Technologies and Their Applications in Multimedia Domains

AI Technology	Application Domain	Specific Use Cases
Machine Learning	Video Production	Automated editing, scene recognition
Natural Language Processing	Interactive Media	Dialogue generation, content personalization
Computer Vision	Graphic Design	Image enhancement, automatic tagging
Deep Learning	Music and Sound Engineering	Music composition, sound effect generation
Generative Adversarial Networks	All Domains	Content generation, style transfer

III. CHALLENGES

A. Technical limitations and challenges

Despite the significant advancements in AI and its application in multimedia content creation, several technical limitations and challenges persist. One of the primary challenges is the requirement for vast amounts of data to train AI models effectively. This dependency not only raises concerns about data privacy and security but also about the bias that might be inherent in the training datasets, potentially leading to biased outputs. Moreover, the interpretability of AI decisions remains a challenge. The "black box" nature of some AI systems makes it difficult for creators to understand how decisions are made, complicating the troubleshooting and refinement processes.

Another technical challenge is the integration of AI technologies with existing content creation tools and workflows. While AI can automate certain tasks, ensuring seamless interaction between AI systems and human creators requires sophisticated interface designs and workflow integrations that are not always straightforward. Additionally, the computational resources required for training and running sophisticated AI models can be substantial, posing a barrier for individuals and smaller organizations.

Furthermore, AI-generated content, particularly in areas requiring high levels of creativity and nuance, like music composition or scriptwriting, sometimes lacks the depth and emotional resonance that human-created content can convey. This highlights a gap in AI's ability to fully replicate the complexities of human creativity and understanding.

After discussing the challenges and considerations in AI-driven multimedia creation, Table 2 presents a comparative analysis of technical limitations and ethical considerations across different AI technologies.

Table 2. Comparative Analysis of Technical Limitations and Ethical Considerations in AI Technologies

AI Technology	Technical Limitations	Ethical Considerations
Machine Learning	Dependency on large datasets, risk of overfitting	Data privacy, potential for bias
Natural Language Processing	Complexity in understanding human language nuances	Misrepresentation, content authenticity
Computer Vision	Challenges in accurate image recognition	Privacy issues in facial recognition
Deep Learning	High computational resource requirements	Transparency, "black box" problem
Generative Adversarial Networks	Potential for generating deceptive content	Misuse in creating deepfakes, copyright issues

B. Intellectual property issues

The emergence of AI in multimedia content creation has sparked complex discussions around intellectual property (IP) rights. One of the main issues is determining the ownership of AI-generated content. The question of whether AI-generated works can be copyrighted, and if so, who holds the copyright—the creator of the AI, the user who provided the input, or the AI itself—remains unresolved. This uncertainty complicates the protection and commercialization of AI-generated content.

Furthermore, the use of copyrighted materials to train AI systems poses another IP challenge. AI models often require large datasets, which may include copyrighted content, to learn and generate new creations. This raises questions about copyright infringement and the fair use doctrine, particularly when AI-generated content closely resembles or is derived from copyrighted works.

IV. CASE STUDIES

Analysis of successful AI-driven multimedia projects

One of the most iconic examples of AI-driven multimedia projects is Google's DeepDream. Initially created to help scientists and engineers understand what a deep neural network is seeing when it looks at an image, DeepDream transforms ordinary images into surreal, dream-like artworks by enhancing and overinterpreting the patterns the network sees. This project not only showcased the potential of AI in creating complex, artistic images but also opened up new avenues for artistic expression, blending the boundaries between technology and art.

Another notable project is the AI Dungeon, an AI-powered text adventure game that uses natural language processing to generate dynamic storylines based on the player's inputs. This game represents an innovative use of AI in interactive media, providing an infinitely replayable experience where no two playthroughs are the same. AI Dungeon demonstrates the potential of AI to create complex, narrative-driven experiences in gaming and interactive media.

In the realm of music, AIVA (Artificial Intelligence Virtual Artist) stands out as a pioneering AI composer that has been recognized by the music industry. AIVA analyzes the work of classical composers to create original pieces of music, showcasing the ability of AI to understand and replicate complex musical compositions. AIVA's success not only highlights the creative potential of AI in music but also raises intriguing questions about creativity and authorship.

V. Conclusion

This paper has explored the multifaceted role of Artificial Intelligence (AI) in multimedia content creation, highlighting current trends, challenges, and future directions. AI's integration into graphic design, video production, music and sound engineering, and interactive media underscores its capacity to enhance creativity, streamline production processes, and personalize user experiences.

REFERENCES

- [1] Anderson, R.E.; Hughes, J.L. "Evolving Multimedia Landscapes: Trends and Innovations." Springer, 2020.

Design and Implementation of a Simulator Platform for the Verification of Medical Device Standardization

Sung-Min Jeon¹, Jung-Hyun Woo¹, Muneeb Muhammad¹, Byung-Seok Seo¹,
Kwang-Man Ko¹

¹Department of Computer Engineering, Sangji University, Wonju, 26399, Republic of Korea
[Email : 2025015104@sj.sangji.ac.kr¹, 2025015001@sj.sangji.ac.kr¹,
2019015001@sj.sangji.ac.kr¹, seobs@sangji.ac.kr¹, kkman@sangji.ac.kr¹]

Abstract

As medical devices increasingly incorporate software, the need for efficient verification frameworks aligned with international standards has grown. However, interpreting and applying diverse regulatory requirements remains a major challenge, often leading to delays, documentation errors, and inconsistent evaluations. To address this, we propose a web-based simulator platform that automates the verification process by mapping product-specific information to standards such as ISO 13485, IEC 62304, and FDA guidelines. The platform features a FastAPI engine, SBOM analysis, and security integration modules to support structured checklist generation and reporting. It also extends to cybersecurity verification for SaMD products. By streamlining documentation and improving regulatory consistency, the platform aims to enhance compliance efficiency and promote global adoption of digital standardization tools.

Index Terms: Medical Device, Standardization, Compliance Verification, Simulator Platform, SaMD

I. INTRODUCTION

The platform aims to support medical device developers and quality managers from early development to regulatory submission by automating standard-based verification. It identifies applicable international standards (e.g., FDA, ISO/IEC, IMDRF) using product-specific inputs and generates structured checklists. Users interact with the checklist through guided responses, while compliance status and historical results are tracked across versions. Evaluation reports can be exported in multiple formats (PDF, Word, Excel) for use in regulatory filings. To ensure modularity and scalability, the platform is organized into four core layers: user workflow, analysis engine, feature modules, and execution layer.

II. Related Work and Background

The transition of the medical device industry toward software-driven systems has underscored the critical importance of verifying quality, safety, and cybersecurity compliance in accordance with international standards. Regulatory authorities such as the

U.S. Food and Drug Administration (FDA), the International Medical Device Regulators Forum (IMDRF), and national agencies like South Korea's Ministry of Food and Drug Safety (MFDS) increasingly emphasize standard-based documentation and validation throughout the product lifecycle [1][2][3]. Key international standards include ISO 13485 for quality management systems, IEC 62304 for software lifecycle processes, ISO/IEC 81001-5-1 for cybersecurity in health software, and the FDA's 2023 Premarket Submission Guidance, which sets forth specific cybersecurity considerations for regulatory approval [1]. The IMDRF has also published a framework that defines cybersecurity best practices across the entire product lifecycle [2]. Complementing these global efforts, the MFDS in South Korea has issued a cybersecurity checklist tailored to medical software, aligning with ISO/IEC 81001-5-1 and supporting national-level regulatory harmonization [3]. Several academic studies have attempted to support the application of these standards. Mahler et al. proposed a risk-based compliance methodology for medical software, drawing from IEC 62304 and focusing on structured security assessments [4]. More recently, Scherb et al. introduced CyMed, a framework for automating cybersecurity evaluations of connected medical devices, which reflects the latest FDA guidance [5]. Despite these efforts, most existing solutions remain limited to specific standards or evaluation stages. In response, this study proposes an integrated platform that addresses the fragmentation by supporting automatic checklist generation, multi-standard mapping, and end-to-end documentation to improve the consistency, speed, and efficiency of compliance verification.

III. System Design and Implementation

This section outlines the design, architecture, and workflow of the proposed simulator platform for verifying medical device standardization. The platform aims to enhance regulatory documentation efficiency by automating the mapping of standards, checklist generation, evaluation, and reporting.

A. Design Objectives and System Requirements

The platform is designed to assist developers and quality managers throughout the development and regulatory submission phases. It automatically extracts applicable international standards (e.g., FDA, ISO/IEC, IMDRF) based on product-specific inputs and generates corresponding checklists. Users can respond to each item with contextual guidance, track results across versions, and visualize compliance status. Reports are generated in PDF, Word, or Excel formats to support regulatory documentation. The platform is structured into four layers: user workflow, analysis engine, feature modules, and execution layer.

B. System Architecture and Structural Layers

The system converts product inputs into checklist items and manages the full evaluation and reporting process. Its four-layer architecture includes:

- **User Scenario Layer:** Handles data input and user interaction.
- **Analysis Engine Layer:** Uses FastAPI to map standards and evaluate inputs.

- **Feature Modules:** Includes SBOM analysis, reporting tools, and external standard integration.
- **Execution Layer:** Manages technical processes like API calls and data storage.

As users submit product information, the system generates checklists, captures responses, and stores versioned results for comparison. External APIs and SBOM analyzers ensure the latest security requirements are reflected in real time.

Table 1. Comparison of Existing Studies on Standardization Verification for Medical Devices

Study Title	Applied Standard	Approach	Automation	Summary of Limitations
Mahler et al. (2020, arXiv)	IEC 62304	Risk-based quantitative	Partially supported	Security-focused, lacks standardized documentation
Scherb et al. (2023, arXiv)	FDA Guidelines	Vulnerability simulation-focused	Test automation	Does not support design documentation or integrated standards
MFDS Cybersecurity Guide (2024)	ISO/IEC 81001-5-1, etc.	Checklist-based guidance	Manual evaluation	Lacks development tools, insufficient documentation automation
IMDRF N60 Commentary paper (2021, KIMDS)	IMDRF N60	Framework-oriented explanation	None	No implementation in real systems
This Study (2025)	ISO 13485, 62304, FDA, etc.	Automated checklist + report generation	Fully automated	Proposes an integrated structure that overcomes existing limitations

C. Technology Stack and Implementation Environment

Built with lightweight open-source tools, the platform uses FastAPI and JSON for checklist logic, Syft and Grype for SBOM and vulnerability analysis, and D3.js/Chart.js for visualization. Reports are produced using python-docx and WeasyPrint. Real-time data is sourced from GitHub Advisory Database and NVD API, and cloud deployment is supported on AWS and GCP.

D. User Workflow and Use Case Scenarios

Users begin by inputting product details or uploading documents. The system identifies relevant standards and generates a structured checklist, where users mark each item as “Applicable,” “Not Applicable,” or “Partially Applicable.” For non-compliant items, the system allows for justification entries. For SaMD products, users can upload an SBOM for automated vulnerability scanning and compliance evaluation. The system outputs a report including compliance summaries,

exception details, and security assessments. All results are version-controlled, supporting comparison across iterations and improving long-term verification productivity.

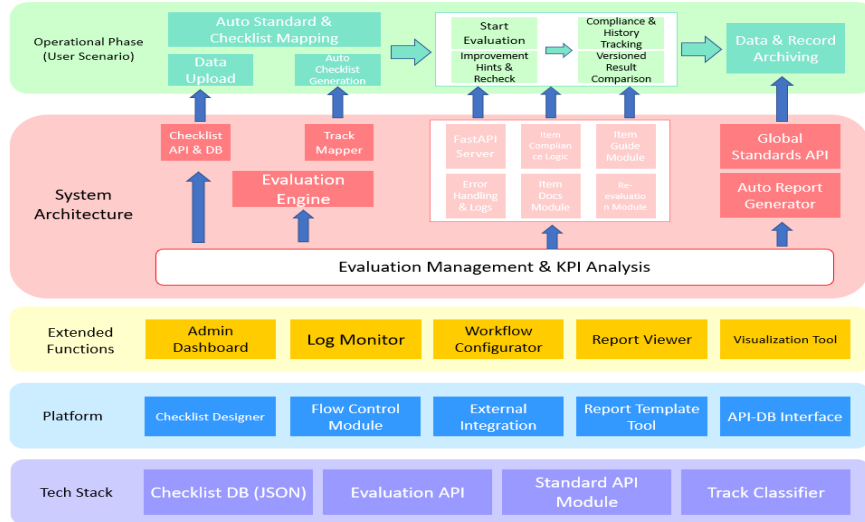


Fig 1. System Architecture of the Medical Devices Standardization Verification Platform

IV. Conclusion and Expected Impact

This study presented the design and implementation of a simulator platform to support systematic verification of medical devices based on international standards. By automatically mapping standards to product-specific information and generating checklist-based evaluations with reporting, the platform reduces documentation workload and improves regulatory efficiency. Its scalable architecture enables repeatable verification processes and includes features for assessing SaMD cybersecurity compliance. Future work will focus on developing a functional prototype and validating its effectiveness through application to real-world product categories.

References

- [1] **U.S. Food and Drug Administration (FDA)**, *Cybersecurity in Medical Devices: Quality System Considerations and Content of Premarket Submissions*, 2023.
- [2] **International Medical Device Regulators Forum (IMDRF)**, *Principles and Practices for Medical Device Cybersecurity*, MDRF/CYBER WG/N60 FINAL:2020, 2020.
- [3] **Ministry of Food and Drug Safety (MFDS)**, *Cybersecurity Checklist for Medical Software*, 2024. (in Korean)
- [4] **Mahler, T., Elovici, Y., & Shahar, Y.**, *A New Methodology for Information Security Risk Assessment for Medical Devices*, arXiv:2002.06938, 2020.
- [5] **Scherb, C., Hadayah, A., Heitz, L. B., et al.**, *CyMed: A Framework for Testing Cybersecurity of Connected Medical Devices*, arXiv:2310.03583, 2023

A Multi-Layered Security Assessment Framework for Kubernetes: Enhancing Cloud Security Assurance Program Criteria in Korea

Jun Heo^{1*}

¹Policy Research Office, Korea Internet & Security Agency, Naju 58324, Republic of Korea
[Email : her_june@naver.com]

Abstract

Kubernetes has become a core component of modern cloud infrastructure, leading to complex security requirements. In Korea, the Cloud Security Assurance Program (CSAP)'s Kubernetes criteria, based on Common Configuration Enumeration (CCE), are limited to static configuration checks, failing to address modern demands like runtime protection, policy control, and supply chain security. This study conducts a comparative analysis of Kubernetes-related CCE items against major global security frameworks, revealing significant limitations in scope and technical depth for addressing real-world threats. This preliminary study articulates these structural limitations within CSAP's Kubernetes criteria. The insights highlight the critical need for a comprehensive enhancement strategy for current CCE items, suggesting a path forward for improving CSAP in alignment with modern Kubernetes security.

Index Terms: Kubernetes security, Cloud Security Assurance Program (CSAP), Common Configuration Enumeration (CCE), CIS Benchmark, MITRE ATT&CK, Runtime Security, Policy Enforcement

I. INTRODUCTION

With the proliferation of cloud-native technologies, Kubernetes has become a core infrastructure globally, leading to increasingly sophisticated security demands. Its complex threat exposure structure, distinct from traditional VM-based models, necessitates effective responses from the global security community through various standards like CIS Kubernetes Benchmark, MITRE ATT&CK, NIST SP 800-190, and NSA-CISA Hardening Guide.

Conversely, Korea's Cloud Security Assurance Program (CSAP), operated by KISA, employs CCE-based items for Kubernetes security assessment. However, these are largely confined to static configuration checks, inadequately addressing modern security needs such as runtime protection, policy-based control, and supply chain security. This limitation can increase vulnerabilities and hinder effective threat response in domestic cloud environments.

This paper aims to clearly analyze the structural limitations of CSAP's Kubernetes CCE criteria in South Korea. We conduct a comparative analysis of current CCE items against global Kubernetes security frameworks to identify limitations in scope and technical sophistication. This study, a **preliminary study**, contributes to understanding these limitations and provides foundational data for future domestic certification standards to align with international requirements. It summarizes key analysis results from a prior KCI-published work and outlines plans for a more comprehensive DevSecOps-based Kubernetes security automation framework..

II. RELATED WORK

Various global security frameworks and guidelines aim to enhance Kubernetes security, serving as crucial comparative benchmarks for diagnosing limitations in domestic certification standards and guiding improvements.

The **CIS Kubernetes Benchmark**, established by CIS, provides minimum security configuration standards for core components like API Server, categorizing items into Level 1 and Level 2. Automated checks are possible using tools like kube-bench, aligning with domestic CCE items as an international standard.

MITRE ATT&CK for Kubernetes is a threat modeling framework that systematizes attacker tactics, techniques, and procedures (TTPs) based on real-world intrusion cases, offering a dedicated Matrix for Kubernetes. Linked with runtime detection tools like Falco and Tetragon, it enables real-time threat detection and analysis.

NIST SP 800-190 provides security guidelines for containerized application environments, including policy-based control measures at the orchestrator level for Kubernetes. It emphasizes a risk-based approach to prioritize security controls, covering aspects like image integrity verification and RBAC policy configuration.

The **NSA-CISA Kubernetes Hardening Guidance** offers practical, operations-focused guidelines for strengthening Kubernetes cluster security. Recommendations include running non-root users, applying read-only filesystems, and implementing RBAC segmentation and NetworkPolicy. These guidelines are highly applicable even in cloud-managed services.

Domestically, the **Current CSAP Kubernetes Security Criteria (CCE)**, overseen by KISA, primarily focus on static configuration checks. These criteria are criticized for

inadequately reflecting modern security demands like runtime security, policy-based control, and supply chain protection. Similarly, domestic ISMS-P lacks sufficient Kubernetes-specific items in these areas, indicating a need for international standard incorporation.

III. Structural Analysis and Implications of Domestic CSAP Kubernetes Security Items (CCE)

The Cloud Security Assurance Program (CSAP) by KISA assesses public cloud service security, providing CCE-based Kubernetes inspection items. This chapter analyzes these items' composition and limitations for a complementary model.

The CCE-based assessment, a static configuration-focused first line of defense, limits reflecting multi-layered cloud-native security. Four key structural limitations are identified via comparison with global frameworks. First, a lack of **runtime security elements** (e.g., eBPF analysis, log-based threat detection) limits real-time response. Second, **insufficient policy-based control** means key Kubernetes mechanisms like RBAC, PodSecurityPolicy, OPA, and NetworkPolicy are excluded, hindering automation. Third, a lack of **supply chain security items** (e.g., image signing, SBOM) impedes secure software supply chain establishment. Fourth, limited **practical effectiveness focus** (e.g., lack of risk info, applicability in managed Kubernetes) hinders intuitive judgment.

This analysis confirms significant structural limitations in the current CCE-based system for reflecting cloud-native security and Kubernetes' complex structure. Future efforts require structural supplementation from a multi-layered security perspective, encompassing runtime, policy, and supply chain protection, aligned with global frameworks and practical applicability.

IV. Policy Improvement Directions for Cloud Security Assurance Program (CSAP) Kubernetes Security Criteria

In this chapter, we propose a comprehensive multi-layered security assessment framework to supplement CCE (Common Configuration Enumeration) limitations in CSAP's Kubernetes criteria, responding to modern cloud security needs. This model enhances effectiveness and automation by including runtime security, policy-based control, and software supply chain security, overcoming static configuration-centric assessment limits.

The framework integrates four key security layers, reflecting global security frameworks and threat landscapes, for a comprehensive approach that internalizes security across the entire Kubernetes cluster lifecycle.

The Configuration Layer is the first line of defense, checking key component security status in the initial cluster state. It verifies core settings based on CCE and CIS Benchmark criteria, with automated checks via kube-bench and Trivy, crucial for robust foundations and preventing initial vulnerabilities.

The Runtime Layer responds to dynamic threats during operation, focusing on real-time detection of anomalous attacks and behaviors (e.g., container escape, suspicious network calls). Tools like Falco and Tetragon, based on eBPF, monitor system calls

and network events for in-depth, threat-centric analysis with MITRE ATT&CK, substantially strengthening security against advanced threats.

The Policy Layer ensures security effectiveness and compliance by defining and enforcing resource access and execution conditions. It inspects resource access and policy application via RBAC, PodSecurityPolicy, OPA, and Kyverno, contributing to compliance and stable operations by verifying policy automation and consistency, with Admission Control central to blocking violations.

The Supply Chain Security Layer is crucial for countering supply chain attacks, ensuring trustworthiness and integrity of deployed application components. Assessment includes verifying image signatures (Cosign), generating SBOMs (Trivy, Grype), and analyzing open-source library vulnerabilities (Trivy, Snyk). These automated processes within CI/CD pipelines proactively block complex attack scenarios, extending existing CSAP inspection items to reflect modern security demands.

The proposed multi-layered framework realizes full-lifecycle security for Kubernetes-based cloud environments, providing 4-layer coverage. It complements CCE limitations, acting as a practical security assessment system covering overall Kubernetes operations and supply chain threats. It offers comprehensive security, enhances assessment effectiveness and automation via open-source tools, aligns with international security frameworks, ensures scalability to diverse Kubernetes environments, and strengthens linkages between domestic certification systems like ISMS-P. This comprehensive approach enhances domestic cloud security efficiency and realism, serving as a strategic alternative for rapidly changing security requirements

V. CONCLUSION

This paper is a preliminary study analyzing CSAP Kubernetes security limitations and proposing a multi-layered framework. Building on this, future research will design and implement a more comprehensive DevSecOps-based Kubernetes security automation framework. Key directions include:

Integrating proposed multi-layered security (configuration, runtime, policy, supply chain) with the DevSecOps development-operation lifecycle (Coding/Build, Build/Test, Deploy, Operate/Monitor) to create a unified, embedded security framework. Specific integration methods for open-source automation tools (e.g., kube-bench, Falco, Kyverno, Cosign, Trivy, Kubescape, Tetragon) within each layer will be explored, including automation script development and feasibility verification. Strategies for expanding to diverse environments (on-premise, managed cloud, hybrid/multi-cluster) will be deepened, with practical security evaluations conducted via PoC or pilot applications. Furthermore, the "never trust, always verify" principle of Zero Trust will be integrated into each security layer to enhance Kubernetes cluster security robustness and responsiveness. Ultimately, this study serves as a theoretical and practical starting point for enhancing the alignment, effectiveness, and international competitiveness of domestic cloud security certification systems.

Deep Learning-Based Detection of Crop Pests and Diseases

Min-hye Lee¹, Sun-Kyoung Kang^{2*}

¹Office of Liberal arts Education, Wonkwang University, Iksan, 54538, Korea

²Department of Computer Software Engineering, Wonkwang University, Iksan, 54538, Korea

[Email : lmh3322@wku.ac.kr, doctor10@wku.ac.kr]

Abstract

This study classified the major pests and diseases of rice crops into four categories—panicle disease, leaf disease, small clustered pests, and individual pests—based on RGB imagery, and analyzed suitable deep learning detection algorithms according to the visual characteristics of each type. Considering object size and distribution density, we compared the architectural features and applicability of models such as YOLOv8, Faster R-CNN, LLA-RCNN, and GoogLeNet, and proposed optimized model configurations tailored to each pest category.

Index Terms: Crop, Deep learning, Object detection, Pest and disease, Precision agriculture

I. INTRODUCTION

Recently, rural areas have been facing increasing challenges in responding promptly to pests and diseases in rice cultivation due to population decline and aging. Moreover, the complexity of crop management has intensified as climate change continues to diversify pest and disease types. In response, AI-based automatic pest detection systems are gaining attention as promising alternatives to enhance both the efficiency and responsiveness of agricultural practices [1].

In particular, image processing technology and deep learning-based object recognition techniques are being effectively employed to automatically assess the pest and disease status of crops. These methods are gradually replacing traditional visual inspection by extracting features of pests and diseases from RGB images and detecting their locations and types. Such agricultural AI technology enables precise analysis of lesion areas, pest appearance, and the extent of crop damage, serving as a key enabler in the advancement of precision agriculture [2].

However, automatic detection in rice pest and disease imagery still poses several technical challenges. Lesions and pests are often small and scattered, making them difficult to detect accurately. In addition, false positives or missed detections frequently occur due to visual similarity with the background and varying lighting conditions in field environments. To address these issues, it is essential to select detection models that are well-suited to the visual characteristics of each pest and disease type [3].

* Corresponding author

II. RESEARCH METHODS

A. Pest and Disease Categories in Rice Crops

Lesions and pests occurring in rice crops can be categorized into four types: panicle disease, leaf disease, small clustered pests, and individual pests.

Panicle diseases typically exhibit colors and shapes similar to the background, and the size of lesions varies considerably. Leaf diseases and small pests are usually small in size, irregularly distributed or densely clustered, and often have unclear boundaries between objects. In contrast, large individual pests are relatively easier to detect due to their distinct shapes and clear spatial separation.

Because each type presents different levels of detection difficulty and unique visual characteristics, it is crucial to select a detection model that reflects these factors. Furthermore, for practical use in field environments, fast and efficient detection is essential. To meet this requirement, lightweight real-time processing models have been analyzed as standard approaches [4,5].

B. Applicable Deep Learning Model Structures

The appropriate deep learning model architecture varies depending on the visual characteristics and detection difficulty associated with each pest or disease type.

Panicle diseases exhibit colors and shapes similar to the background, and lesion sizes are inconsistent. Therefore, a multi-scale feature extraction architecture capable of capturing objects of varying sizes is required. For this purpose, GoogLeNet-based networks or models utilizing BiFPN (Bidirectional Feature Pyramid Network) are suitable choices [4].

Leaf diseases and small clustered pests are typically small and densely distributed within the image, requiring a model with precise small-object detection capability. In such cases, lightweight architectures that combine the MobileNetV3 backbone with attention mechanisms—such as Faster R-CNN or LLA-RCNN (Localization-Aware R-CNN)—have proven effective in addressing these challenges [5].

In contrast, individual large pests usually have well-defined shapes and distinct boundaries, making them easier to detect. One-stage real-time detection models, such as YOLOv8, are well-suited for such tasks. These models provide both high accuracy and fast inference, particularly when the number of targets is limited and object boundaries are clear. Additionally, the YOLO model family is widely adopted due to its reliable accuracy and real-time performance across various agricultural pest detection scenarios [6].

III. RESULTS

In this study, we compared and analyzed the architectural characteristics and applicability of representative detection models for each type of rice pest and disease based on existing literature. Table 1 outlines the key features of each model and their suitability for different pest categories.

Table 1. Pest and disease categories and applicable models.

Category	Representative Examples	Key Characteristics	Deep Learning Model
Panicle Disease	Panicle blast, False smut	- Small, faint lesions on panicles - Similar color to background - Requires high resolution	GoogLeNet, BiFPN-based networks
Leaf Disease	Leaf blast, Bacterial leaf blight	- Spread across leaf surface - Lesions are small and irregular - Needs small-object detection	YOLO model (Pyramid/Attention-enhanced)
Small Clustered Pests	Mites, spider mites	- Very small and clustered - Similar in color to plant surface - Requires fine detection	Faster RCNN, LLA-RCNN
Large Individual Pests	Brown planthopper, Stink bug	- Large and well-defined shape - Requires fast detection - Real-time processing important	YOLO model (YOLOv8s)

IV. DISCUSSION AND CONCLUSIONS

This study classified four major categories of rice pests and diseases—panicle disease, leaf disease, small clustered pests, and individual pests—based on RGB images, and compared the architectural characteristics and applicability of recent deep learning detection models according to the visual features and detection challenges of each type.

The YOLOv8 series demonstrated high overall accuracy and real-time inference capability, making it suitable for various pest detection tasks. LLA-RCNN, designed with a lightweight backbone and attention module, was found to be effective for detecting small, densely clustered pests. A model combining GoogLeNet with multi-scale feature extraction was identified as appropriate for panicle diseases, which require detection of lesions with variable size and low contrast against the background.

While this study focused on analyzing model structures based on prior research, future work will involve training models on actual datasets, validating real-time performance under field conditions, and applying segmentation-based lesion area analysis. These directions are expected to improve the accuracy and practical applicability of rice pest detection systems in the context of smart agriculture.

ACKNOWLEDGMENTS

This work was supported by the Institute of Information & Communications Technology Planning & Evaluation(IITP)-Innovative Human Resource Development for Local Intellectualization program grant funded by the Korea government(MSIT)(IITP-2025-RS-2024-00439292)

REFERENCES

- [1] P. Xiong, C. Zhang, L. He, X. Zhan, and Y. Han, "Deep learning-based rice pest detection research," PLOS ONE, vol. 19, no. 4, e0313387, Apr. 2024. DOI : 10.1371/journal.pone.0313387.
- [2] L. Wu, Z. Cheng, D. Ye, Q. Jin, and Y. Zhang, "UAV ricPlant Methods, vol. 21, no. 1, pp. 10, 2025.
- [3] J. Liu and X. Wang, "Plant diseases and pests detection based on deep learning: a review," Plant Methods, vol. 17, no. 1, pp. 22, 2021.
- [4] L. Wu, Z. Cheng, D. Ye, Q. Jin, and Y. Zhang, "UAV rice panicle blast detection based on enhanced feature representation and optimized attention mechanism," Plant Methods, vol. 21, no. 1, pp. 10, 2025.
- [5] L. Yang, F. Guo, H. Zhang, Y. Zhang, and M. Wang, "Research on lightweight rice false smut disease identification method based on improved YOLOv8n model," Agronomy, vol. 14, no. 9, pp. 1934, 2024.
- [6] S. Liu, Y. Zhang, W. Chen, and X. Zhang, "Rice pest detection based on an improved YOLOv5 model," Applied Sciences, vol. 12, no. 19, pp. 10167, 2022.



THE KOREA INSTITUTE OF INFORMATION
AND COMMUNICATION ENGINEERING

<http://kiice.org>

